



Content: **Tentbook (according to EN 13782)**

Object: **10m stretchtent and alternatives**

Owner object: **Eurostretch B.V.**

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## A. Introduction

Eurostretch B.V. has commissioned Tentech B.V. to perform a static analysis of a stretch tent. Eurostretch B.V. is a Dutch company that specializes in rental and selling of tent structures made of a stretchable membrane, so called Stretch Tents or Bedouin tents. The stretchable membrane enables a freedom of form as there is not a pre-described shape necessary. Depending on the location, the number of poles, length of the poles, and the placement of the poles, number and type of ties can be varied. This results in a custom canopy at each location.

The freedom of form is enabled by a stretchable fabric, as the desired form is stretched from a rectangular fabric. The drawback of this flexibility in shape is the difficulty of investigating all the different possibilities and generalizing them for a static analysis.

This report presents the static analysis of a 10m stretch tent series which consists of different sizes – **10x10, 10x15, 10x20, 10x12 and 9x12m**. The report shows the static analysis of the decisive sizes – 10x10m, 10x15m and 10x20m. The static analysis of tent 10x15m is valid for smaller sizes of 10x12m and 9x12m due to the application of the **scaling principle**. All models and variants are investigated in a “floating” set-up – membrane corners and sides are connected above ground, primarily to poles.

**Scaling principle:** A tent with a similar arrangement of poles and ties but smaller in size or having a more favourable shape is automatically verified when the calculation of the less favourable or bigger tent has been approved, provided that the spacing between the poles and ties of the scaled tent is similar or smaller than the original tent. While constructing smaller tent sizes of different shapes, it is significant to use similar (or more favourable) arrangements of poles and ties based on a ratio of distances (to ensure correct pole and tie spacing).

This document contains the data required for a tent book, according to EN 13782, for the 10m tent series of Eurostretch B.V., including:

- Ownership data;
- Drawings of the different variants of the tent, including dimensions, indications of elements and required anchoring/ballast;
- Permitted live load;
- Maximum wind speeds (according to EN 1991-1-4:2005);
- Structural analysis (according to EN 13782:2015);
- Material certificates (strength properties and fire properties).

Utrecht, S. Banga, 10.08.2026

## B. Contents

|           |                                                                                        |           |
|-----------|----------------------------------------------------------------------------------------|-----------|
| <b>A.</b> | <b>Introduction .....</b>                                                              | <b>3</b>  |
| <b>B.</b> | <b>Contents.....</b>                                                                   | <b>4</b>  |
| <b>C.</b> | <b>Codes and standards.....</b>                                                        | <b>5</b>  |
| <b>D.</b> | <b>Summary .....</b>                                                                   | <b>6</b>  |
| <b>E.</b> | <b>Drawings: main dimensions and anchoring .....</b>                                   | <b>9</b>  |
| <b>F.</b> | <b>Important terms and conditions .....</b>                                            | <b>15</b> |
| <b>G.</b> | <b>Allowed wind speeds .....</b>                                                       | <b>16</b> |
| <b>H.</b> | <b>Static analysis .....</b>                                                           | <b>19</b> |
|           | H.1. Project description .....                                                         | 23        |
|           | H.2. Materials .....                                                                   | 24        |
|           | H.3. Cross sections .....                                                              | 25        |
|           | H.4. Calculation method .....                                                          | 26        |
|           | H.5. Load cases .....                                                                  | 30        |
|           | H.6. Calculation results .....                                                         | 32        |
|           | H.7. Element check .....                                                               | 53        |
|           | H.8. Safety against overturning, sliding and uplifting .....                           | 64        |
| <b>I.</b> | <b>Addition 1 – element check for reduced wind speeds [NL] .....</b>                   | <b>70</b> |
|           | I.1. Summary of results – low wind in Netherlands .....                                | 70        |
|           | I.2. Summary of forces .....                                                           | 72        |
|           | I.3. Element check - low wind in Netherlands .....                                     | 74        |
|           | I.4. Safety against overturning, sliding and uplifting - low wind in Netherlands ..... | 77        |
| <b>J.</b> | <b>Addition 2 – element check for combined wind and snow load [FR] .....</b>           | <b>80</b> |
|           | J.1. Summary of results – snow+wind in France .....                                    | 80        |
|           | J.2. Cross sections .....                                                              | 82        |
|           | J.3. Summary of forces .....                                                           | 82        |
|           | J.4. Element check – snow + wind in France .....                                       | 85        |
| <b>K.</b> | <b>Material certificates.....</b>                                                      | <b>91</b> |
|           | K.1. Fabric specifications .....                                                       | 91        |
|           | K.2. Fabric fire certificate .....                                                     | 92        |
|           | K.3. Capacity of Eye bolts .....                                                       | 94        |
|           | K.4. Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889.....                        | 94        |
|           | K.5. Carabine, DIN 5299C .....                                                         | 95        |
|           | K.6. Rope ratchet .....                                                                | 95        |



### C. Codes and standards

- EN 1990 Eurocode, Basis of structural Design
- EN 1991 Eurocode 1, Part 1-4: General actions - wind actions.
- EN 1995 Eurocode 5, Design of wooden structures
- EN 13782 Temporary Structures – Tents - Safety
- EN 12195-2 Belts made from synthetic fibres
- ISO 1141 Fibre ropes – Polyester
- CTS French standard - Chapiteaux, Tentes, Structures

## D. Summary

|               |                                                                                           |                                                                                    |
|---------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Manufacturer: | Eurostretch B.V.<br>Sporstraat 25<br>7061 ZA Terborg<br>Nederland<br>info@eurostretch.com |  |
|---------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|

This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

| General information: |                               |                                                               |                                                               |                                                               |
|----------------------|-------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| Main dimensions:     |                               | 10x10m                                                        | 10x15m / 10x12m / 9x12m                                       | 10x20m                                                        |
|                      | Width:                        | 10m                                                           | 10m / 9m                                                      | 10m                                                           |
|                      | Length:                       | 10m                                                           | 15m / 12m                                                     | 20m                                                           |
|                      | Side height of membrane:      | 3m                                                            | 3m                                                            | 3m                                                            |
|                      | Max. height:                  | 4m                                                            | 4m                                                            | 4m                                                            |
|                      | Main poles: *                 | 4.0m<br>Ø100 mm [Spruce,C18]                                  | 4.0m<br>Ø100 mm [Spruce,C18]                                  | 4.0m<br>Ø100 mm [Spruce, C18]                                 |
|                      | Side poles: *                 | 3.0m<br>Ø 70 mm [Spruce, C18]                                 | 3.0m<br>Ø 75 mm [Spruce, C18]                                 | 3.0m<br>Ø 75 mm [Spruce, C18]                                 |
|                      | Corner poles: *               | 2.5m<br>Ø 70 mm [Spruce, C18]                                 | 2.5m<br>Ø 75 mm [Spruce, C18]                                 | 2.5m<br>Ø 70 mm [Spruce, C18]                                 |
|                      | Stormbelts:                   | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      |
|                      | Guy ropes:                    | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            |
|                      | Long side – combined:         | 4 sections                                                    | 4 sections                                                    | 4 sections                                                    |
|                      | Long side - V-ropes:          | 3 sections                                                    | 3 sections                                                    | 3 sections                                                    |
|                      | Short side – combined:        | -----                                                         | 4 sections                                                    | 4 sections                                                    |
|                      | Short side - V-ropes:         | -----                                                         | 3 sections                                                    | 3 sections                                                    |
|                      | Belts:                        | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      |
|                      | Fasteners:                    | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN |
|                      | Side & corner pole base plate | 25x25cm                                                       | 25x25cm                                                       | 25x25cm                                                       |
|                      | Main pole base plate          | 35x35cm                                                       | 35x35cm                                                       | 35x35cm                                                       |

|            |                                                                                                                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User load: | A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.                                                                                                                                                                         |
| Snow load: | A snow load of 0.1 kN/m <sup>2</sup> (4 cm) has been considered in accordance with the French CTS standard.                                                                                                                                                                                                |
| Wind load: | According to EN 13782<br>The following pressure values have been used: h < 5m: 0.50kN/m <sup>2</sup><br>This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated. |

\* The average diameter, as a minimum required at the middle of the pole.

\*\* The required breaking load (BL<sub>tot</sub>) may be achieved by using multiple rope sections (n): BL<sub>rope</sub> = BL<sub>tot</sub> / n, see page 57

### Wind speeds for default European terrain categories (not Country specific)

| Out of service <sup>(1)</sup>             |                             |                             |                             |                             |                             |
|-------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
|                                           | Coast                       | Flattened, open area        | Rural area                  | Village                     | City                        |
| 10 min. average wind speed <sup>(2)</sup> | > 17.94 m/s<br>> 64.57 km/h | > 18.89 m/s<br>> 67.99 km/h | > 21.08 m/s<br>> 75.88 km/h | > 24.99 m/s<br>> 89.97 km/h | > 26.08 m/s<br>> 93.89 km/h |
| Beaufort <sup>(3)</sup>                   | > 7 BFT                     | > 7 BFT                     | > 8 BFT                     | > 9 BFT                     | > 9 BFT                     |
| Peak wind speed <sup>(4)</sup>            | > 113 km/h                  | > 113 km/h                  | > 113 km/h                  | > 113 km/h                  | > 113 km/h                  |

(1) 'Out of service' means: above the given wind speed the structure is no longer guaranteed regarding strength and/or stability.

(2) 10min average wind speed at 10m height measured at the nearest weather stations.

(3) wind data in Beaufort (BFT) are indicative values.

(4) 3 second peak wind speed measured on site at 10m height.

### Safety against sliding, overturning and uplifting:

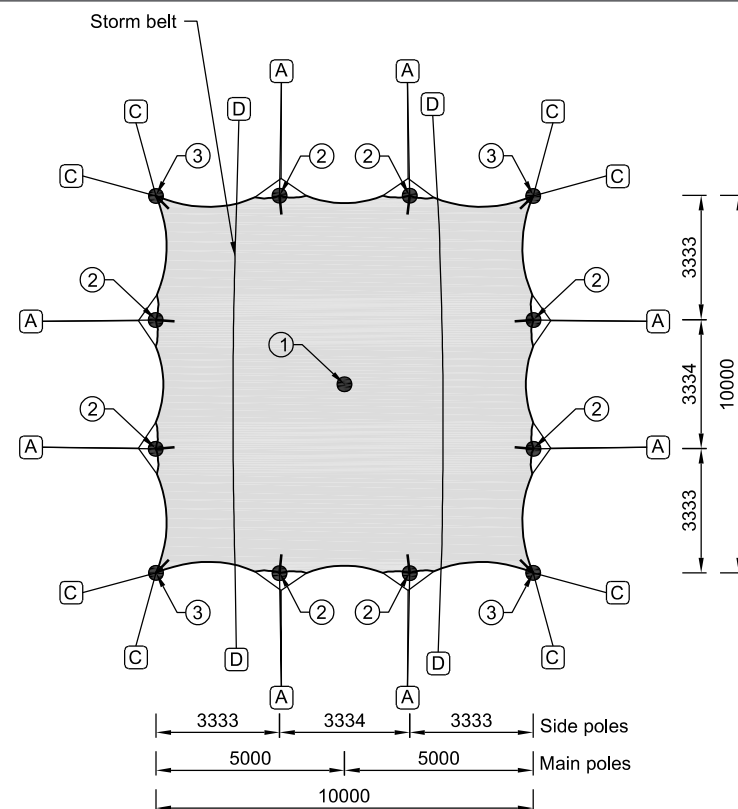
|                   |                                                                                                                                                                      |                   |                                |                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------|-------------------|
| Anchor forces:    | Assumptions: angle of 45 degrees<br>The following design resistance of the anchor forces is required:<br>(See H.8.3, page 67 for Anchor tests according to EN 13782) |                   |                                |                   |
|                   |                                                                                                                                                                      | <b>10x10m</b>     | <b>10x15m / 10x12m / 9x12m</b> | <b>10x20m</b>     |
|                   | Long side guy ropes:                                                                                                                                                 | 6.05 kN (γ = 1.2) | 6.24 kN (γ = 1.2)              | 6.92 kN (γ = 1.2) |
|                   | Short side guy ropes:                                                                                                                                                |                   | 6.28 kN (γ = 1.2)              | 6.70 kN (γ = 1.2) |
|                   | Corner ties:                                                                                                                                                         | 4.73 kN (γ = 1.2) | 5.06 kN (γ = 1.2)              | 5.32 kN (γ = 1.2) |
|                   | Stormbelt:                                                                                                                                                           | 6.74 kN (γ = 1.2) | 7.21 kN (γ = 1.2)              | 7.12 kN (γ = 1.2) |
| Required anchors: | Based on dense, non-cohesive soil (e.g. sandy soils).<br>In the case anchors Ø30mm x 1000mm (effective length) are being used:                                       |                   |                                |                   |
|                   |                                                                                                                                                                      | <b>10x10m</b>     | <b>10x15m / 10x12m / 9x12m</b> | <b>10x20m</b>     |

|                   |                                                                                                                                                                                                                                                     |                      |                            |           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------|-----------|
|                   | Long side guy ropes:                                                                                                                                                                                                                                | 2 anchors            | 2 anchors                  | 2 anchors |
|                   | Short side guy ropes:                                                                                                                                                                                                                               |                      | 2 anchors                  | 2 anchors |
|                   | Corner ties:                                                                                                                                                                                                                                        | 1 anchor             | 1 anchor                   | 1 anchor  |
|                   | Stormbelt:                                                                                                                                                                                                                                          | 2 anchors            | 2 anchors                  | 2 anchors |
| Required ballast: | Ballast calculation is performed for 2 different situations:                                                                                                                                                                                        |                      |                            |           |
|                   | <ul style="list-style-type: none"><li>- Concrete on concrete – concrete ballasts are placed on concrete</li><li>- Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast</li></ul> |                      |                            |           |
|                   | 10x10m                                                                                                                                                                                                                                              |                      |                            |           |
|                   |                                                                                                                                                                                                                                                     | Concrete on concrete | Concrete on wood or rubber |           |
|                   | Long side guy ropes:                                                                                                                                                                                                                                | 1280 kg              | 1140 kg                    |           |
|                   | Corner ties:                                                                                                                                                                                                                                        | 1000 kg              | 890 kg                     |           |
|                   | Stormbelt:                                                                                                                                                                                                                                          | 1430 kg              | 1270 kg                    |           |
|                   | 10x15m / 10x12m / 9x12m                                                                                                                                                                                                                             |                      |                            |           |
|                   |                                                                                                                                                                                                                                                     | Concrete on concrete | Concrete on wood or rubber |           |
|                   | Long side guy ropes:                                                                                                                                                                                                                                | 1320 kg              | 1180 kg                    |           |
|                   | Short side guy ropes:                                                                                                                                                                                                                               | 1330 kg              | 1180 kg                    |           |
|                   | Corner ties:                                                                                                                                                                                                                                        | 1070 kg              | 960 kg                     |           |
|                   | Stormbelt:                                                                                                                                                                                                                                          | 1530 kg              | 1360 kg                    |           |
|                   | 10x20m                                                                                                                                                                                                                                              |                      |                            |           |
|                   |                                                                                                                                                                                                                                                     | Concrete on concrete | Concrete on wood or rubber |           |
|                   | Long side guy ropes:                                                                                                                                                                                                                                | 1470 kg              | 1310 kg                    |           |
|                   | Short side guy ropes:                                                                                                                                                                                                                               | 1420 kg              | 1260 kg                    |           |
|                   | Corner ties:                                                                                                                                                                                                                                        | 1130 kg              | 1000 kg                    |           |
|                   | Stormbelt:                                                                                                                                                                                                                                          | 1510 kg              | 1340 kg                    |           |

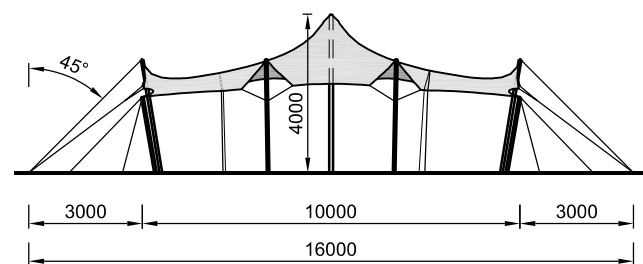
## **E. Drawings: main dimensions and anchoring**

Drawings for the following sizes are presented in this chapter:

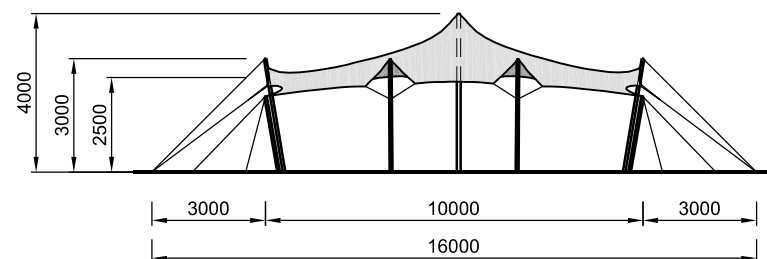
- 10x10m model
- 10x15m model
- 10x20m model
- 10x12m model (based on 10x15m model)
- 9x12m model (based on 10x15m model)



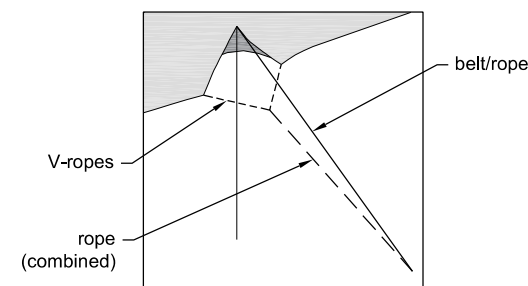
TOP VIEW  
[1:200]



FRONT VIEW  
[1:200]



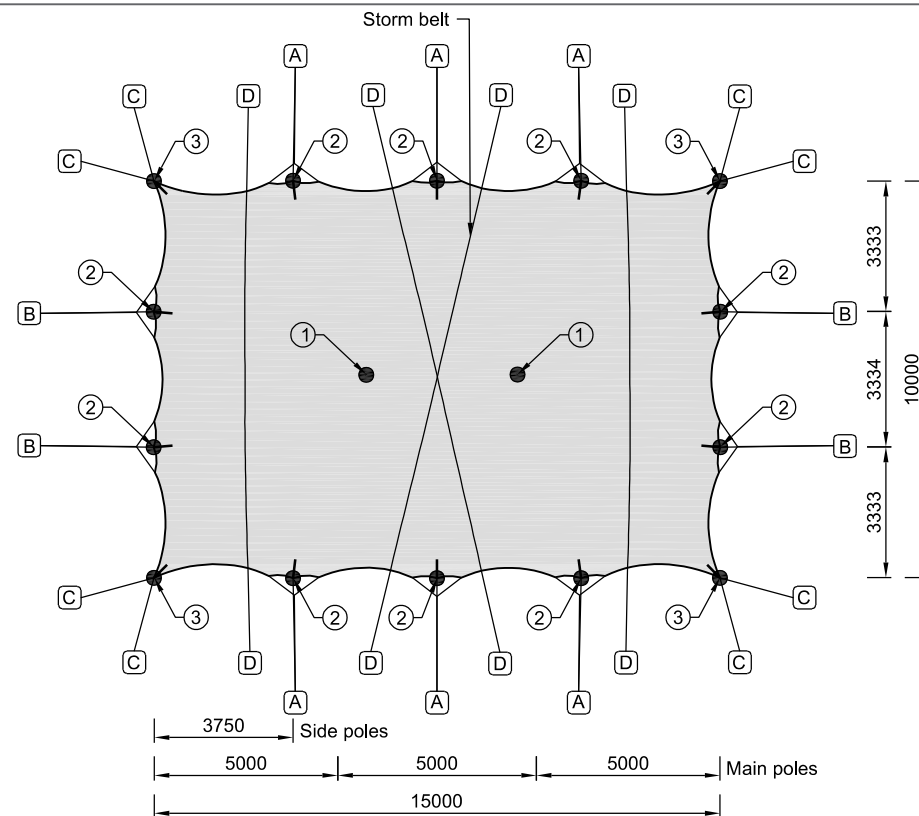
SIDE VIEW  
[1:200]



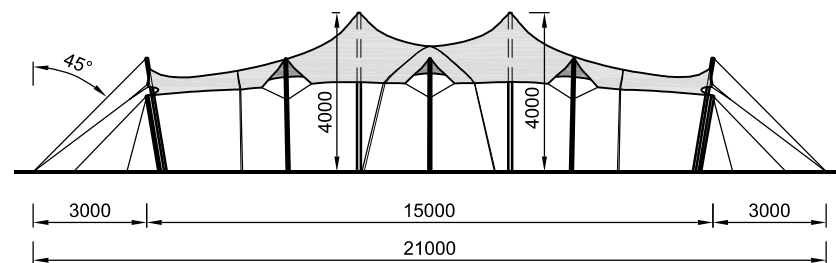
GUYING DIAGRAM  
[1:100]

| Safety against sliding, overturning and uplifting                                                                     |                                                    |                                             |                                                   |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------|---------------------------------------------------|
| Based on dense, non-cohesive soil (e.g. sandy soils).                                                                 |                                                    |                                             |                                                   |
|                                                                                                                       | <u>Anchors</u><br>(Ø30x1000mm)                     | <u>Ballast</u><br>(concrete on<br>concrete) | <u>Ballast</u><br>(concrete on<br>wood or rubber) |
| (A)                                                                                                                   | Guying - long side                                 | 2x anchors                                  | 1280 kg                                           |
| (B)                                                                                                                   | Guying - short side                                | -                                           | -                                                 |
| (C)                                                                                                                   | Guying - corner                                    | 1x anchors                                  | 1000 kg                                           |
| (D)                                                                                                                   | Storm belt ( <i>per end</i> )                      | 2x anchors                                  | 1430 kg                                           |
| All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors. |                                                    |                                             |                                                   |
| Poles                                                                                                                 |                                                    |                                             |                                                   |
| Main poles 4.0m:                                                                                                      |                                                    |                                             |                                                   |
| ① - Ø100mm [C18, Spruce]                                                                                              |                                                    |                                             |                                                   |
| Side poles 3.0m:                                                                                                      |                                                    |                                             |                                                   |
| ② - Ø70mm [C18, Spruce]                                                                                               |                                                    |                                             |                                                   |
| Corner poles 2.5m:                                                                                                    |                                                    |                                             |                                                   |
| ③ - Ø70mm [C18, Spruce]                                                                                               |                                                    |                                             |                                                   |
| Guying                                                                                                                |                                                    |                                             |                                                   |
| Storm belt:                                                                                                           | [PES] BL 3000 daN = LC 1000 daN                    |                                             |                                                   |
| Guying belt:                                                                                                          | [PES] BL 3000 daN = LC 1000 daN                    |                                             |                                                   |
| Guying rope:                                                                                                          | rope: Ø 8mm polyester, breaking strength ≥ 900 daN |                                             |                                                   |
| <u>Combined &amp; v-ropes:</u>                                                                                        |                                                    | <u>Rope</u>                                 |                                                   |
| n - number of sections                                                                                                |                                                    | n                                           |                                                   |
| Guying - long side (combined)                                                                                         |                                                    | 4x                                          |                                                   |
| Guying - long side (v-ropes)                                                                                          |                                                    | 3x                                          |                                                   |
| Guying - short side (combined)                                                                                        |                                                    | -                                           |                                                   |
| Guying - short side (v-ropes)                                                                                         |                                                    | -                                           |                                                   |
| <u>Belts (alternative rope):</u>                                                                                      |                                                    | <u>Belt</u>                                 | <u>Rope</u>                                       |
| n - number of sections                                                                                                |                                                    | (number)                                    | n                                                 |
| Guying - long side                                                                                                    |                                                    | 1x                                          | 1x                                                |
| Guying - short side                                                                                                   |                                                    | 1x                                          | 1x                                                |
| Guying - corner                                                                                                       |                                                    | 1x                                          | 3x                                                |
| Alternative configuration                                                                                             |                                                    |                                             |                                                   |
| 10 x 10 m                                                                                                             | 10 x 12 m                                          |                                             |                                                   |
| 10 x 15 m                                                                                                             | 9 x 12 m                                           |                                             |                                                   |
| 10 x 20 m                                                                                                             |                                                    |                                             |                                                   |
| * Basic configurations                                                                                                |                                                    |                                             |                                                   |

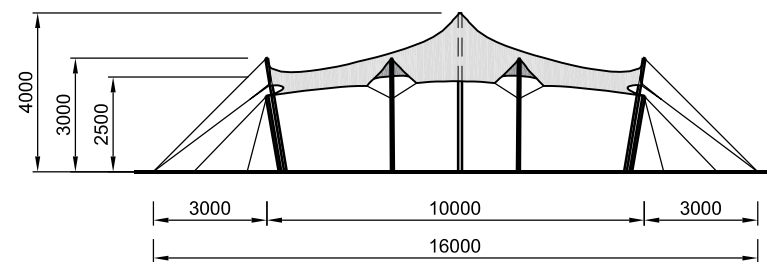
|         |            |                                 |
|---------|------------|---------------------------------|
| EDITOR: | DATE:      | FILENAME:                       |
| CH      | 05.04.2026 | TTP20260008_10x10m_EN_01_REV_00 |



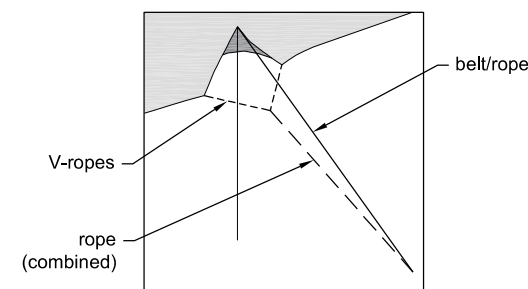
TOP VIEW  
[1:200]



FRONT VIEW  
[1:200]



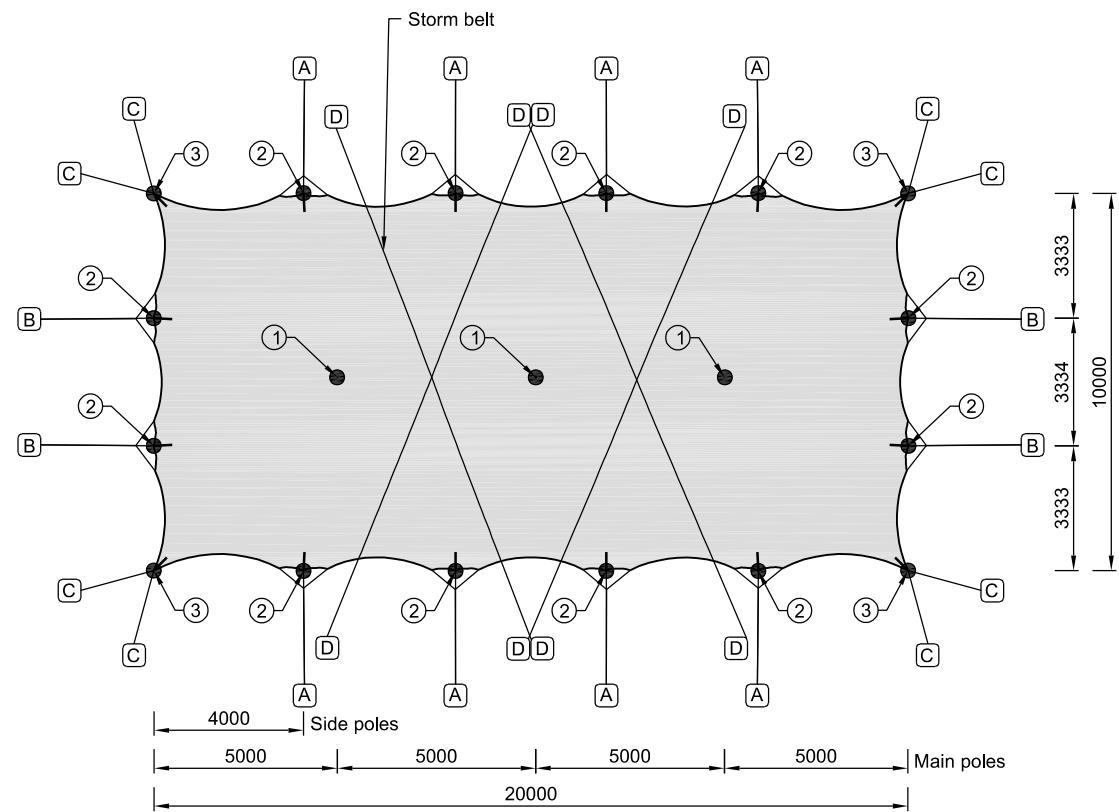
SIDE VIEW  
[1:200]



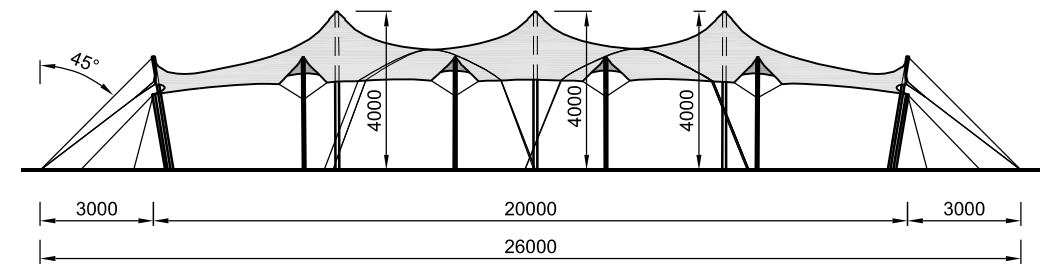
GUYING DIAGRAM  
[1:100]

| Safety against sliding, overturning and uplifting                                                                     |                                                    |                                             |                                                   |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------|---------------------------------------------------|
| Based on dense, non-cohesive soil (e.g. sandy soils).                                                                 |                                                    |                                             |                                                   |
|                                                                                                                       | <u>Anchors</u><br>(Ø30x1000mm)                     | <u>Ballast</u><br>(concrete on<br>concrete) | <u>Ballast</u><br>(concrete on<br>wood or rubber) |
| A                                                                                                                     | Guying - long side                                 | 2x anchors                                  | 1320 kg                                           |
| B                                                                                                                     | Guying - short side                                | 2x anchors                                  | 1180 kg                                           |
| C                                                                                                                     | Guying - corner                                    | 1x anchors                                  | 1070 kg                                           |
| D                                                                                                                     | Storm belt (per end)                               | 2x anchors                                  | 960 kg                                            |
|                                                                                                                       |                                                    | 1530 kg                                     | 1360 kg                                           |
| All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors. |                                                    |                                             |                                                   |
| Poles                                                                                                                 |                                                    |                                             |                                                   |
| Main poles 4.0m:                                                                                                      |                                                    |                                             |                                                   |
| ① - Ø100mm [C18, Spruce]                                                                                              |                                                    |                                             |                                                   |
| Side poles 3.0m:                                                                                                      |                                                    |                                             |                                                   |
| ② - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                             |                                                   |
| Corner poles 2.5m:                                                                                                    |                                                    |                                             |                                                   |
| ③ - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                             |                                                   |
| Guying                                                                                                                |                                                    |                                             |                                                   |
| Storm belt:                                                                                                           | [PES] BL 3000 daN = LC 1000 daN                    |                                             |                                                   |
| Guying belt:                                                                                                          | [PES] BL 3000 daN = LC 1000 daN                    |                                             |                                                   |
| Guying rope:                                                                                                          | rope: Ø 8mm polyester, breaking strength ≥ 900 daN |                                             |                                                   |
| <u>Combined &amp; v-ropes:</u>                                                                                        |                                                    | <u>Rope</u>                                 |                                                   |
| n - number of sections                                                                                                |                                                    | n                                           |                                                   |
| Guying - long side (combined)                                                                                         |                                                    | 4x                                          |                                                   |
| Guying - long side (v-ropes)                                                                                          |                                                    | 3x                                          |                                                   |
| Guying - short side (combined)                                                                                        |                                                    | 4x                                          |                                                   |
| Guying - short side (v-ropes)                                                                                         |                                                    | 3x                                          |                                                   |
| <u>Belts (alternative rope):</u>                                                                                      |                                                    | <u>Belt</u>                                 | <u>Rope</u>                                       |
| n - number of sections                                                                                                |                                                    | (number)                                    | n                                                 |
| Guying - long side                                                                                                    |                                                    | 1x                                          | 1x                                                |
| Guying - short side                                                                                                   |                                                    | 1x                                          | 1x                                                |
| Guying - corner                                                                                                       |                                                    | 1x                                          | 3x                                                |
| Alternative configuration                                                                                             |                                                    |                                             |                                                   |
| 10 x 10 m                                                                                                             | 10 x 12 m                                          |                                             |                                                   |
| 10 x 15 m                                                                                                             | 9 x 12 m                                           |                                             |                                                   |
| 10 x 20 m                                                                                                             |                                                    |                                             |                                                   |
| * Basic configurations                                                                                                |                                                    |                                             |                                                   |

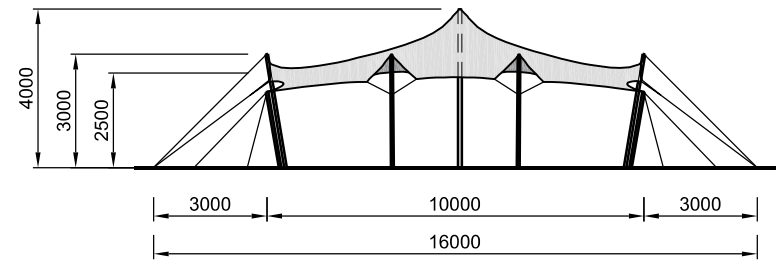
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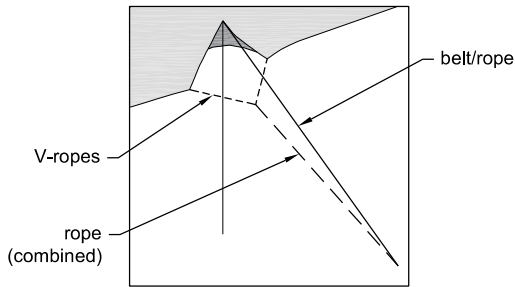
TOP VIEW  
[1:200]



FRONT VIEW  
[1:200]



SIDE VIEW  
[1:200]



GUYING DIAGRAM  
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

|   | Anchors<br>(Ø30x1000mm)            | Ballast<br>(concrete on<br>concrete) | Ballast<br>(concrete on<br>wood or rubber) |
|---|------------------------------------|--------------------------------------|--------------------------------------------|
| A | Guying - long side<br>2x anchors   | 1470 kg                              | 1310 kg                                    |
| B | Guying - short side<br>2x anchors  | 1420 kg                              | 1260 kg                                    |
| C | Guying - corner<br>1x anchors      | 1130 kg                              | 1000 kg                                    |
| D | Storm belt (per end)<br>2x anchors | 1510 kg                              | 1340 kg                                    |

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:  
① - Ø100mm [C18, Spruce]  
Side poles 3.0m:  
② - Ø75mm [C18, Spruce]  
Corner poles 2.5m:  
③ - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN  
Guying belt: [PES] BL 3000 daN = LC 1000 daN  
Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

| Combined & v-ropes:            | Rope |
|--------------------------------|------|
| n - number of sections         | n    |
| Guying - long side (combined)  | 4x   |
| Guying - long side (v-ropes)   | 3x   |
| Guying - short side (combined) | 4x   |
| Guying - short side (v-ropes)  | 3x   |

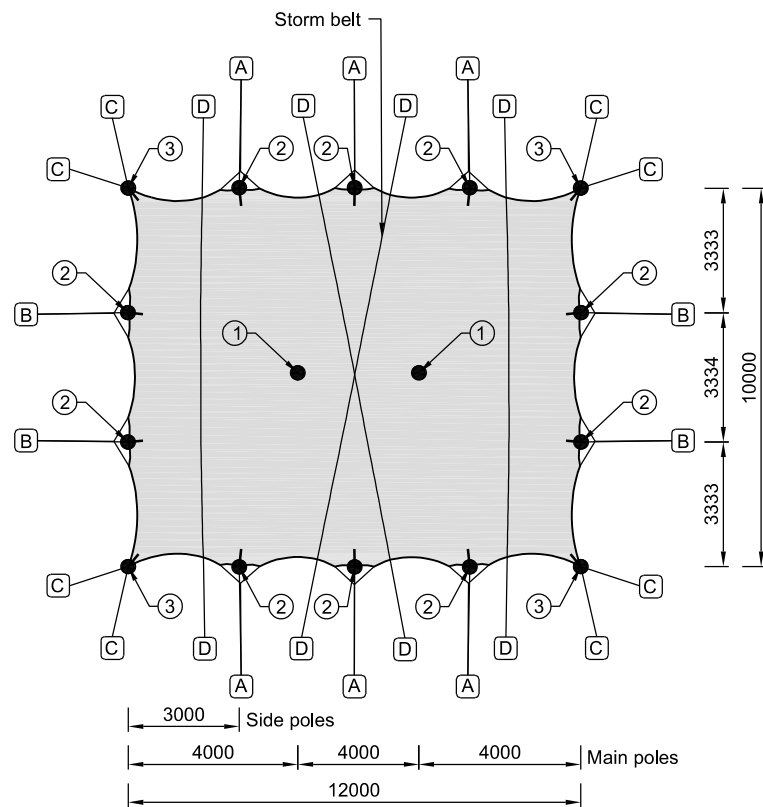
| Belts (alternative rope): | Belt<br>(number) | Rope<br>n |
|---------------------------|------------------|-----------|
| n - number of sections    |                  |           |
| Guying - long side        | 1x               | 1x        |
| Guying - short side       | 1x               | 1x        |
| Guying - corner           | 1x               | 3x        |

Alternative configuration

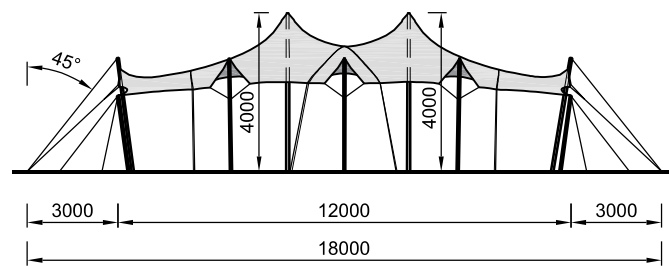
|                        |           |
|------------------------|-----------|
| 10 x 10 m              | 10 x 12 m |
| 10 x 15 m              | 9 x 12 m  |
| 10 x 20 m              |           |
| * Basic configurations |           |

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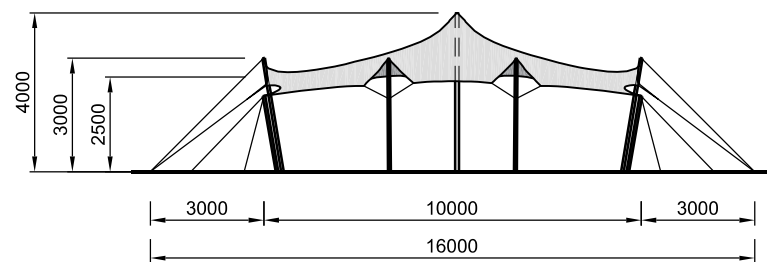




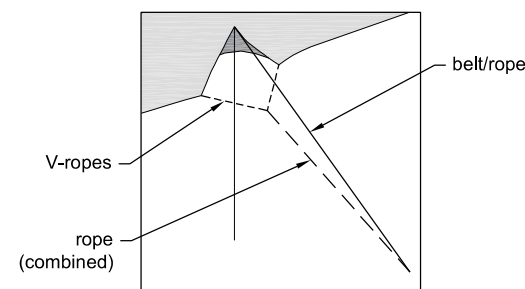
TOP VIEW  
[1:200]



FRONT VIEW  
[1:200]



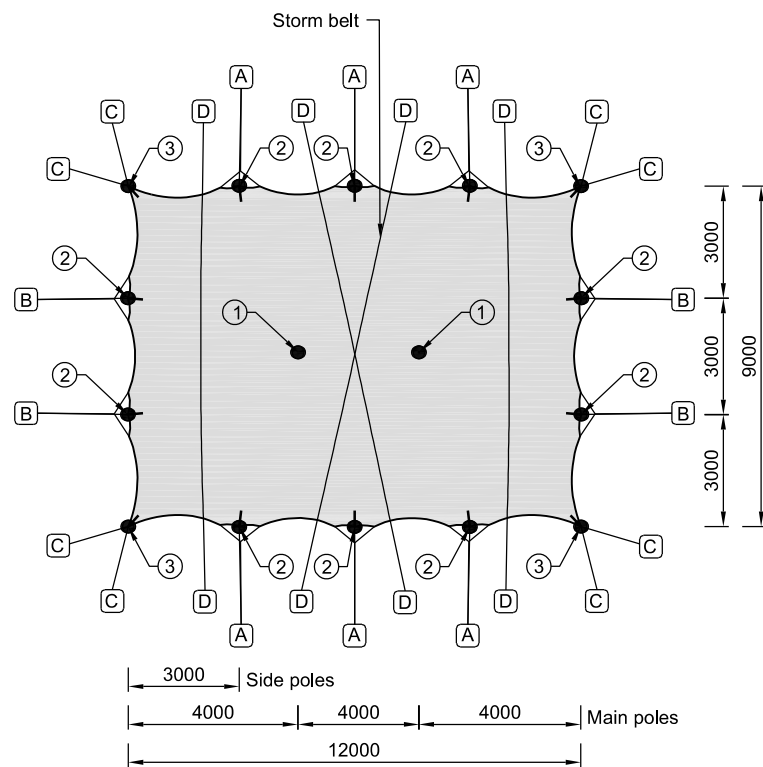
SIDE VIEW  
[1:200]



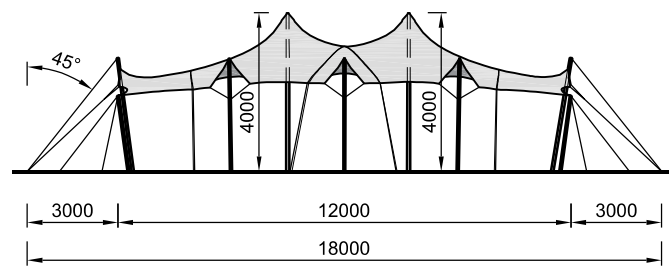
GUYING DIAGRAM  
[1:100]

| Safety against sliding, overturning and uplifting                                                                     |                                                    |                                      |                                            |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------|--------------------------------------------|
| Based on dense, non-cohesive soil (e.g. sandy soils).                                                                 |                                                    |                                      |                                            |
|                                                                                                                       | Anchors<br>(Ø30x1000mm)                            | Ballast<br>(concrete on<br>concrete) | Ballast<br>(concrete on<br>wood or rubber) |
| A                                                                                                                     | Guying - long side                                 | 2x anchors                           | 1320 kg                                    |
| B                                                                                                                     | Guying - short side                                | 2x anchors                           | 1330 kg                                    |
| C                                                                                                                     | Guying - corner                                    | 1x anchors                           | 1070 kg                                    |
| D                                                                                                                     | Storm belt (per end)                               | 2x anchors                           | 1530 kg                                    |
| All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors. |                                                    |                                      |                                            |
| Poles                                                                                                                 |                                                    |                                      |                                            |
| Main poles 4.0m:                                                                                                      |                                                    |                                      |                                            |
| ① - Ø100mm [C18, Spruce]                                                                                              |                                                    |                                      |                                            |
| Side poles 3.0m:                                                                                                      |                                                    |                                      |                                            |
| ② - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                      |                                            |
| Corner poles 2.5m:                                                                                                    |                                                    |                                      |                                            |
| ③ - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                      |                                            |
| Guying                                                                                                                |                                                    |                                      |                                            |
| Storm belt:                                                                                                           | [PES] BL 3000 daN = LC 1000 daN                    |                                      |                                            |
| Guying belt:                                                                                                          | [PES] BL 3000 daN = LC 1000 daN                    |                                      |                                            |
| Guying rope:                                                                                                          | rope: Ø 8mm polyester, breaking strength ≥ 900 daN |                                      |                                            |
| Combined & v-ropes:                                                                                                   |                                                    | Rope                                 |                                            |
| n - number of sections                                                                                                |                                                    | n                                    |                                            |
| Guying - long side (combined)                                                                                         |                                                    | 4x                                   |                                            |
| Guying - long side (v-ropes)                                                                                          |                                                    | 3x                                   |                                            |
| Guying - short side (combined)                                                                                        |                                                    | 4x                                   |                                            |
| Guying - short side (v-ropes)                                                                                         |                                                    | 3x                                   |                                            |
| Belts (alternative rope):                                                                                             |                                                    | Belt<br>(number)                     | Rope<br>n                                  |
| n - number of sections                                                                                                |                                                    |                                      |                                            |
| Guying - long side                                                                                                    |                                                    | 1x                                   | 1x                                         |
| Guying - short side                                                                                                   |                                                    | 1x                                   | 1x                                         |
| Guying - corner                                                                                                       |                                                    | 1x                                   | 3x                                         |
| Alternative configuration                                                                                             |                                                    |                                      |                                            |
| 10 x 10 m                                                                                                             | 10 x 12 m                                          |                                      |                                            |
| 10 x 15 m                                                                                                             | 9 x 12 m                                           |                                      |                                            |
| 10 x 20 m                                                                                                             |                                                    |                                      |                                            |
| * Basic configurations                                                                                                |                                                    |                                      |                                            |

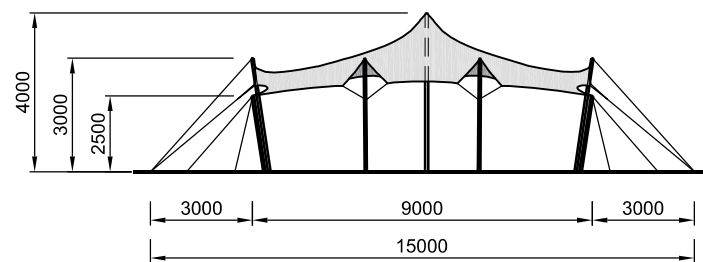
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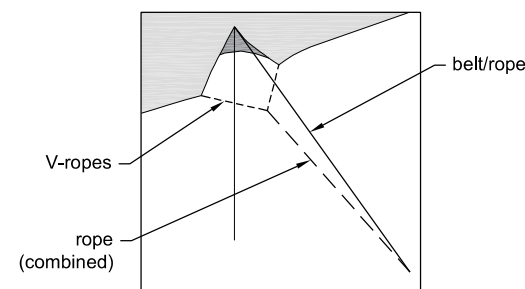
TOP VIEW  
[1:200]



FRONT VIEW  
[1:200]



SIDE VIEW  
[1:200]



GUYING DIAGRAM  
[1:100]

| Safety against sliding, overturning and uplifting                                                                     |                                                    |                                      |                                            |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------|--------------------------------------------|
| Based on dense, non-cohesive soil (e.g. sandy soils).                                                                 |                                                    |                                      |                                            |
|                                                                                                                       | Anchors<br>(Ø30x1000mm)                            | Ballast<br>(concrete on<br>concrete) | Ballast<br>(concrete on<br>wood or rubber) |
| A                                                                                                                     | Guying - long side                                 | 2x anchors                           | 1320 kg                                    |
| B                                                                                                                     | Guying - short side                                | 2x anchors                           | 1330 kg                                    |
| C                                                                                                                     | Guying - corner                                    | 1x anchors                           | 1070 kg                                    |
| D                                                                                                                     | Storm belt (per end)                               | 2x anchors                           | 1530 kg                                    |
| All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors. |                                                    |                                      |                                            |
| Poles                                                                                                                 |                                                    |                                      |                                            |
| Main poles 4.0m:                                                                                                      |                                                    |                                      |                                            |
| ① - Ø100mm [C18, Spruce]                                                                                              |                                                    |                                      |                                            |
| Side poles 3.0m:                                                                                                      |                                                    |                                      |                                            |
| ② - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                      |                                            |
| Corner poles 2.5m:                                                                                                    |                                                    |                                      |                                            |
| ③ - Ø75mm [C18, Spruce]                                                                                               |                                                    |                                      |                                            |
| Guying                                                                                                                |                                                    |                                      |                                            |
| Storm belt:                                                                                                           | [PES] BL 3000 daN = LC 1000 daN                    |                                      |                                            |
| Guying belt:                                                                                                          | [PES] BL 3000 daN = LC 1000 daN                    |                                      |                                            |
| Guying rope:                                                                                                          | rope: Ø 8mm polyester, breaking strength ≥ 900 daN |                                      |                                            |
| Combined & v-ropes:                                                                                                   |                                                    | Rope                                 |                                            |
| n - number of sections                                                                                                |                                                    | n                                    |                                            |
| Guying - long side (combined)                                                                                         |                                                    | 4x                                   |                                            |
| Guying - long side (v-ropes)                                                                                          |                                                    | 3x                                   |                                            |
| Guying - short side (combined)                                                                                        |                                                    | 4x                                   |                                            |
| Guying - short side (v-ropes)                                                                                         |                                                    | 3x                                   |                                            |
| Belts (alternative rope):                                                                                             |                                                    | Belt<br>(number)                     | Rope<br>n                                  |
| n - number of sections                                                                                                |                                                    |                                      |                                            |
| Guying - long side                                                                                                    |                                                    | 1x                                   | 1x                                         |
| Guying - short side                                                                                                   |                                                    | 1x                                   | 1x                                         |
| Guying - corner                                                                                                       |                                                    | 1x                                   | 3x                                         |
| Alternative configuration                                                                                             |                                                    |                                      |                                            |
| 10 x 10 m                                                                                                             | 10 x 12 m                                          |                                      |                                            |
| 10 x 15 m                                                                                                             | 9 x 12 m                                           |                                      |                                            |
| 10 x 20 m                                                                                                             |                                                    |                                      |                                            |
| * Basic configurations                                                                                                |                                                    |                                      |                                            |

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## F. Important terms and conditions

This document applies to the built structure if the following principles and conditions are met:

- The used materials, parts and sections (membrane, poles, ties, anchoring) are in accordance with this document.
- The dimensions of the built structure match the dimensions stated in this document.
- Parts (poles, ties, anchors) may not be removed.
- Obstacles should be placed at least 0.5m from the membrane (measured perpendicular to the fabric); The fabric needs a certain freedom to deform in all directions to prevent damages caused by collision with objects located closely to the fabric (see also EN 13782, article 8.7).
- Above the maximum allowable wind speeds (see D) the structure should be evacuated and access for the public must be denied.
- Only decorations, music- and light installations of less than 25 kg per pole, can be attached to the structure. They must be attached centrally.
- A conventional load of  $0.1 \text{ kN/m}^2$  is considered according to EN 13782, which corresponds with the required snow load (4cm) according to the French CTS.
- Anchoring is based on dense, non-cohesive soil. When soil differs, additional anchoring might be necessary, or anchor tests need to be performed.
- Poles must be stored dry. If they become wet, there needs to be a possibility for them to dry.
- All water accumulation must be removed as soon as possible.

## G. Allowed wind speeds

The maximum wind speed is converted into a basic wind speed for a coastal area, flattened/open area, rural area, village and city according to EN 1991-1-4. Terrain roughness is taken according to the recommended general values for the different terrain categories for Europe. (not country specific) Illustrations of these terrain categories are presented on page 18.

| Full wind load                                |                      |               |
|-----------------------------------------------|----------------------|---------------|
| Wind pressure $p_w$                           | 500 N/m <sup>2</sup> | At 5m height  |
| Corresponding peak wind pressure $p_{w,peak}$ | 605 N/m <sup>2</sup> | At 10m height |

### Peak wind speed at 10m height

Equation:

$$605 = \frac{1}{2} \times \rho \times v^2 = \frac{1}{2} \times 1.25 \times v^2 \rightarrow v = 31.1 \text{ m/s} \rightarrow \pm 113 \text{ km/h}$$

Eq. 4.10 EN 1991-1-4  
Basic wind pressure

### Wind speed coastal area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.003}{0.05}\right)^{0.07} = 0.156$$

Eq. 4.5 EN 1991-1-4  
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.156 \times \ln\left(\frac{4.0}{0.003}\right) = 1.123$$

Eq. 4.4 EN 1991-1-4  
Roughness factor  
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.123 \times V_b$$

Eq. 4.3 EN 1991-1-4  
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.141 \times V_b$$

Eq. 4.6 EN 1991-1-4  
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.156 \times V_b}{1.158 \times V_b} = 0.139$$

Eq. 4.7 EN 1991-1-4  
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.554 \times V_b^2$$

Eq. 4.8 EN 1991-1-4  
Extreme wind pressure

Equation:

$$500 = 1.554 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 17.94 \text{ m/s}$$

Characteristic wind speed

### Wind speed flattened, open area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.01}{0.05}\right)^{0.07} = 0.170$$

Eq. 4.5 EN 1991-1-4  
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.170 \times \ln\left(\frac{4.0}{0.01}\right) = 1.017$$

Eq. 4.4 EN 1991-1-4  
Roughness factor at 3.5m height  
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.017 \times V_b$$

Eq. 4.3 EN 1991-1-4  
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.170 \times V_b$$

Eq. 4.6 EN 1991-1-4  
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.170 \times V_b}{1.017 \times V_b} = 0.167$$

Eq. 4.7 EN 1991-1-4  
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.402 \times V_b^2$$

Eq. 4.8 EN 1991-1-4  
Extreme wind pressure

Equation:

$$500 = 1.402 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 18.89 \text{ m/s}$$

Characteristic wind speed

### Wind speed rural area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.05}{0.05}\right)^{0.07} = 0.190$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.190 \times \ln\left(\frac{4.0}{0.05}\right) = 0.833$$

$$V_m = C_r \times V_b = 0.833 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.190 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.190 \times V_b}{0.833 \times V_b} = 0.228$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.125 \times V_b^2$$

Equation:

$$500 = 1.125 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 21.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4  
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4  
Roughness factor at 4m height  
 $Z = 4.0 > Z_{\min} = 2$

Eq. 4.3 EN 1991-1-4  
Average wind speed at height

Eq. 4.6 EN 1991-1-4  
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4  
Turbulence intensity

Eq. 4.8 EN 1991-1-4  
Extreme wind pressure

Characteristic wind speed

### Wind speed village at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.3}{0.05}\right)^{0.07} = 0.215$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.215 \times \ln\left(\frac{5}{0.3}\right) = 0.606$$

$$V_m = C_r \times V_b = 0.606 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.215 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.215 \times V_b}{0.606 \times V_b} = 0.355$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.801 \times V_b^2$$

Equation:

$$500 = 0.801 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 24.99 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4  
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4  
Roughness factor at 7m height  
 $Z = Z_{\min} = 5$

Eq. 4.3 EN 1991-1-4  
Average wind speed at height

Eq. 4.6 EN 1991-1-4  
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4  
Turbulence intensity

Eq. 4.8 EN 1991-1-4  
Extreme wind pressure

Characteristic wind speed

### Wind speed city at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{1}{0.05}\right)^{0.07} = 0.234$$

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.234 \times \ln\left(\frac{10}{1}\right) = 0.540$$

$$V_m = C_r \times V_b = 0.540 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.234 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.234 \times V_b}{0.540 \times V_b} = 0.434$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.735 \times V_b^2$$

Equation:

$$500 = 0.735 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 26.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4  
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4  
Roughness factor at 7m height  
 $Z = Z_{\min} = 10$

Eq. 4.3 EN 1991-1-4  
Average wind speed at height

Eq. 4.6 EN 1991-1-4  
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4  
Turbulence intensity

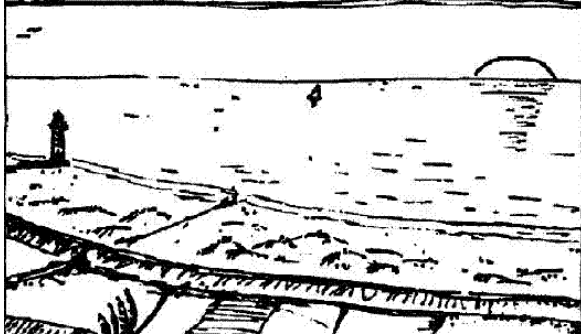
Eq. 4.8 EN 1991-1-4  
Extreme wind pressure

Characteristic wind speed

## Terrain categories

Default European terrain categories (not Country specific)

0: Coastal area:



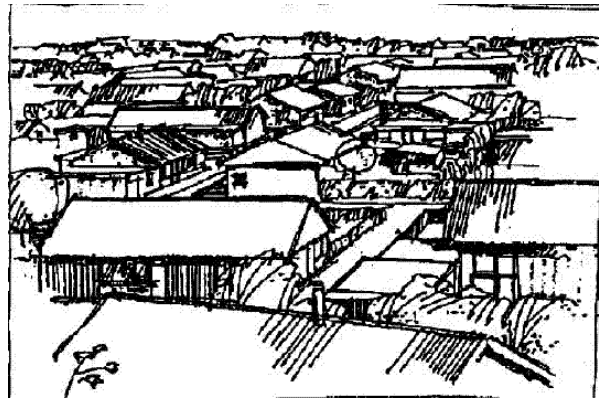
I: Flattened, open area:



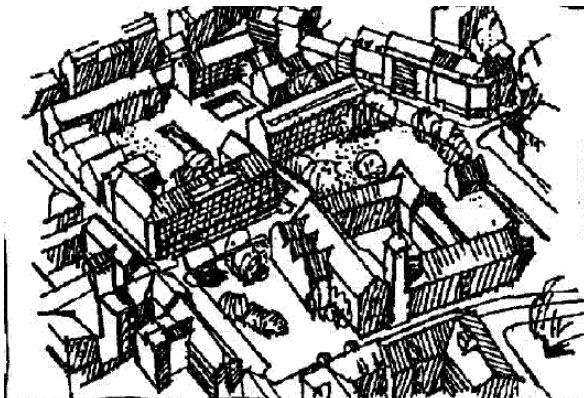
II: Rural area



III: Village



IV: City



## H. Static analysis

|       |                                                        |    |
|-------|--------------------------------------------------------|----|
| H.1.  | Project description.....                               | 23 |
| H.2.  | Materials.....                                         | 24 |
| H.2.1 | Fabric.....                                            | 24 |
| H.2.2 | Guy ropes.....                                         | 24 |
| H.2.3 | Belts.....                                             | 24 |
| H.2.4 | Wood.....                                              | 25 |
| H.3.  | Cross sections.....                                    | 25 |
| H.4.  | Calculation method.....                                | 26 |
| H.4.1 | Modelling.....                                         | 26 |
| H.4.2 | Structural behaviour of membrane structures.....       | 28 |
| H.4.3 | Load combinations.....                                 | 29 |
| H.5.  | Load cases.....                                        | 30 |
| H.5.1 | Own weight.....                                        | 30 |
| H.5.2 | Pretension.....                                        | 30 |
| H.5.3 | Conventional / snow load.....                          | 30 |
| H.5.4 | User load.....                                         | 30 |
| H.5.5 | Wind load.....                                         | 30 |
| H.6.  | Calculation results.....                               | 32 |
| H.6.1 | Listing of calculated load combinations.....           | 32 |
| H.6.2 | Overview: Global results of static analysis.....       | 33 |
| H.6.3 | Summary of forces.....                                 | 51 |
| H.7.  | Element check.....                                     | 53 |
| H.7.1 | Membrane.....                                          | 53 |
| H.7.2 | Main poles.....                                        | 54 |
| H.7.3 | Side poles.....                                        | 55 |
| H.7.4 | Corner poles.....                                      | 56 |
| H.7.5 | Guy ropes.....                                         | 57 |
| H.7.6 | Belts.....                                             | 59 |
| H.7.7 | Stormbelts.....                                        | 60 |
| H.7.8 | Fasteners.....                                         | 61 |
| H.8.  | Safety against overturning, sliding and uplifting..... | 64 |
| H.8.1 | Anchor capacity.....                                   | 64 |
| H.8.2 | Required anchor pins.....                              | 65 |
| H.8.3 | Anchor tests according to BS-EN 13782.....             | 67 |
| H.8.4 | Required ballast.....                                  | 68 |



|                 |                                                                                   |            |
|-----------------|-----------------------------------------------------------------------------------|------------|
| I.1.            | Summary of results – low wind in Netherlands .....                                | 70         |
| I.2.            | Summary of forces .....                                                           | 72         |
| I.2.1           | 10x10m model .....                                                                | 72         |
| I.2.2           | 10x15m model .....                                                                | 73         |
| I.2.3           | 10x20m model .....                                                                | 73         |
| I.3.            | Element check - low wind in Netherlands .....                                     | 74         |
| I.3.1           | Guy ropes .....                                                                   | 74         |
| I.3.2           | Fasteners .....                                                                   | 76         |
| I.4.            | Safety against overturning, sliding and uplifting - low wind in Netherlands ..... | 77         |
| I.4.1           | Required anchor pins .....                                                        | 77         |
| I.4.2           | Required ballast .....                                                            | 78         |
| J.1.            | Summary of results – snow+wind in France .....                                    | 80         |
| J.2.            | Cross sections .....                                                              | 82         |
| J.3.            | Summary of forces .....                                                           | 82         |
| J.3.1           | 10x10m model .....                                                                | 82         |
| J.3.2           | 10x15m model .....                                                                | 83         |
| J.3.3           | 10x20m model .....                                                                | 83         |
| J.4.            | Element check – snow + wind in France .....                                       | 85         |
| J.4.1           | Membrane .....                                                                    | 85         |
| J.4.2           | Main poles .....                                                                  | 86         |
| J.4.3           | Side poles .....                                                                  | 87         |
| J.4.4           | Corner poles .....                                                                | 88         |
| J.4.5           | Belts .....                                                                       | 89         |
| J.4.6           | Fasteners .....                                                                   | 90         |
| K.1.            | Fabric specifications .....                                                       | 91         |
| K.2.            | Fabric fire certificate .....                                                     | 92         |
| K.3.            | Capacity of Eye bolts .....                                                       | 94         |
| K.4.            | Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889 .....                       | 94         |
| K.5.            | Carabine, DIN 5299C .....                                                         | 95         |
| K.6.            | Rope ratchet .....                                                                | 95         |
| <b>Annex A.</b> | <b>Software input (load combination) – 10x10m model .....</b>                     | <b>98</b>  |
| Annex A.1.      | CO1. Own weight + pretension .....                                                | 98         |
| Annex A.2.      | CO2. Own weight + pretension + conventional / snow .....                          | 99         |
| Annex A.3.      | CO3. Own weight + pretension + wind pressure .....                                | 100        |
| Annex A.4.      | CO4. Own weight + pretension + wind suction .....                                 | 101        |
| Annex A.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                    | 102        |
| Annex A.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                    | 103        |
| <b>Annex B.</b> | <b>Software input (load combination) – 10x15m model .....</b>                     | <b>105</b> |
| Annex B.1.      | CO1. Own weight + pretension .....                                                | 105        |
| Annex B.2.      | CO2. Own weight + pretension + conventional / snow .....                          | 106        |



|                 |                                                                                |            |
|-----------------|--------------------------------------------------------------------------------|------------|
| Annex B.3.      | CO3. Own weight + pretension + wind pressure .....                             | 107        |
| Annex B.4.      | CO4. Own weight + pretension + wind suction .....                              | 108        |
| Annex B.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                 | 109        |
| Annex B.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                 | 110        |
| <b>Annex C.</b> | <b>Software input (load combination) – 10x20m model.....</b>                   | <b>112</b> |
| Annex C.1.      | CO1. Own weight + pretension .....                                             | 112        |
| Annex C.2.      | CO2. Own weight + pretension + conventional / snow .....                       | 113        |
| Annex C.3.      | CO3. Own weight + pretension + wind pressure .....                             | 114        |
| Annex C.4.      | CO4. Own weight + pretension + wind suction .....                              | 115        |
| Annex C.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                 | 116        |
| Annex C.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                 | 117        |
| <b>Annex D.</b> | <b>Results per load combination – 10x10m model .....</b>                       | <b>119</b> |
| Annex D.1.      | CO1. Own weight + pretension .....                                             | 119        |
| Annex D.2.      | CO2. Own weight + pretension + snow .....                                      | 123        |
| Annex D.3.      | CO3. Own weight + pretension + wind pressure .....                             | 127        |
| Annex D.4.      | CO4. Own weight + pretension + wind suction .....                              | 131        |
| Annex D.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                 | 135        |
| Annex D.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                 | 139        |
| <b>Annex E.</b> | <b>Results per load combination – 10x15m model .....</b>                       | <b>143</b> |
| Annex E.1.      | CO1. Own weight + pretension .....                                             | 143        |
| Annex E.2.      | CO2. Own weight + pretension + snow .....                                      | 147        |
| Annex E.3.      | CO3. Own weight + pretension + wind pressure .....                             | 151        |
| Annex E.4.      | CO4. Own weight + pretension + wind suction .....                              | 155        |
| Annex E.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                 | 159        |
| Annex E.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                 | 163        |
| <b>Annex F.</b> | <b>Results per load combination – 10x20m model .....</b>                       | <b>167</b> |
| Annex F.1.      | CO1. Own weight + pretension .....                                             | 167        |
| Annex F.2.      | CO2. Own weight + pretension + snow .....                                      | 171        |
| Annex F.3.      | CO3. Own weight + pretension + wind pressure .....                             | 175        |
| Annex F.4.      | CO4. Own weight + pretension + wind suction .....                              | 179        |
| Annex F.5.      | CO5. Own weight + pretension + wind suction reduced [NL] .....                 | 183        |
| Annex F.6.      | CO6. Own weight + pretension + wind pressure + snow [FR] .....                 | 187        |
| <b>Annex G.</b> | <b>Member checks .....</b>                                                     | <b>191</b> |
| Annex G.1.      | Main pole 4.0m – Wood, Spruce C18 .....                                        | 191        |
| Annex G.2.      | Side pole 3m – Wood, Spruce C18 .....                                          | 192        |
| Annex G.3.      | Corner pole 2.5m - Wood, C18 .....                                             | 194        |
| Annex G.4.      | Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR] .....   | 196        |
| Annex G.5.      | Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR] .....   | 197        |
| Annex G.6.      | Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR] ..... | 199        |
| <b>Annex H.</b> | <b>Reaction forces model 10x10m .....</b>                                      | <b>200</b> |
| Annex H.1.      | Point numbers .....                                                            | 200        |
| Annex H.2.      | Reaction forces .....                                                          | 201        |
| <b>Annex I.</b> | <b>Reaction forces model 10x15m .....</b>                                      | <b>202</b> |
| Annex I.1.      | Point numbers .....                                                            | 202        |

|                                                    |            |
|----------------------------------------------------|------------|
| Annex I.2. Reaction forces.....                    | 203        |
| <b>Annex J. Reaction forces model 10x20m .....</b> | <b>204</b> |
| Annex J.1. Point numbers .....                     | 204        |
| Annex J.2. Reaction forces .....                   | 205        |

## H.1. Project description

The principle of a stretchtent is based on a rectangular piece of stretchable membrane with a reinforcement along the boundary. The membrane is supported by poles, both at the edge and under the membrane. The side poles are stabilized by guy ropes and belts. The poles do not require a fixed position which ensures a freedom of shape.

Tentech B.V. has performed a static analysis of three tents – 10x10m, 10x15m and 10x20m always assumed to have stormbelts. Further chapters include material and cross-section description, calculation method, load cases and combinations, results, element checks, anchor and ballast calculations. These chapters present calculation results valid for Europe according to EN 13782.

As an exception, two additions to the static analysis are made (see I and J on pages 70 and 80 respectively):

1. For use only in Netherlands – element check for reduced wind speeds.
2. For use in France or countries following French CTS – element check for combined wind and snow loads

For use in other countries than Netherlands or France, the static analysis must be considered without its additions. When relevant information is mentioned for these specific cases either [NL] or [FR] immediately follows this information. This information is not considered while making the base calculations. Addition J on page 80 must always be considered for use in France.



Figure 1. 10x10m, 10x15m and 10x20m models

## H.2. Materials

### H.2.1 Fabric

|                                               |                       |                     |                  |
|-----------------------------------------------|-----------------------|---------------------|------------------|
| Design tensile strength                       | $f_d$                 | $f_{tk} / \gamma_m$ | art 8.6 EN 13782 |
| Characteristic tensile strength (warp)        | $f_{tk, \text{warp}}$ |                     |                  |
| Characteristic tensile strength (weft)        | $f_{tk, \text{weft}}$ |                     |                  |
| Material factor – global, permanent load      | $\gamma_m$            | 2.5                 | tbl 4. EN13782   |
| Material factor – global, short duration load | $\gamma_m$            | 2.0                 | tbl 4. EN13782   |

Table 1. Used symbols, codes and standard for fabric materials

| Material     | Type | Weight                       | $f_{rd; \text{warp}; \text{perm}}$ | $f_{rd; \text{weft}; \text{perm}}$ | $f_{rd; \text{warp}; \text{short}}$ | $f_{rd; \text{weft}; \text{short}}$ |
|--------------|------|------------------------------|------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|
| Sioen RP5639 | -    | $\approx 730 \text{ gr/m}^2$ | 9.6 kN/m                           | 7.2 kN/m                           | 12.0 kN/m                           | 9.0 kN/m                            |

Table 2. Used fabrics

### H.2.2 Guy ropes

|                                 |                             |                     |                   |
|---------------------------------|-----------------------------|---------------------|-------------------|
| Design resistance               | $F_{rd}$                    | $R_m / \gamma_{m1}$ | art 10.2. EN13782 |
| Characteristic tensile strength | $R_m$                       |                     | art 10.2. EN13782 |
| Material factor                 | $\gamma_{m1} < 12\text{mm}$ | 4.0                 | art 10.3. EN13782 |
|                                 | $\gamma_{m1} > 12\text{mm}$ | 3.3                 | art 10.3. EN13782 |

Table 3. Used symbols, codes and standard for rope materials

| Material                 | Cross section              | Breaking strength      | $R_m$  | $F_{rd}$ |
|--------------------------|----------------------------|------------------------|--------|----------|
| 8mm Polyester Braid rope | $\varnothing 8 \text{ mm}$ | $\geq 900 \text{ daN}$ | 9.0 kN | 2.25 kN  |

Table 4. Used guy ropes

### H.2.3 Belts

|                                  |               |                         |                    |
|----------------------------------|---------------|-------------------------|--------------------|
| Design resistance                | $F_{rd}$      | $R_m / \gamma_{m1}$     | art 10.2. EN13782  |
| Characteristic breaking strength | $R_m$         | $LC \times \gamma_{m2}$ | art 10.2. EN13782  |
| Lashing capacity                 | LC            |                         | Conform EN 12195-2 |
| Material factor                  | $\gamma_{m1}$ | 2.0                     | art 10.2. EN13782  |
| Material factor                  | $\gamma_{m2}$ | 3.0                     | EN1492-1           |

Table 5. Used symbols, codes and standard for belt materials

| Material                    | LC                | $R_m$             | $F_{rd}$ |
|-----------------------------|-------------------|-------------------|----------|
| Belt [PES] EN 12195-2, 50mm | 1000 daN<br>10 kN | 3000 daN<br>30 kN | 15 kN    |

Table 6. Used belts

## H.2.4 Wood

|          |               |     |                       |
|----------|---------------|-----|-----------------------|
| Material | $\gamma_{m1}$ | 1.3 | tbl. 2.3. EN 1995-1-1 |
|----------|---------------|-----|-----------------------|

Table 7. Used material factors

| Material          | Weight                | $E_{0.05}$            | $F_{c0k}$            | $F_{mk}$             |
|-------------------|-----------------------|-----------------------|----------------------|----------------------|
| Wood, Spruce, C18 | 320 kg/m <sup>3</sup> | 6.0 kN/m <sup>2</sup> | 18 N/mm <sup>2</sup> | 18 N/mm <sup>2</sup> |

Table 8. Used wooden materials

## H.3. Cross sections

| Profile                        | Material | $\varnothing$<br>mm | G<br>kg/m | A<br>mm <sup>2</sup> | $I_y$<br>mm <sup>4</sup> | $W_{el,y}$<br>mm <sup>3</sup> |
|--------------------------------|----------|---------------------|-----------|----------------------|--------------------------|-------------------------------|
| $\varnothing \approx 100$ mm * | Wood C18 | 100                 | 2.51      | 7854                 | 49087839                 | 98175                         |
| $\varnothing \approx 75$ mm *  | Wood C18 | 75                  | 1.41      | 4418                 | 1553156                  | 41417                         |
| $\varnothing \approx 70$ mm *  | Wood C18 | 70                  | 1.23      | 3848                 | 1178588                  | 33674                         |

Table 9. Used cross sections

\* the average diameter, as a minimum required at the middle of the pole.

## H.4. Calculation method

### H.4.1 Modelling

The analysis of the structure is performed with the software package EASY FCS supplied by TECHNET GmbH, Berlin. This software is specially developed for structures with large deformability, such as membrane structures. The performed analysis is a full non-linear second order analysis.

The membrane structure is modelled in 3D. The membrane is modelled as a cable net structure and is supported by poles. These centre poles will be stabilized by the tensioned membrane. The side poles are stabilized and tied down by tension belts or ropes, which are attached to ground anchors or ballast.

The membrane edges are reinforced. They consist of multiple layers of fabric about 10cm wide with additional patches where the lugs are placed. Storm belts are required constantly and placed in the valleys between the centre poles.

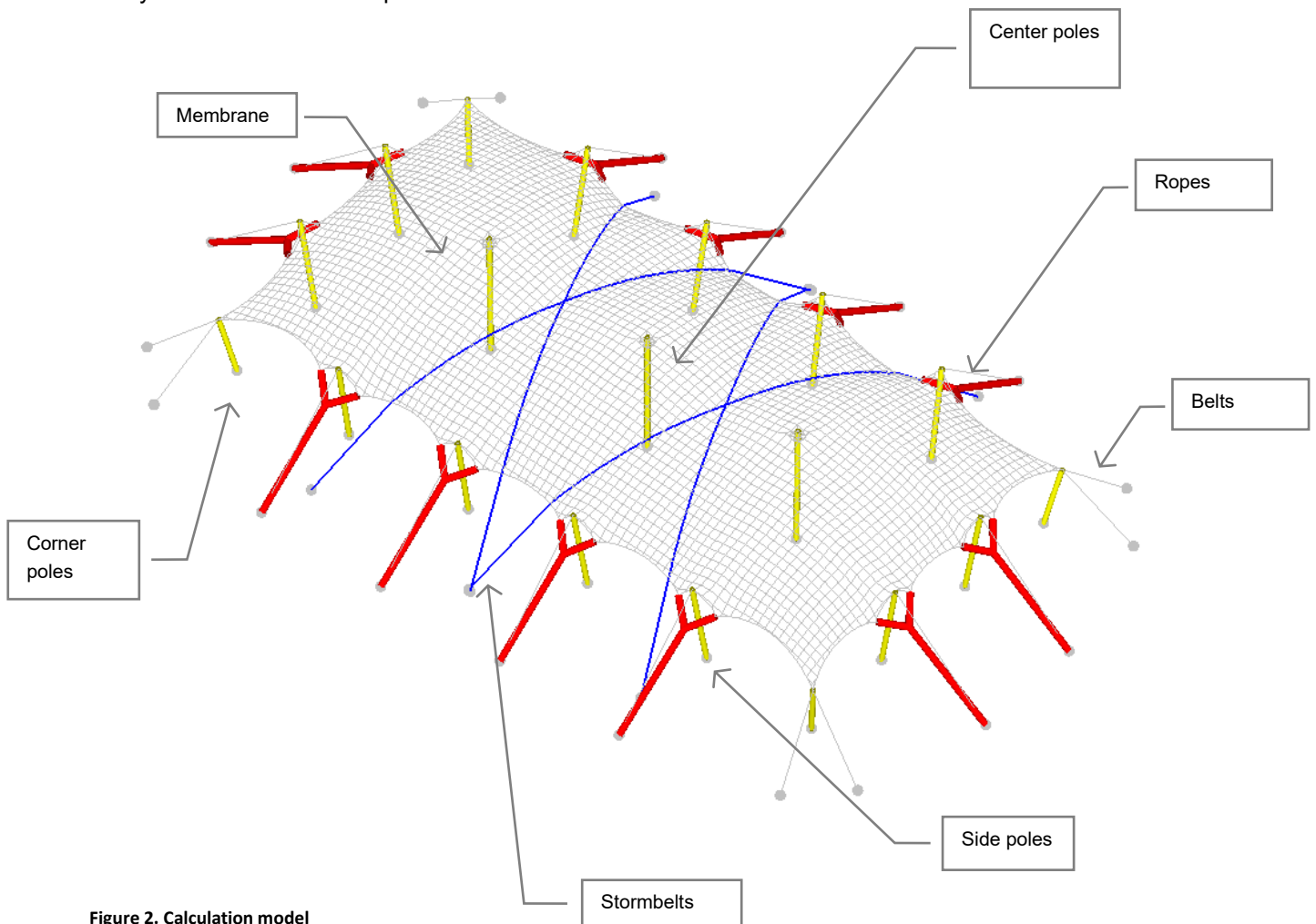
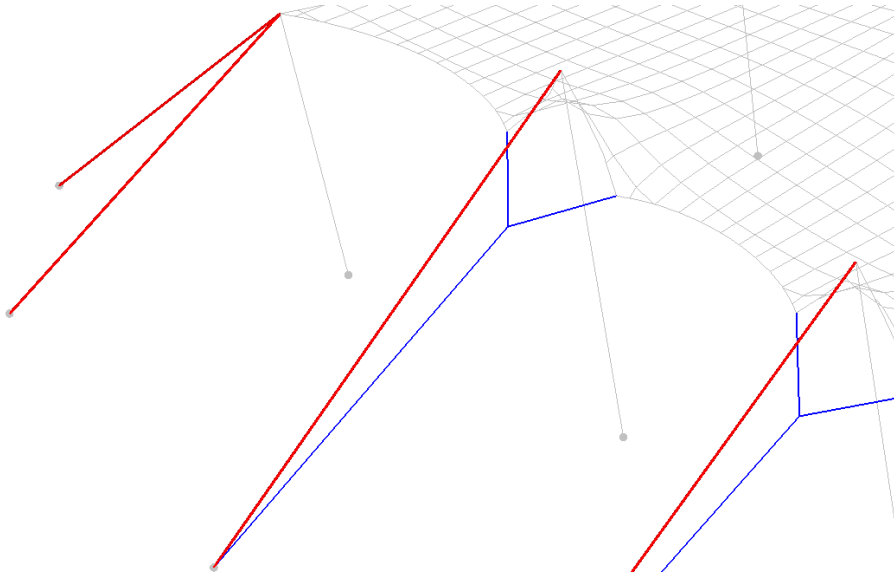


Figure 2. Calculation model



**Figure 3. Principle of ties**

Ties stabilize the tent and create the shape of the stretchable membrane. Guy ropes (see blue above) transfer the forces to the ground when wind suction is present. Belts (see red above) help transfer the pressure forces.

#### H.4.2 Structural behaviour of membrane structures

Since the fabric is a highly deformable material, it is only possible to calculate stresses and deformations with a non-linear method. FEM-software EASY is used to perform these calculations. Because of the non-linearity of the calculations the partial safety factors are not applied beforehand, since the deformations will be greater due to these safety factors, resulting in lower stresses in the fabric. See figure below.

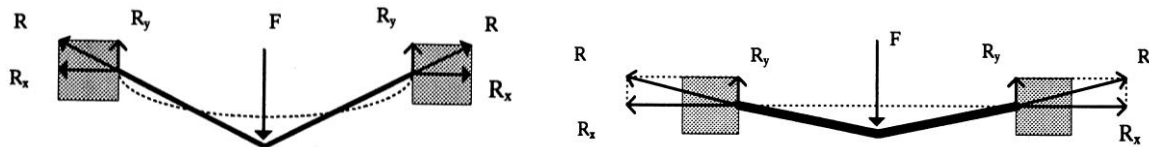


Figure 4. membrane behavior

As a membrane structure is a form-active structure, article 6.3 (4) b) of the EN-1990:2002 applies:

*When the action effect increases less than the action, the partial factor  $\gamma_F$  should be applied to the action effect of the representative value of the action.*

This means that no load factors are applied on the load beforehand, but afterwards.



### H.4.3 Load combinations

#### H.4.3.1 Fundamental - Ultimate limit state

For determination of strength and to check elements and connections ultimate limit state is used.

|                            | One variable load              | Multiple variable loads                |
|----------------------------|--------------------------------|----------------------------------------|
| Unfavorable permanent load | $1.35 \times G + 1.5 \times Q$ | $1.35 \times G + \sum 1.35 \times Q_i$ |
| Favorable permanent load   | $1.0 \times G + 1.5 \times Q$  | $1.0 \times G + \sum 1.35 \times Q_i$  |

**Table 10. Load combinations according to EN 13782**

This means that the following load combinations will be checked/calculated:

1. 1.35 x Own weight + 1.5 x Wind load
2. 1.35 x Own weight + 1.5 x Snow load
3. 1.35 x Own weight + 1.35 x Wind load + 1.35 x Snow load [FR]

#### H.4.3.2 Safety against overturning, sliding and uplifting - Ultimate limit state

For determination and check of the needed ballast weight and/or anchors ultimate limit state is used

|                            | One or multiple variable loads                             |
|----------------------------|------------------------------------------------------------|
| Unfavorable permanent load | $1.1 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$ |
| Favorable permanent load   | $1.0 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$ |

**Table 11. Load combinations according to EN 13782**

This means the following load combinations will be checked/calculated:

1. 1.0 x Own weight + 1.2 x Wind load
2. 1.0 x Own weight + 1.2 x Wind load + 1.3 x Snow load [FR]

## H.5. Load cases

### H.5.1 Own weight

The own weight of the fabric is  $730 \text{ g/m}^2 = 0.0073 \text{ kN/m}^2$  and is considered automatically within EASY software.

### H.5.2 Pretension

The structure will be pretensioned with guy ropes and belts. With a ratchet it is possible to achieve 2-3 kN force in corner belts. Approximately 2kN force is achieved in side ropes. The tensile force in ropes and belts results in a membrane stress of approximately  $0.50 \text{ kN/m}^2$ . Higher membrane forces can be present near connection points due to the stretchable nature of the fabric.

### H.5.3 Conventional / snow load

Conventional load according to EN 13782: The stability shall be checked with a conventional vertical load of  $0.1 \text{ kN/m}^2$ . This can be seen as a snow load of  $0.1 \text{ kN/m}^2$  (4cm) according to the French CTS, where snow load must be combined with wind load. This condition is tackled in J. Addition 2 – element check for combined wind and snow load on page 80.

### H.5.4 User load

A user defined load (for light, sound and/or decoration purposes) is 25 kg per pole and is added after the analysis while performing element checks.

### H.5.5 Wind load

#### H.5.5.1 Wind pressure

Wind load according to EN 13782, 7.4.2.2:

$q_p(z_e)$  of  $0.50 \text{ kN/m}^2$  if the height of the structure does not exceed 5m.

Wind load can be reduced for use in the Netherlands. A reduction factor  $\alpha=0.7$  is applied, which leads to  $q_p(z_e)$  of  $0.25 \text{ kN/m}^2$  [NL], this is tackled in I. Addition 1 – element check for reduced wind speeds [NL]

#### H.5.5.2 Wind shape values (Cp-factors)

Two different wind situations are reviewed for the membrane:

1. The whole tent is subjected to wind suction (Cp values given in EN 13782)
2. The whole tent is subjected to wind pressure (Cp values given in EN 13782)

#### Wind suction

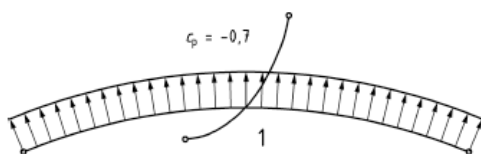


Figure 5.  $C_p$  values for rectangle structures EN 13782

| $P_w$                | $p_{w,rep}$                                               |
|----------------------|-----------------------------------------------------------|
| 500 N/m <sup>2</sup> | $-0.7 \times 0.500 = \mathbf{-0.350 \text{ kN/m}^2}$      |
| 350 N/m <sup>2</sup> | $-0.7 \times 0.250 = \mathbf{-0.175 \text{ kN/m}^2}$ [NL] |

### Wind pressure

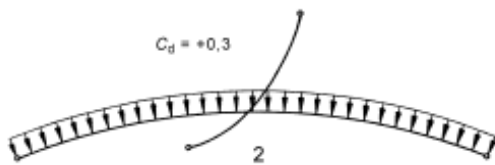


Figure 6.  $C_p$  values for rectangle structures EN 13782

| $P_w$                | $p_{w,rep}$                                        |
|----------------------|----------------------------------------------------|
| 500 N/m <sup>2</sup> | $0.3 \times 0.500 = \mathbf{0.150 \text{ kN/m}^2}$ |

## H.6. Calculation results

### H.6.1 Listing of calculated load combinations

LC1 = Pretension

LC2 = Own weight

LC3 = Snow load

LC4 = Wind pressure

LC5 = Wind suction

LC6 = Wind suction reduced

The following load combinations are considered:

*partial safety factors are added after the static analysis (see H.4.2).*

|          | LC1 | LC2 | LC3 | LC4 | LC5 | LC6 [NL] |
|----------|-----|-----|-----|-----|-----|----------|
| CO1      | 1 x | 1 x |     |     |     |          |
| CO2      | 1 x | 1 x | 1 x |     |     |          |
| CO3      | 1 x | 1 x |     | 1 x |     |          |
| CO4      | 1 x | 1 x |     |     | 1 x |          |
| CO5 [NL] | 1 x | 1 x |     |     |     | 1 x      |
| CO6 [FR] | 1 x | 1 x | 1 x | 1 x |     |          |

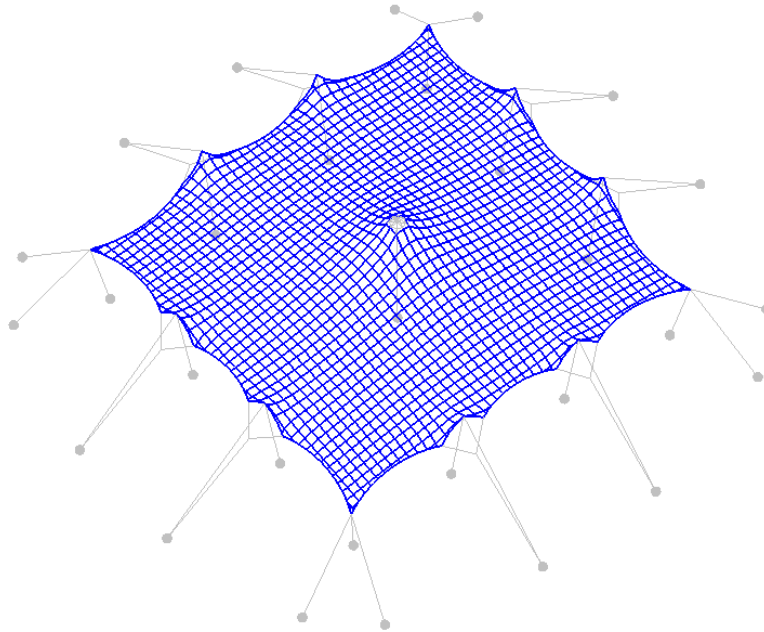
**Table 12. Combinations (CO)**

The results for load combinations CO5 and CO6 are mentioned in the upcoming chapter but considered in calculations only in chapters I and J on pages 70 and 80 respectively.

## H.6.2 Overview: Global results of static analysis

### H.6.2.1 Membrane

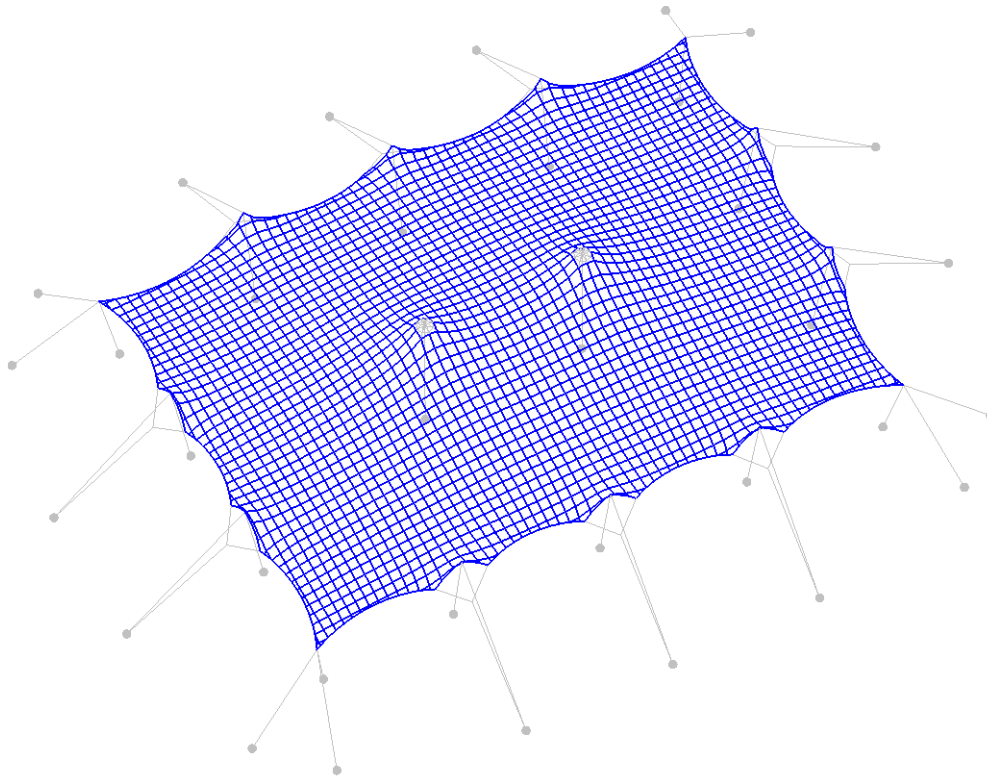
#### H.6.2.1.1. 10x10m model



|      | Load combination                                         | $F_{rep}$ | Pag. |
|------|----------------------------------------------------------|-----------|------|
| Warp | CO1. Own weight + pretension                             | 0.95 kN/m | 119  |
|      | CO2. Own weight + pretension + snow                      | 2.42 kN/m | 123  |
|      | CO3. Own weight + pretension + wind pressure             | 3.09 kN/m | 127  |
|      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.39 kN/m | 131  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.49 kN/m | 135  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.55 kN/m | 139  |
| Weft | CO1. Own weight + pretension                             | 0.91 kN/m | 119  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.11 kN/m | 123  |
|      | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 2.62 kN/m | 127  |
|      | CO4. Own weight + pretension + wind suction              | 2.18 kN/m | 131  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 1.99 kN/m | 135  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.79 kN/m | 139  |
| Edge | CO1. Own weight + pretension                             | 1.76 kN   | 119  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.13 kN   | 123  |
|      | CO3. Own weight + pretension + wind pressure             | 2.44 kN   | 127  |
|      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.41 kN   | 131  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.68 kN   | 135  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.85 kN   | 139  |

Table 13. Leading forces in membrane – 10x10m

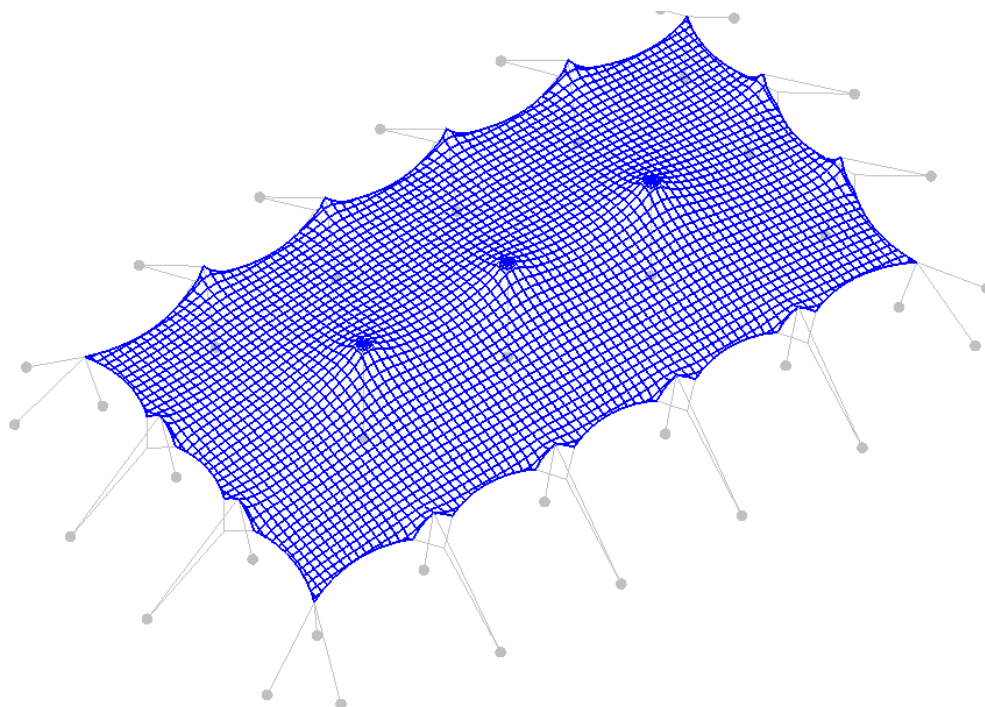
#### H.6.2.1.2. 10x15m model



|      | Load combination                                         | $F_{rep}$ | Pag. |
|------|----------------------------------------------------------|-----------|------|
| Warp | CO1. Own weight + pretension                             | 0.91 kN/m | 143  |
|      | CO2. Own weight + pretension + snow                      | 2.59 kN/m | 147  |
|      | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 3.35 kN/m | 151  |
|      | CO4. Own weight + pretension + wind suction              | 3.13 kN/m | 155  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.51 kN/m | 159  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.97 kN/m | 163  |
| Weft | CO1. Own weight + pretension                             | 0.89 kN/m | 143  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.18 kN/m | 147  |
|      | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 2.93 kN/m | 151  |
|      | CO4. Own weight + pretension + wind suction              | 2.27 kN/m | 155  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.48 kN/m | 159  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.24 kN/m | 163  |
| Edge | CO1. Own weight + pretension                             | 1.90 kN   | 143  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.35 kN   | 147  |
|      | CO3. Own weight + pretension + wind pressure             | 2.74 kN   | 151  |
|      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.55 kN   | 155  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 3.17 kN   | 159  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.19 kN   | 163  |

Table 14. Leading forces in membrane – 10x15m

## H.6.2.1.3. 10x20m model

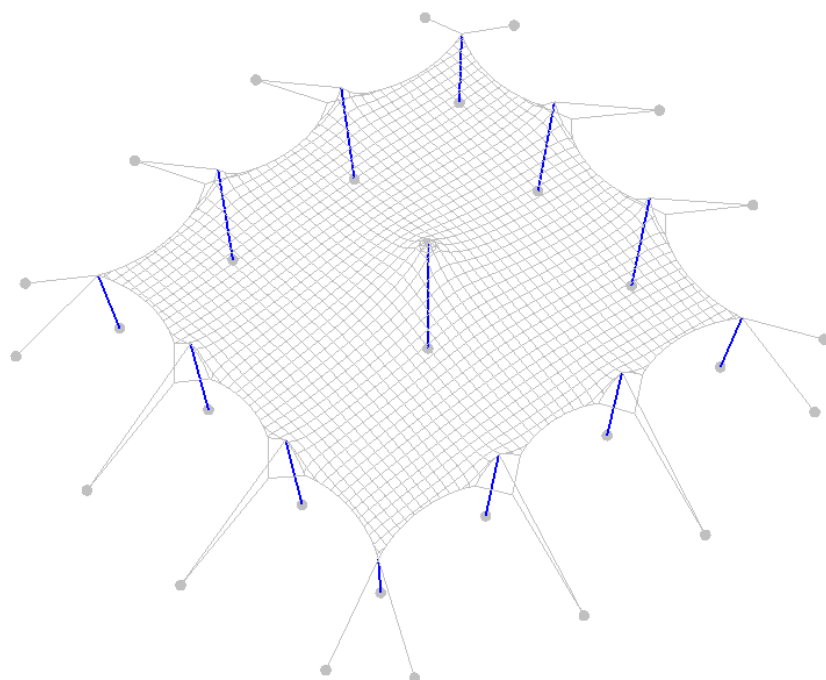


|      | Load combination                                         | F <sub>rep</sub> | Pag. |
|------|----------------------------------------------------------|------------------|------|
| Warp | CO1. Own weight + pretension                             | 0.95 kN/m        | 167  |
|      | CO2. Own weight + pretension + snow                      | 2.63 kN/m        | 171  |
|      | CO3. Own weight + pretension + wind pressure             | 3.41 kN/m        | 175  |
|      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.45 kN/m        | 179  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.53 kN/m        | 183  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 5.01 kN/m        | 187  |
| Weft | CO1. Own weight + pretension                             | 0.88 kN/m        | 167  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.16 kN/m        | 171  |
|      | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 2.94 kN/m        | 175  |
|      | CO4. Own weight + pretension + wind suction              | 2.89 kN/m        | 179  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.68 kN/m        | 183  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.20 kN/m        | 187  |
| Edge | CO1. Own weight + pretension                             | 1.80 kN          | 167  |
|      | CO2. Own weight + pretension + conventional / snow       | 2.26 kN          | 171  |
|      | CO3. Own weight + pretension + wind pressure             | 2.67 kN          | 175  |
|      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.43 kN          | 179  |
|      | CO5. Own weight + pretension + wind suction reduced [NL] | 3.43 kN          | 183  |
|      | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.16 kN          | 187  |

Table 15. Leading forces in membrane – 10x20m

## H.6.2.2 Poles

### H.6.2.2.1. 10x10m model

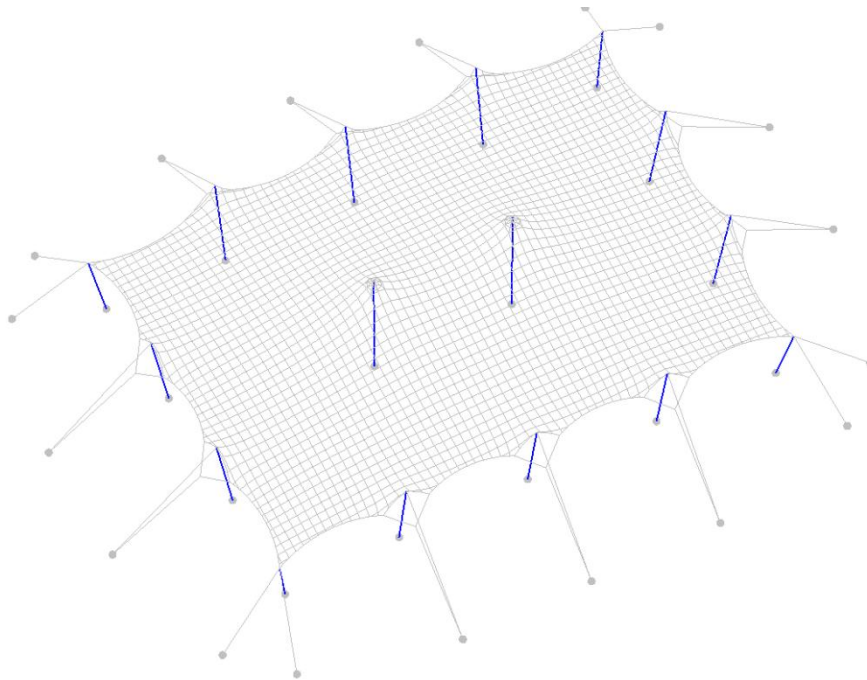


|              | Load combination                                         | $F_{rep}$ | Pag. |
|--------------|----------------------------------------------------------|-----------|------|
| Main poles   | CO1. Own weight + pretension                             | -2.12 kN  | 119  |
|              | CO2. Own weight + pretension + snow                      | -5.41 kN  | 123  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -6.95 kN  | 127  |
|              | CO4. Own weight + pretension + wind suction              | -0.05 kN  | 131  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -0.08 kN  | 135  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -10.32 kN | 139  |
| Side poles   | CO1. Own weight + pretension                             | -1.30 kN  | 119  |
|              | CO2. Own weight + pretension + conventional / snow       | -2.43 kN  | 123  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -3.13 kN  | 127  |
|              | CO4. Own weight + pretension + wind suction              | -2.02 kN  | 131  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -0.83 kN  | 135  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -4.33 kN  | 139  |
| Corner poles | CO1. Own weight + pretension                             | -2.64 kN  | 119  |
|              | CO2. Own weight + pretension + conventional / snow       | -3.38 kN  | 123  |
|              | CO3. Own weight + pretension + wind pressure             | -3.95 kN  | 127  |
|              | <b>Max</b> CO4. Own weight + pretension + wind suction   | -4.52 kN  | 131  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -3.43 kN  | 135  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -4.75 kN  | 139  |

Table 16. Leading forces in poles – 10x10m model



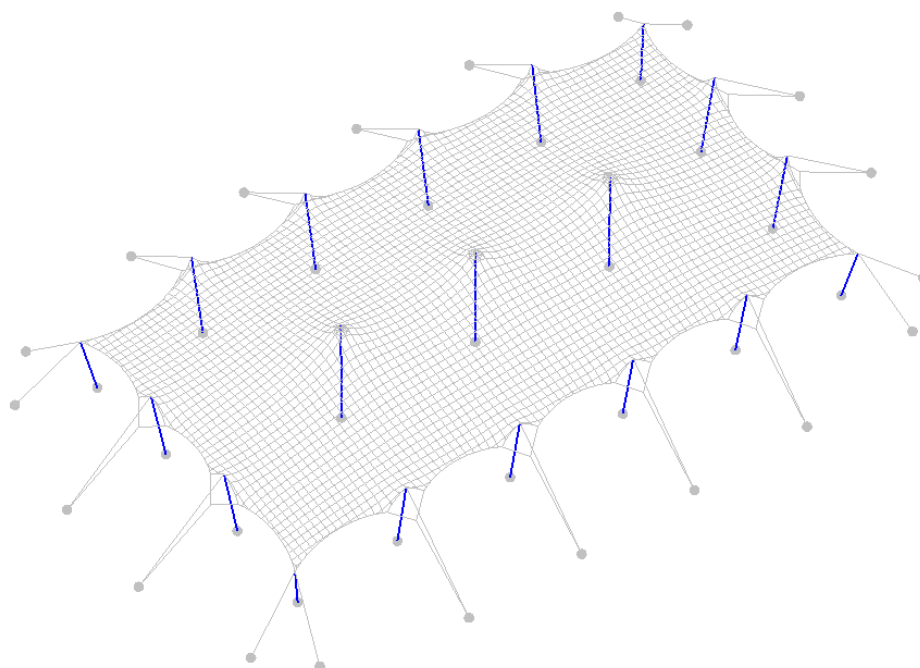
#### H.6.2.2.2. 10x15m model



|              | Load combination                                         | F <sub>rep</sub> | Pag. |
|--------------|----------------------------------------------------------|------------------|------|
| Main poles   | CO1. Own weight + pretension                             | -1.74 kN         | 143  |
|              | CO2. Own weight + pretension + snow                      | -4.96 kN         | 147  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -6.49 kN         | 151  |
|              | CO4. Own weight + pretension + wind suction              | 0.06 kN          | 155  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -0.02 kN         | 159  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -9.78 kN         | 163  |
| Side poles   | CO1. Own weight + pretension                             | -1.37 kN         | 143  |
|              | CO2. Own weight + pretension + conventional / snow       | -2.74 kN         | 147  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -3.60 kN         | 151  |
|              | CO4. Own weight + pretension + wind suction              | -2.47 kN         | 155  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -0.84 kN         | 159  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -4.99 kN         | 163  |
| Corner poles | CO1. Own weight + pretension                             | -2.86 kN         | 143  |
|              | CO2. Own weight + pretension + conventional / snow       | -3.81 kN         | 147  |
|              | CO3. Own weight + pretension + wind pressure             | -4.56 kN         | 151  |
|              | <b>Max</b> CO4. Own weight + pretension + wind suction   | -4.77 kN         | 155  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -3.87 kN         | 159  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -5.50 kN         | 163  |

Table 17. Leading forces in poles – 10x15m model

#### H.6.2.2.3. 10x20m model

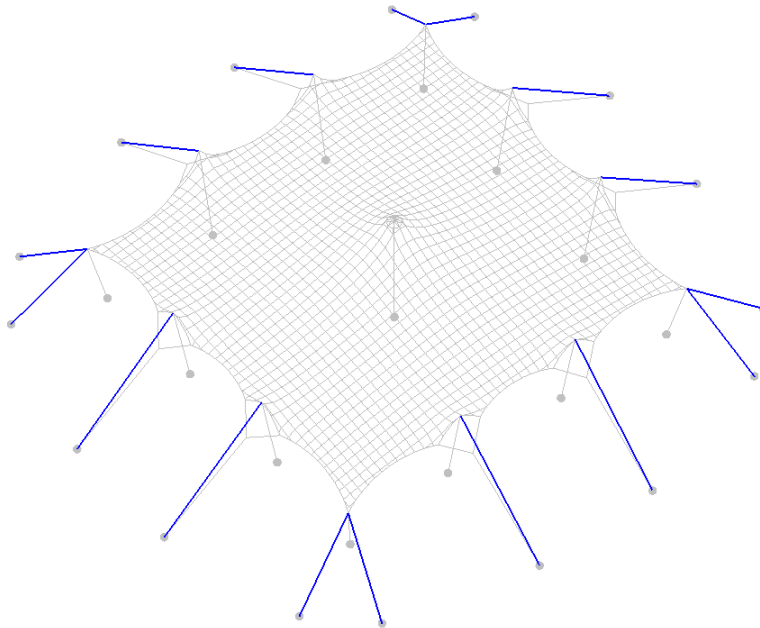


|              | Load combination                                         | F <sub>rep</sub> | Pag. |
|--------------|----------------------------------------------------------|------------------|------|
| Main poles   | CO1. Own weight + pretension                             | -1.84 kN         | 167  |
|              | CO2. Own weight + pretension + snow                      | -5.15 kN         | 171  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -6.75 kN         | 175  |
|              | CO4. Own weight + pretension + wind suction              | -0.06 kN         | 179  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | 0.03 kN          | 183  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -10.07 kN        | 187  |
| Side poles   | CO1. Own weight + pretension                             | -1.35 kN         | 167  |
|              | CO2. Own weight + pretension + conventional / snow       | -2.69 kN         | 171  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -3.55 kN         | 175  |
|              | CO4. Own weight + pretension + wind suction              | -2.21 kN         | 179  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -0.80 kN         | 183  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -4.88 kN         | 187  |
| Corner poles | CO1. Own weight + pretension                             | -2.73 kN         | 167  |
|              | CO2. Own weight + pretension + conventional / snow       | -3.67 kN         | 171  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | -4.43 kN         | 175  |
|              | CO4. Own weight + pretension + wind suction              | -4.31 kN         | 179  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | -3.63 kN         | 183  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | -5.40 kN         | 187  |

Table 18. Leading forces in poles – 10x20m model

### H.6.2.3 Belts

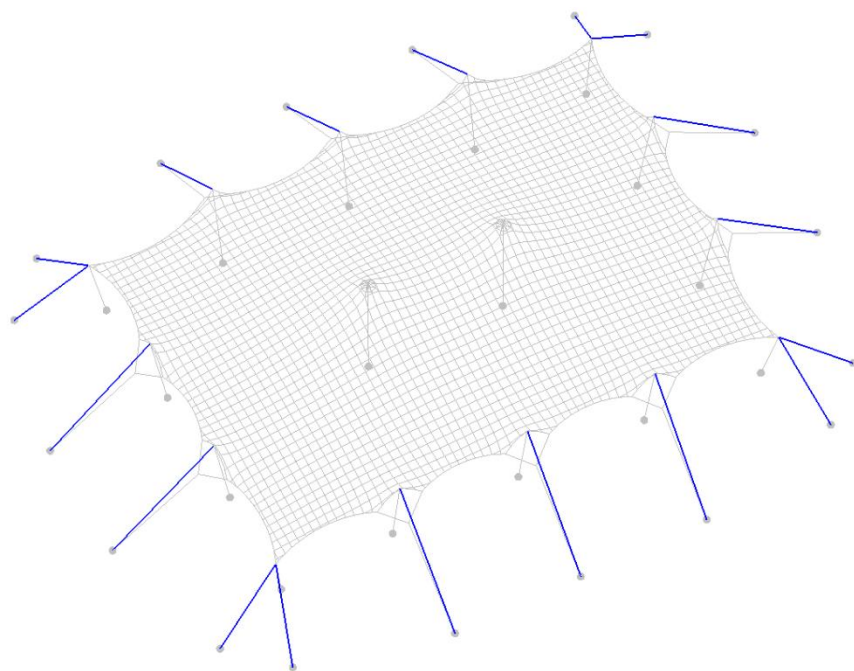
#### H.6.2.3.1. 10x10m model



|              | Load combination                                         | $F_{rep}$ | Pag. |
|--------------|----------------------------------------------------------|-----------|------|
| Corner belts | CO1. Own weight + pretension                             | 2.10 kN   | 119  |
|              | CO2. Own weight + pretension + snow                      | 2.50 kN   | 123  |
|              | CO3. Own weight + pretension + wind pressure             | 2.86 kN   | 127  |
|              | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.94 kN   | 131  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | 3.19 kN   | 135  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.33 kN   | 139  |
| Belts – side | CO1. Own weight + pretension                             | 0.12 kN   | 119  |
|              | CO2. Own weight + pretension + conventional / snow       | 0.47 kN   | 123  |
|              | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 0.70 kN   | 127  |
|              | CO4. Own weight + pretension + wind suction              | 0.35 kN   | 131  |
|              | CO5. Own weight + pretension + wind suction reduced [NL] | 0.16 kN   | 135  |
|              | CO6. Own weight + pretension + wind pressure + snow [FR] | 1.13 kN   | 139  |

Table 19. Leading forces in belts – 10x10m model

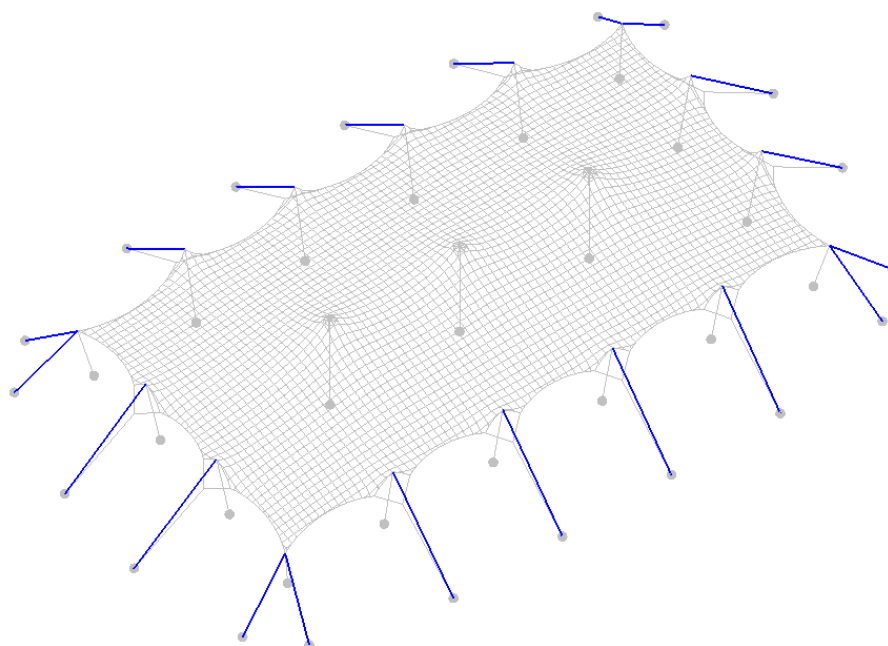
#### H.6.2.3.2. 10x15m model



|                    | Load combination                                         | F <sub>rep</sub> | Pag. |
|--------------------|----------------------------------------------------------|------------------|------|
| Corner belts       | CO1. Own weight + pretension                             | 2.31 kN          | 143  |
|                    | CO2. Own weight + pretension + snow                      | 2.85 kN          | 147  |
|                    | CO3. Own weight + pretension + wind pressure             | 3.28 kN          | 151  |
|                    | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.22 kN          | 155  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 3.51 kN          | 159  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.81 kN          | 163  |
| Belts – long side  | CO1. Own weight + pretension                             | 0.14 kN          | 143  |
|                    | CO2. Own weight + pretension + conventional / snow       | 0.51 kN          | 147  |
|                    | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 0.81 kN          | 151  |
|                    | CO4. Own weight + pretension + wind suction              | 0.50 kN          | 155  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 0.18 kN          | 159  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 1.30 kN          | 163  |
| Belts – short side | CO1. Own weight + pretension                             | 0.12 kN          | 143  |
|                    | CO2. Own weight + pretension + conventional / snow       | 0.56 kN          | 147  |
|                    | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 0.84 kN          | 151  |
|                    | CO4. Own weight + pretension + wind suction              | 0.22 kN          | 155  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 0.18 kN          | 159  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 1.37 kN          | 163  |

Table 20. Leading forces in belts – 10x15m model

#### H.6.2.3.3. 10x20m model

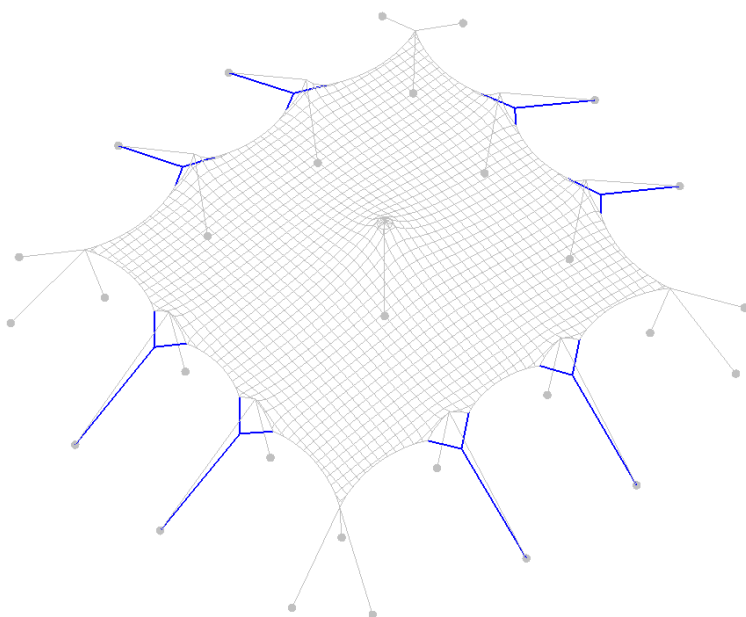


|                    | Load combination                                         | F <sub>rep</sub> | Pag. |
|--------------------|----------------------------------------------------------|------------------|------|
| Corner belts       | CO1. Own weight + pretension                             | 2.33 kN          | 167  |
|                    | CO2. Own weight + pretension + snow                      | 2.95 kN          | 171  |
|                    | CO3. Own weight + pretension + wind pressure             | 3.44 kN          | 175  |
|                    | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.43 kN          | 179  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 3.44 kN          | 183  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.01 kN          | 187  |
| Belts – long side  | CO1. Own weight + pretension                             | 0.22 kN          | 167  |
|                    | CO2. Own weight + pretension + conventional / snow       | 0.70 kN          | 171  |
|                    | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 1.01 kN          | 175  |
|                    | CO4. Own weight + pretension + wind suction              | 0.57 kN          | 179  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 0.26 kN          | 183  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 1.50 kN          | 187  |
| Belts – short side | CO1. Own weight + pretension                             | 0.13 kN          | 167  |
|                    | CO2. Own weight + pretension + conventional / snow       | 0.55 kN          | 171  |
|                    | <b>Max</b> CO3. Own weight + pretension + wind pressure  | 0.85 kN          | 175  |
|                    | CO4. Own weight + pretension + wind suction              | 0.13 kN          | 179  |
|                    | CO5. Own weight + pretension + wind suction reduced [NL] | 0.13 kN          | 183  |
|                    | CO6. Own weight + pretension + wind pressure + snow [FR] | 1.35 kN          | 187  |

Table 21. Leading forces in belts – 10x20m model

## H.6.2.4 Guy ropes

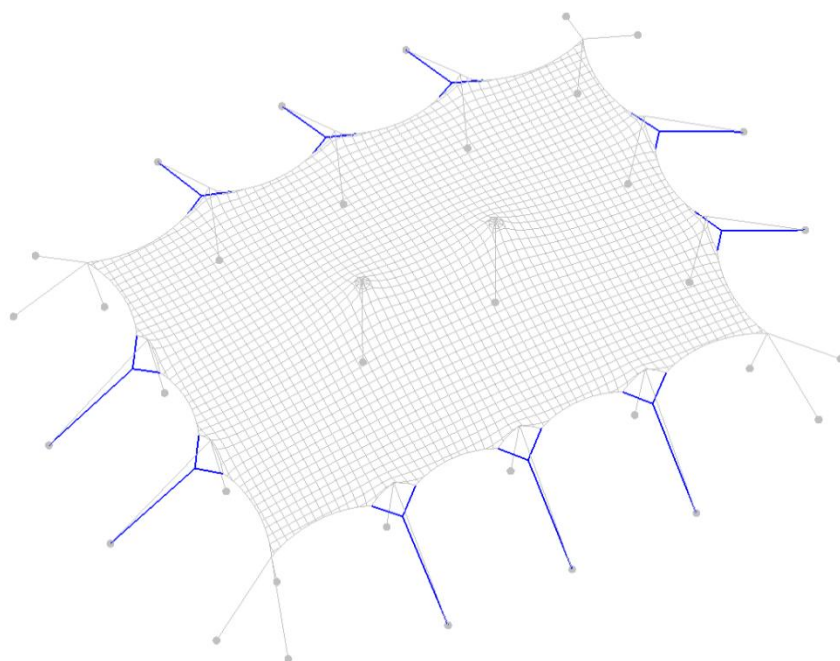
### H.6.2.4.1. 10x10m model



|                      | Load combination                                         | F <sub>rep</sub> | Pag. |
|----------------------|----------------------------------------------------------|------------------|------|
| Combined rope – side | CO1. Own weight + pretension                             | 1.83 kN          | 119  |
|                      | CO2. Own weight + pretension + snow                      | 2.12 kN          | 123  |
|                      | CO3. Own weight + pretension + wind pressure             | 2.41 kN          | 127  |
|                      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.83 kN          | 131  |
|                      | CO5. Own weight + pretension + wind suction reduced [NL] | 3.64 kN          | 135  |
|                      | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.76 kN          | 139  |
| V-ropes –side        | CO1. Own weight + pretension                             | 1.40 kN          | 119  |
|                      | CO2. Own weight + pretension + conventional / snow       | 1.63 kN          | 123  |
|                      | CO3. Own weight + pretension + wind pressure             | 1.84 kN          | 127  |
|                      | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.51 kN          | 131  |
|                      | CO5. Own weight + pretension + wind suction reduced [NL] | 2.72 kN          | 135  |
|                      | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.10 kN          | 139  |

Table 22. Leading forces in ropes – 10x10m model

#### H.6.2.4.2. 10x15m model

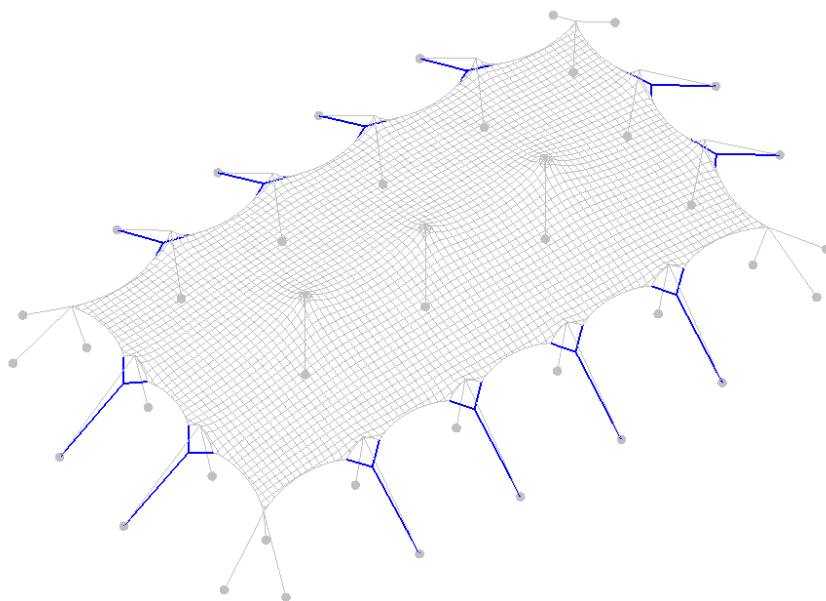


|                            | Load combination                                         | F <sub>rep</sub> | Pag. |
|----------------------------|----------------------------------------------------------|------------------|------|
| Combined rope – long side  | CO1. Own weight + pretension                             | 2.11 kN          | 143  |
|                            | CO2. Own weight + pretension + snow                      | 2.59 kN          | 147  |
|                            | CO3. Own weight + pretension + wind pressure             | 3.04 kN          | 151  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.70 kN          | 155  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 4.66 kN          | 159  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.50 kN          | 163  |
| V-ropes – long side        | CO1. Own weight + pretension                             | 1.53 kN          | 143  |
|                            | CO2. Own weight + pretension + conventional / snow       | 1.83 kN          | 147  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.11 kN          | 151  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.38 kN          | 155  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.30 kN          | 159  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.39 kN          | 163  |
| Combined rope – short side | CO1. Own weight + pretension                             | 1.92 kN          | 143  |
|                            | CO2. Own weight + pretension + conventional / snow       | 2.38 kN          | 147  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.77 kN          | 151  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.01 kN          | 155  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 4.04 kN          | 159  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.21 kN          | 163  |
| V-ropes – short side       | CO1. Own weight + pretension                             | 1.45 kN          | 143  |
|                            | CO2. Own weight + pretension + conventional / snow       | 1.80 kN          | 147  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.09 kN          | 151  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.64 kN          | 155  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 2.94 kN          | 159  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.41 kN          | 163  |

Table 23. Leading forces in ropes – 10x15m model



#### H.6.2.4.3. 10x20m model



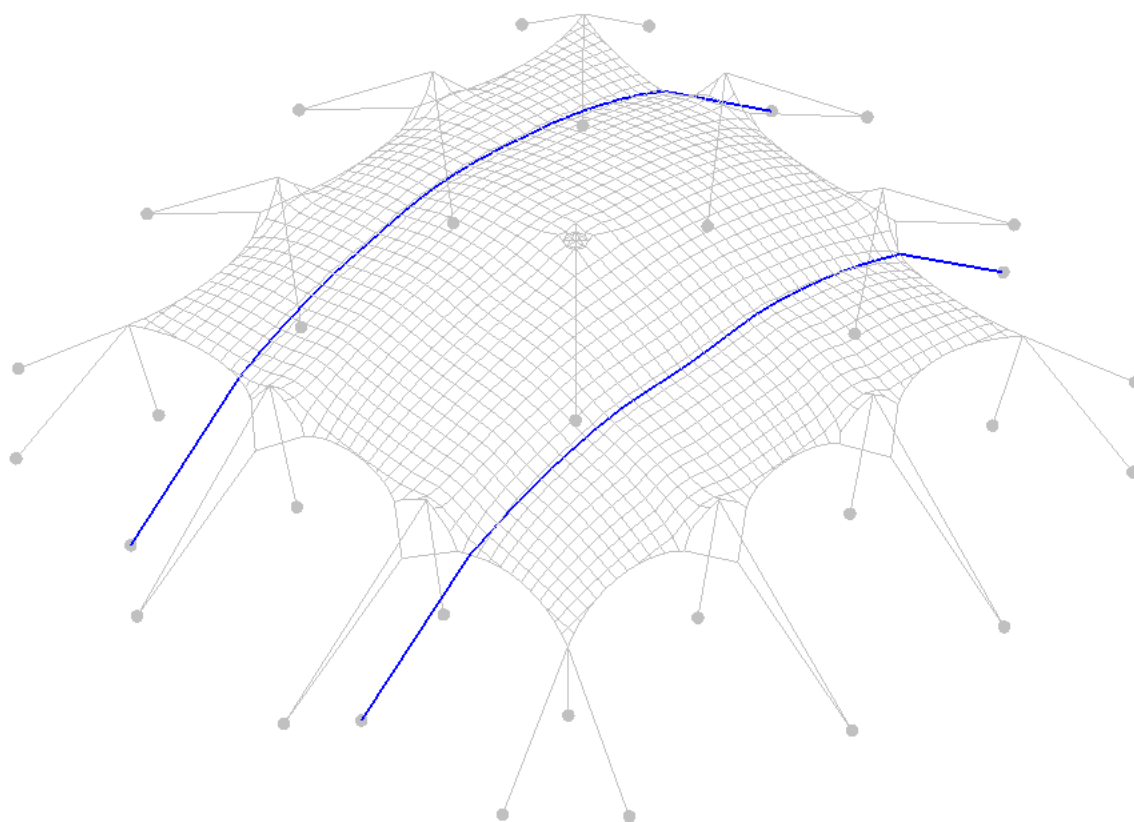
|                            | Load combination                                         | F <sub>rep</sub> | Pag. |
|----------------------------|----------------------------------------------------------|------------------|------|
| Combined rope – long side  | CO1. Own weight + pretension                             | 2.16 kN          | 167  |
|                            | CO2. Own weight + pretension + snow                      | 2.48 kN          | 171  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.91 kN          | 175  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.43 kN          | 179  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 5.46 kN          | 183  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.34 kN          | 187  |
| V-ropes – long side        | CO1. Own weight + pretension                             | 1.50 kN          | 167  |
|                            | CO2. Own weight + pretension + conventional / snow       | 1.76 kN          | 171  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.05 kN          | 175  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.91 kN          | 179  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.68 kN          | 183  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.28 kN          | 187  |
| Combined rope – short side | CO1. Own weight + pretension                             | 1.87 kN          | 167  |
|                            | CO2. Own weight + pretension + conventional / snow       | 2.42 kN          | 171  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.89 kN          | 175  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.46 kN          | 179  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.52 kN          | 183  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.39 kN          | 187  |
| V-ropes – short side       | CO1. Own weight + pretension                             | 1.37 kN          | 167  |
|                            | CO2. Own weight + pretension + conventional / snow       | 1.80 kN          | 171  |
|                            | CO3. Own weight + pretension + wind pressure             | 2.14 kN          | 175  |
|                            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.83 kN          | 179  |
|                            | CO5. Own weight + pretension + wind suction reduced [NL] | 2.54 kN          | 183  |
|                            | CO6. Own weight + pretension + wind pressure + snow [FR] | 2.51 kN          | 187  |

Table 24. Leading forces in ropes – 10x20m model



### H.6.2.5 Stormbelts

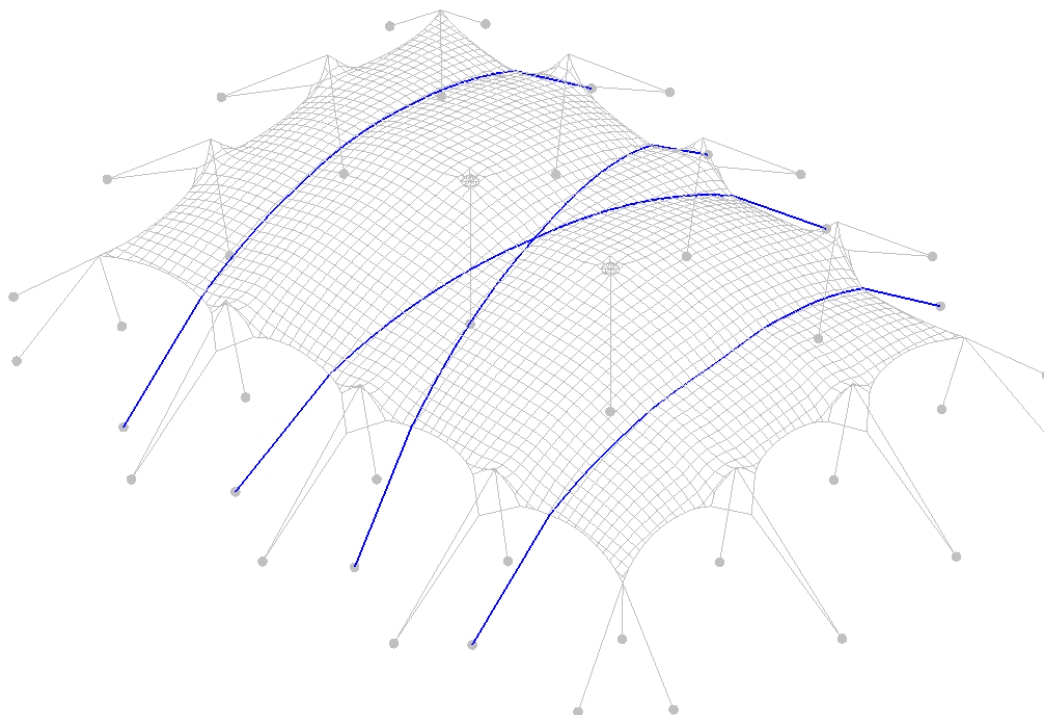
#### H.6.2.5.1. 10x10m model



|            | Load combination                                         | $F_{rep}$ | Pag. |
|------------|----------------------------------------------------------|-----------|------|
| Stormbelts | CO1. Own weight + pretension                             | 0.00 kN   | 119  |
|            | CO2. Own weight + pretension + snow                      | 0.00 kN   | 123  |
|            | CO3. Own weight + pretension + wind pressure             | 0.00 kN   | 127  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.62 kN   | 131  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 0.00 kN   | 135  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 0.00 kN   | 139  |

Table 25. Leading forces in stormbelts – 10x10m model

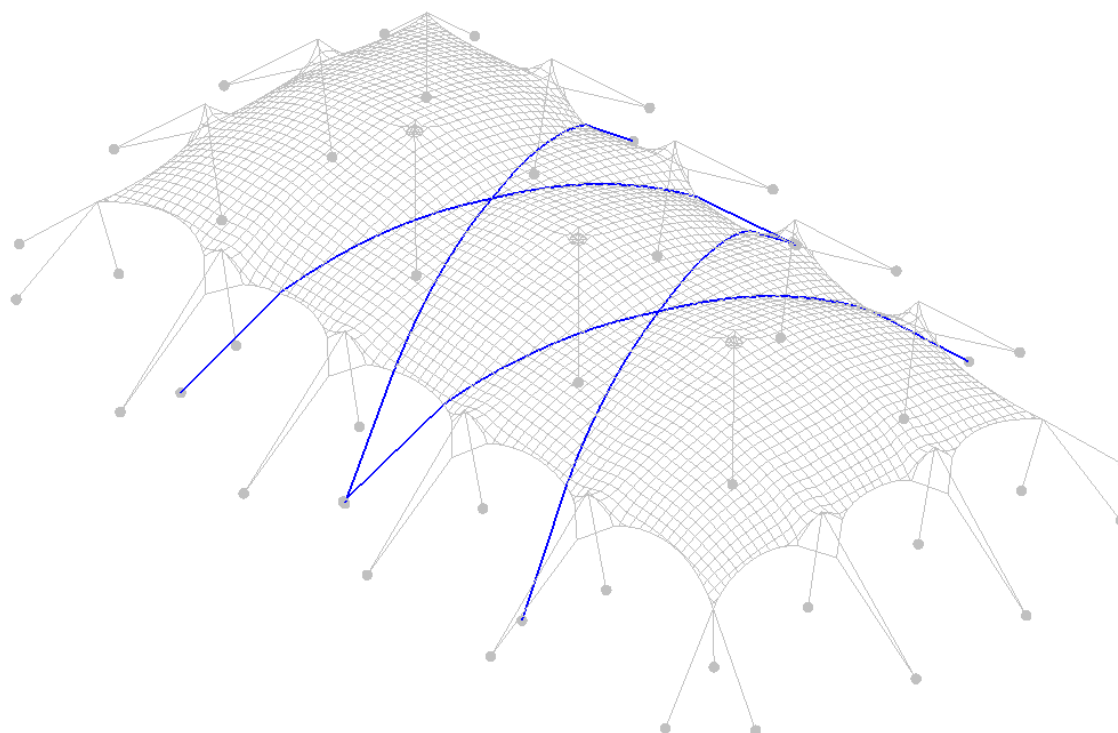
#### H.6.2.5.2. 10x15m model



|            | Load combination                                         | $F_{rep}$ | Pag. |
|------------|----------------------------------------------------------|-----------|------|
| Stormbelts | CO1. Own weight + pretension                             | 0.00 kN   | 143  |
|            | CO2. Own weight + pretension + snow                      | 0.00 kN   | 147  |
|            | CO3. Own weight + pretension + wind pressure             | 0.00 kN   | 151  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 6.01 kN   | 155  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 0.00 kN   | 159  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 0.00 kN   | 163  |

**Table 26. Leading forces in stormbelts – 10x15m model**

#### H.6.2.5.3. 10x20m model

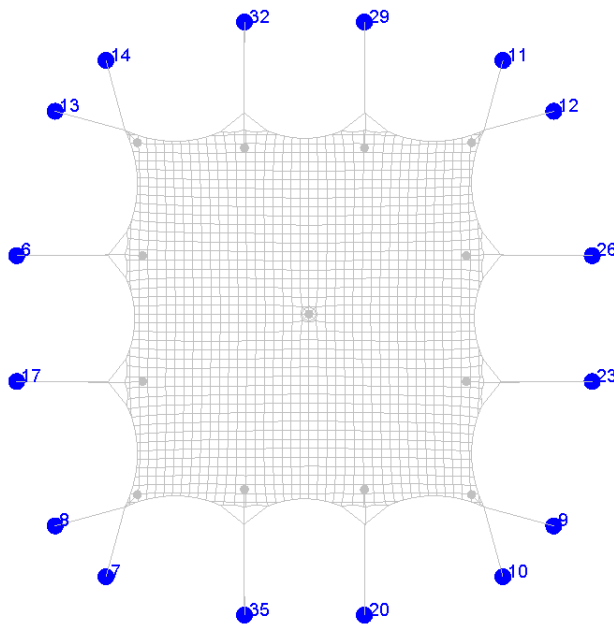


|            | Load combination                                         | $F_{rep}$ | Pag. |
|------------|----------------------------------------------------------|-----------|------|
| Stormbelts | CO1. Own weight + pretension                             | 0.00 kN   | 167  |
|            | CO2. Own weight + pretension + snow                      | 0.00 kN   | 171  |
|            | CO3. Own weight + pretension + wind pressure             | 0.00 kN   | 175  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.93 kN   | 179  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 0.00 kN   | 183  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 0.00 kN   | 187  |

**Table 27. Leading forces in stormbelts – 10x20m model**

## H.6.2.6 Reaction forces

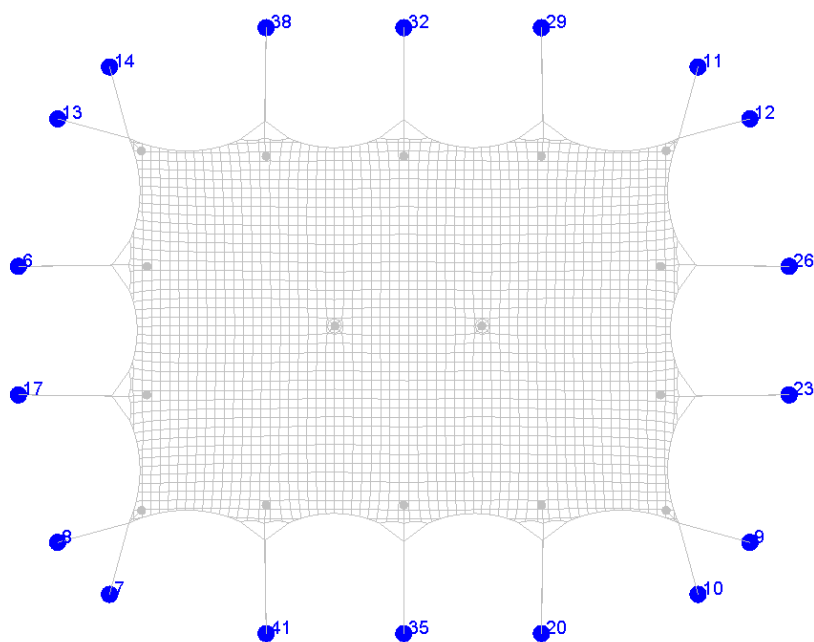
### H.6.2.6.1. 10x10m model



|        | Load combination                                         | F <sub>rep</sub> | Pag. |
|--------|----------------------------------------------------------|------------------|------|
| Side   | CO1. Own weight + pretension                             | 1.94 kN          | 119  |
|        | CO2. Own weight + pretension + snow                      | 2.59 kN          | 123  |
|        | CO3. Own weight + pretension + wind pressure             | 3.10 kN          | 127  |
|        | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.04 kN          | 131  |
|        | CO5. Own weight + pretension + wind suction reduced [NL] | 3.80 kN          | 135  |
|        | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.94 kN          | 139  |
| Corner | CO1. Own weight + pretension                             | 2.10 kN          | 119  |
|        | CO2. Own weight + pretension + conventional / snow       | 2.50 kN          | 123  |
|        | CO3. Own weight + pretension + wind pressure             | 2.86 kN          | 127  |
|        | <b>Max</b> CO4. Own weight + pretension + wind suction   | 3.94 kN          | 131  |
|        | CO5. Own weight + pretension + wind suction reduced [NL] | 3.19 kN          | 135  |
|        | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.36 kN          | 139  |

Table 28. Leading reaction forces – 10x10m model

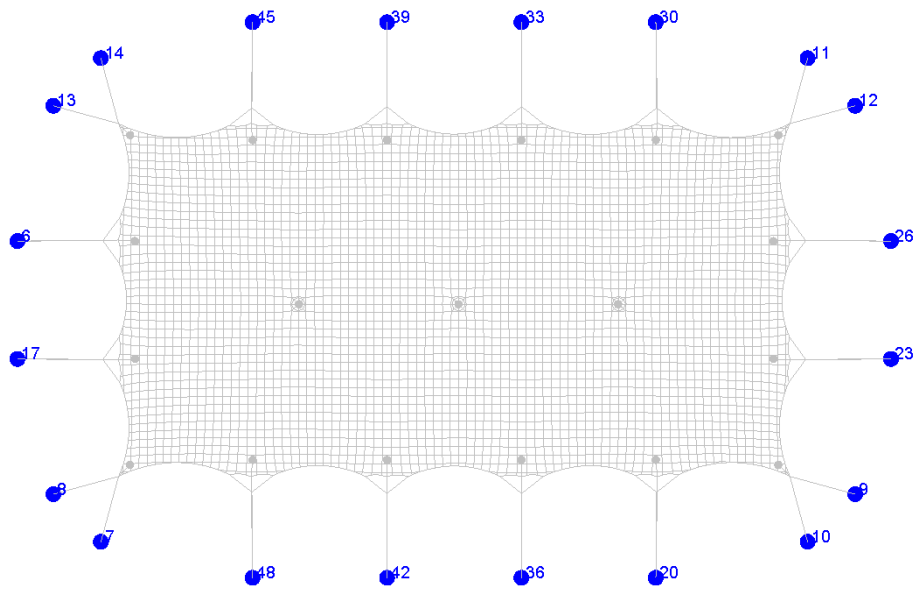
#### H.6.2.6.2. 10x15m model



|            | Load combination                                         | F <sub>rep</sub> | Pag. |
|------------|----------------------------------------------------------|------------------|------|
| Long side  | CO1. Own weight + pretension                             | 2.22 kN          | 143  |
|            | CO2. Own weight + pretension + snow                      | 3.08 kN          | 147  |
|            | CO3. Own weight + pretension + wind pressure             | 3.83 kN          | 151  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.20 kN          | 155  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 4.82 kN          | 159  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.83 kN          | 163  |
| Short side | CO1. Own weight + pretension                             | 2.04 kN          | 143  |
|            | CO2. Own weight + pretension + conventional / snow       | 2.92 kN          | 147  |
|            | CO3. Own weight + pretension + wind pressure             | 3.60 kN          | 151  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.23 kN          | 155  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 4.22 kN          | 159  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.62 kN          | 163  |
| Corner     | CO1. Own weight + pretension                             | 2.31 kN          | 143  |
|            | CO2. Own weight + pretension + conventional / snow       | 2.85 kN          | 147  |
|            | CO3. Own weight + pretension + wind pressure             | 3.28 kN          | 151  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.22 kN          | 155  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.51 kN          | 159  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 3.85 kN          | 163  |

Table 29. Leading reaction forces – 10x15m model

#### H.6.2.6.3. 10x20m model



|            | Load combination                                         | F <sub>rep</sub> | Pag. |
|------------|----------------------------------------------------------|------------------|------|
| Long side  | CO1. Own weight + pretension                             | 2.31 kN          | 167  |
|            | CO2. Own weight + pretension + snow                      | 3.16 kN          | 171  |
|            | CO3. Own weight + pretension + wind pressure             | 3.87 kN          | 175  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.77 kN          | 179  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 5.71 kN          | 183  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.82 kN          | 187  |
| Short side | CO1. Own weight + pretension                             | 1.99 kN          | 167  |
|            | CO2. Own weight + pretension + conventional / snow       | 2.95 kN          | 171  |
|            | CO3. Own weight + pretension + wind pressure             | 3.72 kN          | 175  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 5.58 kN          | 179  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.65 kN          | 183  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.81 kN          | 187  |
| Corner     | CO1. Own weight + pretension                             | 2.33 kN          | 167  |
|            | CO2. Own weight + pretension + conventional / snow       | 2.95 kN          | 171  |
|            | CO3. Own weight + pretension + wind pressure             | 3.44 kN          | 175  |
|            | <b>Max</b> CO4. Own weight + pretension + wind suction   | 4.43 kN          | 179  |
|            | CO5. Own weight + pretension + wind suction reduced [NL] | 3.44 kN          | 183  |
|            | CO6. Own weight + pretension + wind pressure + snow [FR] | 4.05 kN          | 187  |

**Table 30. Leading reaction forces – 10x20m model**

### H.6.3 Summary of forces

#### H.6.3.1 10x10m model

| kN or kN/m                | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|---------------------------|-------|-------|-------|-------|----------|----------|
| Warp                      | 0.95  | 2.42  | 3.09  | 3.39  | 2.49     | 4.55     |
| Weft                      | 0.91  | 2.11  | 2.62  | 2.18  | 1.99     | 3.79     |
| Edge                      | 1.76  | 2.13  | 2.44  | 3.41  | 2.68     | 2.85     |
| Main pole                 | -2.12 | -5.41 | -6.95 | 0.05  | -0.08    | -10.32   |
| Side pole                 | -1.30 | -2.43 | -3.13 | -2.02 | -0.83    | -4.33    |
| Corner pole               | -2.64 | -3.38 | -3.95 | -4.52 | -3.43    | -4.75    |
| Corner belt               | 2.10  | 2.50  | 2.86  | 3.94  | 3.19     | 3.33     |
| Belts - long side         | 0.12  | 0.47  | 0.70  | 0.35  | 0.16     | 1.13     |
| Stormbelts                | 0.00  | 0.00  | 0.00  | 5.62  | 0.00     | 0.00     |
| Combined rope - long side | 1.83  | 2.12  | 2.41  | 4.83  | 3.64     | 2.76     |
| V-ropes - long side       | 1.40  | 1.63  | 1.84  | 3.51  | 2.72     | 2.10     |
| Reaction - long side      | 1.94  | 2.59  | 3.10  | 5.04  | 3.80     | 3.94     |
| Reaction - corner         | 2.10  | 2.50  | 2.86  | 3.94  | 3.19     | 3.36     |

Table 31. Summary of forces for 10x10m model

#### H.6.3.2 10x15m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.91  | 2.59  | 3.35  | 3.13  | 2.51     | 4.97     |
| Weft                       | 0.89  | 2.18  | 2.93  | 2.27  | 2.48     | 4.24     |
| Edge                       | 1.90  | 2.35  | 2.74  | 3.55  | 3.17     | 3.19     |
| Main pole                  | -1.74 | -4.96 | -6.49 | 0.06  | -0.02    | -9.78    |
| Side pole                  | -1.37 | -2.74 | -3.60 | -2.47 | -0.84    | -4.99    |
| Corner pole                | -2.86 | -3.81 | -4.56 | -4.77 | -3.87    | -5.50    |
| Corner belt                | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.81     |
| Belts - long side          | 0.14  | 0.51  | 0.81  | 0.50  | 0.18     | 1.30     |
| Belts - short side         | 0.12  | 0.56  | 0.84  | 0.22  | 0.18     | 1.37     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 6.01  | 0.00     | 0.00     |
| Combined rope - long side  | 2.11  | 2.59  | 3.04  | 4.70  | 4.66     | 3.50     |
| V-ropes - long side        | 1.53  | 1.83  | 2.11  | 3.38  | 3.30     | 2.39     |
| Combined rope - short side | 1.92  | 2.38  | 2.77  | 5.01  | 4.04     | 3.21     |
| V-ropes short side         | 1.45  | 1.80  | 2.09  | 3.64  | 2.94     | 2.41     |
| Reaction - long side       | 2.22  | 3.08  | 3.83  | 5.20  | 4.82     | 4.83     |
| Reaction - short side      | 2.04  | 2.92  | 3.60  | 5.23  | 4.22     | 4.62     |
| Reaction - corner          | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.85     |

Table 32. Summary of forces for 10x15m model

### H.6.3.3 10x20m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.95  | 2.63  | 3.41  | 4.45  | 2.53     | 5.01     |
| Weft                       | 0.88  | 2.16  | 2.94  | 2.89  | 2.68     | 4.20     |
| Edge                       | 1.80  | 2.26  | 2.67  | 4.43  | 3.43     | 3.16     |
| Main pole                  | -1.84 | -5.15 | -6.75 | 0.06  | 0.03     | -10.07   |
| Side pole                  | -1.35 | -2.69 | -3.55 | -2.21 | -0.80    | -4.88    |
| Corner pole                | -2.73 | -3.67 | -4.43 | -4.31 | -3.63    | -5.40    |
| Corner belt                | 2.33  | 2.95  | 3.44  | 4.43  | 3.44     | 4.01     |
| Belts - long side          | 0.22  | 0.70  | 1.01  | 0.57  | 0.26     | 1.50     |
| Belts - short side         | 0.13  | 0.55  | 0.85  | 0.13  | 0.13     | 1.35     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 5.93  | 0.00     | 0.00     |
| Combined rope - long side  | 2.16  | 2.48  | 2.91  | 5.43  | 5.46     | 3.34     |
| V-ropes - long side        | 1.50  | 1.76  | 2.05  | 3.91  | 3.68     | 2.28     |
| Combined rope - short side | 1.87  | 2.42  | 2.89  | 5.46  | 3.52     | 3.39     |
| V-ropes short side         | 1.37  | 1.80  | 2.14  | 3.83  | 2.54     | 2.51     |
| Reaction - long side       | 2.31  | 3.16  | 3.87  | 5.77  | 5.71     | 4.82     |
| Reaction - short side      | 1.99  | 2.95  | 3.72  | 5.58  | 3.65     | 4.81     |
| Reaction - corner          | 2.33  | 2.95  | 3.44  | 4.43  | 3.44     | 4.05     |

Table 33. Summary of forces for 10x20m model



## H.7. Element check

### H.7.1 Membrane

|        | Load combination                             | Element         | Representative stress | Design value stress           | Pag. |
|--------|----------------------------------------------|-----------------|-----------------------|-------------------------------|------|
| 10x10m | CO1. Own weight + pretension                 | Long term warp  | 0.95 kN/m             | 1.28 kN/m ( $\gamma = 1.35$ ) | 51   |
| 10x10m | CO1. Own weight + pretension                 | Long term weft  | 0.91 kN/m             | 1.23 kN/m ( $\gamma = 1.35$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction  | Short term warp | 4.45 kN/m             | 6.68 kN/m ( $\gamma = 1.5$ )  | 52   |
| 10x20m | CO3. Own weight + pretension + wind pressure | Short term weft | 2.94 kN/m             | 4.41 kN/m ( $\gamma = 1.5$ )  | 52   |

Sioen RP5639 fabric is being used.

|       |                 |                       |                          |    |
|-------|-----------------|-----------------------|--------------------------|----|
| UC.1a | Long term warp  | $S_{Ed} / S_{rd} < 1$ | $1.28 / 9.6 = 0.13 < 1$  | OK |
| UC.1b | Long term weft  | $S_{Ed} / S_{rd} < 1$ | $1.23 / 7.2 = 0.17 < 1$  | OK |
| UC.1c | Short term warp | $S_{Ed} / S_{rd} < 1$ | $6.68 / 12.0 = 0.56 < 1$ | OK |
| UC.1d | Short term weft | $S_{Ed} / S_{rd} < 1$ | $4.41 / 9.0 = 0.49 < 1$  | OK |

For capacity of membrane see H.2.1, page 24

## H.7.2 Main poles

|        | Load combinations                            | Element        | Representative force | Design value force           | Pag. |
|--------|----------------------------------------------|----------------|----------------------|------------------------------|------|
| 10x10m | CO3. Own weight + pretension + wind pressure | Main pole 4.0m | -6.95 kN             | -10.43 kN ( $\gamma = 1.5$ ) | 51   |

User load of max. 25 kg is applied, loaded centrically.

4.0m pole  $N_{ed} = -6.95 \times 1.50 + (-0.25 \times 1.35) = -10.76 \text{ kN}$

### H.7.2.1 4.0m pole - Wood, Spruce C18

|         |   |                                                                                                                 |
|---------|---|-----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 100 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 4.00m                                                                                                      |
| Quality | = | C18                                                                                                             |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |      |                                                                                          |    |
|------|------|------------------------------------------------------------------------------------------|----|
| UC.2 | 4.0m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.37 / (12.46 \times 0.120) = 0.92 \leq 1$ | OK |
|------|------|------------------------------------------------------------------------------------------|----|

See Annex G.1 Main pole 4.0m – Wood, on page 191 for the elaborate check

### H.7.3 Side poles

#### H.7.3.1 10x10m model

|        | Load combinations                            | Element        | Representative force | Design value force          | Pag. |
|--------|----------------------------------------------|----------------|----------------------|-----------------------------|------|
| 10x10m | CO3. Own weight + pretension + wind pressure | Side pole 3.0m | -3.13 kN             | -4.70 kN ( $\gamma = 1.5$ ) | 51   |

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -3.13 \times 1.50 + (-0.25 \times 1.35) = -5.03 \text{ kN}$$

H.7.3.1.1. 3m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 70 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 3.0m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |    |                                                                                          |    |
|------|----|------------------------------------------------------------------------------------------|----|
| UC.3 | 3m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.31 / (12.46 \times 0.105) = 1.00 \leq 1$ | OK |
|------|----|------------------------------------------------------------------------------------------|----|

See Annex G.2 Side pole 3m – Wood, Spruce C18, on page 192 for the elaborate check

#### H.7.3.2 10x15m and 10x20m model

|        | Load combinations                            | Element        | Representative force | Design value force          | Pag. |
|--------|----------------------------------------------|----------------|----------------------|-----------------------------|------|
| 10x15m | CO3. Own weight + pretension + wind pressure | Side pole 3.0m | -3.60 kN             | -5.40 kN ( $\gamma = 1.5$ ) | 51   |

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -3.60 \times 1.50 + (-0.25 \times 1.35) = -5.74 \text{ kN}$$

H.7.3.2.1. 3m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 75 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 3.0m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |    |                                                                                          |    |
|------|----|------------------------------------------------------------------------------------------|----|
| UC.3 | 3m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.30 / (12.46 \times 0.120) = 0.87 \leq 1$ | OK |
|------|----|------------------------------------------------------------------------------------------|----|

See Annex G.2 Side pole 3m – Wood, Spruce C18, on page 192 for the elaborate check

## H.7.4 Corner poles

### H.7.4.1 10x10 and 10x20m model

|        | Load combinations                           | Element          | Representative force | Design value force          | Pag. |
|--------|---------------------------------------------|------------------|----------------------|-----------------------------|------|
| 10x10m | CO4. Own weight + pretension + wind suction | Corner pole 2.5m | -4.52 kN             | -6.78 kN ( $\gamma = 1.5$ ) | 51   |

User load of max. 25 kg is applied, loaded centrally.

2.5m corner pole  $N_{ed} = -4.52 \times 1.50 + (-0.25 \times 1.35) = -7.12 \text{ kN}$

H.7.4.1.1. 2.5m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 70 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 2.5m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |      |                                                                                          |    |
|------|------|------------------------------------------------------------------------------------------|----|
| UC.4 | 2.5m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.85 / (12.46 \times 0.149) = 1.00 \leq 1$ | OK |
|------|------|------------------------------------------------------------------------------------------|----|

See Annex G.3 Corner pole 2.5m - Wood, C18, on page 194 for the elaborate check

### H.7.4.2 10x15m model

|        | Load combinations                           | Element          | Representative force | Design value force          | Pag. |
|--------|---------------------------------------------|------------------|----------------------|-----------------------------|------|
| 10x15m | CO4. Own weight + pretension + wind suction | Corner pole 2.5m | -4.77 kN             | -7.16 kN ( $\gamma = 1.5$ ) | 51   |

User load of max. 25 kg is applied, loaded centrally.

2.5m corner pole  $N_{ed} = -4.77 \times 1.50 + (-0.25 \times 1.35) = -7.49 \text{ kN}$

H.7.4.2.1. 2.5m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 75 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 2.5m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |      |                                                                                          |    |
|------|------|------------------------------------------------------------------------------------------|----|
| UC.4 | 2.5m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.70 / (12.46 \times 0.170) = 0.80 \leq 1$ | OK |
|------|------|------------------------------------------------------------------------------------------|----|

See Annex G.3 Corner pole 2.5m - Wood, C18, on page 194 for the elaborate check

### H.7.5 Guy ropes

|        | Load combination                            | Element               | Representative force | Design value force         | Pag. |
|--------|---------------------------------------------|-----------------------|----------------------|----------------------------|------|
| 10x10m | CO4. Own weight + pretension + wind suction | side - combined       | 4.83 kN              | 7.25 kN ( $\gamma = 1.5$ ) | 51   |
| 10x15m | CO4. Own weight + pretension + wind suction | long side - combined  | 4.70 kN              | 7.05 kN ( $\gamma = 1.5$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | long side - combined  | 5.43 kN              | 8.15 kN ( $\gamma = 1.5$ ) | 52   |
| 10x10m | CO4. Own weight + pretension + wind suction | side – V-ropes        | 3.51 kN              | 5.27 kN ( $\gamma = 1.5$ ) | 51   |
| 10x15m | CO4. Own weight + pretension + wind suction | long side – V-ropes   | 3.38 kN              | 5.07 kN ( $\gamma = 1.5$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | long side – V-ropes   | 3.91 kN              | 5.87 kN ( $\gamma = 1.5$ ) | 52   |
| 10x15m | CO4. Own weight + pretension + wind suction | short side - combined | 5.01 kN              | 7.52 kN ( $\gamma = 1.5$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | short side - combined | 5.46 kN              | 8.19 kN ( $\gamma = 1.5$ ) | 52   |
| 10x15m | CO4. Own weight + pretension + wind suction | short side – V-ropes  | 3.64 kN              | 5.46 kN ( $\gamma = 1.5$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | short side – V-ropes  | 3.83 kN              | 5.75 kN ( $\gamma = 1.5$ ) | 52   |

Synthetic rope Ø8mm, BL 900 daN. Is being used  $\rightarrow F_{rd,8mm} = 2.25$  kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

#### 10x10m model

|       |                 |                                                  |                    |                                              |    |
|-------|-----------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5a | side - combined | 1 individual rope made of <u>4 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $7.25 / (1 \times 4 \times 2.25) = 0.81 < 1$ | OK |
| UC.5b | side – V-ropes  | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.27 / (1 \times 3 \times 2.25) = 0.78 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

#### 10x15m model

|       |                      |                                                  |                    |                                              |    |
|-------|----------------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5e | long side - combined | 1 individual rope made of <u>4 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $7.05 / (1 \times 4 \times 2.25) = 0.78 < 1$ | OK |
| UC.5f | long side – V-ropes  | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.07 / (1 \times 3 \times 2.25) = 0.75 < 1$ | OK |

|       |                       |                                                  |                    |                                              |    |
|-------|-----------------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5g | short side - combined | 1 individual rope made of <u>4 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $7.52 / (1 \times 4 \times 2.25) = 0.84 < 1$ | OK |
| UC.5h | short side – V-ropes  | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.46 / (1 \times 3 \times 2.25) = 0.81 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

#### 10x20m model

|       |                       |                                                  |                    |                                              |    |
|-------|-----------------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5e | long side - combined  | 1 individual rope made of <u>4 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $8.15 / (1 \times 4 \times 2.25) = 0.91 < 1$ | OK |
| UC.5f | long side – V-ropes   | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.87 / (1 \times 3 \times 2.25) = 0.87 < 1$ | OK |
| UC.5g | short side - combined | 1 individual rope made of <u>4 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $8.19 / (1 \times 4 \times 2.25) = 0.91 < 1$ | OK |
| UC.5h | short side – V-ropes  | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.75 / (1 \times 3 \times 2.25) = 0.85 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength ( $BL_{rope}$ ) and less rope sections (n), as long as  $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

#### 10x10m model

Side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2900 \text{ kg}$

Side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2108 \text{ kg}$

#### 10x15m model

Long side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2820 \text{ kg}$

Long side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2028 \text{ kg}$

Short side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3008 \text{ kg}$

Short side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2184 \text{ kg}$

#### 10x20m model

Long side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3260 \text{ kg}$

Long side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2348 \text{ kg}$

Short side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3276 \text{ kg}$

Short side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2300 \text{ kg}$

## H.7.6 Belts

|        | Load combination                             | Element           | Representative force | Design value force         | Pag. |
|--------|----------------------------------------------|-------------------|----------------------|----------------------------|------|
| 10x20m | CO4. Own weight + pretension + wind suction  | Corner belt       | 4.43 kN              | 6.65 kN ( $\gamma = 1.5$ ) | 52   |
| 10x20m | CO3. Own weight + pretension + wind pressure | Belt – long side  | 1.01 kN              | 1.52 kN ( $\gamma = 1.5$ ) | 52   |
| 10x20m | CO3. Own weight + pretension + wind pressure | Belt – short side | 0.85 kN              | 1.28 kN ( $\gamma = 1.5$ ) | 52   |

### H.7.6.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

|       |                   |                                      |                    |                                   |    |
|-------|-------------------|--------------------------------------|--------------------|-----------------------------------|----|
| UC.6a | Corner belt       | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $6.65 / (1 \times 15) = 0.44 < 1$ | OK |
| UC.6b | Belt – long side  | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $1.52 / (1 \times 15) = 0.10 < 1$ | OK |
| UC.6c | Belt – short side | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $1.28 / (1 \times 15) = 0.09 < 1$ | OK |

For capacity of belts see H.2.3, page 24

### H.7.6.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used  $\rightarrow F_{rd,8mm} = 2.25$  kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

|       |                   |                                                  |                    |                                              |    |
|-------|-------------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.6d | Corner rope       | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $6.65 / (1 \times 3 \times 2.25) = 0.99 < 1$ | OK |
| UC.6e | Rope – long side  | 1 individual rope made of <u>1 section</u> Ø8mm  | $F_d / F_{rd} < 1$ | $1.52 / (1 \times 1 \times 2.25) = 0.68 < 1$ | OK |
| UC.6f | Rope – short side | 1 individual rope made of <u>1 section</u> Ø8mm  | $F_d / F_{rd} < 1$ | $1.28 / (1 \times 1 \times 2.25) = 0.57 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength ( $BL_{rope}$ ) and less rope sections (n), as long as  $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2660$  kg

Rope – long side:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 608$  kg

Rope – long side:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 512$  kg

### H.7.7 Stormbelts

|        | Load combination                            | Element   | Representative force | Design value force         | Pag. |
|--------|---------------------------------------------|-----------|----------------------|----------------------------|------|
| 10x15m | CO4. Own weight + pretension + wind suction | Stormbelt | 6.01 kN              | 9.02 kN ( $\gamma = 1.5$ ) | 51   |

PES belt with a breaking strength of 3000kg is being used.

|      |            |                                      |                    |                                     |    |
|------|------------|--------------------------------------|--------------------|-------------------------------------|----|
| UC.7 | Storm belt | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $9.02 / (1 \times 15) = 0.60 < 1.0$ | OK |
|------|------------|--------------------------------------|--------------------|-------------------------------------|----|

For capacity of belts see H.2.3, page 24



## H.7.8 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

### H.7.8.1 Corner belt connection

H.7.8.1.1. M16 Eye bolt, DIN580

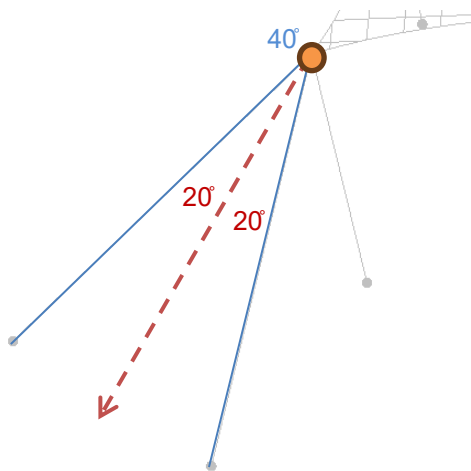
For  $\leq 45^\circ$ :

M16, WLL 700kg  $\rightarrow$  BL = 6 x 7 kN = 42 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 42 / 2 = 21 kN

$F_d = 2 \times F_{\text{corner belt}} \times \cos 20^\circ = 2 \times 6.65 \times \cos 20^\circ = 12.65 \text{ kN}$

|       |                 |                    |                           |    |
|-------|-----------------|--------------------|---------------------------|----|
| UC.8a | Corner eye bolt | $F_d / F_{rd} < 1$ | $12.65 / 21 = 0.60 < 1.0$ | OK |
|-------|-----------------|--------------------|---------------------------|----|

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.



H.7.8.1.2. Corner belt connection

A similar connection can be used if the capacity of it is at least 12.7 kN.

Orange Pin D-sluiting, WLL 1.0 t  $\rightarrow$  BL = 6 x 10 kN = 60 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 60 kN / 2 = 30 kN

|       |            |                    |                           |    |
|-------|------------|--------------------|---------------------------|----|
| UC.8b | D-sluiting | $F_d / F_{rd} < 1$ | $12.65 / 30 = 0.42 < 1.0$ | OK |
|-------|------------|--------------------|---------------------------|----|

For capacity of D-sluiting see K.4 Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889 on page 94.

H.7.8.1.3. Rope ratchet – for alternative corner connection with ropes

Ratchet, BL = 3.5 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 3.5 kN / 1.5 = 2.33 kN

$F_d = 6.65 \text{ kN}$  (Instead of corner belts corner ropes are used).

|       |                            |                    |                                   |    |
|-------|----------------------------|--------------------|-----------------------------------|----|
| UC.8c | Corner ropes (alternative) | $F_d / F_{rd} < 1$ | $6.65 / 3 \times 2.33 = 0.95 < 1$ | OK |
|-------|----------------------------|--------------------|-----------------------------------|----|

For capacity of rope ratchet see K.6 Rope ratchet, page 95

\* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

### H.7.8.2 Side belt connection

H.7.8.2.1. M12 Eye bolt, DIN580

For  $\leq 45^\circ$ :

M12, WLL 340kg  $\rightarrow$  BL = 6 x 3.4 kN = 20.4 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 20.4 / 2 = 10.2 kN

F<sub>d</sub>=1.52 kN

|       |                 |                    |                            |    |
|-------|-----------------|--------------------|----------------------------|----|
| UC.8d | Corner eye bolt | $F_d / F_{rd} < 1$ | $1.52 / 10.2 = 0.15 < 1.0$ | OK |
|-------|-----------------|--------------------|----------------------------|----|

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.

H.7.8.2.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg  $\rightarrow$  BL = 2 x 5.8 kN = 11.6 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 11.6 kN / 2 = 5.8 kN

F<sub>d</sub>=1.52 kN

|       |  |                    |                          |    |
|-------|--|--------------------|--------------------------|----|
| UC.8e |  | $F_d / F_{rd} < 1$ | $1.52 / 5.80 = 0.26 < 1$ | OK |
|-------|--|--------------------|--------------------------|----|

For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

### H.7.8.3 Side rope connection

H.7.8.3.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg  $\rightarrow$  BL = 2 x 5.8 kN = 11.6 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 11.6 kN / 2 = 5.8 kN

F<sub>d</sub>=5.87 kN

|       |  |                    |                             |            |
|-------|--|--------------------|-----------------------------|------------|
| UC.8f |  | $F_d / F_{rd} < 1$ | $5.87 / 5.80 = 1.01 \sim 1$ | Acceptable |
|-------|--|--------------------|-----------------------------|------------|

For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

H.7.8.3.2. Rope ratchet

Ratchet, BL = 3.5 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 3.5 kN / 1.5 = 2.33 kN

|        |       |                            |                    |                                   |    |
|--------|-------|----------------------------|--------------------|-----------------------------------|----|
| 10x10m | UC.8g | Side – combined rope       | $F_d / F_{rd} < 1$ | $7.25 / 4 \times 2.33 = 0.78 < 1$ | OK |
| 10x15m | UC.8h | Long side – combined rope  | $F_d / F_{rd} < 1$ | $7.05 / 4 \times 2.33 = 0.76 < 1$ | OK |
| 10x20m | UC.8i | Long side – combined rope  | $F_d / F_{rd} < 1$ | $8.15 / 4 \times 2.33 = 0.87 < 1$ | OK |
| 10x15m | UC.8j | Short side – combined rope | $F_d / F_{rd} < 1$ | $7.52 / 4 \times 2.33 = 0.81 < 1$ | OK |
| 10x20m | UC.8k | Short side – combined rope | $F_d / F_{rd} < 1$ | $8.19 / 4 \times 2.33 = 0.88 < 1$ | OK |

For capacity of rope ratchet see K.6 Rope ratchet, page 95

\* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

#### H.7.8.4 Base plate

##### H.7.8.4.1. Side pole & corner pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm,  $p_{\text{soil}} = 125 \text{ kN/m}^2$ )

|       |      |                            |                                             |    |
|-------|------|----------------------------|---------------------------------------------|----|
| UC.8l | Soil | $\frac{F_d}{A} / p \leq 1$ | $7.49 / (0.25 \cdot 0.25) / 125 = 0.96 < 1$ | OK |
|-------|------|----------------------------|---------------------------------------------|----|

##### H.7.8.4.2. Main pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 35x35 cm,  $p_{\text{soil}} = 175 \text{ kN/m}^2$ )

|       |      |                            |                                              |    |
|-------|------|----------------------------|----------------------------------------------|----|
| UC.8m | Soil | $\frac{F_d}{A} / p \leq 1$ | $10.76 / (0.35 \cdot 0.35) / 175 = 0.50 < 1$ | OK |
|-------|------|----------------------------|----------------------------------------------|----|

## H.8. Safety against overturning, sliding and uplifting

The calculations in this chapter provide a guideline for dense cohesionless soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

### H.8.1 Anchor capacity

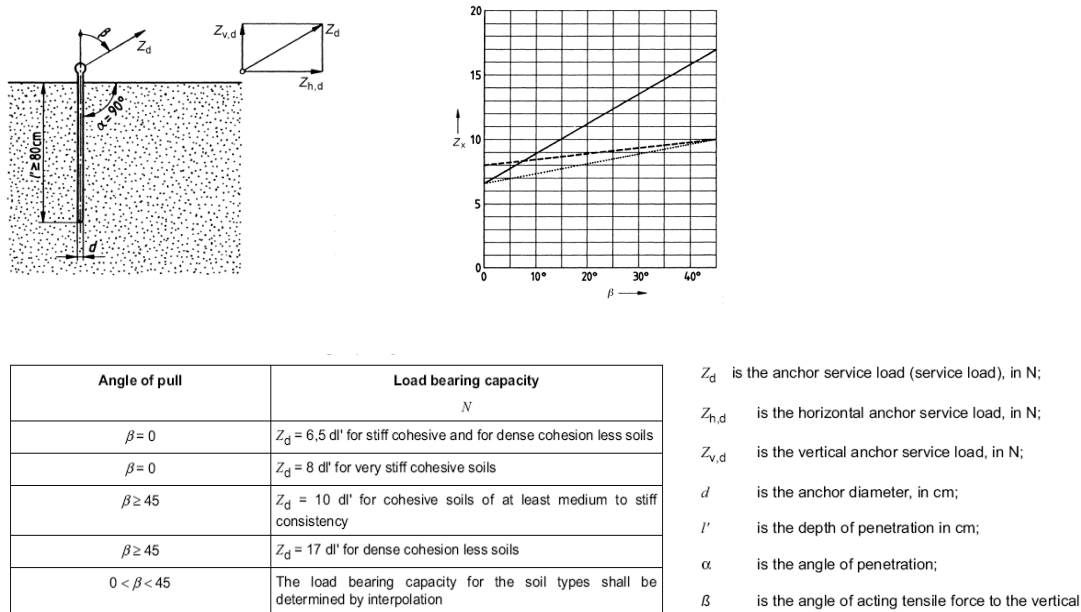


figure 7. Taken from EN 13782: Figures 4 & 5, table 5

Anchors of  $\varnothing 30\text{mm}$  are used, considering an **effective length of at least 1000 mm**. If multiple anchors are placed at the same location, the anchors need to be at least  $5 \times \varnothing$  apart to use the full capacity of the anchors.

Anchor capacity for anchors based in dense cohesionless soil (sandy soil)

| <b><math>\varnothing 30 \times 1000 \text{ mm}</math></b> |         |           |
|-----------------------------------------------------------|---------|-----------|
| Angle                                                     | $\beta$ | $\geq 45$ |
| Effective length anchor                                   | $l'$    | 100 cm    |
| Diameter anchor                                           | $d$     | 3.0 cm    |
| Anchor capacity*                                          | $Z_d$   | 5.10 kN   |

## H.8.2 Required anchor pins

|        | Load combination                            | Element                 | Representative force | Design value force         | Pag. |
|--------|---------------------------------------------|-------------------------|----------------------|----------------------------|------|
| 10x10m | CO4. Own weight + pretension + wind suction | Side anchor point       | 5.04 kN              | 6.05 kN ( $\gamma = 1.2$ ) | 51   |
| 10x15m | CO4. Own weight + pretension + wind suction | Long side anchor point  | 5.20 kN              | 6.24 kN ( $\gamma = 1.2$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | Long side anchor point  | 5.77 kN              | 6.92 kN ( $\gamma = 1.2$ ) | 52   |
| 10x15m | CO4. Own weight + pretension + wind suction | Short side anchor point | 5.23 kN              | 6.28 kN ( $\gamma = 1.2$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | Short side anchor point | 5.58 kN              | 6.70 kN ( $\gamma = 1.2$ ) | 52   |
| 10x10m | CO4. Own weight + pretension + wind suction | Corner anchor point     | 3.94 kN              | 4.73 kN ( $\gamma = 1.2$ ) | 51   |
| 10x15m | CO4. Own weight + pretension + wind suction | Corner anchor point     | 4.22 kN              | 5.06 kN ( $\gamma = 1.2$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | Corner anchor point     | 4.43 kN              | 5.32 kN ( $\gamma = 1.2$ ) | 52   |
| 10x10m | CO4. Own weight + pretension + wind suction | Stormbelt               | 5.62 kN              | 6.74 kN ( $\gamma = 1.2$ ) | 51   |
| 10x15m | CO4. Own weight + pretension + wind suction | Stormbelt               | 6.01 kN              | 7.21 kN ( $\gamma = 1.2$ ) | 51   |
| 10x20m | CO4. Own weight + pretension + wind suction | Stormbelt               | 5.93 kN              | 7.12 kN ( $\gamma = 1.2$ ) | 52   |

Required anchor amount for dense cohesionless soil – **10x10m model**

|                |    |               |                                                       |    |
|----------------|----|---------------|-------------------------------------------------------|----|
| Side guy ropes | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.05 / (2 \times 5.10) = 0.59 \leq 1$ | OK |
| Corner ties    | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 4.73 / (1 \times 5.10) = 0.93 < 1$    | OK |
| Stormbelt      | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.74 / (2 \times 5.10) = 0.66 < 1$    | OK |

Required anchor amount for dense cohesionless soil – **10x15m model**

|                      |    |               |                                                    |    |
|----------------------|----|---------------|----------------------------------------------------|----|
| Long side guy ropes  | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.24 / (2 \times 5.10) = 0.61 < 1$ | OK |
| Short side guy ropes | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.28 / (2 \times 5.10) = 0.62 < 1$ | OK |
| Corner ties          | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 5.06 / (1 \times 5.10) = 0.99 < 1$ | OK |
| Stormbelts           | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 7.21 / (2 \times 5.10) = 0.71 < 1$ | OK |

Required anchor amount for dense cohesionless soil – **10x20m model**

|                      |    |               |                                                       |            |
|----------------------|----|---------------|-------------------------------------------------------|------------|
| Long side guy ropes  | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.92 / (2 \times 5.10) = 0.68 < 1$    | OK         |
| Short side guy ropes | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.70 / (2 \times 5.10) = 0.66 < 1$    | OK         |
| Corner ties          | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 5.32 / (1 \times 5.10) = 1.04 \sim 1$ | Acceptable |
| Stormbelt            | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 7.12 / (2 \times 5.10) = 0.70 < 1$    | OK         |

### H.8.3 Anchor tests according to BS-EN 13782

It is advised to perform anchor test on location when there is a reason to doubt the “pull-out force” of the anchors, which could be when ground conditions differ from dense, non-cohesive soil.

Anchor tests should be carried out according to the following procedure:

Three anchors spread throughout the terrain should be put perpendicular into the ground. The anchors should be pulled out with the aid of a spring balance in the direction of the force acting on the anchor. The lowest of the three measured values should be used.

A safety factor of  $v = 1.6$  regarding ultimate limit load is to apply for the lowest test value to determine the load bearing capacity in subsequent calculation. The load bearing capacity determined in this manner shall not result in anchor movement which would result in stresses, deformations or instability inadmissible for the structure.

If the foundation conditions are comparable, test loadings carried out in another location may be adduced for substantiation purposes.

For example:

Force in belts:  $F_{rep} = 5.20 \text{ kN}$

$F_{sd,belt} = 1.2 \times F_{rep} = 1.2 \times 5.20 = 6.24 \text{ kN}$

The partial safety factor  $\gamma = 1.6$  is applied on the ultimate limit load:

$Z_{u,d,test} > 1.6 \times F_{sd} = 1.6 \times 6.24 = 9.98 \text{ kN}$

If for example the anchor test point has a minimal anchor capacity of 5.5 kN (550 kg), then 2 anchors are needed:  $2 \times 5.5 = 11 \text{ kN} > Z_{u,d,test}$

#### H.8.4 Required ballast

*The safety against sliding shall be calculated from:*

$$\sum \mu * \gamma * N_k \geq \sum \gamma * H_k \quad (\text{eq. 5. NEN-EN 13782})$$

where

$\mu$  the coefficient of friction according to table 3

$\gamma$  The safety factor according to Table 2,  $\gamma = 1.2$  (unfavorable wind load)

$N$  the vertical load components (characteristic value), corresponds to the required ballast

$H$  the horizontal load components (characteristic value).

*The safety against lifting shall be calculated from:*

$$\sum \gamma * N_{Rk, stb} \geq \sum \gamma * N_{Ek, dst} \quad (\text{eq. 7. NEN-EN 13782})$$

where

$N_{Rk, stb}$  are the vertical stabilizing load components (characteristic value);

$N_{Ek, dst}$  are the vertical lifting destabilizing load components (characteristic value).

For load calculations against sliding and lifting,  $F_{rep}$  is split into the horizontal ( $F_h$ ) and vertical ( $F_v$ ) forces. If the separate reaction force factors are not known:

$$F_h = F_{rep} \times \cos \alpha \quad \text{and} \quad F_v = F_{rep} \times \sin \alpha$$

Where  $\alpha$  is equal to the angle between  $F_{rep}$  and the ground plane.

It follows from the above equation:

$$\text{Required load (against sliding):} \quad B_h = (\gamma F_h) / \mu$$

$$\text{Required load (against lifting):} \quad B_v = \gamma F_v$$

$$B_{tot} = B_h + B_v$$

$B_{tot}$  = total required load

$B_h$  = required load due to horizontal forces

$B_v$  = required load due to vertical forces



### 10x10m model

| Location            | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|---------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                     |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                     |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                     | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes | 6.05                            | 4.28  | 4.28  | 45       | 1280                   | 1140                         |
| Corner ties         | 4.73                            | 3.34  | 3.34  | 45       | 1000                   | 890                          |
| Stormbelts          | 6.74                            | 4.77  | 4.77  | 45       | 1430                   | 1270                         |

### 10x15m model

| Location             | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|----------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                      |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                      |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                      | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes  | 6.24                            | 4.41  | 4.41  | 45       | 1320                   | 1180                         |
| Short side guy ropes | 6.28                            | 4.44  | 4.44  | 45       | 1330                   | 1180                         |
| Corner ties          | 5.06                            | 3.58  | 3.58  | 45       | 1070                   | 960                          |
| Stormbelts           | 7.21                            | 5.10  | 5.10  | 45       | 1530                   | 1360                         |

### 10x20m model

| Location             | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|----------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                      |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                      |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                      | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes  | 6.92                            | 4.90  | 4.90  | 45       | 1470                   | 1310                         |
| Short side guy ropes | 6.70                            | 4.73  | 4.73  | 45       | 1420                   | 1260                         |
| Corner ties          | 5.32                            | 3.76  | 3.76  | 45       | 1130                   | 1000                         |
| Stormbelts           | 7.12                            | 5.03  | 5.03  | 45       | 1510                   | 1340                         |

## I. Addition 1 – element check for reduced wind speeds [NL]

For stretchtent use in the Netherlands, a reduction of maximal allowable wind speeds is specified. As a result, the stretchtent is calculated for two wind scenarios – full wind and reduced wind. Maximal wind speeds are specified and if the structure exceeds them, the tent must be evacuated and access must be denied. For these wind speeds no stormbelts are necessary.

A reduction factor  $\alpha=0.7$  is applied, which leads to  $q_p(z_e)$  of **0.25 kN/m<sup>2</sup> [NL]**. **No stormbelts are required.**

### I.1. Summary of results – low wind in Netherlands

This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

Changes that come from lower wind and no stormbelts are highlighted in orange.

| General information:     |  |                                                               |                                                               |                                                               |
|--------------------------|--|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| Main dimensions:         |  | 10x10m                                                        | 10x15m / 10x12m / 9x12m                                       | 10x20m                                                        |
| Width:                   |  | 10m                                                           | 10m / 9m                                                      | 10m                                                           |
| Length:                  |  | 10m                                                           | 15m / 12m                                                     | 20m                                                           |
| Side height of membrane: |  | 3m                                                            | 3m                                                            | 3m                                                            |
| Max. height:             |  | 4m                                                            | 4m                                                            | 4m                                                            |
| Main poles: *            |  | 4.0m<br>Ø100 mm [Spruce,C18]                                  | 4.0m<br>Ø100 mm [Spruce,C18]                                  | 4.0m<br>Ø100 mm [Spruce, C18]                                 |
| Side poles: *            |  | 3.0m<br>Ø 70 mm [Spruce, C18]                                 | 3.0m<br>Ø 75 mm [Spruce, C18]                                 | 3.0m<br>Ø 75 mm [Spruce, C18]                                 |
| Corner poles: *          |  | 2.5m<br>Ø 70 mm [Spruce, C18]                                 | 2.5m<br>Ø 75 mm [Spruce, C18]                                 | 2.5m<br>Ø 70 mm [Spruce, C18]                                 |
| Guy ropes:               |  | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            | Ø 8mm polyester,<br>breaking strength ≥ 900 daN **            |
| Long side – combined:    |  | <b>3 sections</b>                                             | <b>3 sections</b>                                             | 4 sections                                                    |
| Long side - V-ropes:     |  | <b>2 sections</b>                                             | 3 sections                                                    | 3 sections                                                    |
| Short side – combined:   |  | -----                                                         | <b>3 sections</b>                                             | <b>3 sections</b>                                             |
| Short side - V-ropes:    |  | -----                                                         | <b>2 sections</b>                                             | <b>2 sections</b>                                             |
| Belts:                   |  | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                      |
| Fasteners:               |  | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M12 Eye bolt<br>Ratchet, BL = 3.5 kN |

|            |                                                                                                                                    |         |         |         |
|------------|------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|
|            | Side & corner pole base plate                                                                                                      | 25x25cm | 25x25cm | 25x25cm |
|            | Main pole base plate                                                                                                               | 35x35cm | 35x35cm | 35x35cm |
| User load: | A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally. |         |         |         |
| Snow load: | A snow load of 0.1 kN/m <sup>2</sup> (4 cm) has been considered in accordance with the French CTS standard.                        |         |         |         |
| Wind load: | According to EN 13782<br>The following pressure values have been used: h < 5m: 0.25kN/m <sup>2</sup>                               |         |         |         |

\* The average diameter, as a minimum required at the middle of the pole.

\*\* The required breaking load (BL<sub>tot</sub>) may be achieved by using multiple rope sections (n): BL<sub>rope</sub> = BL<sub>tot</sub> / n, see page 74

#### Safety against sliding, overturning and uplifting:

|                   |                                                                                                                                                                                                                                                                |                      |                            |                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------|-------------------|
| Anchor forces:    | Assumptions: angle of 45 degrees<br>The following design resistance of the anchor forces is required:<br>(See H.8.3, page 67 for Anchor tests according to EN 13782)                                                                                           |                      |                            |                   |
|                   |                                                                                                                                                                                                                                                                | 10x10m               | 10x15m / 10x12m / 9x12m    | 10x20m            |
|                   | Long side guy ropes:                                                                                                                                                                                                                                           | 4.56 kN (γ = 1.2)    | 5.78 kN (γ = 1.2)          | 6.85 kN (γ = 1.2) |
|                   | Short side guy ropes:                                                                                                                                                                                                                                          | 5.06 kN (γ = 1.2)    | 4.38 kN (γ = 1.2)          | 4.13 kN (γ = 1.2) |
| Required anchors: | Corner ties:                                                                                                                                                                                                                                                   | 3.83 kN (γ = 1.2)    | 4.21 kN (γ = 1.2)          | 4.13 kN (γ = 1.2) |
|                   | Based on dense, non-cohesive soil (e.g. sandy soils).<br>In the case anchors Ø30mm x 1000mm (effective length) are being used:                                                                                                                                 |                      |                            |                   |
|                   |                                                                                                                                                                                                                                                                | 10x10m               | 10x15m / 10x12m / 9x12m    | 10x20m            |
|                   | Long side guy ropes:                                                                                                                                                                                                                                           | 1 anchor             | 2 anchors                  | 2 anchors         |
| Required ballast: | Short side guy ropes:                                                                                                                                                                                                                                          | 1 anchor             | 1 anchor                   | 1 anchor          |
|                   | Corner ties:                                                                                                                                                                                                                                                   | 1 anchor             | 1 anchor                   | 1 anchor          |
|                   | Ballast calculation is performed for 2 different situations:<br>- Concrete on concrete – concrete ballasts are placed on concrete<br>- Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast |                      |                            |                   |
|                   | 10x10m                                                                                                                                                                                                                                                         |                      |                            |                   |
|                   |                                                                                                                                                                                                                                                                | Concrete on concrete | Concrete on wood or rubber |                   |
|                   | Long side guy ropes:                                                                                                                                                                                                                                           | 970 kg               | 860 kg                     |                   |
|                   | Corner ties:                                                                                                                                                                                                                                                   | 810 kg               | 720 kg                     |                   |
|                   | 10x15m / 10x12m / 9x12m                                                                                                                                                                                                                                        |                      |                            |                   |

|                       | Concrete on<br>concrete | Concrete on<br>wood or rubber |
|-----------------------|-------------------------|-------------------------------|
| Long side guy ropes:  | <b>1230 kg</b>          | <b>1090 kg</b>                |
| Short side guy ropes: | <b>1070 kg</b>          | <b>960 kg</b>                 |
| Corner ties:          | <b>980 kg</b>           | <b>790 kg</b>                 |
| <b>10x20m</b>         |                         |                               |
|                       | Concrete on<br>concrete | Concrete on<br>wood or rubber |
| Long side guy ropes:  | <b>1450 kg</b>          | <b>1290 kg</b>                |
| Short side guy ropes: | <b>950 kg</b>           | <b>840 kg</b>                 |
| Corner ties:          | <b>880 kg</b>           | <b>780 kg</b>                 |

## I.2. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO5 [NL] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

### I.2.1 10x10m model

| kN or kN/m           | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------|-------|-------|-------|-------|----------|----------|
| Warp                 | 0.95  | 2.42  | 3.09  | 3.39  | 2.49     | 4.55     |
| Weft                 | 0.91  | 2.11  | 2.62  | 2.18  | 1.99     | 3.79     |
| Edge                 | 1.76  | 2.13  | 2.44  | 3.41  | 2.68     | 2.85     |
| Main pole            | -2.12 | -5.41 | -6.95 | 0.05  | -0.08    | -10.32   |
| Side pole            | -1.30 | -2.43 | -3.13 | -2.02 | -0.83    | -4.33    |
| Corner pole          | -2.64 | -3.38 | -3.95 | -4.52 | -3.43    | -4.75    |
| Corner belt          | 2.10  | 2.50  | 2.86  | 3.94  | 3.19     | 3.33     |
| Belts - side         | 0.12  | 0.47  | 0.70  | 0.35  | 0.16     | 1.13     |
| Stormbelts           | 0.00  | 0.00  | 0.00  | 5.62  | 0.00     | 0.00     |
| Combined rope - side | 1.83  | 2.12  | 2.41  | 4.83  | 3.64     | 2.76     |
| V-ropes - side       | 1.40  | 1.63  | 1.84  | 3.51  | 2.72     | 2.10     |
| Reaction - side      | 1.94  | 2.59  | 3.10  | 5.04  | 3.80     | 3.94     |
| Reaction - corner    | 2.10  | 2.50  | 2.86  | 3.94  | 3.19     | 3.36     |

Table 34. Summary of forces for 10x10m model

### I.2.2 10x15m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.91  | 2.59  | 3.35  | 3.13  | 2.51     | 4.97     |
| Weft                       | 0.89  | 2.18  | 2.93  | 2.27  | 2.48     | 4.24     |
| Edge                       | 1.90  | 2.35  | 2.74  | 3.55  | 3.17     | 3.19     |
| Main pole                  | -1.74 | -4.96 | -6.49 | 0.06  | -0.02    | -9.78    |
| Side pole                  | -1.37 | -2.74 | -3.60 | -2.47 | -0.84    | -4.99    |
| Corner pole                | -2.86 | -3.81 | -4.56 | -4.77 | -3.87    | -5.50    |
| Corner belt                | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.81     |
| Belts - long side          | 0.14  | 0.51  | 0.81  | 0.50  | 0.18     | 1.30     |
| Belts - short side         | 0.12  | 0.56  | 0.84  | 0.22  | 0.18     | 1.37     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 6.01  | 0.00     | 0.00     |
| Combined rope - long side  | 2.11  | 2.59  | 3.04  | 4.70  | 4.66     | 3.50     |
| V-ropes - long side        | 1.53  | 1.83  | 2.11  | 3.38  | 3.30     | 2.39     |
| Combined rope - short side | 1.92  | 2.38  | 2.77  | 5.01  | 4.04     | 3.21     |
| V-ropes short side         | 1.45  | 1.80  | 2.09  | 3.64  | 2.94     | 2.41     |
| Reaction - long side       | 2.22  | 3.08  | 3.83  | 5.20  | 4.82     | 4.83     |
| Reaction - short side      | 2.04  | 2.92  | 3.60  | 5.23  | 4.22     | 4.62     |
| Reaction - corner          | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.85     |

Table 35. Summary of forces for 10x15m model

### I.2.3 10x20m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.95  | 2.63  | 3.41  | 4.45  | 2.53     | 5.01     |
| Weft                       | 0.88  | 2.16  | 2.94  | 2.89  | 2.68     | 4.20     |
| Edge                       | 1.80  | 2.26  | 2.67  | 4.43  | 3.43     | 3.16     |
| Main pole                  | -1.84 | -5.15 | -6.75 | 0.06  | 0.03     | -10.07   |
| Side pole                  | -1.35 | -2.69 | -3.55 | -2.21 | -0.80    | -4.88    |
| Corner pole                | -2.73 | -3.67 | -4.43 | -4.31 | -3.63    | -5.40    |
| Corner belt                | 2.33  | 2.95  | 3.44  | 4.43  | 3.44     | 4.01     |
| Belts - long side          | 0.22  | 0.70  | 1.01  | 0.57  | 0.26     | 1.50     |
| Belts - short side         | 0.13  | 0.55  | 0.85  | 0.13  | 0.13     | 1.35     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 5.93  | 0.00     | 0.00     |
| Combined rope - long side  | 2.16  | 2.48  | 2.91  | 5.43  | 5.46     | 3.34     |
| V-ropes - long side        | 1.50  | 1.76  | 2.05  | 3.91  | 3.68     | 2.28     |
| Combined rope - short side | 1.87  | 2.42  | 2.89  | 5.46  | 3.52     | 3.39     |
| V-ropes short side         | 1.37  | 1.80  | 2.14  | 3.83  | 2.54     | 2.51     |
| Reaction - long side       | 2.31  | 3.16  | 3.87  | 5.77  | 5.71     | 4.82     |
| Reaction - short side      | 1.99  | 2.95  | 3.72  | 5.58  | 3.65     | 4.81     |
| Reaction - corner          | 2.33  | 2.95  | 3.44  | 4.43  | 3.44     | 4.05     |

Table 36. Summary of forces for 10x20m model

### I.3. Element check - low wind in Netherlands

#### I.3.1 Guy ropes

|        | Load combination                                    | Element               | Representative force | Design value force         | Pag. |
|--------|-----------------------------------------------------|-----------------------|----------------------|----------------------------|------|
| 10x10m | CO5. Own weight + pretension + wind suction reduced | side - combined       | 3.64 kN              | 5.46 kN ( $\gamma = 1.5$ ) | 72   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | long side - combined  | 4.66 kN              | 6.99 kN ( $\gamma = 1.5$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | long side - combined  | 5.46 kN              | 8.19 kN ( $\gamma = 1.5$ ) | 73   |
| 10x10m | CO5. Own weight + pretension + wind suction reduced | side – V-ropes        | 2.72 kN              | 4.08 kN ( $\gamma = 1.5$ ) | 72   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | long side – V-ropes   | 3.30 kN              | 4.95 kN ( $\gamma = 1.5$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | long side – V-ropes   | 3.68 kN              | 5.52 kN ( $\gamma = 1.5$ ) | 73   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | short side - combined | 4.04 kN              | 6.06 kN ( $\gamma = 1.5$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | short side - combined | 3.52 kN              | 5.28 kN ( $\gamma = 1.5$ ) | 73   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | short side – V-ropes  | 2.94 kN              | 4.41 kN ( $\gamma = 1.5$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | short side – V-ropes  | 2.54 kN              | 3.76 kN ( $\gamma = 1.5$ ) | 73   |

Synthetic rope Ø8mm, BL 900 daN. Is being used →  $F_{rd,8mm} = 2.25$  kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

#### 10x10m model

|       |                 |                                                  |                    |                                              |    |
|-------|-----------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5a | side - combined | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $5.46 / (1 \times 3 \times 2.25) = 0.81 < 1$ | OK |
| UC.5b | side – V-ropes  | 1 individual rope made of <u>2 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $4.08 / (1 \times 2 \times 2.25) = 0.91 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

#### 10x15m model

|       |                       |                                           |                    |                                                 |    |
|-------|-----------------------|-------------------------------------------|--------------------|-------------------------------------------------|----|
| UC.5e | long side - combined  | 1 individual rope made of 3 sections Ø8mm | $F_d / F_{rd} < 1$ | $6.99 / (1 \times 3 \times 2.25) = 1.04 \sim 1$ | OK |
| UC.5f | long side – V-ropes   | 1 individual rope made of 3 sections Ø8mm | $F_d / F_{rd} < 1$ | $4.95 / (1 \times 3 \times 2.25) = 0.73 < 1$    | OK |
| UC.5g | short side - combined | 1 individual rope made of 3 sections Ø8mm | $F_d / F_{rd} < 1$ | $6.06 / (1 \times 3 \times 2.25) = 0.90 < 1$    | OK |
| UC.5h | short side – V-ropes  | 1 individual rope made of 2 sections Ø8mm | $F_d / F_{rd} < 1$ | $4.41 / (1 \times 2 \times 2.25) = 0.98 < 1$    | OK |

For capacity of ropes see H.2.2, page 24

#### 10x20m model

|       |                       |                                           |                    |                                              |    |
|-------|-----------------------|-------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5e | long side - combined  | 1 individual rope made of 4 sections Ø8mm | $F_d / F_{rd} < 1$ | $8.19 / (1 \times 4 \times 2.25) = 0.91 < 1$ | OK |
| UC.5f | long side – V-ropes   | 1 individual rope made of 3 sections Ø8mm | $F_d / F_{rd} < 1$ | $5.52 / (1 \times 3 \times 2.25) = 0.82 < 1$ | OK |
| UC.5g | short side - combined | 1 individual rope made of 3 sections Ø8mm | $F_d / F_{rd} < 1$ | $5.28 / (1 \times 3 \times 2.25) = 0.78 < 1$ | OK |
| UC.5h | short side – V-ropes  | 1 individual rope made of 2 sections Ø8mm | $F_d / F_{rd} < 1$ | $3.76 / (1 \times 2 \times 2.25) = 0.84 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength ( $BL_{rope}$ ) and less rope sections (n), as long as  $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

#### 10x10m model

Side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2184 \text{ kg}$

Side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1632 \text{ kg}$

#### 10x15m model

Long side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2796 \text{ kg}$

Long side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1900 \text{ kg}$

Short side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2424 \text{ kg}$

Short side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1764 \text{ kg}$

#### 10x20m model

Long side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3276 \text{ kg}$

Long side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2208 \text{ kg}$

Short side - combined:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2112 \text{ kg}$

Short side – V-ropes:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1504 \text{ kg}$

### I.3.2 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapter I.3.1.

#### I.3.2.1 Side rope connection

##### I.3.2.1.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg  $\rightarrow$  BL = 2 x 5.8 kN = 11.6 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 11.6 kN / 2 = 5.8 kN  
F<sub>d</sub>=5.52 kN

|              |                                              |                              |           |
|--------------|----------------------------------------------|------------------------------|-----------|
| <b>UC.3a</b> | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>5.52 / 5.8 = 0.95 ≤ 1</b> | <b>OK</b> |
|--------------|----------------------------------------------|------------------------------|-----------|

For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

##### I.3.2.1.2. Rope ratchet

Ratchet, BL = 3.5 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 3.5 kN / 1.5 = 2.33 kN

|        |              |                            |                                              |                                    |           |
|--------|--------------|----------------------------|----------------------------------------------|------------------------------------|-----------|
| 10x10m | <b>UC.3b</b> | Side – combined rope       | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>5.46 / 3*x2.33= 0.78 &lt; 1</b> | <b>OK</b> |
| 10x15m | <b>UC.3c</b> | Long side – combined rope  | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>6.99 / 3*x2.33= 1.00 &lt; 1</b> | <b>OK</b> |
| 10x20m | <b>UC.3d</b> | Long side – combined rope  | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>8.19 / 4*x2.33= 0.88 &lt; 1</b> | <b>OK</b> |
| 10x15m | <b>UC.3e</b> | Short side – combined rope | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>6.06 / 3*x2.33= 0.87 &lt; 1</b> | <b>OK</b> |
| 10x20m | <b>UC.3f</b> | Short side – combined rope | <b>F<sub>d</sub> / F<sub>rd</sub> &lt; 1</b> | <b>5.28 / 3*x2.33= 0.76 &lt; 1</b> | <b>OK</b> |

For capacity of rope ratchet see K.6 Rope ratchet, page 95

\* Number of cross sections making a rope. Ratchet is applied only on one cross-section.



## I.4. Safety against overturning, sliding and uplifting - low wind in Netherlands

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity  $F_{rd}=5.1 \text{ kN}$  (see H.8.1)

### I.4.1 Required anchor pins

|        | Load combination                                    | Element                 | Representative force | Design value force         | Pag. |
|--------|-----------------------------------------------------|-------------------------|----------------------|----------------------------|------|
| 10x10m | CO5. Own weight + pretension + wind suction reduced | Side anchor point       | 3.80 kN              | 4.56 kN ( $\gamma = 1.2$ ) | 72   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | Long side anchor point  | 4.82 kN              | 5.78 kN ( $\gamma = 1.2$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | Long side anchor point  | 5.71 kN              | 6.85 kN ( $\gamma = 1.2$ ) | 73   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | Short side anchor point | 4.22 kN              | 5.06 kN ( $\gamma = 1.2$ ) | 73   |
| 10x20m | CO3. Own weight + pretension + wind pressure        | Short side anchor point | 3.72 kN              | 4.46 kN ( $\gamma = 1.2$ ) | 73   |
| 10x10m | CO5. Own weight + pretension + wind suction reduced | Corner anchor point     | 3.19 kN              | 3.83 kN ( $\gamma = 1.2$ ) | 72   |
| 10x15m | CO5. Own weight + pretension + wind suction reduced | Corner anchor point     | 3.51 kN              | 4.21 kN ( $\gamma = 1.2$ ) | 73   |
| 10x20m | CO5. Own weight + pretension + wind suction reduced | Corner anchor point     | 3.44 kN              | 4.13 kN ( $\gamma = 1.2$ ) | 73   |

Required anchor amount for dense cohesionless soil – **10x10m model**

|                |    |               |                                                       |    |
|----------------|----|---------------|-------------------------------------------------------|----|
| Side guy ropes | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 4.56 / (1 \times 5.10) = 0.89 \leq 1$ | OK |
| Corner ties    | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 3.83 / (1 \times 5.10) = 0.75 < 1$    | OK |

Required anchor amount for dense cohesionless soil – **10x15m model**

|                      |    |               |                                                    |    |
|----------------------|----|---------------|----------------------------------------------------|----|
| Long side guy ropes  | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 5.78 / (2 \times 5.10) = 0.57 < 1$ | OK |
| Short side guy ropes | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 5.06 / (1 \times 5.10) = 0.99 < 1$ | OK |
| Corner ties          | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 4.21 / (1 \times 5.10) = 0.83 < 1$ | OK |

Required anchor amount for dense cohesionless soil – **10x20m model**

|                      |    |               |                                                    |    |
|----------------------|----|---------------|----------------------------------------------------|----|
| Long side guy ropes  | 2x | Ø30 x 1000 mm | $F_d / F_{rd} = 6.85 / (2 \times 5.10) = 0.67 < 1$ | OK |
| Short side guy ropes | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 4.46 / (1 \times 5.10) = 0.87 < 1$ | OK |
| Corner ties          | 1x | Ø30 x 1000 mm | $F_d / F_{rd} = 4.13 / (1 \times 5.10) = 0.81 < 1$ | OK |

#### I.4.2 Required ballast

For calculation method see H.8.4.

##### 10x10m model

| Location            | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|---------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                     |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                     |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                     | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes | 4.56                            | 3.22  | 3.22  | 45       | 970                    | 860                          |
| Corner ties         | 3.83                            | 2.71  | 2.71  | 45       | 810                    | 720                          |

##### 10x15m model

| Location             | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|----------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                      |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                      |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                      | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes  | 5.78                            | 4.09  | 4.09  | 45       | 1230                   | 1090                         |
| Short side guy ropes | 5.06                            | 3.58  | 3.58  | 45       | 1070                   | 960                          |
| Corner ties          | 4.21                            | 2.98  | 2.98  | 45       | 890                    | 790                          |

### 10x20m model

| Location             | $F_{rep}$<br>( $\gamma = 1.2$ ) | $F_H$ | $F_V$ | $\alpha$ | Ballast for            | Ballast for                  |
|----------------------|---------------------------------|-------|-------|----------|------------------------|------------------------------|
|                      |                                 |       |       |          | $\mu = 0.5$            | $\mu = 0.6$                  |
|                      |                                 |       |       |          | (concrete on concrete) | (concrete on rubber or wood) |
|                      | (kN)                            | (kN)  | (kN)  | °        | (KG)                   | (KG)                         |
| Long side guy ropes  | 6.85                            | 4.85  | 4.85  | 45       | 1450                   | 1290                         |
| Short side guy ropes | 4.46                            | 3.10  | 3.10  | 45       | 930                    | 830                          |
| Corner ties          | 4.13                            | 2.92  | 2.92  | 45       | 880                    | 780                          |

## J. Addition 2 – element check for combined wind and snow load [FR]

For stretchtent use in France, the static analysis must include a load combination including snow and wind loads, corresponding to article 7 of CTS. This addition represents changes in element checks and anchor and ballast determination if this load combination is considered. Stormbelts are always required.

### J.1. Summary of results – snow+wind in France

This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

Changes that come from snow and wind combination are highlighted in orange.

| General information:     |  |                                                              |                                                              |                                                              |
|--------------------------|--|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| Main dimensions:         |  | 10x10m                                                       | 10x15m / 10x12m / 9x12m                                      | 10x20m                                                       |
| Width:                   |  | 10m                                                          | 10m / 9m                                                     | 10m                                                          |
| Length:                  |  | 10m                                                          | 15m / 12m                                                    | 20m                                                          |
| Side height of membrane: |  | 3m                                                           | 3m                                                           | 3m                                                           |
| Max. height:             |  | 4m                                                           | 4m                                                           | 4m                                                           |
| Main poles: *            |  | 4.0m<br>Ø105 mm [Spruce,C18]                                 | 4.0m<br>Ø105 mm [Spruce,C18]                                 | 4.0m<br>Ø105 mm [Spruce, C18]                                |
| Side poles: *            |  | 3.0m<br>Ø 75 mm [Spruce, C18]                                | 3.0m<br>Ø 80 mm [Spruce, C18]                                | 3.0m<br>Ø 80 mm [Spruce, C18]                                |
| Corner poles: *          |  | 2.5m<br>Ø 75 mm [Spruce, C18]                                | 2.5m<br>Ø 75 mm [Spruce, C18]                                | 2.5m<br>Ø 70 mm [Spruce, C18]                                |
| Stormbelts:              |  | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     |
| Guy ropes:               |  | Ø 8mm polyester,<br>breaking strength ≥ 900<br>daN **        | Ø 8mm polyester,<br>breaking strength ≥ 900<br>daN **        | Ø 8mm polyester,<br>breaking strength ≥ 900<br>daN **        |
| Long side – combined:    |  | 4 sections                                                   | 4 sections                                                   | 4 sections                                                   |
| Long side - V-ropes:     |  | 3 sections                                                   | 3 sections                                                   | 3 sections                                                   |
| Short side – combined:   |  | -----                                                        | 4 sections                                                   | 4 sections                                                   |
| Short side - V-ropes:    |  | -----                                                        | 3 sections                                                   | 3 sections                                                   |
| Belts:                   |  | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     | 50mm PES<br>[EN 12195-2],<br>LC=1000 daN                     |
| Fasteners:               |  | Carabine 180x14<br>M16 & M8 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M8 Eye bolt<br>Ratchet, BL = 3.5 kN | Carabine 180x14<br>M16 & M8 Eye bolt<br>Ratchet, BL = 3.5 kN |

|            |                                                                                                                                                                                                                                                                                                            |         |         |         |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|
|            | Side & corner pole base plate                                                                                                                                                                                                                                                                              | 25x25cm | 25x25cm | 25x25cm |
|            | Main pole base plate                                                                                                                                                                                                                                                                                       | 35x35cm | 35x35cm | 35x35cm |
| User load: | A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.                                                                                                                                                                         |         |         |         |
| Snow load: | A snow load of 0.1 kN/m <sup>2</sup> (4 cm) has been considered in accordance with the French CTS standard.                                                                                                                                                                                                |         |         |         |
| Wind load: | According to EN 13782<br>The following pressure values have been used: h < 5m: 0.50kN/m <sup>2</sup><br>This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated. |         |         |         |

\* The average diameter, as a minimum required at the middle of the pole.

\*\* The required breaking load (BL<sub>tot</sub>) may be achieved by using multiple rope sections (n): BL<sub>rope</sub> = BL<sub>tot</sub> / n, see page 57

#### Safety against sliding, overturning and uplifting:

|                   |                                                                                                                                                                                                                                                        |                      |                                |                   |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------|-------------------|
| Anchor forces:    | Assumptions: angle of 45 degrees<br>The following design resistance of the anchor forces is required:<br>(See H.8.3, page 67 for Anchor tests according to EN 13782)                                                                                   |                      |                                |                   |
|                   |                                                                                                                                                                                                                                                        | <b>10x10m</b>        | <b>10x15m / 10x12m / 9x12m</b> | <b>10x20m</b>     |
|                   | Long side guy ropes:                                                                                                                                                                                                                                   | 6.05 kN (γ = 1.2)    | 6.24 kN (γ = 1.2)              | 6.92 kN (γ = 1.2) |
|                   | Short side guy ropes:                                                                                                                                                                                                                                  |                      | 6.28 kN (γ = 1.2)              | 6.70 kN (γ = 1.2) |
|                   | Corner ties:                                                                                                                                                                                                                                           | 4.73 kN (γ = 1.2)    | 5.06 kN (γ = 1.2)              | 5.32 kN (γ = 1.2) |
| Required anchors: | Stormbelt:                                                                                                                                                                                                                                             | 6.74 kN (γ = 1.2)    | 7.21 kN (γ = 1.2)              | 7.12 kN (γ = 1.2) |
|                   | Based on dense, non-cohesive soil (e.g. sandy soils).<br>In the case anchors <b>Ø30mm x 1000mm</b> (effective length) are being used:                                                                                                                  |                      |                                |                   |
|                   |                                                                                                                                                                                                                                                        | <b>10x10m</b>        | <b>10x15m / 10x12m / 9x12m</b> | <b>10x20m</b>     |
|                   | Long side guy ropes:                                                                                                                                                                                                                                   | 2 anchors            | 2 anchors                      | 2 anchors         |
|                   | Short side guy ropes:                                                                                                                                                                                                                                  |                      | 2 anchors                      | 2 anchors         |
| Required ballast: | Corner ties:                                                                                                                                                                                                                                           | 1 anchor             | 1 anchor                       | 1 anchor          |
|                   | Stormbelt:                                                                                                                                                                                                                                             | 2 anchors            | 2 anchors                      | 2 anchors         |
|                   | Ballast calculation is performed for 2 different situations:                                                                                                                                                                                           |                      |                                |                   |
|                   | <ul style="list-style-type: none"> <li>- Concrete on concrete – concrete ballasts are placed on concrete</li> <li>- Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast</li> </ul> |                      |                                |                   |
|                   | <b>10x10m</b>                                                                                                                                                                                                                                          |                      |                                |                   |
|                   |                                                                                                                                                                                                                                                        | Concrete on concrete | Concrete on wood or rubber     |                   |

|                                |                      |                            |
|--------------------------------|----------------------|----------------------------|
| Long side guy ropes:           | 1280 kg              | 1140 kg                    |
| Corner ties:                   | 1000 kg              | 890 kg                     |
| Stormbelt:                     | 1430 kg              | 1270 kg                    |
| <b>10x15m / 10x12m / 9x12m</b> |                      |                            |
|                                | Concrete on concrete | Concrete on wood or rubber |
| Long side guy ropes:           | 1320 kg              | 1180 kg                    |
| Short side guy ropes:          | 1330 kg              | 1180 kg                    |
| Corner ties:                   | 1070 kg              | 960 kg                     |
| Stormbelt:                     | 1530 kg              | 1360 kg                    |
| <b>10x20m</b>                  |                      |                            |
|                                | Concrete on concrete | Concrete on wood or rubber |
| Long side guy ropes:           | 1470 kg              | 1310 kg                    |
| Short side guy ropes:          | 1420 kg              | 1260 kg                    |
| Corner ties:                   | 1130 kg              | 1000 kg                    |
| Stormbelt:                     | 1510 kg              | 1340 kg                    |

## J.2. Cross sections

| Profile      | Material | Ø<br>mm | G<br>kg/m | A<br>mm <sup>2</sup> | I <sub>y</sub><br>mm <sup>4</sup> | W <sub>el,y</sub><br>mm <sup>3</sup> |
|--------------|----------|---------|-----------|----------------------|-----------------------------------|--------------------------------------|
| Ø ≈ 105 mm * | Wood C18 | 105     | 2.77      | 8659                 | 5966602                           | 113649                               |
| Ø ≈ 80 mm *  | Wood C18 | 80      | 1.61      | 5026                 | 2010619                           | 50265                                |
| Ø ≈ 75 mm *  | Wood C18 | 75      | 1.41      | 4418                 | 1553156                           | 41417                                |

Table 37. Used cross sections

\* The average diameter, as a minimum required at the middle of the pole.

## J.3. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO6 [FR] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

### J.3.1 10x10m model

| kN or kN/m  | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|-------------|-------|-------|-------|-------|----------|----------|
| Warp        | 0.95  | 2.42  | 3.09  | 3.39  | 2.49     | 4.55     |
| Weft        | 0.91  | 2.11  | 2.62  | 2.18  | 1.99     | 3.79     |
| Edge        | 1.76  | 2.13  | 2.44  | 3.41  | 2.68     | 2.85     |
| Main pole   | -2.12 | -5.41 | -6.95 | 0.05  | -0.08    | -10.32   |
| Side pole   | -1.30 | -2.43 | -3.13 | -2.02 | -0.83    | -4.33    |
| Corner pole | -2.64 | -3.38 | -3.95 | -4.52 | -3.43    | -4.75    |
| Corner belt | 2.10  | 2.50  | 2.86  | 3.94  | 3.19     | 3.33     |

|                      |      |      |      |      |      |      |
|----------------------|------|------|------|------|------|------|
| Belts - side         | 0.12 | 0.47 | 0.70 | 0.35 | 0.16 | 1.13 |
| Stormbelts           | 0.00 | 0.00 | 0.00 | 5.62 | 0.00 | 0.00 |
| Combined rope - side | 1.83 | 2.12 | 2.41 | 4.83 | 3.64 | 2.76 |
| V-ropes - side       | 1.40 | 1.63 | 1.84 | 3.51 | 2.72 | 2.10 |
| Reaction - side      | 1.94 | 2.59 | 3.10 | 5.04 | 3.80 | 3.94 |
| Reaction - corner    | 2.10 | 2.50 | 2.86 | 3.94 | 3.19 | 3.36 |

Table 38. Summary of forces for 10x10m model

### J.3.2 10x15m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.91  | 2.59  | 3.35  | 3.13  | 2.51     | 4.97     |
| Weft                       | 0.89  | 2.18  | 2.93  | 2.27  | 2.48     | 4.24     |
| Edge                       | 1.90  | 2.35  | 2.74  | 3.55  | 3.17     | 3.19     |
| Main pole                  | -1.74 | -4.96 | -6.49 | 0.06  | -0.02    | -9.78    |
| Side pole                  | -1.37 | -2.74 | -3.60 | -2.47 | -0.84    | -4.99    |
| Corner pole                | -2.86 | -3.81 | -4.56 | -4.77 | -3.87    | -5.50    |
| Corner belt                | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.81     |
| Belts - long side          | 0.14  | 0.51  | 0.81  | 0.50  | 0.18     | 1.30     |
| Belts - short side         | 0.12  | 0.56  | 0.84  | 0.22  | 0.18     | 1.37     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 6.01  | 0.00     | 0.00     |
| Combined rope - long side  | 2.11  | 2.59  | 3.04  | 4.70  | 4.66     | 3.50     |
| V-ropes - long side        | 1.53  | 1.83  | 2.11  | 3.38  | 3.30     | 2.39     |
| Combined rope - short side | 1.92  | 2.38  | 2.77  | 5.01  | 4.04     | 3.21     |
| V-ropes short side         | 1.45  | 1.80  | 2.09  | 3.64  | 2.94     | 2.41     |
| Reaction - long side       | 2.22  | 3.08  | 3.83  | 5.20  | 4.82     | 4.83     |
| Reaction - short side      | 2.04  | 2.92  | 3.60  | 5.23  | 4.22     | 4.62     |
| Reaction - corner          | 2.31  | 2.85  | 3.28  | 4.22  | 3.51     | 3.85     |

Table 39. Summary of forces for 10x15m model

### J.3.3 10x20m model

| kN or kN/m                 | CO1   | CO2   | CO3   | CO4   | CO5 [NL] | CO6 [FR] |
|----------------------------|-------|-------|-------|-------|----------|----------|
| Warp                       | 0.95  | 2.63  | 3.41  | 4.45  | 2.53     | 5.01     |
| Weft                       | 0.88  | 2.16  | 2.94  | 2.89  | 2.68     | 4.20     |
| Edge                       | 1.80  | 2.26  | 2.67  | 4.43  | 3.43     | 3.16     |
| Main pole                  | -1.84 | -5.15 | -6.75 | 0.06  | 0.03     | -10.07   |
| Side pole                  | -1.35 | -2.69 | -3.55 | -2.21 | -0.80    | -4.88    |
| Corner pole                | -2.73 | -3.67 | -4.43 | -4.31 | -3.63    | -5.40    |
| Corner belt                | 2.33  | 2.95  | 3.44  | 4.43  | 3.44     | 4.01     |
| Belts - long side          | 0.22  | 0.70  | 1.01  | 0.57  | 0.26     | 1.50     |
| Belts - short side         | 0.13  | 0.55  | 0.85  | 0.13  | 0.13     | 1.35     |
| Stormbelts                 | 0.00  | 0.00  | 0.00  | 5.93  | 0.00     | 0.00     |
| Combined rope - long side  | 2.16  | 2.48  | 2.91  | 5.43  | 5.46     | 3.34     |
| V-ropes - long side        | 1.50  | 1.76  | 2.05  | 3.91  | 3.68     | 2.28     |
| Combined rope - short side | 1.87  | 2.42  | 2.89  | 5.46  | 3.52     | 3.39     |
| V-ropes short side         | 1.37  | 1.80  | 2.14  | 3.83  | 2.54     | 2.51     |
| Reaction - long side       | 2.31  | 3.16  | 3.87  | 5.77  | 5.71     | 4.82     |

|                       |      |      |      |      |      |      |
|-----------------------|------|------|------|------|------|------|
| Reaction - short side | 1.99 | 2.95 | 3.72 | 5.58 | 3.65 | 4.81 |
| Reaction - corner     | 2.33 | 2.95 | 3.44 | 4.43 | 3.44 | 4.05 |

**Table 40. Summary of forces for 10x20m model**



## J.4. Element check – snow + wind in France

### J.4.1 Membrane

|        | Load combination                                    | Element         | Representative stress | Design value stress           | Pag. |
|--------|-----------------------------------------------------|-----------------|-----------------------|-------------------------------|------|
| 10x10m | CO1. Own weight + pretension                        | Long term warp  | 0.95 kN/m             | 1.28 kN/m ( $\gamma = 1.35$ ) | 82   |
| 10x10m | CO1. Own weight + pretension                        | Long term weft  | 0.91 kN/m             | 1.23 kN/m ( $\gamma = 1.35$ ) | 82   |
| 10x20m | CO6. Own weight + pretension + wind pressure + snow | Short term warp | 5.01 kN/m             | 6.76 kN/m ( $\gamma = 1.35$ ) | 83   |
| 10x15m | CO6. Own weight + pretension + wind pressure + snow | Short term weft | 4.24 kN/m             | 5.72 kN/m ( $\gamma = 1.35$ ) | 83   |

Sioen RP5639 fabric is being used.

|       |                 |                       |                          |    |
|-------|-----------------|-----------------------|--------------------------|----|
| UC.1a | Long term warp  | $S_{Ed} / S_{rd} < 1$ | $1.28 / 9.6 = 0.13 < 1$  | OK |
| UC.1b | Long term weft  | $S_{Ed} / S_{rd} < 1$ | $1.23 / 7.2 = 0.17 < 1$  | OK |
| UC.1c | Short term warp | $S_{Ed} / S_{rd} < 1$ | $6.76 / 12.0 = 0.56 < 1$ | OK |
| UC.1d | Short term weft | $S_{Ed} / S_{rd} < 1$ | $5.72 / 9.0 = 0.64 < 1$  | OK |

For capacity of membrane see H.2.1, page 24

#### J.4.2 Main poles

|        | Load combinations                                   | Element        | Representative force | Design value force            | Pag. |
|--------|-----------------------------------------------------|----------------|----------------------|-------------------------------|------|
| 10x10m | CO6. Own weight + pretension + wind pressure + snow | Main pole 4.0m | -10.32 kN            | -13.93 kN ( $\gamma = 1.35$ ) | 82   |

User load of max. 25 kg is applied, loaded centrally.

4.0m pole  $N_{ed} = -10.32 \times 1.35 + (-0.25 \times 1.35) = -14.27 \text{ kN}$

##### J.4.2.1 4.0m pole - Wood, Spruce C18

|         |   |                                                                                                                 |
|---------|---|-----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 105 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 4.00m                                                                                                      |
| Quality | = | C18                                                                                                             |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |      |                                                                                          |            |
|------|------|------------------------------------------------------------------------------------------|------------|
| UC.2 | 4.0m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.65 / (12.46 \times 0.132) = 1.01 \sim 1$ | Acceptable |
|------|------|------------------------------------------------------------------------------------------|------------|

See Annex G.4 Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 196 for the elaborate check

### J.4.3 Side poles

#### J.4.3.1 10x10m model

|        | Load combinations                                   | Element        | Representative force | Design value force          | Pag. |
|--------|-----------------------------------------------------|----------------|----------------------|-----------------------------|------|
| 10x10m | CO6. Own weight + pretension + wind pressure + snow | Side pole 3.0m | -4.33 kN             | -5.85 kN ( $\gamma = 1.5$ ) | 83   |

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -4.33 \times 1.35 + (-0.25 \times 1.35) = -6.18 \text{ kN}$$

##### J.4.3.1.1. 3m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 75 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 3.0m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |    |                                                                                          |    |
|------|----|------------------------------------------------------------------------------------------|----|
| UC.3 | 3m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.40 / (12.46 \times 0.120) = 0.94 \leq 1$ | OK |
|------|----|------------------------------------------------------------------------------------------|----|

See Annex G.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 197 for the elaborate check

#### J.4.3.2 10x15m and 10x20m model

|        | Load combinations                                   | Element        | Representative force | Design value force          | Pag. |
|--------|-----------------------------------------------------|----------------|----------------------|-----------------------------|------|
| 10x15m | CO6. Own weight + pretension + wind pressure + snow | Side pole 3.0m | -4.99 kN             | -6.74 kN ( $\gamma = 1.5$ ) | 83   |

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -4.99 \times 1.35 + (-0.25 \times 1.35) = -7.07 \text{ kN}$$

##### J.4.3.2.1. 3m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 80 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 3.0m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |    |                                                                                          |    |
|------|----|------------------------------------------------------------------------------------------|----|
| UC.3 | 3m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.41 / (12.46 \times 0.136) = 0.83 \leq 1$ | OK |
|------|----|------------------------------------------------------------------------------------------|----|

See Annex G.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 197 for the elaborate check

#### J.4.4 Corner poles

|        | Load combinations                                   | Element          | Representative force | Design value force           | Pag. |
|--------|-----------------------------------------------------|------------------|----------------------|------------------------------|------|
| 10x15m | CO6. Own weight + pretension + wind pressure + snow | Corner pole 2.5m | -5.50 kN             | -7.43 kN ( $\gamma = 1.35$ ) | 83   |

User load of max. 25 kg is applied, loaded centrically.

2.5m corner pole  $N_{ed} = -5.50 \times 1.35 + (-0.25 \times 1.35) = -7.76 \text{ kN}$

##### J.4.4.1 2.5m pole - Wood, Spruce C18

|         |   |                                                                                                                |
|---------|---|----------------------------------------------------------------------------------------------------------------|
| Profile | = | Pole, $\varnothing \approx 75 \text{ mm}$<br>average diameter, as a minimum required at the middle of the pole |
| Length  | = | max. 2.5m                                                                                                      |
| Quality | = | C18                                                                                                            |

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

|      |      |                                                                                       |    |
|------|------|---------------------------------------------------------------------------------------|----|
| UC.4 | 2.5m | $\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.76 / (12.46 \times 0.170) = 0.80 < 1$ | OK |
|------|------|---------------------------------------------------------------------------------------|----|

See Annex G.6 Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR], on page 199 for the elaborate check

#### J.4.5 Belts

|        | Load combination                                    | Element           | Representative force | Design value force          | Pag. |
|--------|-----------------------------------------------------|-------------------|----------------------|-----------------------------|------|
| 10x20m | CO4. Own weight + pretension + wind suction         | Corner belt       | 4.43 kN              | 6.65 kN ( $\gamma = 1.5$ )  | 83   |
| 10x20m | CO6. Own weight + pretension + wind pressure + snow | Belt – long side  | 1.50 kN              | 2.03 kN ( $\gamma = 1.35$ ) | 83   |
| 10x15m | CO6. Own weight + pretension + wind pressure + snow | Belt – short side | 1.37 kN              | 1.85 kN ( $\gamma = 1.35$ ) | 83   |

##### J.4.5.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

|       |                   |                                      |                    |                                   |    |
|-------|-------------------|--------------------------------------|--------------------|-----------------------------------|----|
| UC.5a | Corner belt       | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $6.65 / (1 \times 15) = 0.44 < 1$ | OK |
| UC.5b | Belt – long side  | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $2.03 / (1 \times 15) = 0.14 < 1$ | OK |
| UC.5c | Belt – short side | 1 PES belt with minimum BL = 3000 kg | $F_d / F_{rd} < 1$ | $1.85 / (1 \times 15) = 0.12 < 1$ | OK |

For capacity of belts see H.2.3, page 24

##### J.4.5.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used →  $F_{rd,8mm} = 2.25$  kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

|       |                   |                                                  |                    |                                              |    |
|-------|-------------------|--------------------------------------------------|--------------------|----------------------------------------------|----|
| UC.5d | Corner rope       | 1 individual rope made of <u>3 sections</u> Ø8mm | $F_d / F_{rd} < 1$ | $6.65 / (1 \times 3 \times 2.25) = 0.99 < 1$ | OK |
| UC.5e | Rope – long side  | 1 individual rope made of <u>1 section</u> Ø8mm  | $F_d / F_{rd} < 1$ | $2.13 / (1 \times 1 \times 2.25) = 0.95 < 1$ | OK |
| UC.5f | Rope – short side | 1 individual rope made of <u>1 section</u> Ø8mm  | $F_d / F_{rd} < 1$ | $1.85 / (1 \times 1 \times 2.25) = 0.82 < 1$ | OK |

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength ( $BL_{rope}$ ) and less rope sections (n), as long as  $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2660$  kg

Rope – long side:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 852$  kg

Rope – long side:  $BL_{rope} \times n_{sections} \times n_{ropes} \geq 740$  kg

#### J.4.6 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

##### J.4.6.1 Side belt connection

J.4.6.1.1. M12 Eye bolt, DIN580

For  $\leq 45^\circ$ :

M12, WLL 340kg  $\rightarrow$  BL = 6 x 3.4 kN = 20.4 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 20.4 / 2 = 10.2 kN

F<sub>d</sub>=2.13 kN

|       |                 |                    |                            |    |
|-------|-----------------|--------------------|----------------------------|----|
| UC.6a | Corner eye bolt | $F_d / F_{rd} < 1$ | $2.13 / 10.2 = 0.21 < 1.0$ | OK |
|-------|-----------------|--------------------|----------------------------|----|

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.

J.4.6.1.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg  $\rightarrow$  BL = 2 x 5.8 kN = 11.6 kN  $\rightarrow$  Frd = BL /  $\gamma_m$  = 11.6 kN / 2 = 5.8 kN

F<sub>d</sub>=2.13 kN

|       |                    |                          |    |
|-------|--------------------|--------------------------|----|
| UC.6b | $F_d / F_{rd} < 1$ | $2.13 / 5.80 = 0.78 < 1$ | OK |
|-------|--------------------|--------------------------|----|

For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

##### J.4.6.2 Base plate

J.4.6.2.1. Side pole & corner pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm,  $p_{soil} = 125 \text{ kN/m}^2$ )

|       |      |                            |                                             |    |
|-------|------|----------------------------|---------------------------------------------|----|
| UC.6c | Soil | $\frac{F_d}{A} / p \leq 1$ | $7.76 / (0.25 \cdot 0.25) / 125 = 0.99 < 1$ | OK |
|-------|------|----------------------------|---------------------------------------------|----|

J.4.6.2.2. Main pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 35x35 cm,  $p_{soil} = 175 \text{ kN/m}^2$ )

|       |      |                            |                                              |    |
|-------|------|----------------------------|----------------------------------------------|----|
| UC.6d | Soil | $\frac{F_d}{A} / p \leq 1$ | $14.27 / (0.35 \cdot 0.35) / 175 = 0.63 < 1$ | OK |
|-------|------|----------------------------|----------------------------------------------|----|

## K. Material certificates

### K.1. Fabric specifications

# SIOEN FABRICS

Z.I. du Blanc Ballot, Avenue Urbino 6, B-7700 Mouscron - Tel +32(0)56 85 68 80 - Fax +32(0)56 34 61 31  
sioenfabrics@sioen.be - <http://www.sioen.com>

## Technical Data Sheet

### RP5639

FLEXITENT SINGLE M2 (fiche provisoire) - Width 153 cm

|                                                              |                               |                    |
|--------------------------------------------------------------|-------------------------------|--------------------|
| Total weight / Poids total                                   | 730 [+/-10%] g/m <sup>2</sup> | ISO 3801/5 1977    |
| Weight of fabric [PES] / Poids tissu [PES]                   | 280 [+/-10] g/m <sup>2</sup>  | ISO 3801/5 1977    |
| Weight of coating [PU/PVC] / Poids enduction [PU/PVC]        | 450 [+/-10%] g/m <sup>2</sup> | ISO 3801/5 1977    |
| Breaking strength warp<br>Force de rupture chaîne            | 1200 N/5cm                    | ISO 13934 1999     |
| Breaking strength weft<br>Force de rupture trame             | 900 N/5cm                     | ISO 13934 1999     |
| Breaking extension warp<br>Allongement à la rupture chaîne   | 110 %                         | ISO 13934 1999     |
| Breaking extension weft<br>Allongement à la rupture trame    | 110 %                         | ISO 13934 1999     |
| Tear strenght warp<br>Résistance à la déchirure chaîne       | 100 N                         | ISO 4674A1 1977    |
| Tear strenght weft<br>Résistance à la déchirure trame        | 180 N                         | ISO 4674A1 1977    |
| Adhesion warp / Adhérence chaîne                             | 23 N/5cm                      | D41 1015           |
| Adhesion weft / Adhérence trame                              | 23 N/5cm                      | D41 1015           |
| Waterproofness before washing<br>Imperméabilité avant lavage | OK:2000 G M D mm              | ISO 811 1981       |
| Waterproofness after washing<br>Imperméabilité après lavage  | OK:2000 G M D mm              | ISO 811 1981       |
| Flammability / Inflammabilité                                | OK:M2 -                       | NF P 92-507 (2004) |



May be cleaned with mild, PH neutral, non-ignogenic detergents. Certainly avoid solvent containing cleaning agents (alcohols, polar solvents, etc.). Testing is always advised before using cleaning agents.  
After surface cleaning, please let air dry.

All our technical characteristics are indicative  
Rev. 10/2012

## K.2. Fabric fire certificate



INSTITUT TECHNOLOGIQUE

### INSTITUT TECHNOLOGIQUE

FCBA

Laboratoire d'essais feu

10 rue Galilée - 77420 Champs-sur-Marne

Laboratoire agréé par le Ministère de l'Intérieur et de l'Aménagement du Territoire

JO du 10 février 2007

### PROCES-VERBAL DE CLASSEMENT DE REACTION AU FEU

D'UN MATERIAU PREVU A L'ARTICLE 5

DE L'ARRETE DU 21 NOVEMBRE 2002

**N° CM - 21 - P - 096**

et 1 annexe de 4 pages.

Valable 5 ans à compter du mardi 13 juillet 2021

**Matériau présenté par : SIOEN FABRICS SA**

Avenue Urbino 6

7700 MOUSCRON

**Marque commerciale : F5639**

#### Description sommaire:

**Type de textile enduit** tricot polyester enduit de PVC ignifugé

**Composition** support textile/ fabric 100% polyester (280 g/m<sup>2</sup>) + enduction/ coating PVC ignifugé (450 g/m<sup>2</sup>)

**Masse surfacique** 730 g/m<sup>2</sup>

**Epaisseur** 0,75 mm

**Coloris** tous coloris/ all colors

**Nature de l'essai :** Essai au brûleur électrique  
Essai de persistance de flamme  
Essai de gouttes

**Classement : M2**

**Observations :** /

**Durabilité du classement (l'annexe 22) :** Non limitée a priori

**Usage :** Revêtement de murs tendus, voilages, rideaux,... **Conditions d'entretien :** selon préconisations du client

**Compte tenu des critères résultant des essais annexés décrits dans le(s) rapport(s) d'essais N° : FE 21-0216-1182**

Ce procès-verbal atteste uniquement des caractéristiques de l'échantillon soumis aux essais et ne préjuge pas des caractéristiques de produits similaires. Il ne constitue donc pas une certification de produits au sens de l'article L.115-27 du code de la consommation et de la loi du 3 juin 1994.

NOTA : Seules sont autorisées les reproductions intégrales et par photocopies du présent procès-verbal de classement ou de l'ensemble du procès-verbal de classement et rapport(s) d'essais annexé(s). Ces conclusions ne portent que sur les performances de résistance / réaction au feu de l'élément objet du présent procès verbal de classement. Elles ne préjugent, en aucun cas, des autres performances liées à son incorporation à un ouvrage"

**"Non valable pour toute application couverte par l'article AM 18 (Arrêté du 6 mars 2006) concernant les sièges rembourrés"**

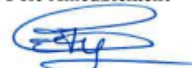
**Champs-sur-Marne, le vendredi 23 juillet 2021**

Technicien d'essais du Laboratoire Feu



Bilel HAMILA

Responsable Technique du laboratoire Feu  
du Pôle Ameublement



Clotilde ETIEMVRE

Siège social  
10 rue Galilée  
77420 Champs-sur-Marne  
Tél + 33 (0)1 72 84 97 84  
www.fcba.fr  
Siret 775 680 903 00132  
APE 7219 Z  
Code TVA CEE : FR 14 775 680 903



Institut technologique FCBA : Forêt, Cellulose, Bois - construction, Ameublement



Laboratory approved by the Ministry for the Interior  
JO of February 10, 2007  
**OFFICIAL REPORT OF CLASSIFICATION OF FIRE PERFORMANCE**  
**Of a MATERIAL ENVISAGED IN ARTICLE 5**  
**DECREE OF November 21, 2002**

N° **CM - 21 - P - 096**

Page 1

et 1 annexe de 4 pages.

Valid 5 years as from mardi 13 juillet 2021

Material presents by : **SIOEN FABRICS SA**

Avenue Urbino 6  
7700 MOUSCRON

Trade mark: **F5639**

**Summary description**

Kind of coating fabric **tricot polyester enduit de PVC ignifugé**  
Composition **support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)**  
Mass per unit area **730 g/m²**  
Thickness **0,75 mm**  
Color **tous coloris/ all colors**

Nature of the test : **Electrical burner test**  
**Flame persistence**  
**Dripping test**

Classification : **M2**

Observations :

**Durability of the classification (Article 5 de l'annexe 2) Not limited**

Usual term **Walls hung covering, sheers, curtains,...**

Cleaning conditions **Recommendation of Customer**

**Taking into account the criteria resulting from the tests annexed described in the test report N°FE 21-0216-1182**

This official report attests only characteristics of the sample submitted for testing and does not prejudice similar characteristics of products. It thus does not constitute a certification of products within the meaning of the L.115-27 article of the code of the consumption and the law of June 3rd, 1994.  
NOTE: The integral reproductions and by photocopy of this official report of classification are only authorized or the official report unit of classification and report of annexed test. These conclusions relate only to the resistance / reaction to fire performance of the element that is the subject of the present classification report. They do not prejudice, in any case, other performances related to its incorporation to a work \*

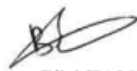
**"Not legally acceptable for any use under the pretence of the article AM 18 (Decree of March 6th, 2006) for upholstered seat"**

**Champs-sur-Marne, le vendredi 23 juillet 2021**

Technicien d'essais du Laboratoire Feu

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Code TVA CEE : FR 14 775 680 903



Bilal HAMILA

Only the french version is valid  
Institut technologique FCBA : Forêt, Bois, Industrie - construction, Ameublement



Clotilde ETIEMVRE



## K.5. Carabine, DIN 5299C

# Karta Techniczna Nr 10GM/KAR

1. Nazwa dostawcy: „GÓRALMET” M. i J. Góral Sp. J.

Adres dostawcy: ul. Krakowska 68, 32-860 Czchów, tel/fax 014 6635260

2. Nazwa wyrobu: Karabińczyk do lin 5299C

3. Opis i przeznaczenie: Zatrzaśniki stalowe, służące jako elementy łączące w zakończeniach lin, łańcuchów gospodarczych, oraz do innych zastosowań ogólnych. Do stosowania w zakresie parametrów technicznych podanych w punkcie 5.

4. Karabińczyki do lin nie mogą być użyte w zastosowaniach posiadających inne - uregulowane prawem czy normami wymagania bezpieczeństwa, np. do podnoszenia ładunków, jako środek do ochrony indywidualnej przed upadkiem z wysokości, lub osprzęt alpinistyczny. Nie przeciążać ponad wartość WLL (kG) podaną w kolumnie 13.

5. Parametry techniczne wyrobu:

| LP. | Rozmiar [mm] | Wymiary nominalne [mm] |            |        |            |        |        |     |     |     |       | Masa [kg/szt] | WLL Obciążenie robocze [kG] | BF Siła niszcząca [min. kG] | Materiał              | Powłoki |
|-----|--------------|------------------------|------------|--------|------------|--------|--------|-----|-----|-----|-------|---------------|-----------------------------|-----------------------------|-----------------------|---------|
|     |              | A [mm]                 | C min [mm] | D [mm] | E min [mm] | Ø [mm] | L [mm] | --- | --- | --- | ---   |               |                             |                             |                       |         |
| 1   | 2            | 3                      | 4          | 5      | 6          | 7      | 8      | 9   | 10  | 11  | 12    | 13            | 14                          | 15                          | 16                    |         |
| 1   | 4            | 19,5                   | 4,5        | 6      | 4,5        | 3,9    | 40     | --- | --- | --- | 0,008 | 70            | 140                         | C10 ~ C15                   | Ocynk elektrolityczny |         |
| 2   | 5            | 24                     | 5,5        | 7,5    | 5,5        | 4,8    | 50     | --- | --- | --- | 0,015 | 100           | 200                         |                             |                       |         |
| 3   | 6            | 29                     | 8          | 8,5    | 6          | 5,5    | 60     | --- | --- | --- | 0,023 | 120           | 240                         |                             |                       |         |
| 4   | 7            | 34                     | 9          | 10,5   | 8,5        | 6,7    | 70     | --- | --- | --- | 0,040 | 180           | 360                         |                             |                       |         |
| 5   | 8            | 39                     | 10         | 11,5   | 9          | 7,4    | 80     | --- | --- | --- | 0,056 | 230           | 460                         |                             |                       |         |
| 6   | 9            | 44                     | 11         | 13,5   | 10         | 8,4    | 90     | --- | --- | --- | 0,083 | 250           | 500                         |                             |                       |         |
| 7   | 10           | 48                     | 11,5       | 14,5   | 11         | 9,4    | 100    | --- | --- | --- | 0,115 | 350           | 700                         |                             |                       |         |
| 8   | 11           | 57                     | 17         | 17,5   | 13         | 10,7   | 120    | --- | --- | --- | 0,179 | 400           | 800                         |                             |                       |         |
| 9   | 12           | 67                     | 22         | 20     | 15         | 11,5   | 140    | --- | --- | --- | 0,246 | 450           | 900                         |                             |                       |         |
| 10  | 13           | 80                     | 26         | 21     | 17         | 12,6   | 160    | --- | --- | --- | 0,343 | 530           | 1060                        |                             |                       |         |
| 11  | 14           | 87                     | 28         | 24     | 20         | 13,6   | 180    | --- | --- | --- | 0,443 | 580           | 1160                        |                             |                       |         |
| 12  | 15           | 98                     | 32         | 24     | 20         | 14,7   | 200    | --- | --- | --- | 0,563 | 700           | 1400                        |                             |                       |         |

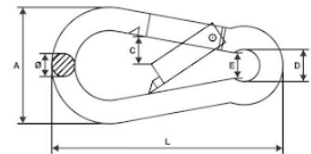
6. Informacje dodatkowe:

➤ Współczynnik bezpieczeństwa - 2

➤ BF = 2,0 x WLL

➤ Cechowanie: symbol producenta – GM lub GMG; kod rozmiaru

➤ Informacje dodatkowe – etykieta: nazwa wyrobu; rozmiar; ilość szt.; numer dostawy; dane producenta i adres strony internetowej.



## K.6. Rope ratchet



### 3/8" Rope Ratchet

Carolina North Manufacturing  
10022

\$10.89 ~~\$12.89~~

**Maximum load limit 250 lbs.** Outer casings are made from #6-33% glass filled nylon with interior mechanisms of die cast zinc. Perfect for large jobs:

- heavy loads for transport
- hoisting items
- straightening items
- securing objects under tension
- product weight 0.16 lbs

Product is compatible with our 3/8" S-hooks and 3/8" solid braided rope.

Bulk purchasing options available.

ADD TO CART

BUY IT NOW

Share:    

Collections: Bulk Rope Ratchets

Type: Bulk Ratchet

## Annex

|                                                                           |            |
|---------------------------------------------------------------------------|------------|
| <b>Annex A. Software input (load combination) – 10x10m model.....</b>     | <b>98</b>  |
| Annex A.1. CO1. Own weight + pretension .....                             | 98         |
| Annex A.2. CO2. Own weight + pretension + conventional / snow .....       | 99         |
| Annex A.3. CO3. Own weight + pretension + wind pressure .....             | 100        |
| Annex A.4. CO4. Own weight + pretension + wind suction .....              | 101        |
| Annex A.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 102        |
| Annex A.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 103        |
| <b>Annex B. Software input (load combination) – 10x15m model.....</b>     | <b>105</b> |
| Annex B.1. CO1. Own weight + pretension .....                             | 105        |
| Annex B.2. CO2. Own weight + pretension + conventional / snow .....       | 106        |
| Annex B.3. CO3. Own weight + pretension + wind pressure .....             | 107        |
| Annex B.4. CO4. Own weight + pretension + wind suction .....              | 108        |
| Annex B.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 109        |
| Annex B.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 110        |
| <b>Annex C. Software input (load combination) – 10x20m model.....</b>     | <b>112</b> |
| Annex C.1. CO1. Own weight + pretension .....                             | 112        |
| Annex C.2. CO2. Own weight + pretension + conventional / snow .....       | 113        |
| Annex C.3. CO3. Own weight + pretension + wind pressure .....             | 114        |
| Annex C.4. CO4. Own weight + pretension + wind suction .....              | 115        |
| Annex C.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 116        |
| Annex C.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 117        |
| <b>Annex D. Results per load combination – 10x10m model .....</b>         | <b>119</b> |
| Annex D.1. CO1. Own weight + pretension .....                             | 119        |
| Annex D.2. CO2. Own weight + pretension + snow .....                      | 123        |
| Annex D.3. CO3. Own weight + pretension + wind pressure .....             | 127        |
| Annex D.4. CO4. Own weight + pretension + wind suction .....              | 131        |
| Annex D.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 135        |
| Annex D.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 139        |
| <b>Annex E. Results per load combination – 10x15m model .....</b>         | <b>143</b> |
| Annex E.1. CO1. Own weight + pretension .....                             | 143        |
| Annex E.2. CO2. Own weight + pretension + snow .....                      | 147        |
| Annex E.3. CO3. Own weight + pretension + wind pressure .....             | 151        |
| Annex E.4. CO4. Own weight + pretension + wind suction .....              | 155        |
| Annex E.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 159        |
| Annex E.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 163        |
| <b>Annex F. Results per load combination – 10x20m model .....</b>         | <b>167</b> |
| Annex F.1. CO1. Own weight + pretension .....                             | 167        |
| Annex F.2. CO2. Own weight + pretension + snow .....                      | 171        |
| Annex F.3. CO3. Own weight + pretension + wind pressure .....             | 175        |
| Annex F.4. CO4. Own weight + pretension + wind suction .....              | 179        |
| Annex F.5. CO5. Own weight + pretension + wind suction reduced [NL] ..... | 183        |
| Annex F.6. CO6. Own weight + pretension + wind pressure + snow [FR] ..... | 187        |

|                                                                                     |            |
|-------------------------------------------------------------------------------------|------------|
| <b>Annex G. Member checks .....</b>                                                 | <b>191</b> |
| Annex G.1. Main pole 4.0m – Wood, Spruce C18.....                                   | 191        |
| Annex G.2. Side pole 3m – Wood, Spruce C18 .....                                    | 192        |
| Annex G.3. Corner pole 2.5m - Wood, C18.....                                        | 194        |
| Annex G.4. Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR]   | 196        |
| Annex G.5. Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR]   | 197        |
| Annex G.6. Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR] | 199        |
| <b>Annex H. Reaction forces model 10x10m .....</b>                                  | <b>200</b> |
| Annex H.1. Point numbers.....                                                       | 200        |
| Annex H.2. Reaction forces .....                                                    | 201        |
| <b>Annex I. Reaction forces model 10x15m .....</b>                                  | <b>202</b> |
| Annex I.1. Point numbers .....                                                      | 202        |
| Annex I.2. Reaction forces.....                                                     | 203        |
| <b>Annex J. Reaction forces model 10x20m .....</b>                                  | <b>204</b> |
| Annex J.1. Point numbers .....                                                      | 204        |
| Annex J.2. Reaction forces .....                                                    | 205        |

## Annex A. Software input (load combination) – 10x10m model

### Annex A.1. CO1. Own weight + pretension

| EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)<br>ORDERED BY LOADGROUPS |            |        |                  |        |        |         |             |
|----------------------------------------------------------------|------------|--------|------------------|--------|--------|---------|-------------|
| LOADGROUP                                                      | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
| STRUTS                                                         | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES                                                         | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0291 | 0.000371    |
| MEM-LINKS1000                                                  | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000                                                  | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM                                                            |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |
| EXTERNAL LOADS (AREA-DEPENDENT)<br>ORDERED BY LOADGROUPS       |            |        |                  |        |        |         |             |
| LOADGROUP                                                      | LOADMODE   | LOAD   | FACTOR           | SUM_X  | SUM_Y  | SUM_Z   | LOADED AREA |
| SUM                                                            |            |        |                  | 0.0000 | 0.0000 | 0.0000  |             |
| EXTERNAL LOADS (AREA-DEPENDENT)<br>ORDERED BY LOADMODES        |            |        |                  |        |        |         |             |
|                                                                | LOADMODE   |        |                  | SUM_X  | SUM_Y  | SUM_Z   |             |
| SUM                                                            | AREA-LOADS |        |                  | 0.0000 | 0.0000 | 0.0000  |             |
| EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS                      |            |        |                  |        |        |         |             |
|                                                                |            |        |                  | SUM_X  | SUM_Y  | SUM_Z   |             |
|                                                                |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |

Inner Energy= 1.574062

StatikD: Sum of external loads: 0.000000 0.000000 -1.710614  
 StatikD: Sum of support forces: 0.000004 0.000011 1.710763  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\C01\_OW+VO  
 StatikD: Time of calculation: 0.24s  
 StatikD: Version 2025.38 done.

## Annex A.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0291 | 0.000371    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X  | SUM_Y  | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|--------|--------|---------|-------------|
| 1         | SCHNEE   | 0.1000 | 1.00   | 0.0000 | 0.0000 | -9.4121 | 94.12       |
| SUM       |          |        |        | 0.0000 | 0.0000 | -9.4121 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z   |
|-----|------------|--------|--------|---------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -9.4121 |
| SUM | AREA-LOADS | 0.0000 | 0.0000 | -9.4121 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0000 | 0.0000 | -11.1227 |

Inner Energy= 2.607481

StatikD: Sum of external loads: 0.000000 0.000000 -11.122765  
 StatikD: Sum of support forces: 0.000003 0.000012 11.124678  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO2\_OW+VO+SNOW  
 StatikD: Time of calculation: 0.27s  
 StatikD: Version 2025.38 done.



### Annex A.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0291 | 0.000371    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X   | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|---------|--------|----------|-------------|
| 1         | WIND     | -0.1500 | 1.00   | -0.0000 | 0.0000 | -14.1181 | 98.07       |
| SUM       |          |         |        | -0.0000 | 0.0000 | -14.1181 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y  | SUM_Z    |
|-----|------------|---------|--------|----------|
| SUM | WIND       | -0.0000 | 0.0000 | -14.1181 |
| SUM | AREA-LOADS | -0.0000 | 0.0000 | -14.1181 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y  | SUM_Z    |
|---------|--------|----------|
| -0.0000 | 0.0000 | -15.8288 |

Inner Energy= 3.515106

StatikD: Sum of external loads: -0.000001 0.000001 -15.828783  
 StatikD: Sum of support forces: 0.000007 0.000009 15.829106  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO3\_OW+V0+WD  
 StatikD: Time of calculation: 0.28s  
 StatikD: Version 2025.38 done.



## Annex A.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0385 | 0.000491    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -1.7200 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X   | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|---------|---------|---------|-------------|
| 1         | WIND     | 0.3500 | 1.00   | -0.0013 | -0.0041 | 32.9139 | 98.05       |
| SUM       |          |        |        | -0.0013 | -0.0041 | 32.9139 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y   | SUM_Z   |
|-----|------------|---------|---------|---------|
| SUM | WIND       | -0.0013 | -0.0041 | 32.9139 |
| SUM | AREA-LOADS | -0.0013 | -0.0041 | 32.9139 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y   | SUM_Z   |
|---------|---------|---------|
| -0.0013 | -0.0041 | 31.1939 |

Inner Energy= 7.087801

StatikD: Sum of external loads: -0.001328 -0.004069 31.193895  
 StatikD: Sum of support forces: 0.001270 0.004033 -31.194731  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO4\_OW+VO+WU  
 StatikD: Time of calculation: 0.99s  
 StatikD: Version 2025.38 done.

## Annex A.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0291 | 0.000371    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X  | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|--------|---------|---------|-------------|
| 1         | WIND     | 0.1750 | 1.00   | 0.0000 | -0.0000 | 16.4711 | 98.07       |
| SUM       |          |        |        | 0.0000 | -0.0000 | 16.4711 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y   | SUM_Z   |
|-----|------------|--------|---------|---------|
| SUM | WIND       | 0.0000 | -0.0000 | 16.4711 |
| SUM | AREA-LOADS | 0.0000 | -0.0000 | 16.4711 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y   | SUM_Z   |
|--------|---------|---------|
| 0.0000 | -0.0000 | 14.7605 |

Inner Energy= 4.086116

StatikD: Sum of external loads: -0.000003 -0.000004 14.760512  
 StatikD: Sum of support forces: 0.000000 0.000005 -14.760568  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO5\_OW+VO+WU\_red  
 StatikD: Time of calculation: 1.98s  
 StatikD: Version 2025.38 done.

## Annex A.6. CO6. Own weight + pretension + wind pressure + snow [FR]

### Annex A.6.1. For element check

#### EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.7115 | 0.009063    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0291 | 0.000371    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4864 | 97.285966   |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.4836 | 96.722595   |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -1.7107 |             |

#### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X   | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|---------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1000  | 1.00   | 0.0000  | 0.0000 | -9.4121  | 94.12       |
| 1         | WIND     | -0.1500 | 1.00   | -0.0000 | 0.0000 | -14.1181 | 98.07       |
| SUM       |          |         |        | -0.0000 | 0.0000 | -23.5302 |             |

#### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y  | SUM_Z    |
|-----|------------|---------|--------|----------|
| SUM | SCHNEE     | 0.0000  | 0.0000 | -9.4121  |
| SUM | WIND       | -0.0000 | 0.0000 | -14.1181 |
| SUM | AREA-LOADS | -0.0000 | 0.0000 | -23.5302 |

#### EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y  | SUM_Z    |
|---------|--------|----------|
| -0.0000 | 0.0000 | -25.2409 |

Inner Energy= 5.265614

StatikD: Sum of external loads: -0.000001 0.000001 -25.240880  
 StatikD: Sum of support forces: 0.000009 0.000008 25.240553  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO6\_OW+VO+WD+SNOW  
 StatikD: Time of calculation: 0.30s  
 StatikD: Version 2025.38 done.

## Annex A.6.2. For anchor and ballast calculation

| EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)<br>ORDERED BY LOADGROUPS |                                                                                                |           |                  |            |        |          |             |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|------------------|------------|--------|----------|-------------|
| LOADGROUP                                                      | LOADMODE                                                                                       | FACTOR    | VOLUME/AREA(P.U) | SUM_X      | SUM_Y  | SUM_Z    | VOLUME/AREA |
| STRUTS                                                         | SELFWEIGHT                                                                                     | 1.00      | 78.500000        | 0.0000     | 0.0000 | -0.7115  | 0.009063    |
| CABLES                                                         | SELFWEIGHT                                                                                     | 1.00      | 78.500000        | 0.0000     | 0.0000 | -0.0291  | 0.000371    |
| MEM-LINKS1000                                                  | SELFWEIGHT                                                                                     | 1.00      | 0.005000         | 0.0000     | 0.0000 | -0.4864  | 97.285966   |
| MEM-LINKS2000                                                  | SELFWEIGHT                                                                                     | 1.00      | 0.005000         | 0.0000     | 0.0000 | -0.4836  | 96.722595   |
| SUM                                                            |                                                                                                |           |                  | 0.0000     | 0.0000 | -1.7107  |             |
| EXTERNAL LOADS (AREA-DEPENDENT)<br>ORDERED BY LOADGROUPS       |                                                                                                |           |                  |            |        |          |             |
| LOADGROUP                                                      | LOADMODE                                                                                       | LOAD      | FACTOR           | SUM_X      | SUM_Y  | SUM_Z    | LOADED AREA |
| 1                                                              | SCHNEE                                                                                         | 0.1083    | 1.00             | 0.0000     | 0.0000 | -10.1961 | 94.12       |
| 1                                                              | WIND                                                                                           | -0.1500   | 1.00             | -0.0000    | 0.0000 | -14.1181 | 98.07       |
| SUM                                                            |                                                                                                |           |                  | -0.0000    | 0.0000 | -24.3142 |             |
| EXTERNAL LOADS (AREA-DEPENDENT)<br>ORDERED BY LOADMODES        |                                                                                                |           |                  |            |        |          |             |
|                                                                | LOADMODE                                                                                       |           |                  | SUM_X      | SUM_Y  | SUM_Z    |             |
| SUM                                                            | SCHNEE                                                                                         |           |                  | 0.0000     | 0.0000 | -10.1961 |             |
| SUM                                                            | WIND                                                                                           |           |                  | -0.0000    | 0.0000 | -14.1181 |             |
| SUM                                                            | AREA-LOADS                                                                                     |           |                  | -0.0000    | 0.0000 | -24.3142 |             |
| EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS                      |                                                                                                |           |                  |            |        |          |             |
|                                                                |                                                                                                |           |                  | SUM_X      | SUM_Y  | SUM_Z    |             |
|                                                                |                                                                                                |           |                  | -0.0000    | 0.0000 | -26.0249 |             |
| Inner Energy= 5.420759                                         |                                                                                                |           |                  |            |        |          |             |
| StatikD:                                                       | Sum of external loads:                                                                         | -0.000001 | 0.000001         | -26.024883 |        |          |             |
| StatikD:                                                       | Sum of support forces:                                                                         | 0.000009  | 0.000009         | 26.026117  |        |          |             |
| StatikD:                                                       | P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\C06a_OW+VO+WD+SNOW_anchors |           |                  |            |        |          |             |
| StatikD:                                                       | Time of calculation:                                                                           | 0.24s     |                  |            |        |          |             |
| StatikD:                                                       | Version 2025.38 done.                                                                          |           |                  |            |        |          |             |

## Annex B. Software input (load combination) – 10x15m model

### Annex B.1. CO1. Own weight + pretension

|                                           |            |        |                  |        |         |         |             |
|-------------------------------------------|------------|--------|------------------|--------|---------|---------|-------------|
| EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)     |            |        |                  |        |         |         |             |
| ORDERED BY LOADGROUPS                     |            |        |                  |        |         |         |             |
| LOADGROUP                                 | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y   | SUM_Z   | VOLUME/AREA |
| STRUTS                                    | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000  | -0.9460 | 0.012051    |
| CABLES                                    | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000  | -0.0393 | 0.000500    |
| MEM-LINKS1000                             | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000  | -0.7673 | 153.469253  |
| MEM-LINKS2000                             | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000  | -0.7628 | 152.565787  |
| SUM                                       |            |        |                  | 0.0000 | 0.0000  | -2.5155 |             |
| EXTERNAL LOADS (AREA-DEPENDENT)           |            |        |                  |        |         |         |             |
| ORDERED BY LOADGROUPS                     |            |        |                  |        |         |         |             |
| LOADGROUP                                 | LOADMODE   | LOAD   | FACTOR           | SUM_X  | SUM_Y   | SUM_Z   | LOADED AREA |
| SUM                                       |            |        |                  | 0.0000 | 0.0000  | 0.0000  |             |
| EXTERNAL LOADS (AREA-DEPENDENT)           |            |        |                  |        |         |         |             |
| ORDERED BY LOADMODES                      |            |        |                  |        |         |         |             |
| LOADMODE                                  |            |        | SUM_X            | SUM_Y  | SUM_Z   |         |             |
| SUM                                       | AREA-LOADS |        | 0.0000           | 0.0000 | 0.0000  |         |             |
| EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS |            |        |                  |        |         |         |             |
|                                           |            |        | SUM_X            | SUM_Y  | SUM_Z   |         |             |
|                                           |            |        | 0.0000           | 0.0000 | -2.5155 |         |             |

Inner Energy= 2.345706

StatikD: Sum of external loads: 0.000000 0.000000 -2.515503  
 StatikD: Sum of support forces: -0.002219 -0.001612 2.515072  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO1\_OW+VO  
 StatikD: Time of calculation: 0.68s  
 StatikD: Version 2025.38 done.

## Annex B.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0393 | 0.000500    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5155 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|--------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1000 | 1.00   | 0.0000 | 0.0000 | -14.8990 | 148.99      |
| SUM       |          |        |        | 0.0000 | 0.0000 | -14.8990 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -14.8990 |
| SUM | AREA-LOADS | 0.0000 | 0.0000 | -14.8990 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0000 | 0.0000 | -17.4145 |

Inner Energy= 4.245182

StatikD: Sum of external loads: 0.000000 0.000000 -17.414544  
 StatikD: Sum of support forces: -0.002619 0.001603 17.415259  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO2\_OW+VO+SNOW  
 StatikD: Time of calculation: 1.13s  
 StatikD: Version 2025.38 done.

### Annex B.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0393 | 0.000500    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5155 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | WIND     | -0.1500 | 1.00   | 0.0042 | 0.0084 | -22.3486 | 154.64      |
| SUM       |          |         |        | 0.0042 | 0.0084 | -22.3486 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | WIND       | 0.0042 | 0.0084 | -22.3486 |
| SUM | AREA-LOADS | 0.0042 | 0.0084 | -22.3486 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0042 | 0.0084 | -24.8641 |

Inner Energy= 5.886284

StatikD: Sum of external loads: 0.004168 0.008447 -24.864059  
 StatikD: Sum of support forces: -0.005342 -0.006802 24.865402  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO3\_OW+VO+WD  
 StatikD: Time of calculation: 1.23s  
 StatikD: Version 2025.38 done.

## Annex B.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0586 | 0.000747    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5349 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X   | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|---------|---------|---------|-------------|
| 1         | WIND     | 0.3500 | 1.00   | -0.0091 | -0.0203 | 52.0521 | 154.58      |
| SUM       |          |        |        | -0.0091 | -0.0203 | 52.0521 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y   | SUM_Z   |
|-----|------------|---------|---------|---------|
| SUM | WIND       | -0.0091 | -0.0203 | 52.0521 |
| SUM | AREA-LOADS | -0.0091 | -0.0203 | 52.0521 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y   | SUM_Z   |
|---------|---------|---------|
| -0.0091 | -0.0203 | 49.5172 |

Inner Energy= 11.014835

StatikD: Sum of external loads: -0.009041 -0.020279 49.517238  
 StatikD: Sum of support forces: 0.009277 0.019985 -49.516573  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO4\_OW+VO+WU  
 StatikD: Time of calculation: 2.25s  
 StatikD: Version 2025.38 done.



## Annex B.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0393 | 0.000500    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5155 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X   | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|---------|---------|---------|-------------|
| 1         | WIND     | 0.1750 | 1.00   | -0.0049 | -0.0098 | 26.0733 | 154.64      |
| SUM       |          |        |        | -0.0049 | -0.0098 | 26.0733 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y   | SUM_Z   |
|-----|------------|---------|---------|---------|
| SUM | WIND       | -0.0049 | -0.0098 | 26.0733 |
| SUM | AREA-LOADS | -0.0049 | -0.0098 | 26.0733 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y   | SUM_Z   |
|---------|---------|---------|
| -0.0049 | -0.0098 | 23.5579 |

Inner Energy= 7.521792

StatikD: Sum of external loads: -0.004882 -0.009859 23.557848  
 StatikD: Sum of support forces: 0.006997 0.018128 -23.596153  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO5\_OW+VO+WU\_red  
 StatikD: Time of calculation: 2.12s  
 StatikD: Version 2025.38 done.

## Annex B.6. CO6. Own weight + pretension + wind pressure + snow [FR]

### Annex B.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0393 | 0.000500    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5155 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1000  | 1.00   | 0.0000 | 0.0000 | -14.8990 | 148.99      |
| 1         | WIND     | -0.1500 | 1.00   | 0.0042 | 0.0084 | -22.3486 | 154.64      |
| SUM       |          |         |        | 0.0042 | 0.0084 | -37.2476 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -14.8990 |
| SUM | WIND       | 0.0042 | 0.0084 | -22.3486 |
| SUM | AREA-LOADS | 0.0042 | 0.0084 | -37.2476 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0042 | 0.0084 | -39.7631 |

Inner Energy= 8.975232

StatikD: Sum of external loads: 0.004168 0.008447 -39.763095  
 StatikD: Sum of support forces: -0.006145 -0.008567 39.762601  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO6\_OW+VO+WD+SNOW  
 StatikD: Time of calculation: 1.43s  
 StatikD: Version 2025.38 done.

## Annex B.6.2. For anchor and ballast calculation

### EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.9460 | 0.012051    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0393 | 0.000500    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7673 | 153.469253  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.7628 | 152.565787  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -2.5155 |             |

### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1083  | 1.00   | 0.0000 | 0.0000 | -16.1401 | 148.99      |
| 1         | WIND     | -0.1500 | 1.00   | 0.0042 | 0.0084 | -22.3486 | 154.64      |
| SUM       |          |         |        | 0.0042 | 0.0084 | -38.4887 |             |

### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -16.1401 |
| SUM | WIND       | 0.0042 | 0.0084 | -22.3486 |
| SUM | AREA-LOADS | 0.0042 | 0.0084 | -38.4887 |

### EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0042 | 0.0084 | -41.0042 |

Inner Energy= 9.248682

StatikD: Sum of external loads: 0.004168 0.008447 -41.004176  
 StatikD: Sum of support forces: -0.005850 -0.008439 41.004555  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\C06a\_OW+VO+WD+SNOW\_anchors  
 StatikD: Time of calculation: 1.46s  
 StatikD: Version 2025.38 done.

## Annex C. Software input (load combination) – 10x20m model

### Annex C.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD | FACTOR | SUM_X  | SUM_Y  | SUM_Z  | LOADED AREA |
|-----------|----------|------|--------|--------|--------|--------|-------------|
| SUM       |          |      |        | 0.0000 | 0.0000 | 0.0000 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

| LOADMODE |            | SUM_X  | SUM_Y  | SUM_Z  |
|----------|------------|--------|--------|--------|
| SUM      | AREA-LOADS | 0.0000 | 0.0000 | 0.0000 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

|        |        |         |
|--------|--------|---------|
| SUM_X  | SUM_Y  | SUM_Z   |
| 0.0000 | 0.0000 | -3.1683 |

Inner Energy= 2.667523

StatikD: Sum of external loads: 0.000000 0.000000 -3.168216  
 StatikD: Sum of support forces: -0.001067 -0.000072 3.167956  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\C01\_OW+VO  
 StatikD: Time of calculation: 0.93s  
 StatikD: Version 2025.38 done.

## Annex C.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|--------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1000 | 1.00   | 0.0000 | 0.0000 | -18.8287 | 188.29      |
| SUM       |          |        |        | 0.0000 | 0.0000 | -18.8287 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -18.8287 |
| SUM | AREA-LOADS | 0.0000 | 0.0000 | -18.8287 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0000 | 0.0000 | -21.9970 |

Inner Energy= 4.932091

StatikD: Sum of external loads: 0.000000 0.000000 -21.996996  
 StatikD: Sum of support forces: -0.003667 -0.001001 21.997637  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO2\_OW+VO+SNOW  
 StatikD: Time of calculation: 1.35s  
 StatikD: Version 2025.38 done.

### Annex C.3. CO3. Own weight + pretension + wind pressure

#### EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

#### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | WIND     | -0.1500 | 1.00   | 0.0004 | 0.0148 | -28.2431 | 195.79      |
| SUM       |          |         |        | 0.0004 | 0.0148 | -28.2431 |             |

#### EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | WIND       | 0.0004 | 0.0148 | -28.2431 |
| SUM | AREA-LOADS | 0.0004 | 0.0148 | -28.2431 |

#### EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0004 | 0.0148 | -31.4113 |

Inner Energy= 6.986727

StatikD: Sum of external loads: 0.000460 0.014788 -31.411354  
 StatikD: Sum of support forces: -0.002743 -0.011967 31.411551  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO3\_OW+VO+WD  
 StatikD: Time of calculation: 1.54s  
 StatikD: Version 2025.38 done.

## Annex C.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0688 | 0.000877    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1877 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X   | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|---------|---------|---------|-------------|
| 1         | WIND     | 0.3500 | 1.00   | -0.0032 | -0.0323 | 65.8053 | 195.75      |
| SUM       |          |        |        | -0.0032 | -0.0323 | 65.8053 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y   | SUM_Z   |
|-----|------------|---------|---------|---------|
| SUM | WIND       | -0.0032 | -0.0323 | 65.8053 |
| SUM | AREA-LOADS | -0.0032 | -0.0323 | 65.8053 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y   | SUM_Z   |
|---------|---------|---------|
| -0.0032 | -0.0323 | 62.6176 |

Inner Energy= 14.312429

StatikD: Sum of external loads: -0.003195 -0.032267 62.617612  
 StatikD: Sum of support forces: -0.001255 0.032404 -62.605597  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO4\_OW+VO+WU  
 StatikD: Time of calculation: 2.71s  
 StatikD: Version 2025.38 done.

## Annex C.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD   | FACTOR | SUM_X   | SUM_Y   | SUM_Z   | LOADED AREA |
|-----------|----------|--------|--------|---------|---------|---------|-------------|
| 1         | WIND     | 0.1750 | 1.00   | -0.0005 | -0.0172 | 32.9503 | 195.79      |
| SUM       |          |        |        | -0.0005 | -0.0172 | 32.9503 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X   | SUM_Y   | SUM_Z   |
|-----|------------|---------|---------|---------|
| SUM | WIND       | -0.0005 | -0.0172 | 32.9503 |
| SUM | AREA-LOADS | -0.0005 | -0.0172 | 32.9503 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X   | SUM_Y   | SUM_Z   |
|---------|---------|---------|
| -0.0005 | -0.0172 | 29.7820 |

Inner Energy= 9.377589

StatikD: Sum of external loads: -0.000487 -0.017266 29.782032  
 StatikD: Sum of support forces: -0.001501 0.018529 -29.791202  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20 - final\CO5\_OW+VO+WU\_red  
 StatikD: Time of calculation: 3.07s  
 StatikD: Version 2025.38 done.



## Annex C.6. CO6. Own weight + pretension + wind pressure + snow [FR]

### Annex C.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1000  | 1.00   | 0.0000 | 0.0000 | -18.8287 | 188.29      |
| 1         | WIND     | -0.1500 | 1.00   | 0.0004 | 0.0148 | -28.2431 | 195.79      |
| SUM       |          |         |        | 0.0004 | 0.0148 | -47.0718 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -18.8287 |
| SUM | WIND       | 0.0004 | 0.0148 | -28.2431 |
| SUM | AREA-LOADS | 0.0004 | 0.0148 | -47.0718 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0004 | 0.0148 | -50.2401 |

Inner Energy= 10.819541

StatikD: Sum of external loads: 0.000460 0.014788 -50.240093  
 StatikD: Sum of support forces: -0.004591 -0.012770 50.240338  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO6\_OW+VO+WD+SNOW  
 StatikD: Time of calculation: 1.56s  
 StatikD: Version 2025.38 done.

## Annex C.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)  
ORDERED BY LOADGROUPS

| LOADGROUP     | LOADMODE   | FACTOR | VOLUME/AREA(P.U) | SUM_X  | SUM_Y  | SUM_Z   | VOLUME/AREA |
|---------------|------------|--------|------------------|--------|--------|---------|-------------|
| STRUTS        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -1.1806 | 0.015039    |
| CABLES        | SELFWEIGHT | 1.00   | 78.500000        | 0.0000 | 0.0000 | -0.0494 | 0.000629    |
| MEM-LINKS1000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9720 | 194.398401  |
| MEM-LINKS2000 | SELFWEIGHT | 1.00   | 0.005000         | 0.0000 | 0.0000 | -0.9663 | 193.267579  |
| SUM           |            |        |                  | 0.0000 | 0.0000 | -3.1683 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADGROUPS

| LOADGROUP | LOADMODE | LOAD    | FACTOR | SUM_X  | SUM_Y  | SUM_Z    | LOADED AREA |
|-----------|----------|---------|--------|--------|--------|----------|-------------|
| 1         | SCHNEE   | 0.1083  | 1.00   | 0.0000 | 0.0000 | -20.3972 | 188.29      |
| 1         | WIND     | -0.1500 | 1.00   | 0.0004 | 0.0148 | -28.2431 | 195.79      |
| SUM       |          |         |        | 0.0004 | 0.0148 | -48.6403 |             |

EXTERNAL LOADS (AREA-DEPENDENT)  
ORDERED BY LOADMODES

|     | LOADMODE   | SUM_X  | SUM_Y  | SUM_Z    |
|-----|------------|--------|--------|----------|
| SUM | SCHNEE     | 0.0000 | 0.0000 | -20.3972 |
| SUM | WIND       | 0.0004 | 0.0148 | -28.2431 |
| SUM | AREA-LOADS | 0.0004 | 0.0148 | -48.6403 |

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

| SUM_X  | SUM_Y  | SUM_Z    |
|--------|--------|----------|
| 0.0004 | 0.0148 | -51.8085 |

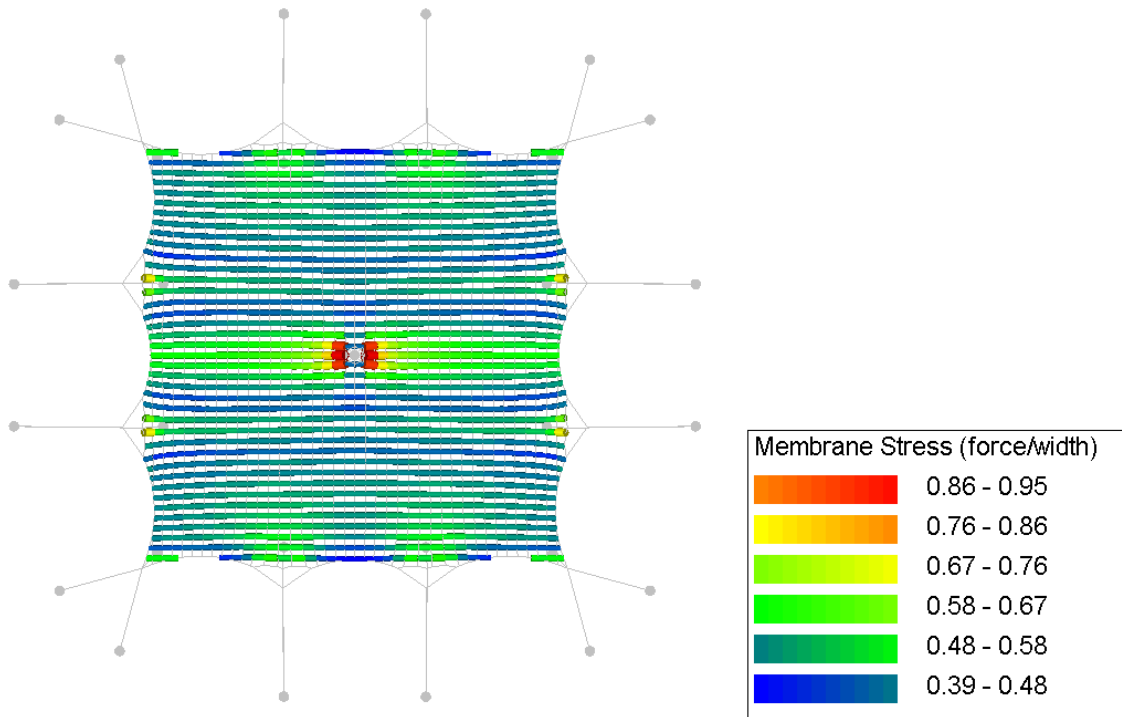
Inner Energy= 11.160164

StatikD: Sum of external loads: 0.000460 0.014788 -51.808511  
 StatikD: Sum of support forces: -0.005047 -0.013400 51.808345  
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\C06a\_OW+VO+WD+SNOW\_anchors  
 StatikD: Time of calculation: 1.55s  
 StatikD: Version 2025.38 done.

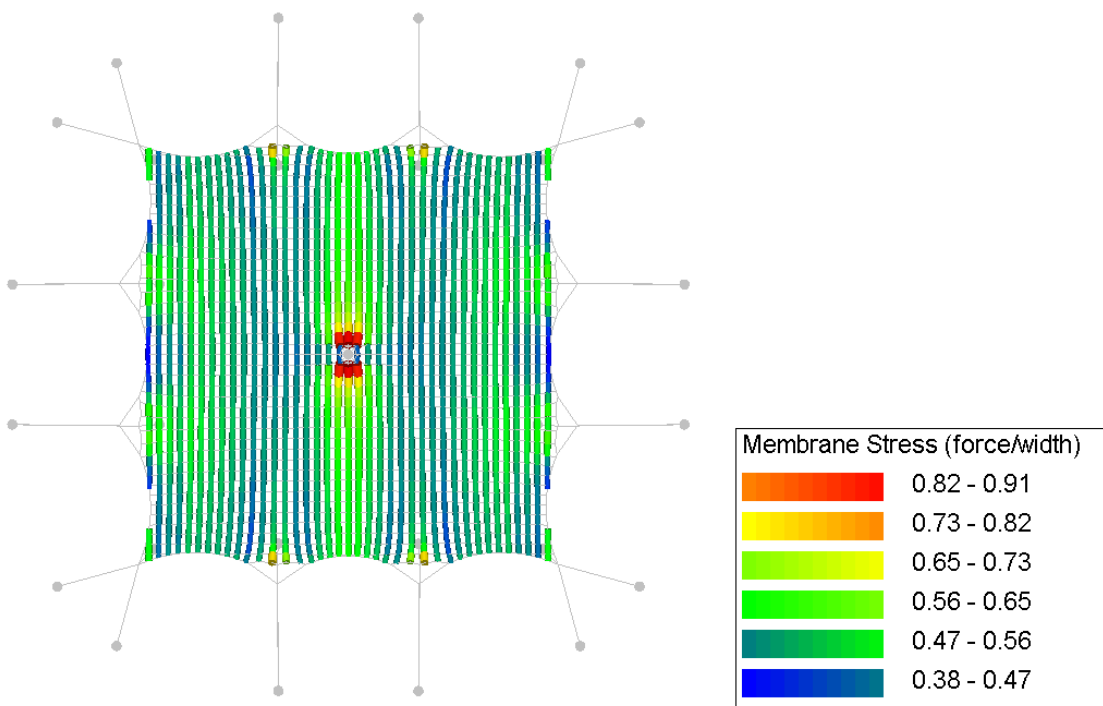
## Annex D. Results per load combination – 10x10m model

### Annex D.1. CO1. Own weight + pretension

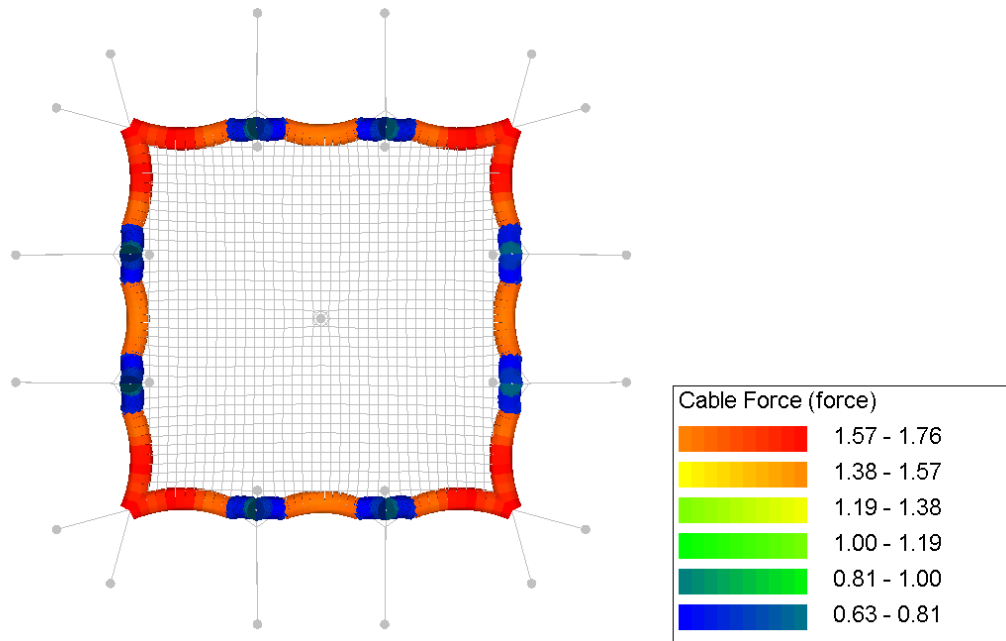
#### Annex D.1.1. Membrane stress (warp)



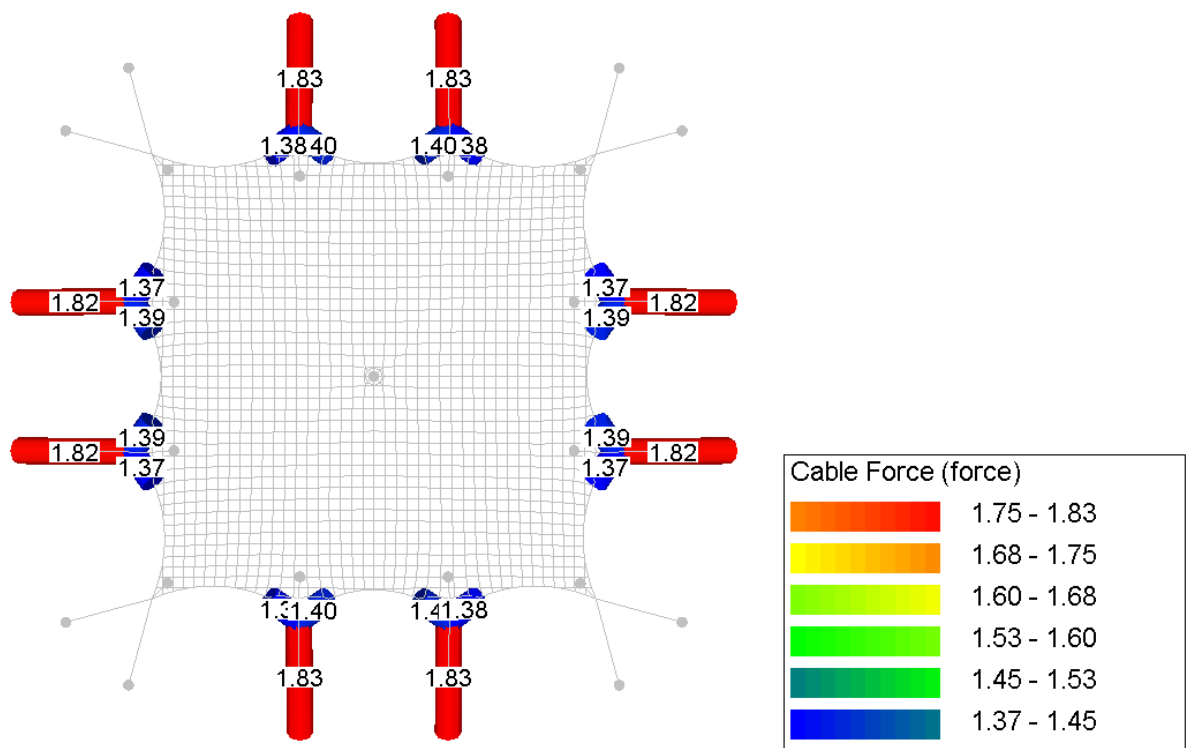
#### Annex D.1.2. Membrane stress (weft)



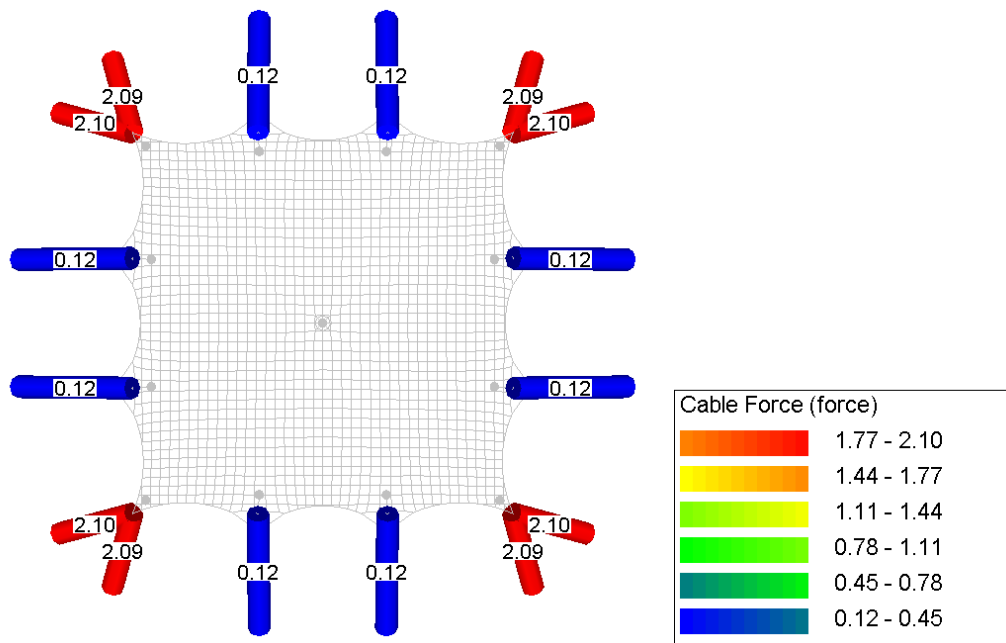
### Annex D.1.3. Circumference rope forces



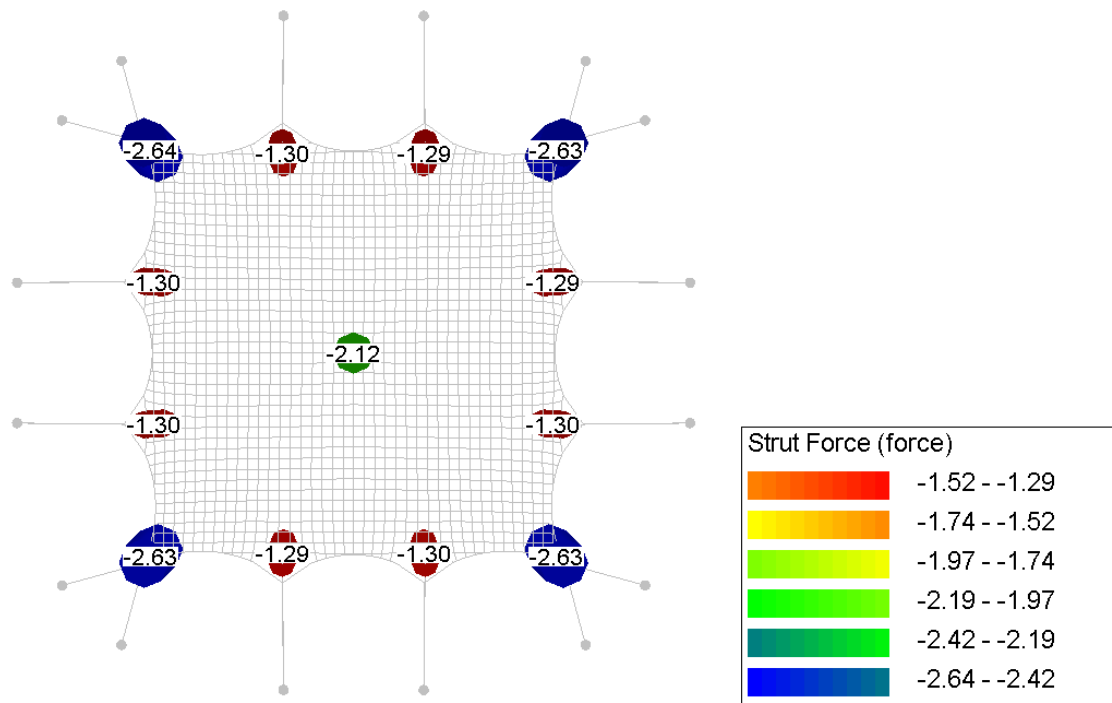
### Annex D.1.4. Rope forces



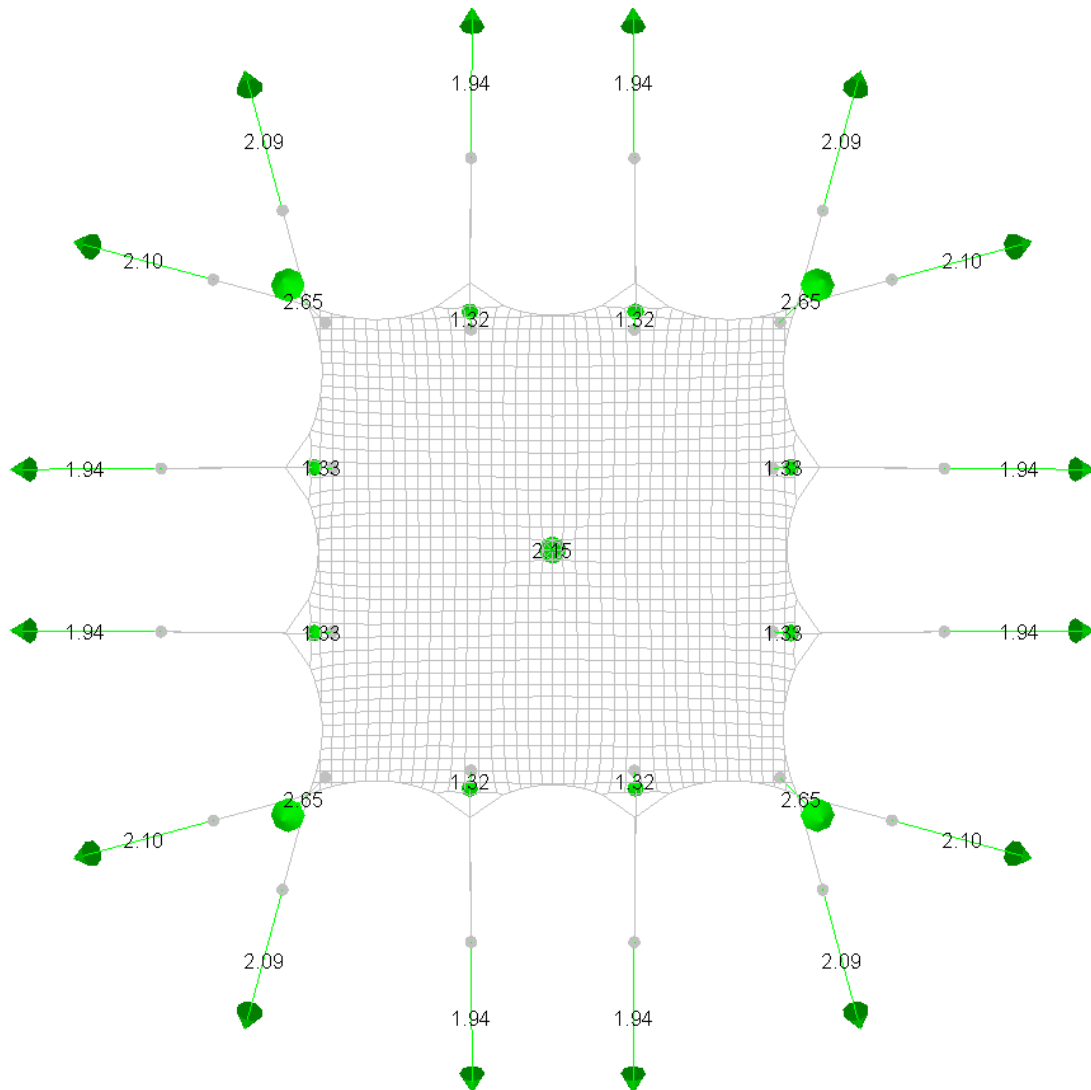
### Annex D.1.5. Belt forces



### Annex D.1.6. Pole forces

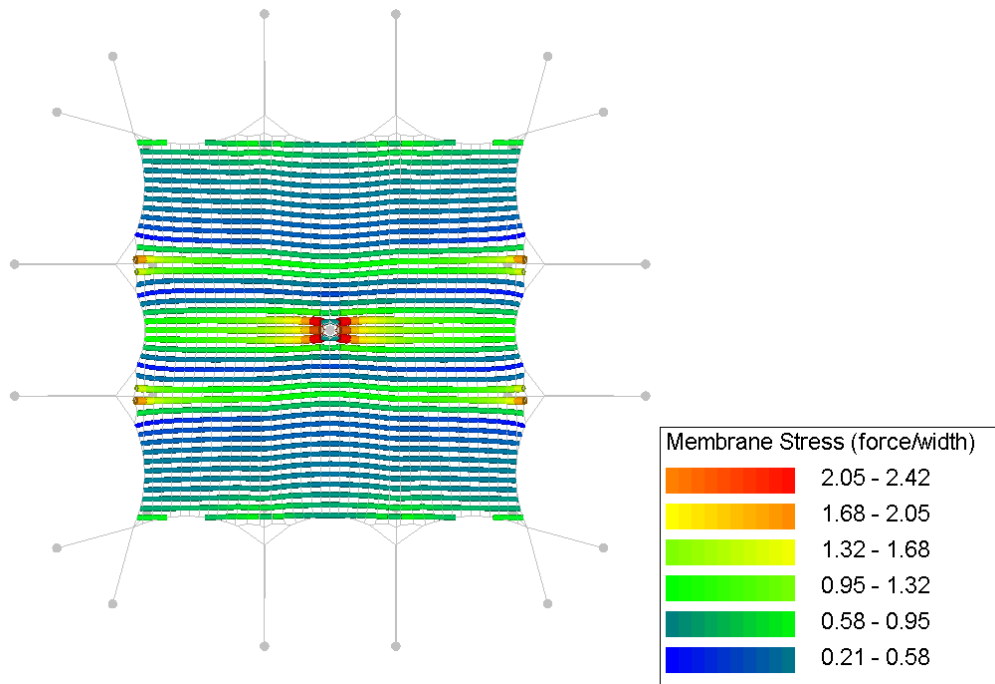


## Annex D.1.7. Reaction forces

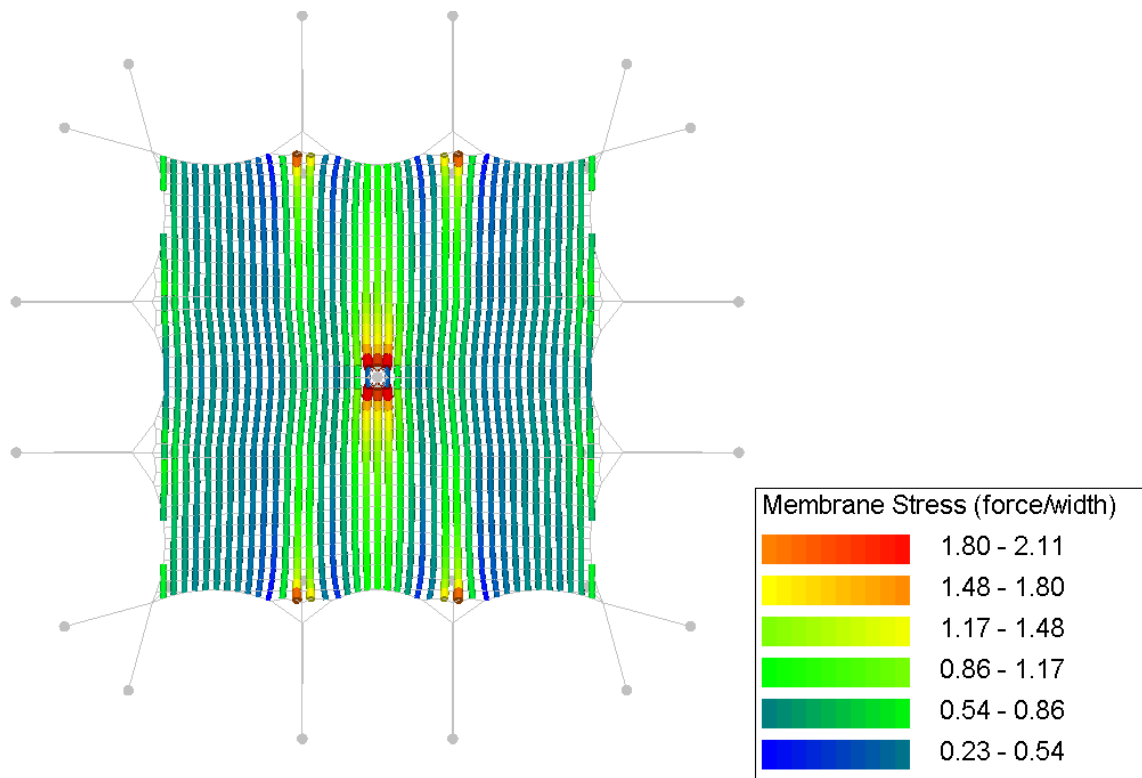


## Annex D.2. CO2. Own weight + pretension + snow

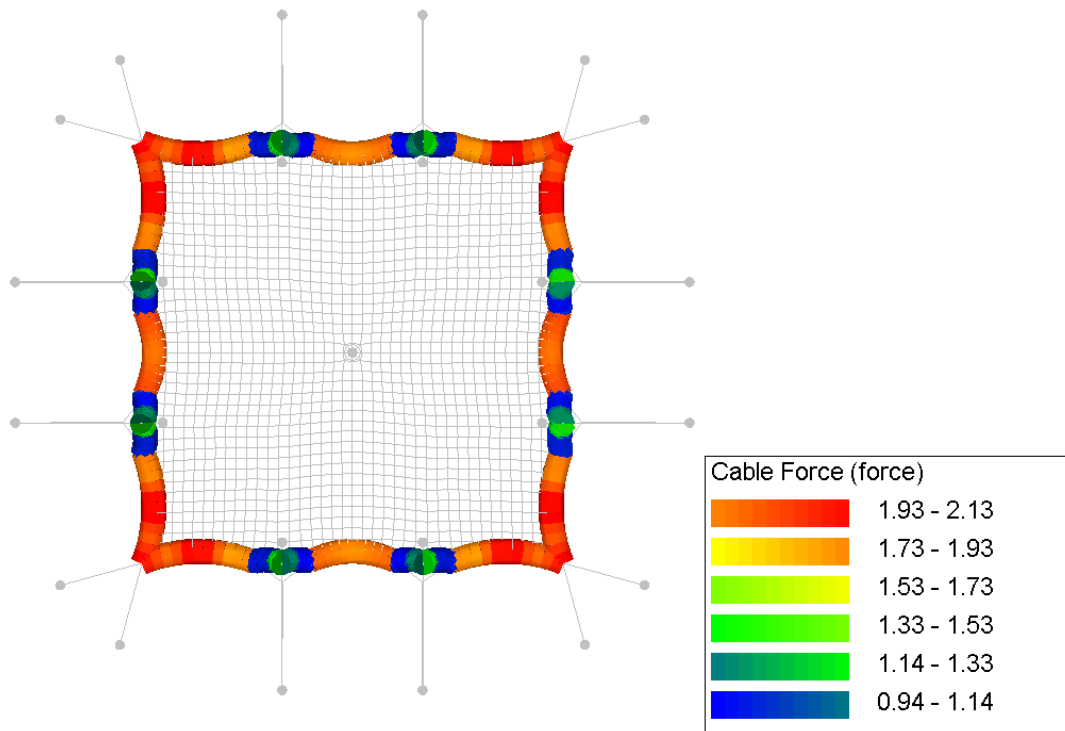
### Annex D.2.1. Membrane stress (warp)



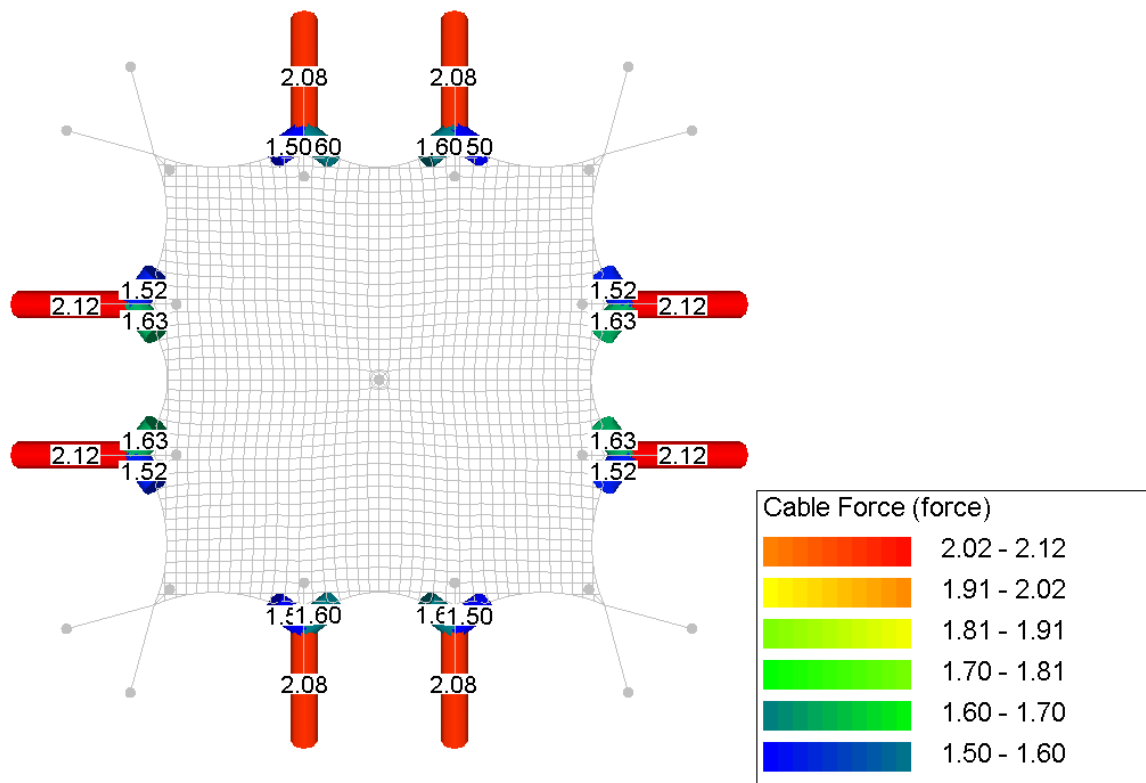
### Annex D.2.2. Membrane stress (weft)



### Annex D.2.3. Circumference rope forces

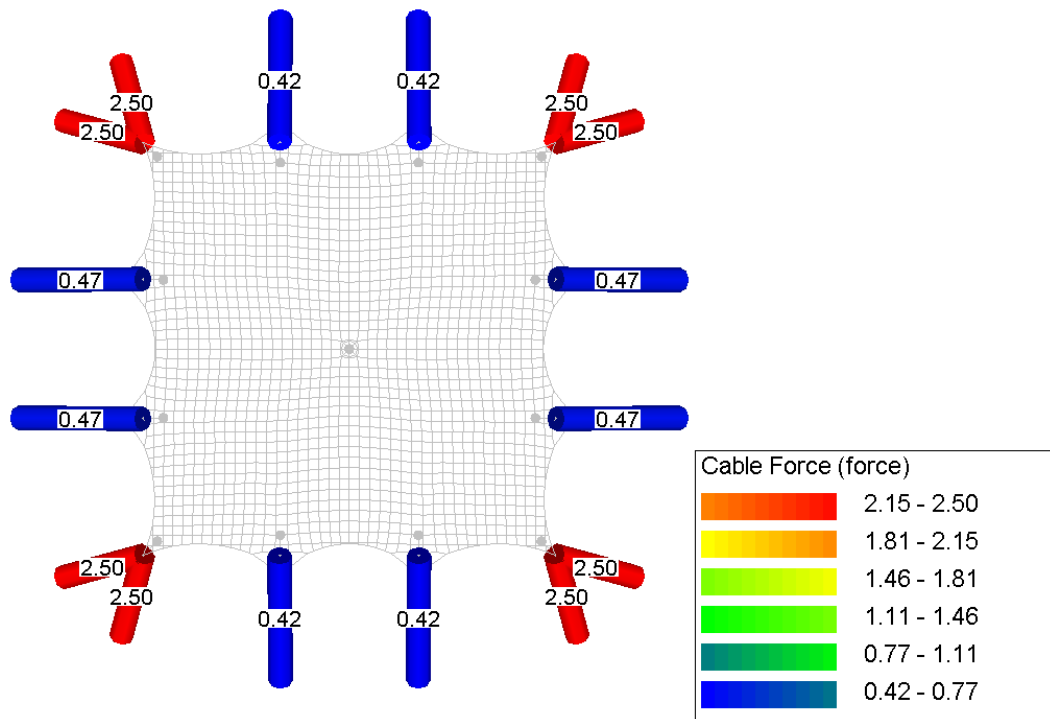


### Annex D.2.4. Rope forces

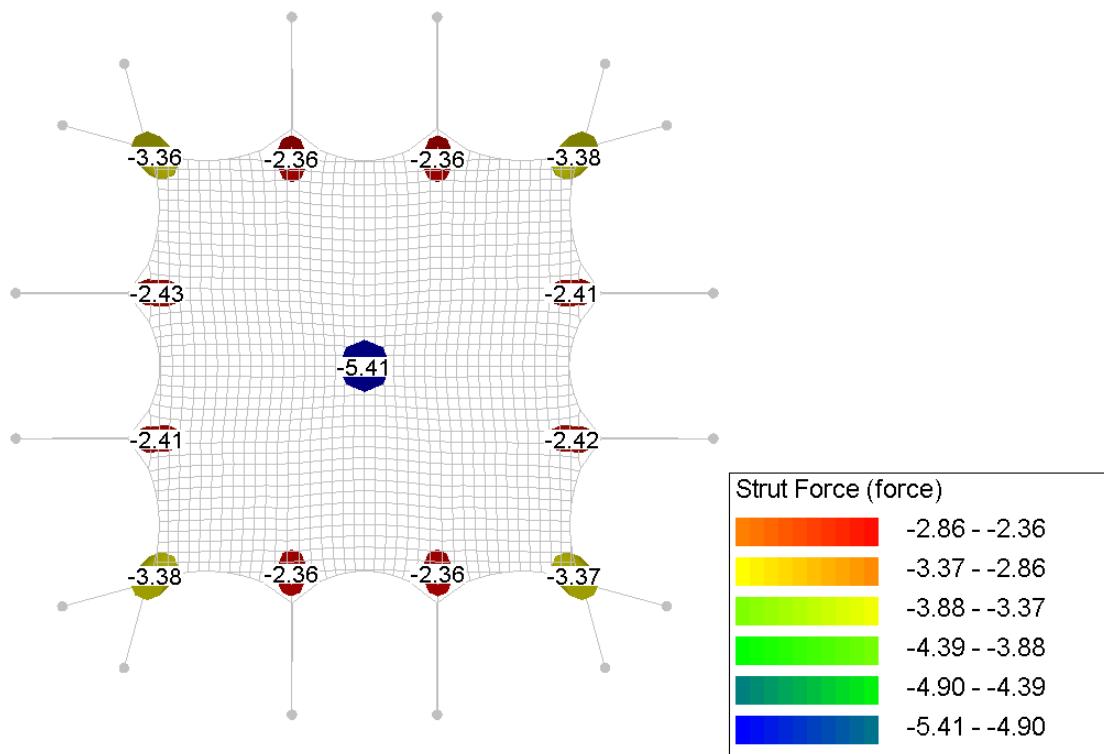




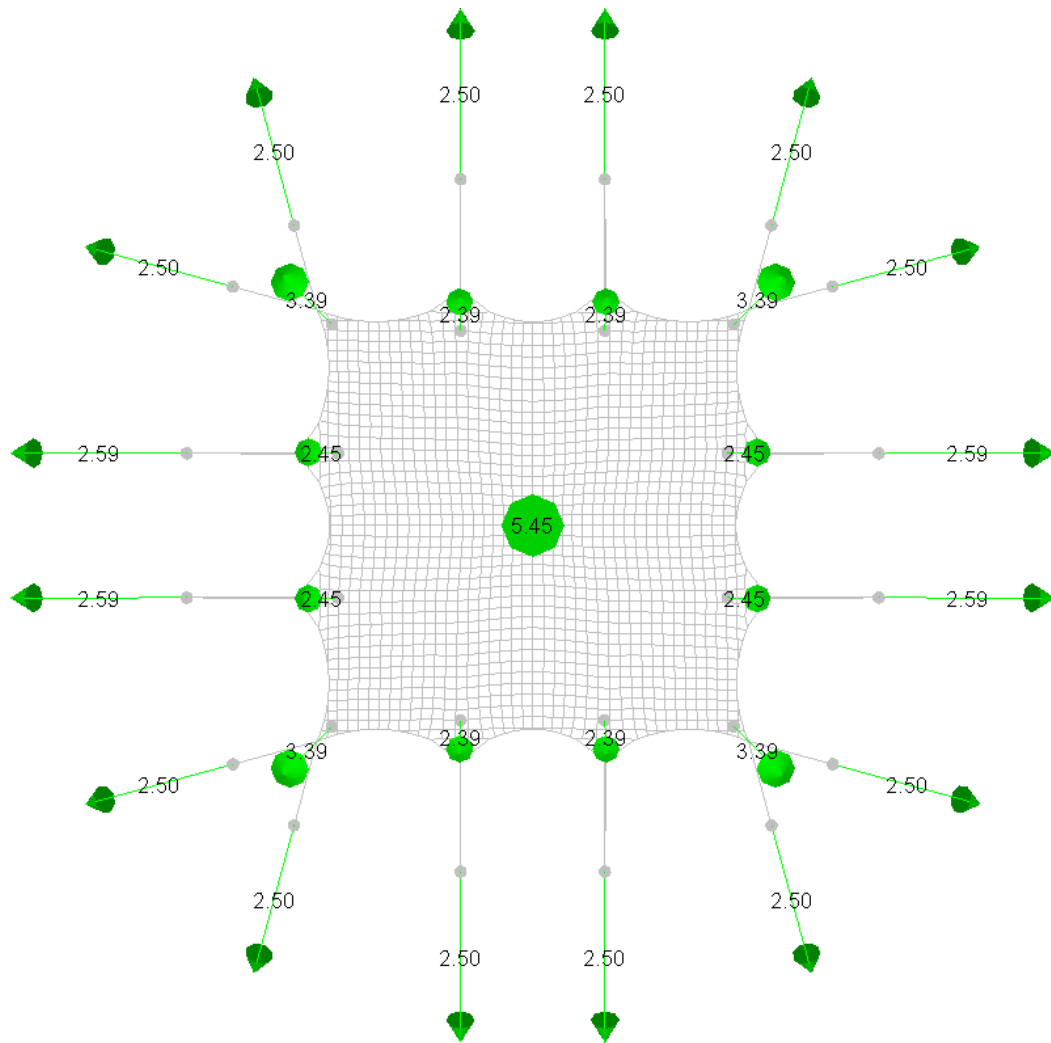
### Annex D.2.5. Belt forces



### Annex D.2.6. Pole forces

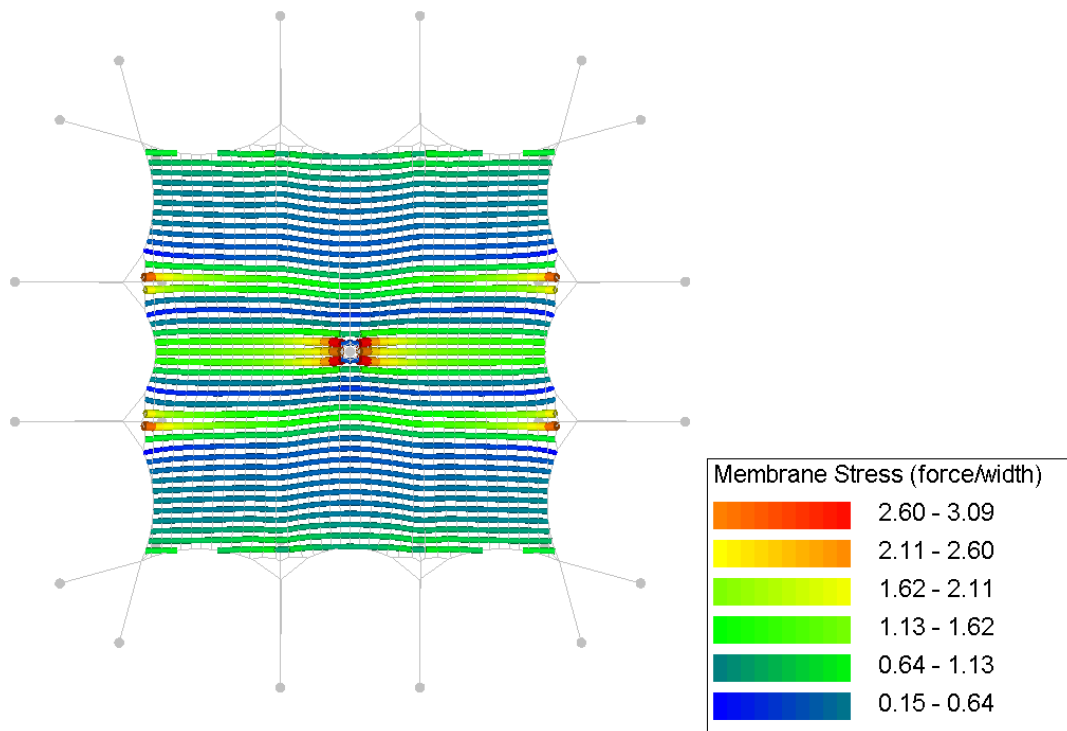


## Annex D.2.7. Anchor forces

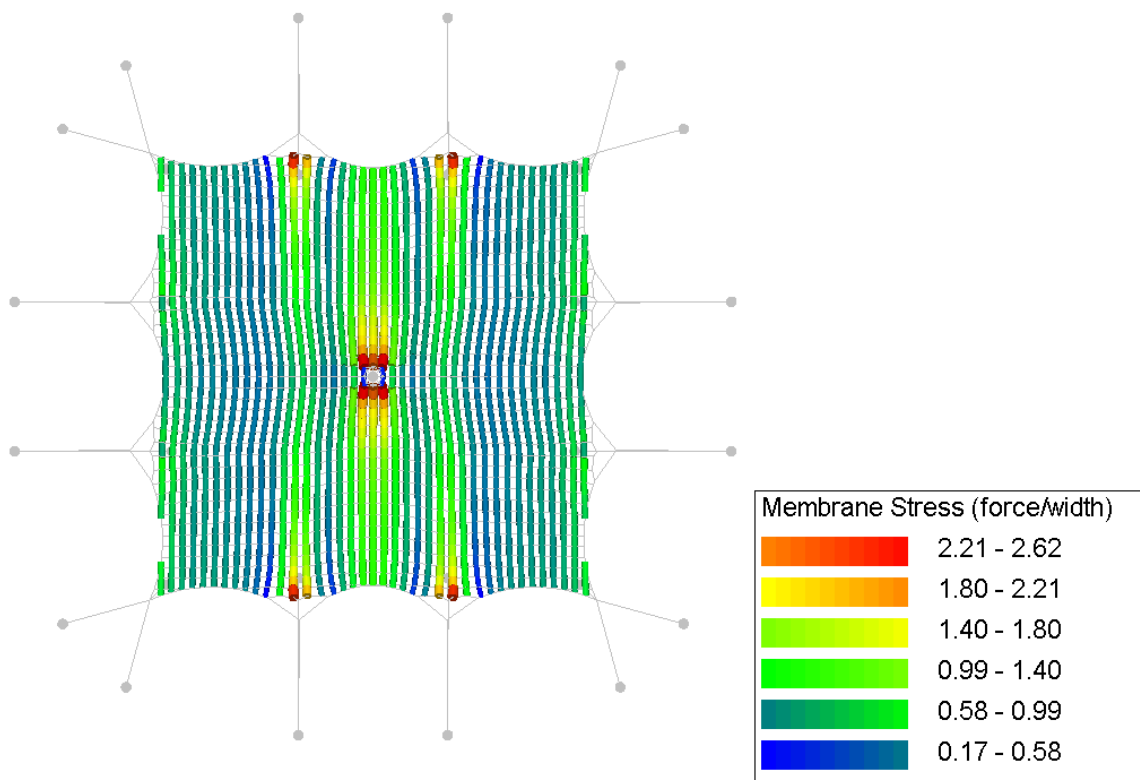


## Annex D.3. CO3. Own weight + pretension + wind pressure

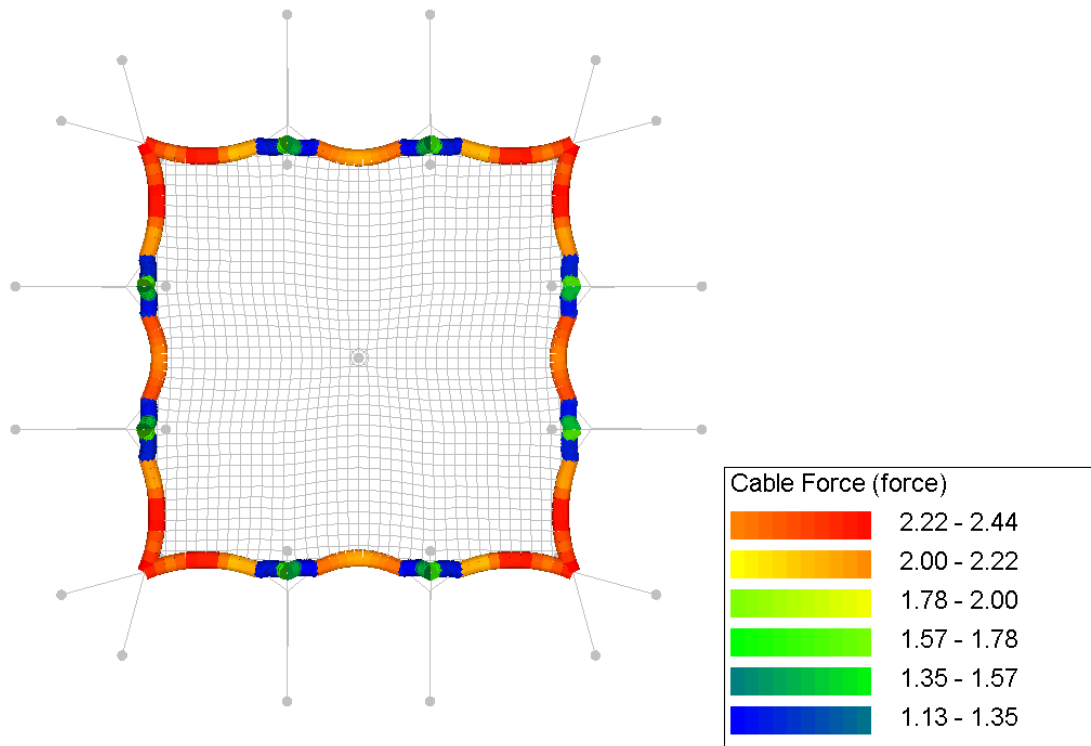
### Annex D.3.1. Membrane stress (warp)



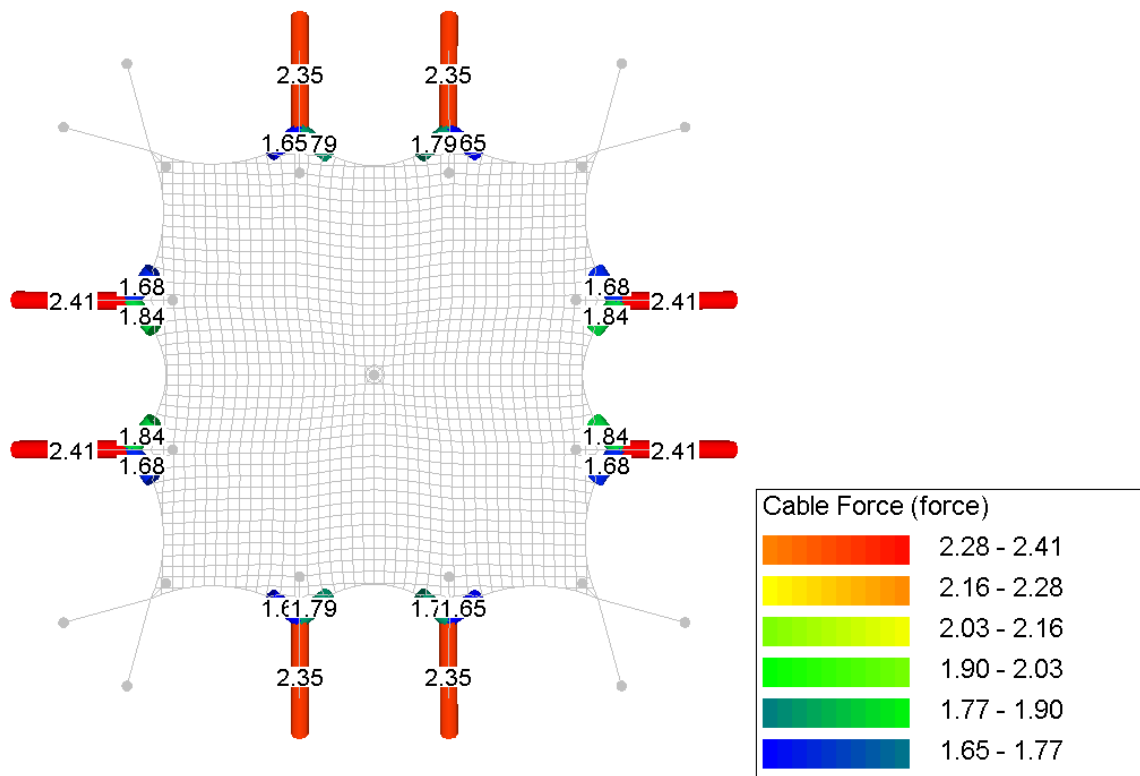
### Annex D.3.2. Membrane stress (weft)



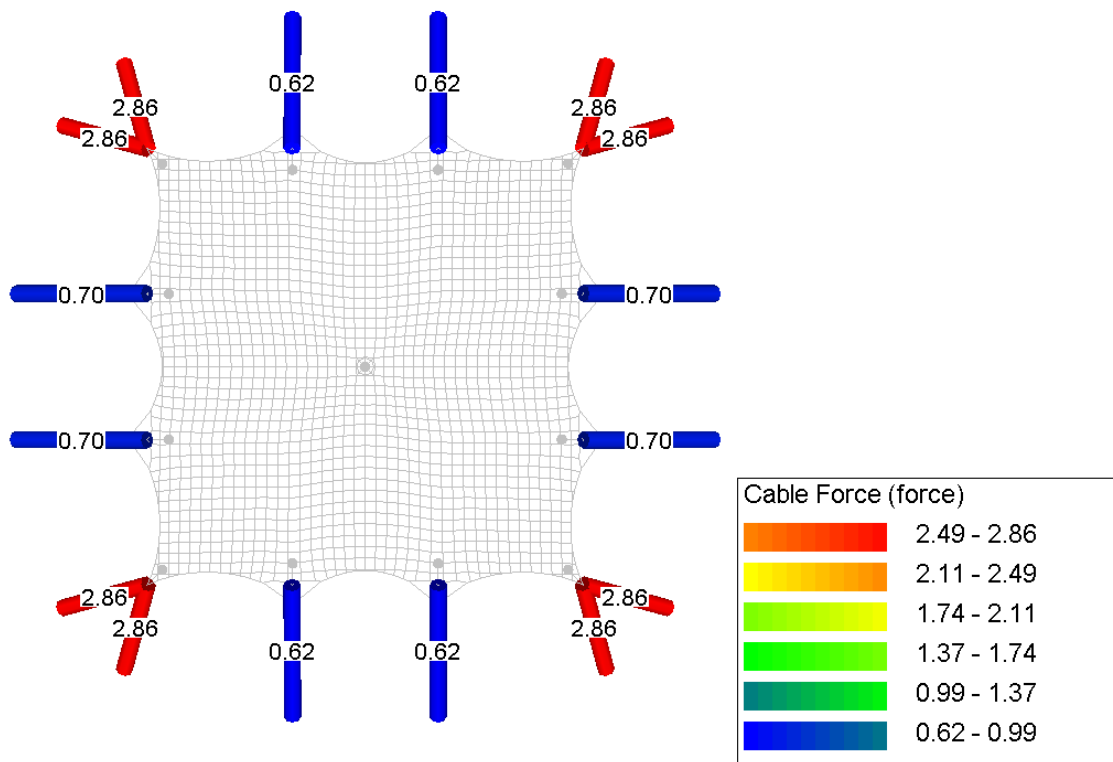
### Annex D.3.3. Circumference rope forces



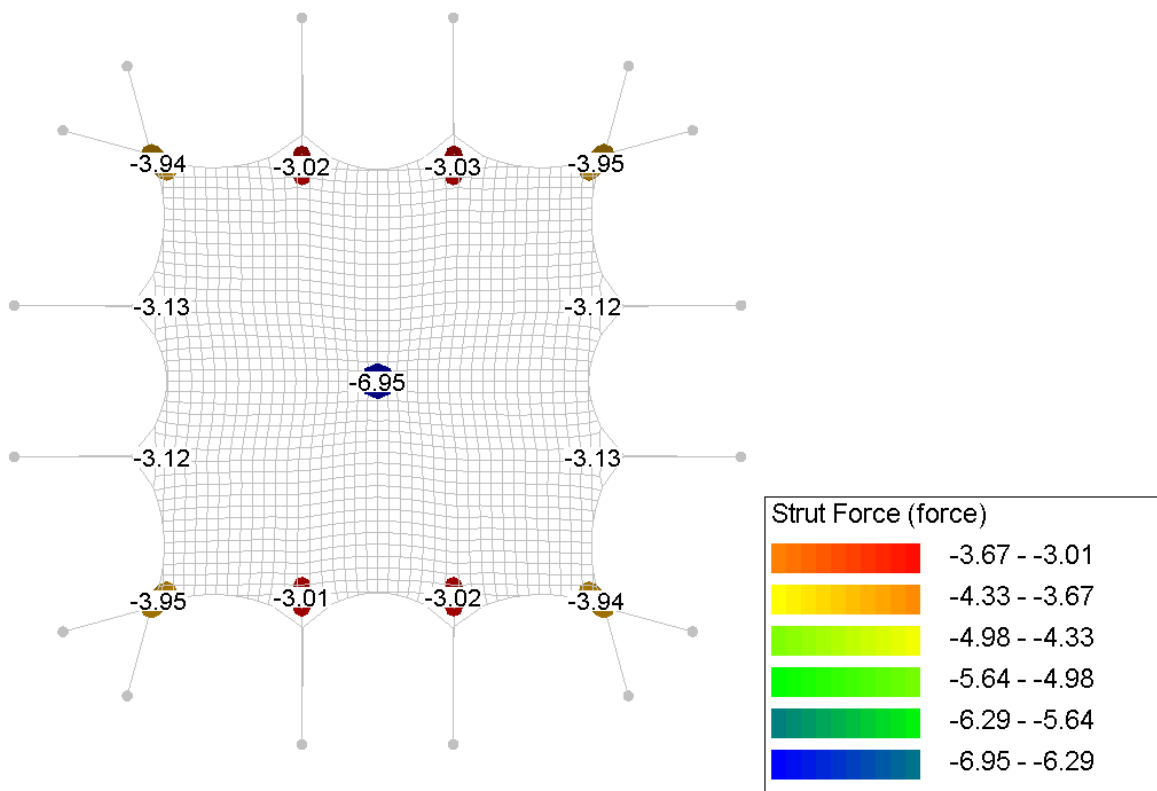
### Annex D.3.4. Rope forces



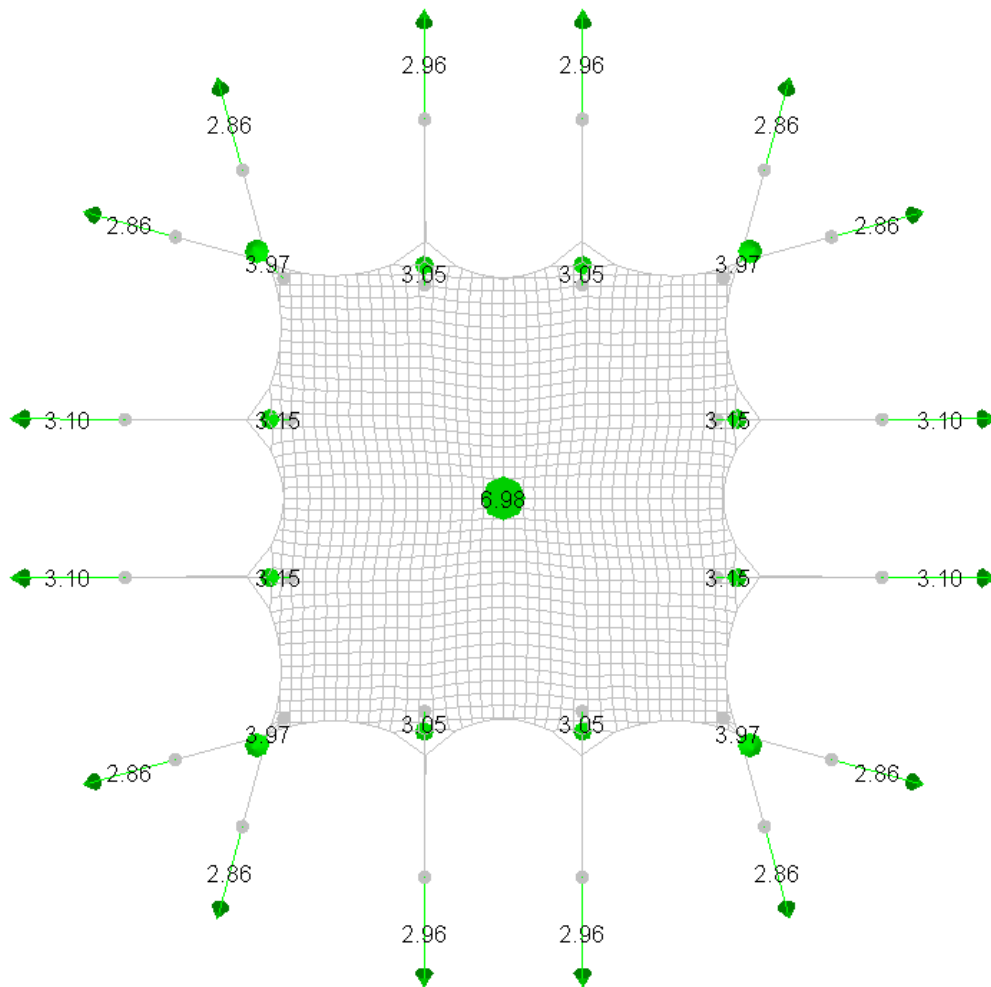
### Annex D.3.5. Belt forces



### Annex D.3.6. Poles forces

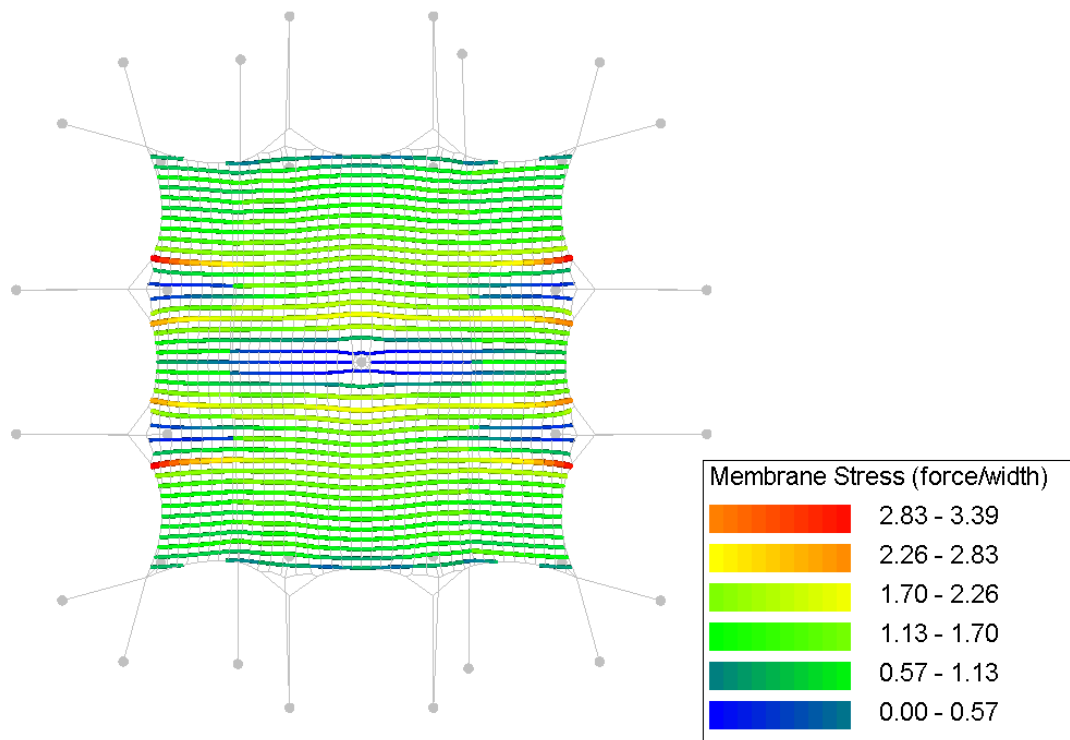


### Annex D.3.7. Reaction forces

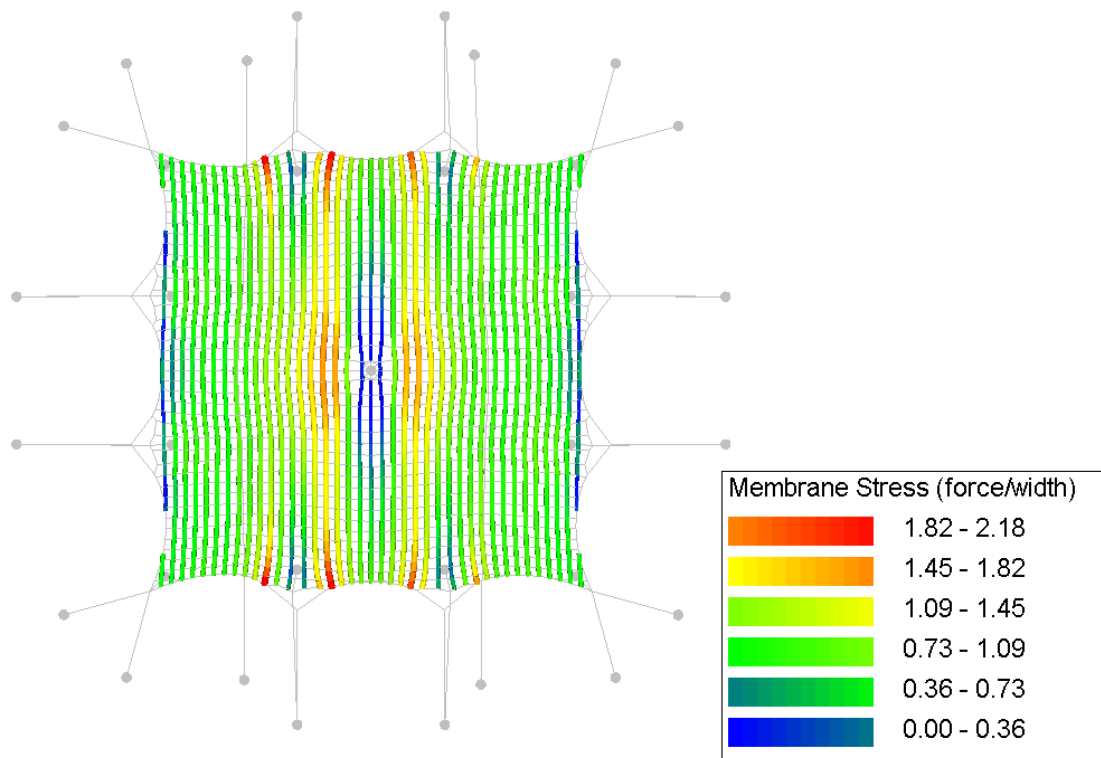


## Annex D.4. CO4. Own weight + pretension + wind suction

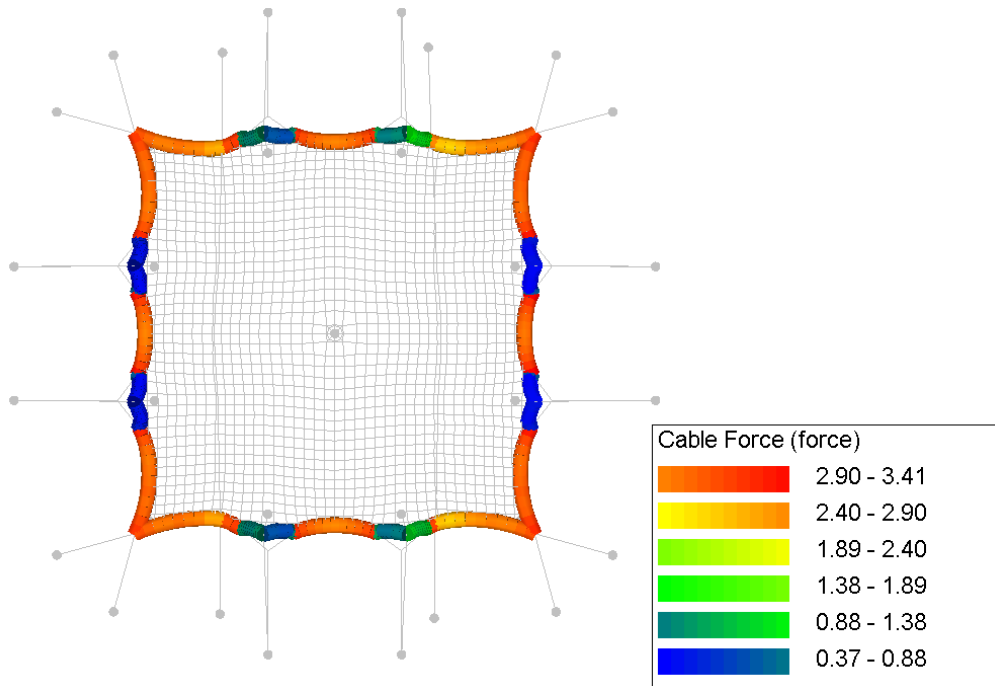
### Annex D.4.1. Membrane stress (warp)



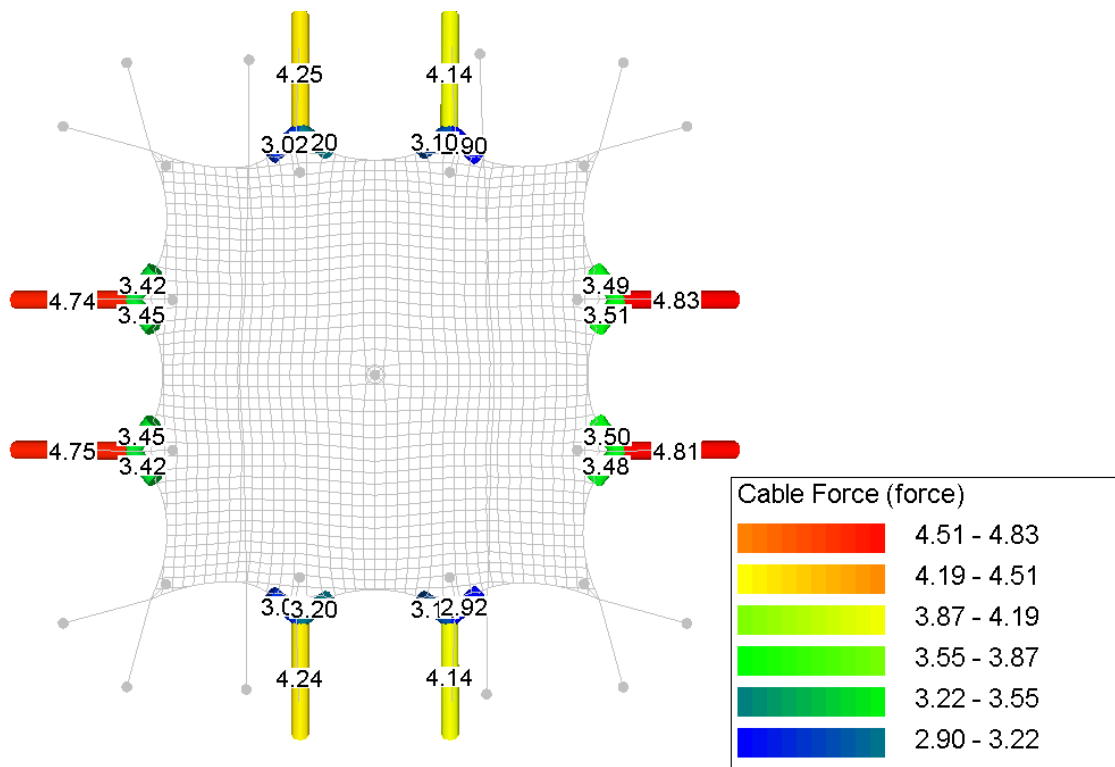
### Annex D.4.2. Membrane stress (weft)



#### Annex D.4.3. Circumference rope forces

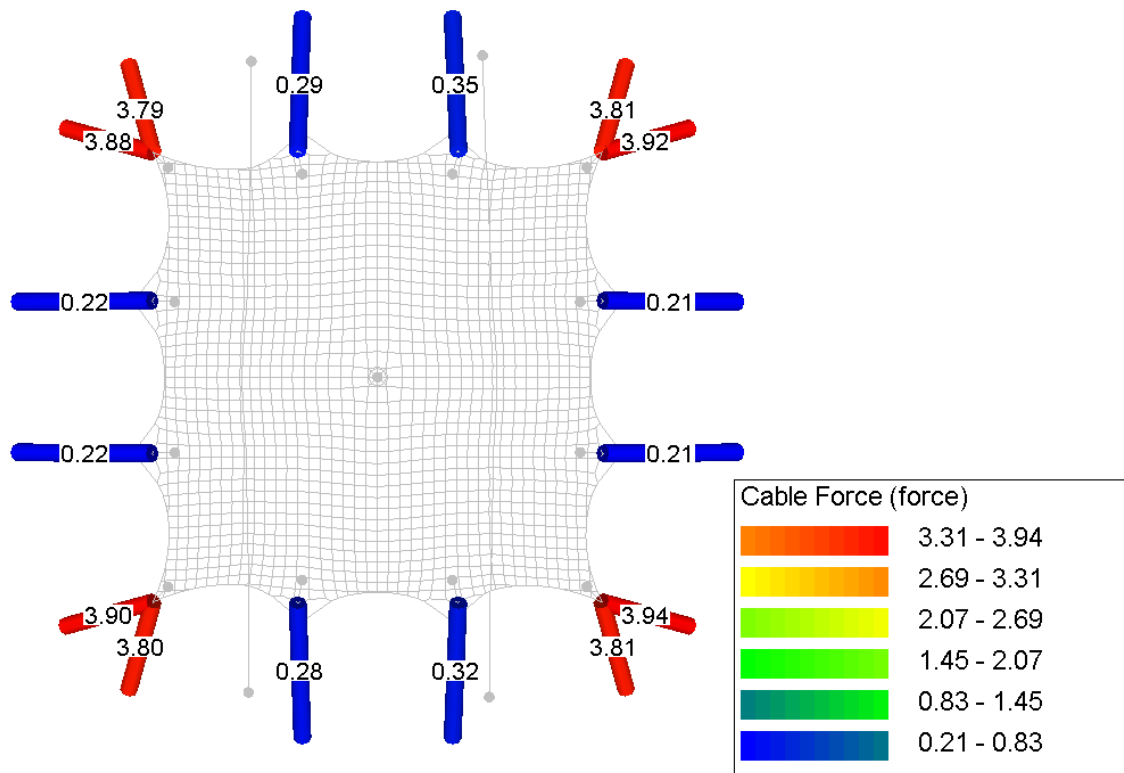


#### Annex D.4.4. Guy rope forces

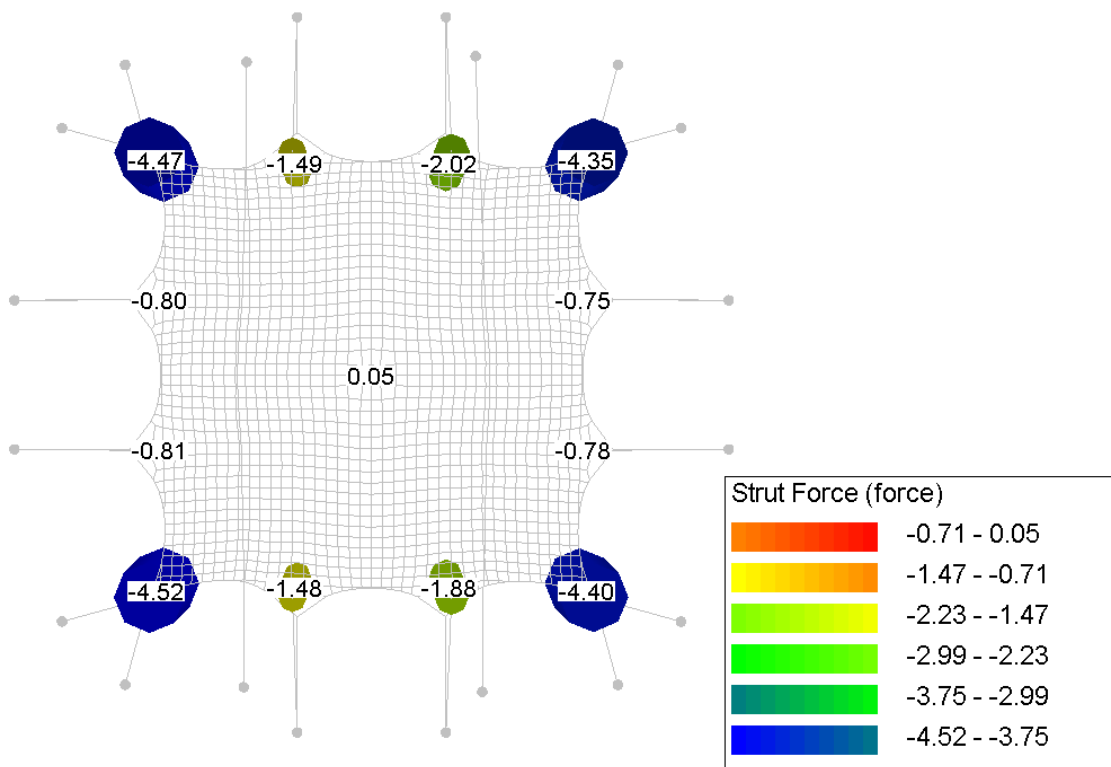




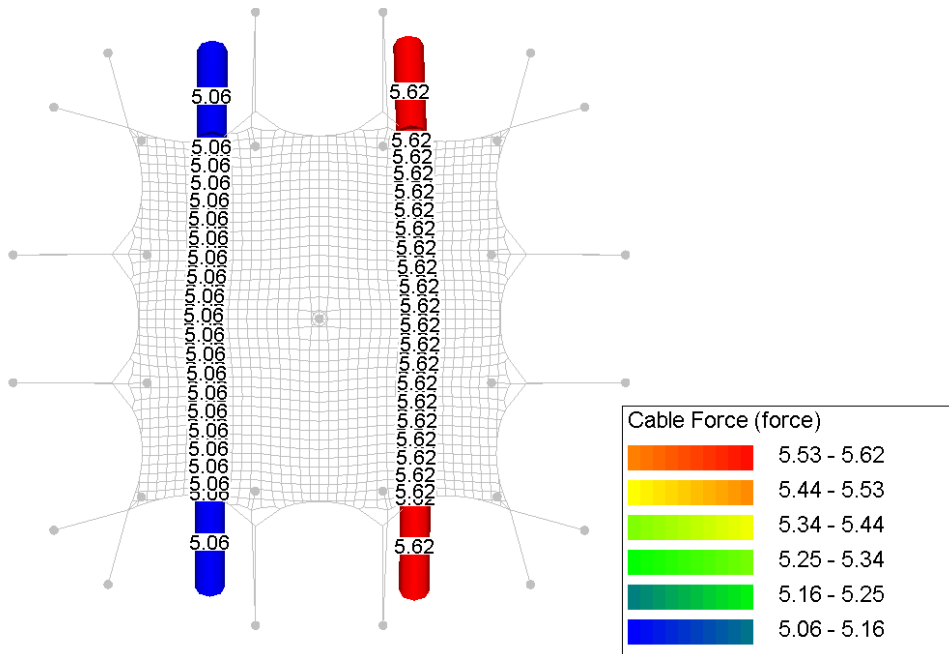
#### Annex D.4.5. Belt forces



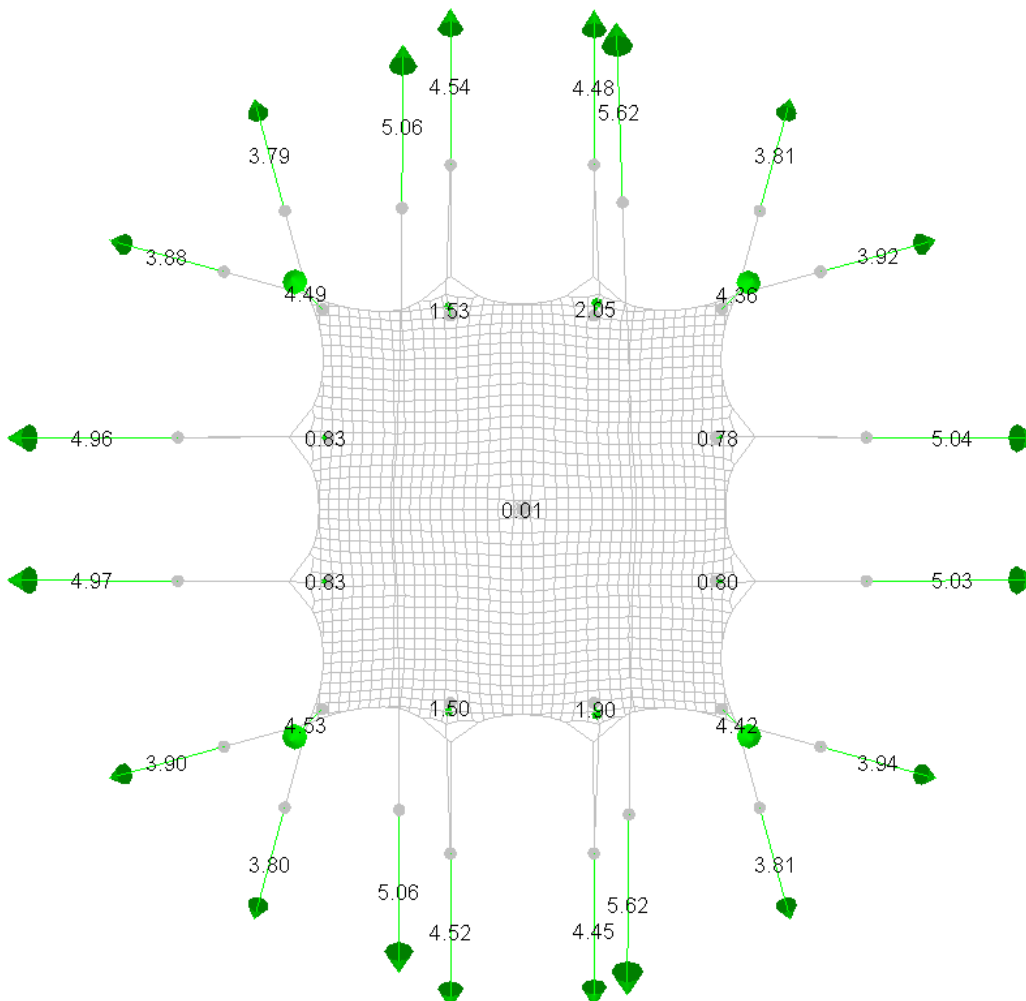
#### Annex D.4.6. Poles forces



#### Annex D.4.7. Storm belt forces

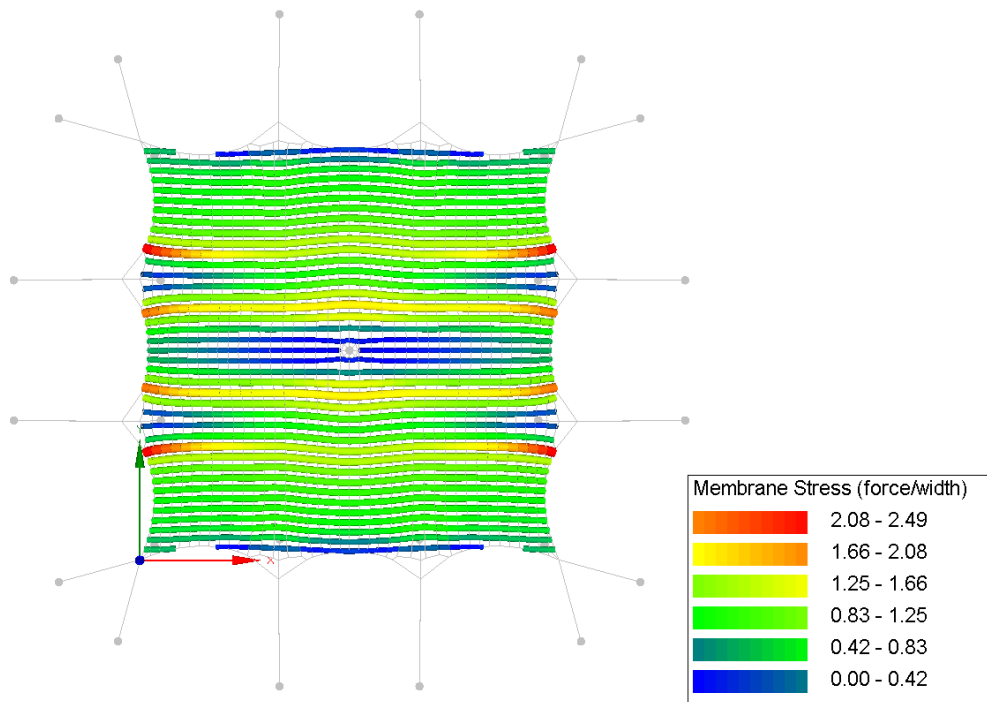


#### Annex D.4.8. Reaction forces

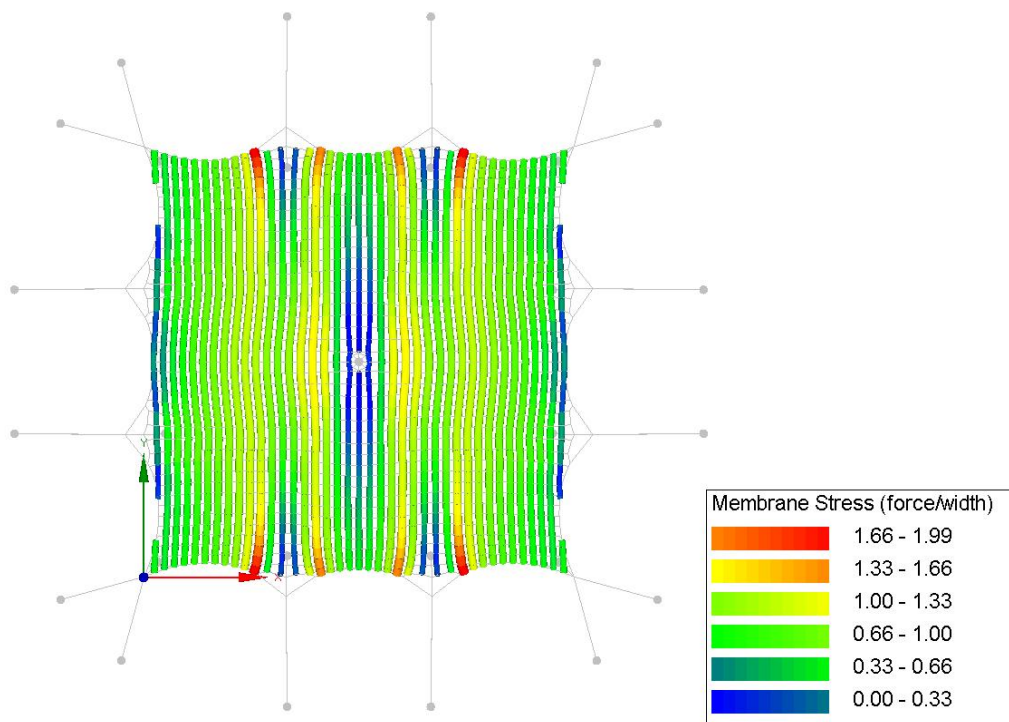


## Annex D.5. CO5. Own weight + pretension + wind suction reduced [NL]

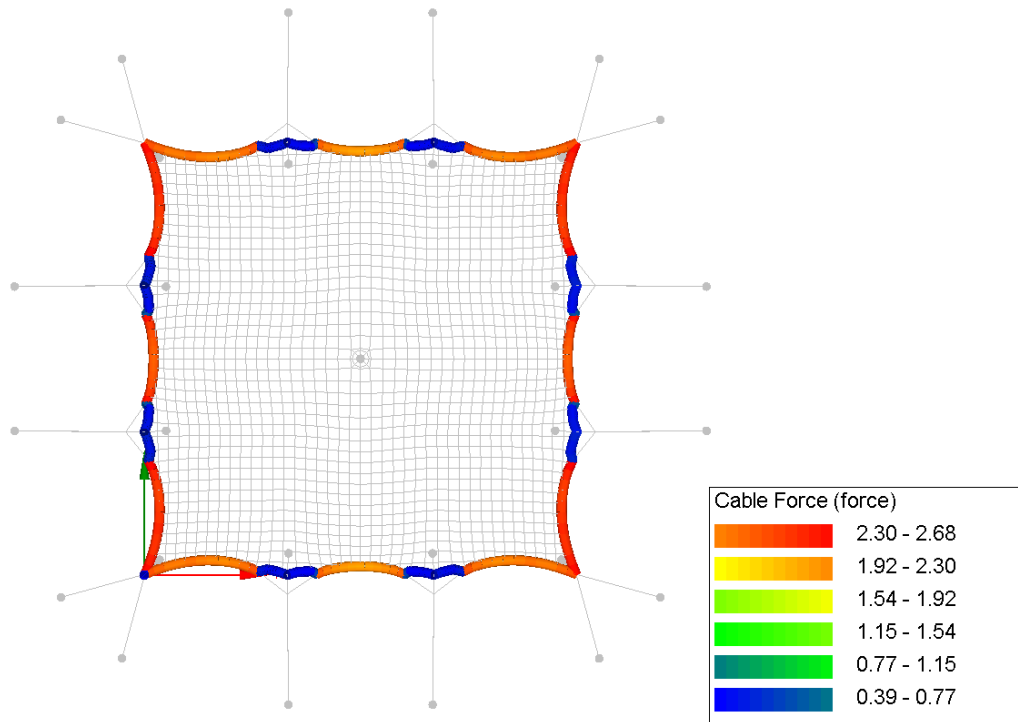
### Annex D.5.1. Membrane stress (warp)



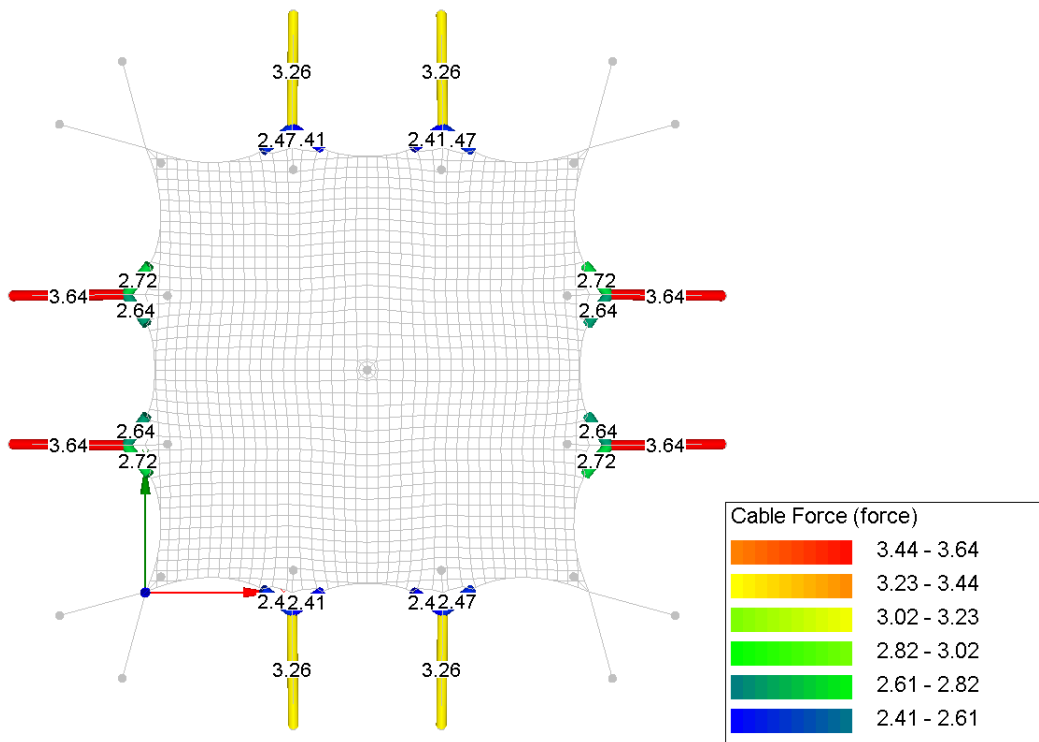
### Annex D.5.2. Membrane stress (weft)



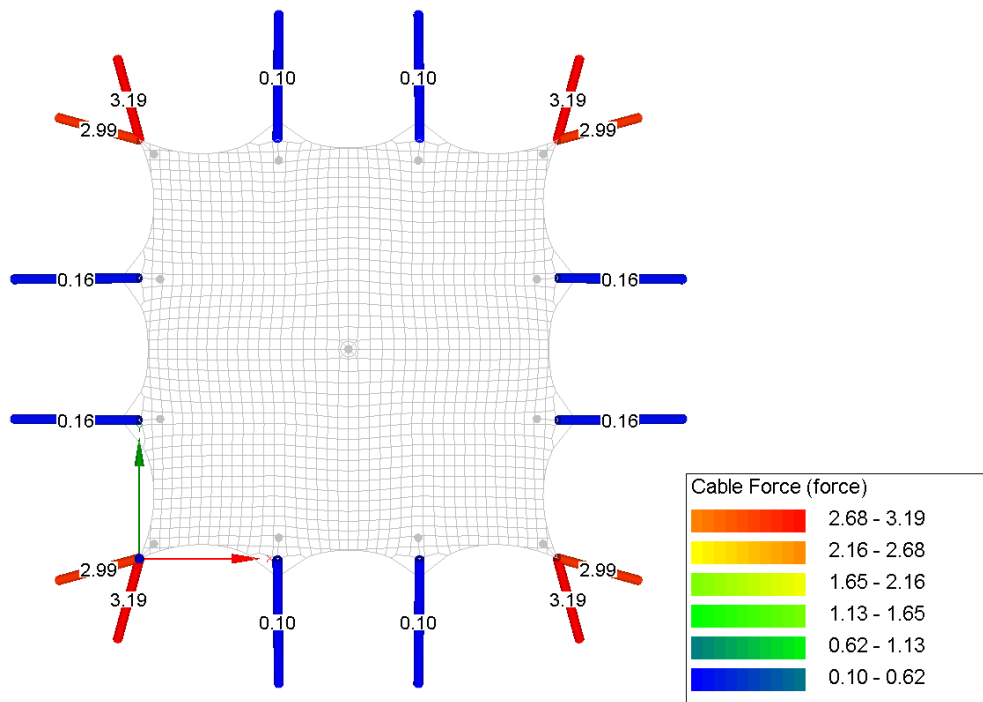
### Annex D.5.3. Circumference rope forces



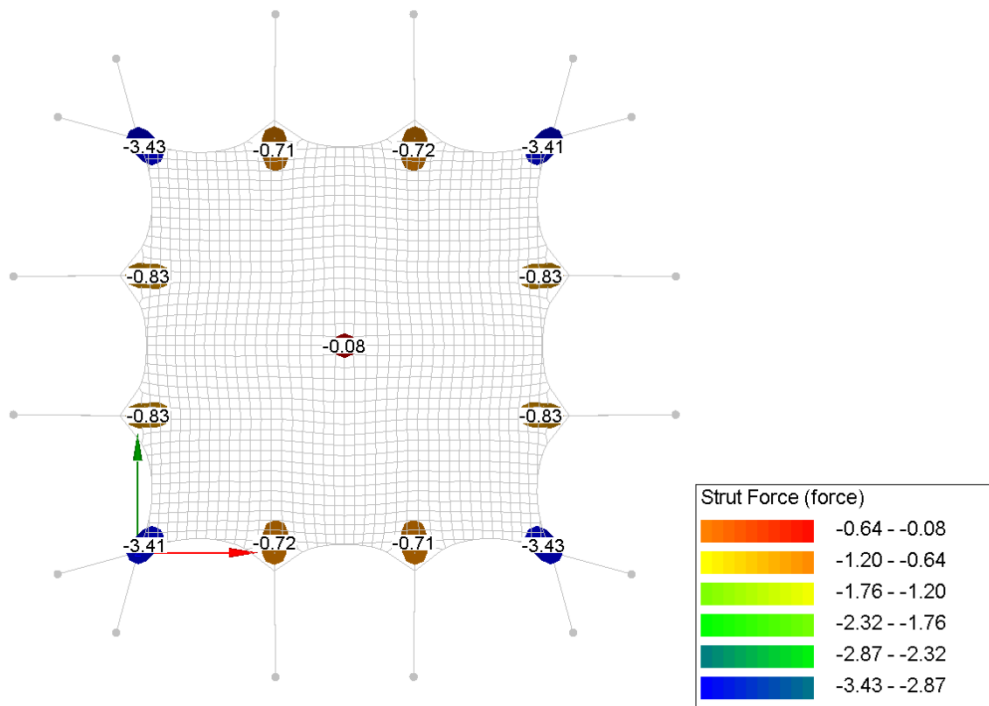
### Annex D.5.4. Rope forces



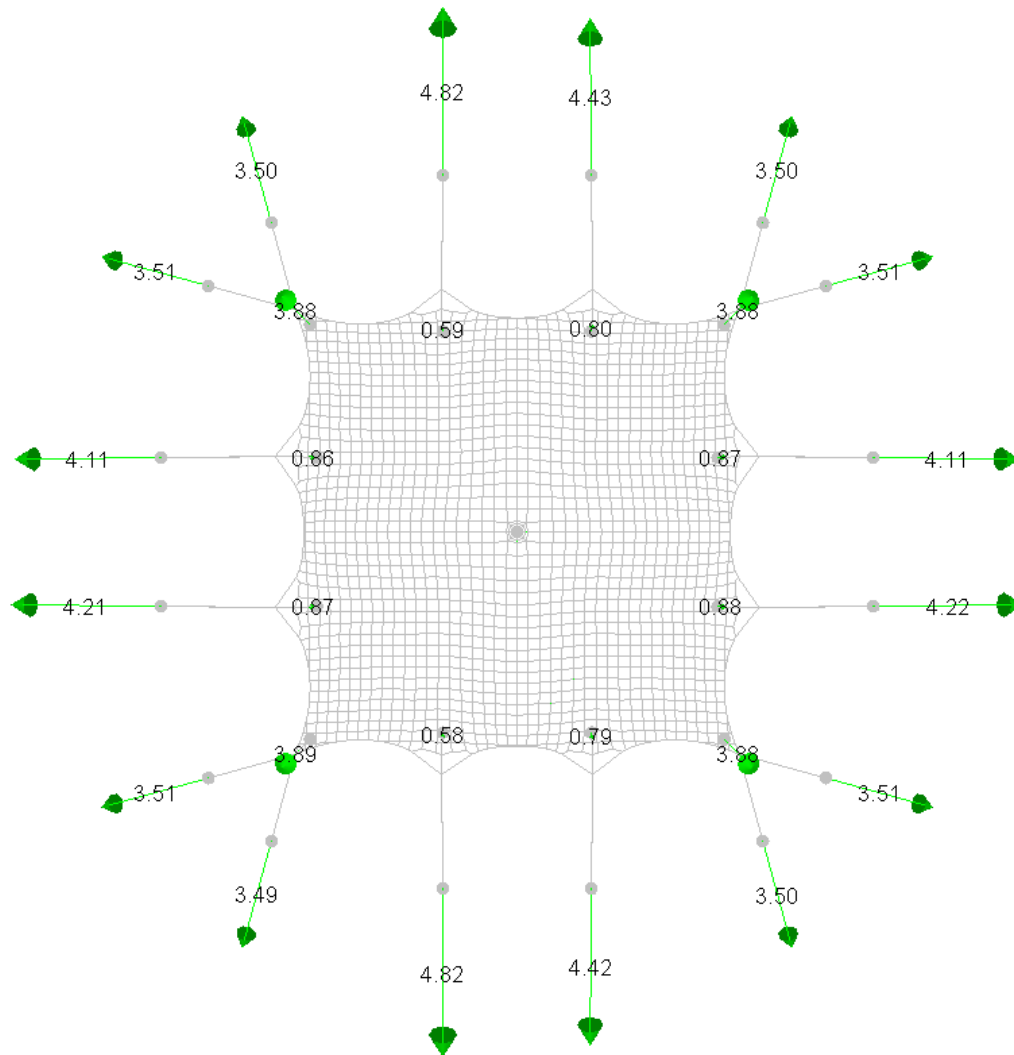
### Annex D.5.5. Belt forces



### Annex D.5.6. Poles forces

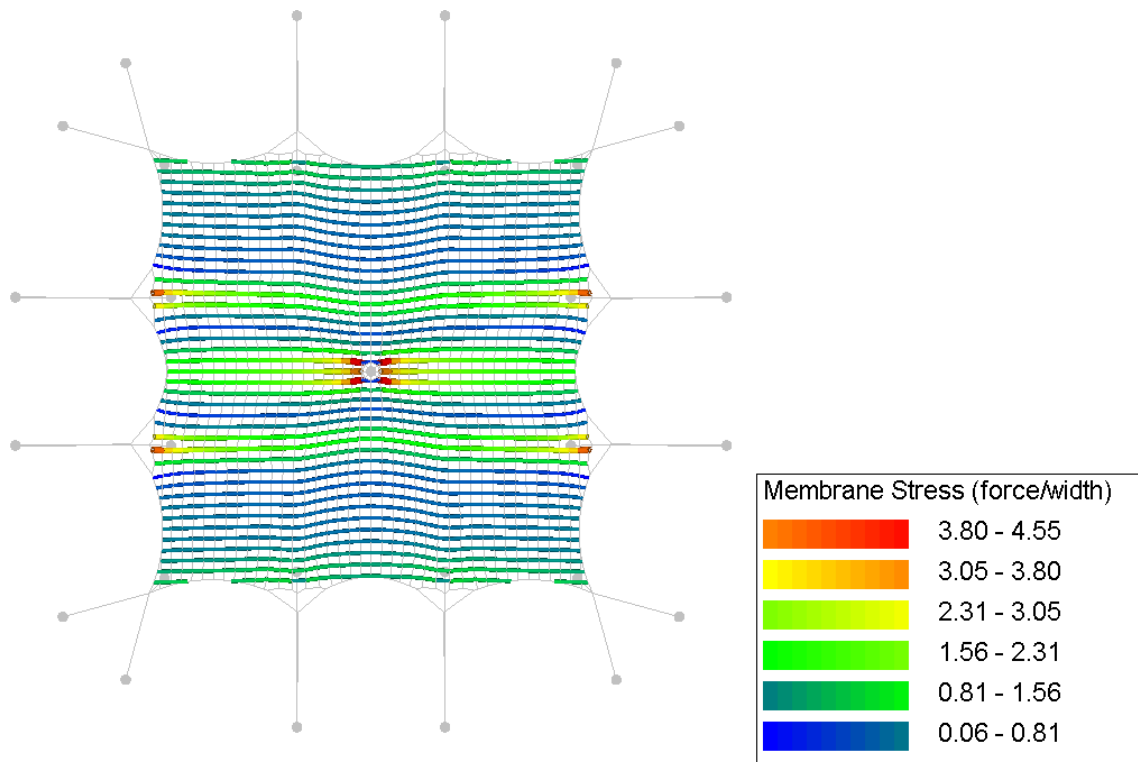


## Annex D.5.7. Reaction forces

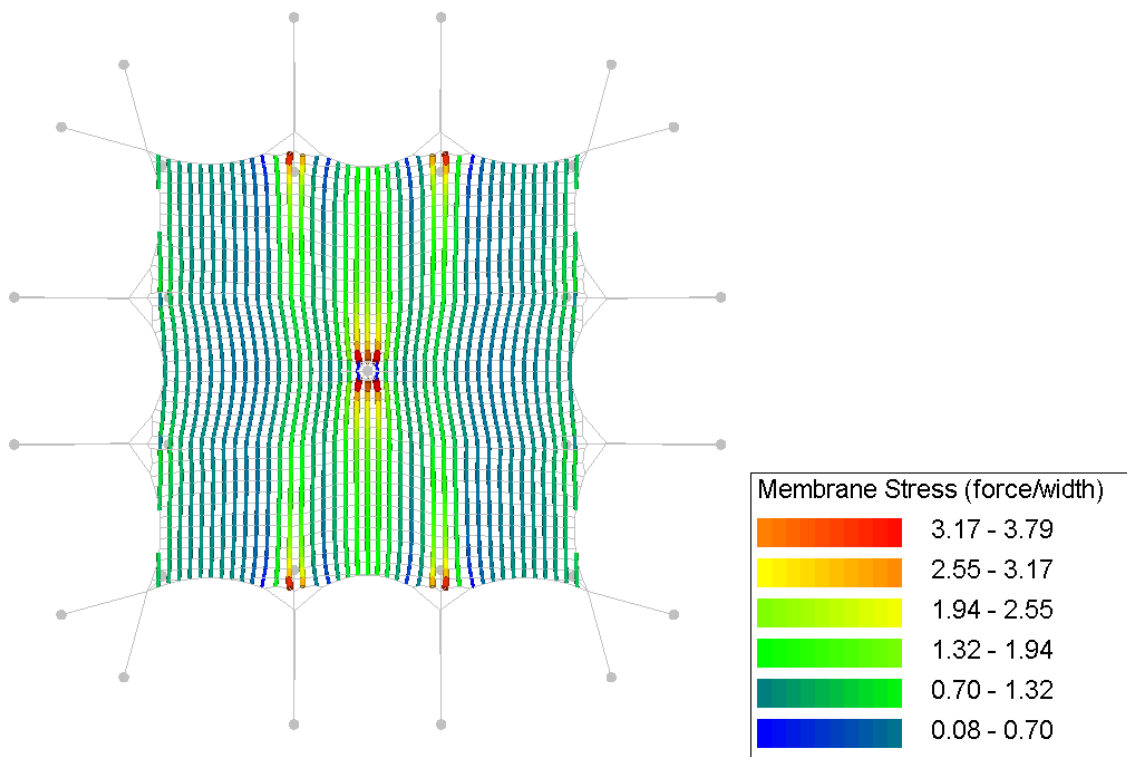


## Annex D.6. CO6. Own weight + pretension + wind pressure + snow [FR]

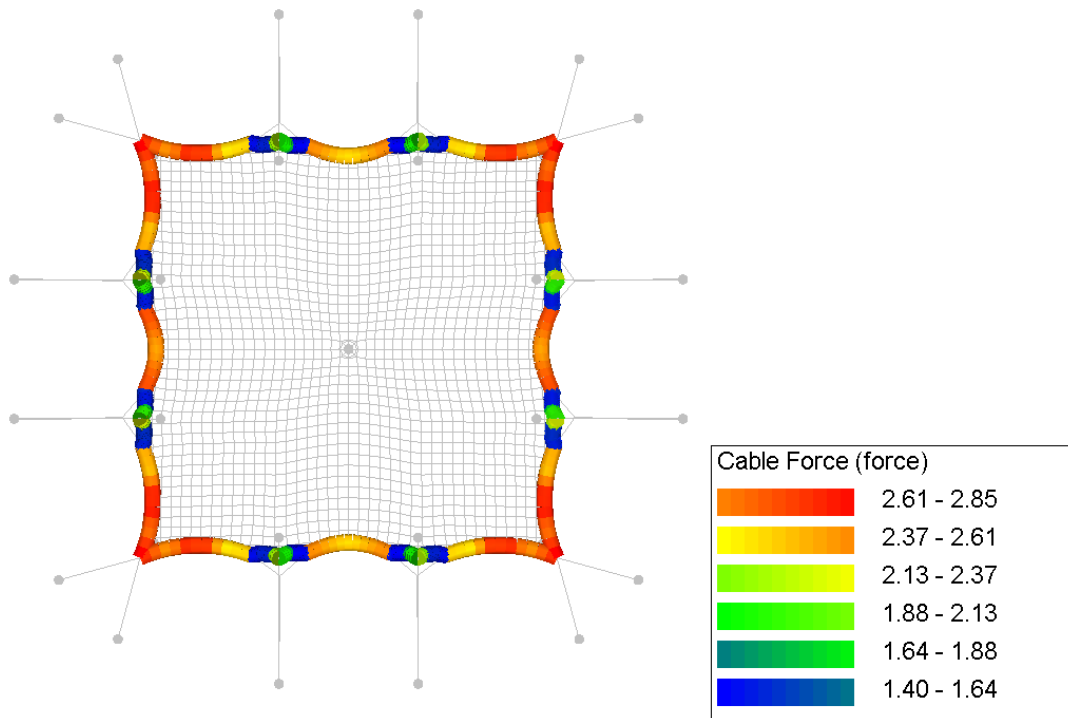
### Annex D.6.1. Membrane stress (warp)



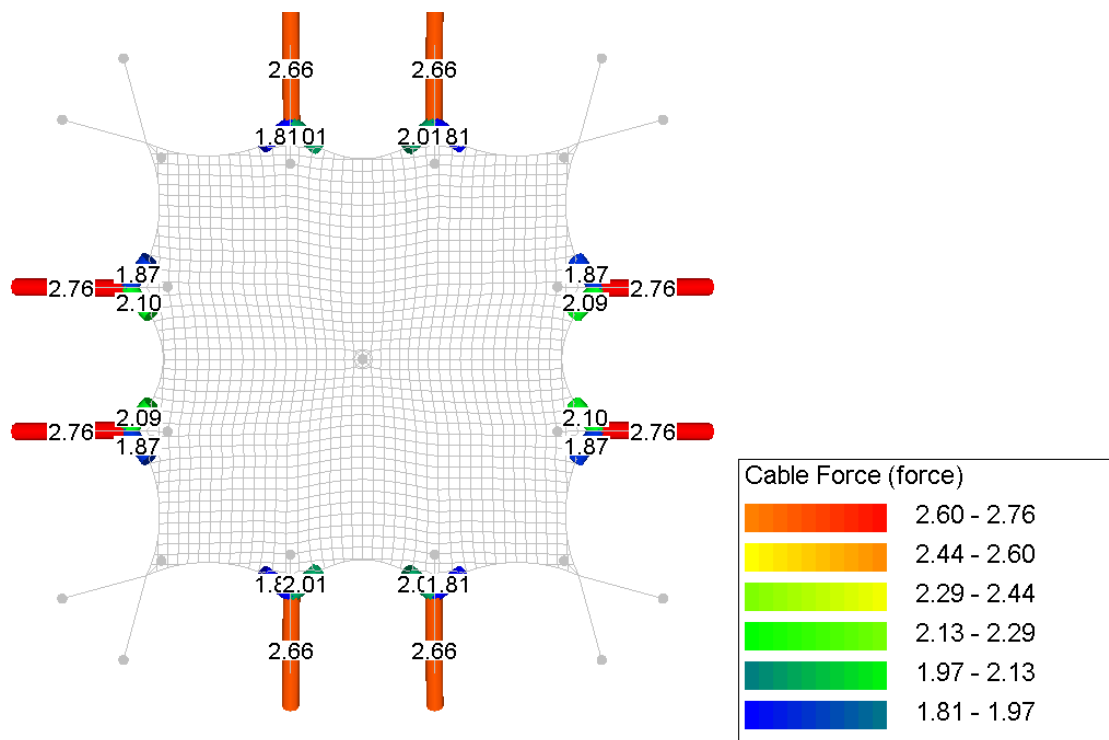
### Annex D.6.2. Membrane stress (weft)



### Annex D.6.3. Circumference rope forces

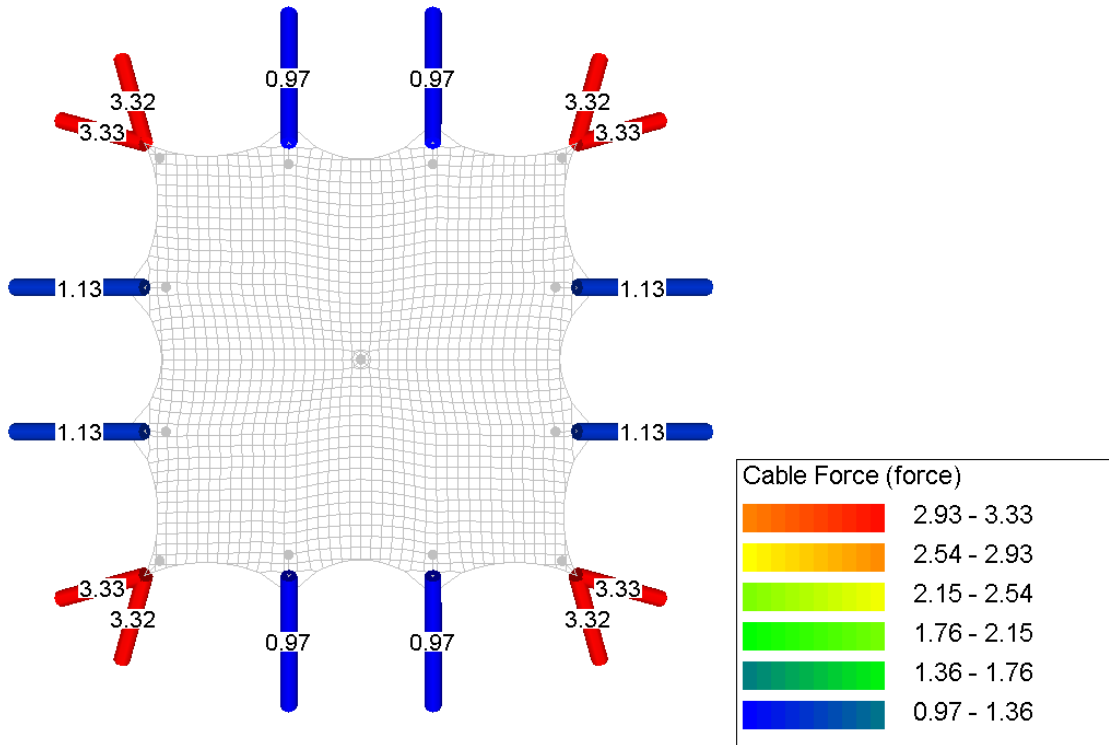


### Annex D.6.4. Guy rope forces

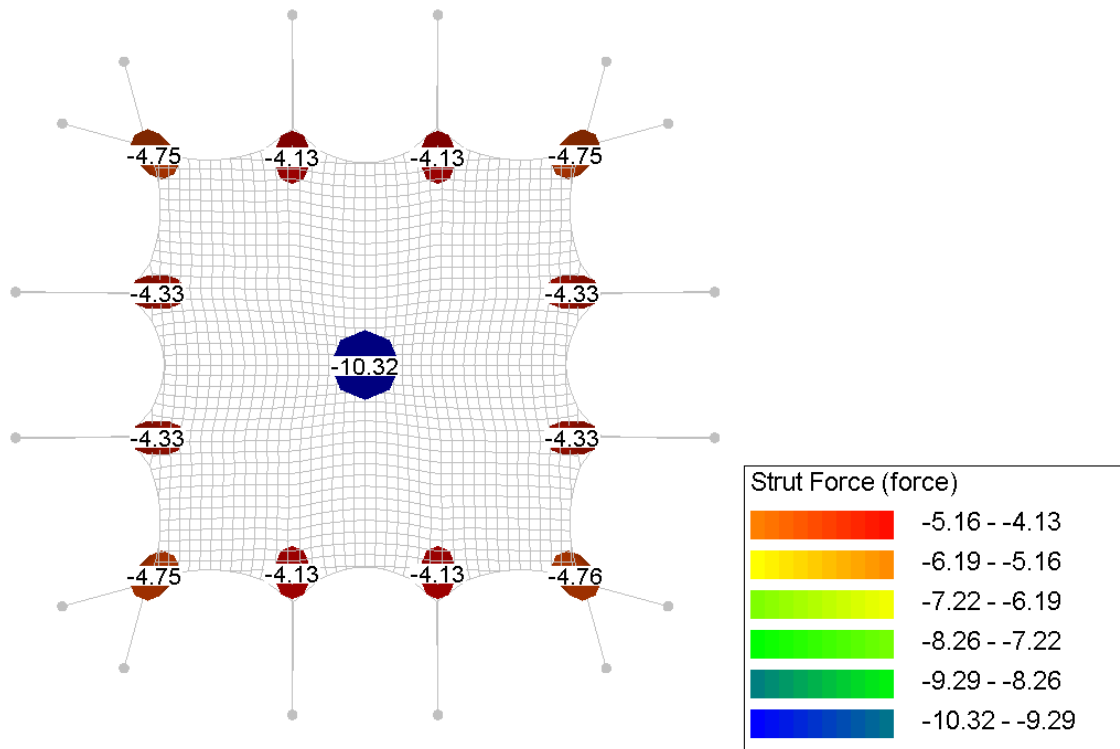




#### Annex D.6.5. Belt forces

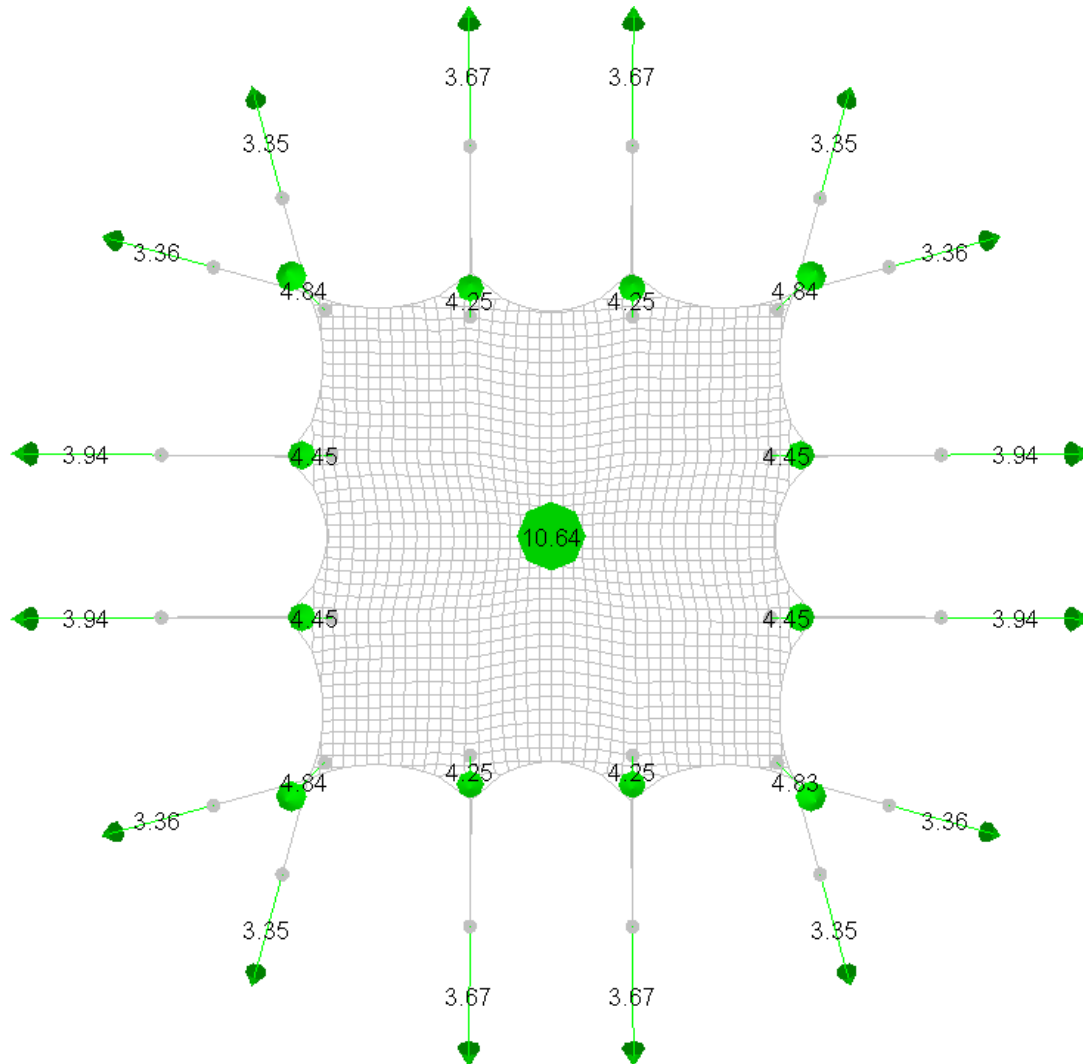


#### Annex D.6.6. Poles forces



### Annex D.6.7. Reaction forces

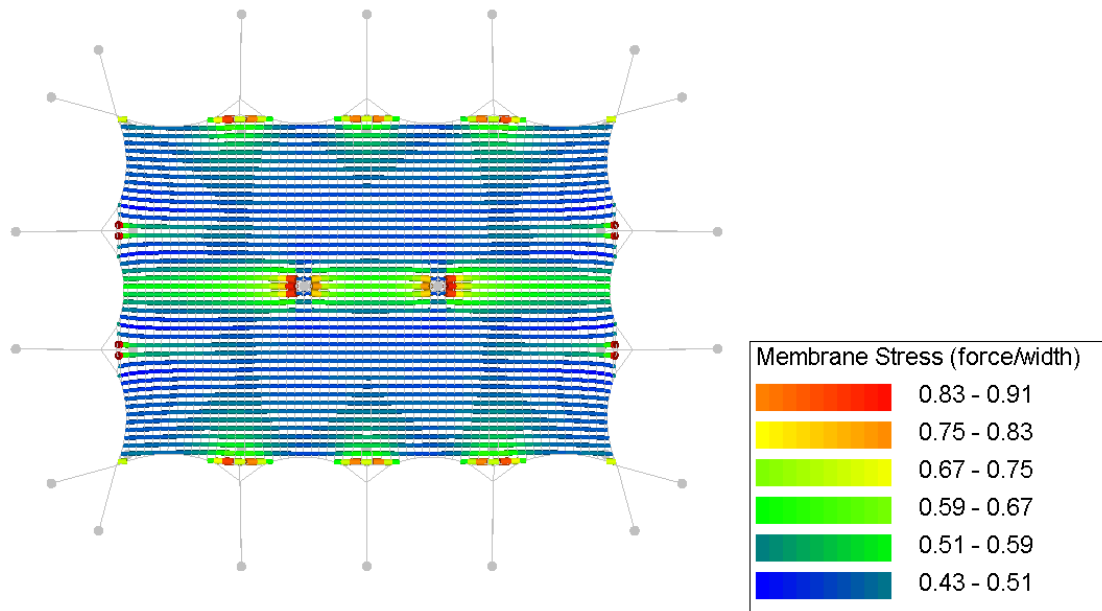
Increased snow load of  $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$  is applied for ballast and anchor calculation



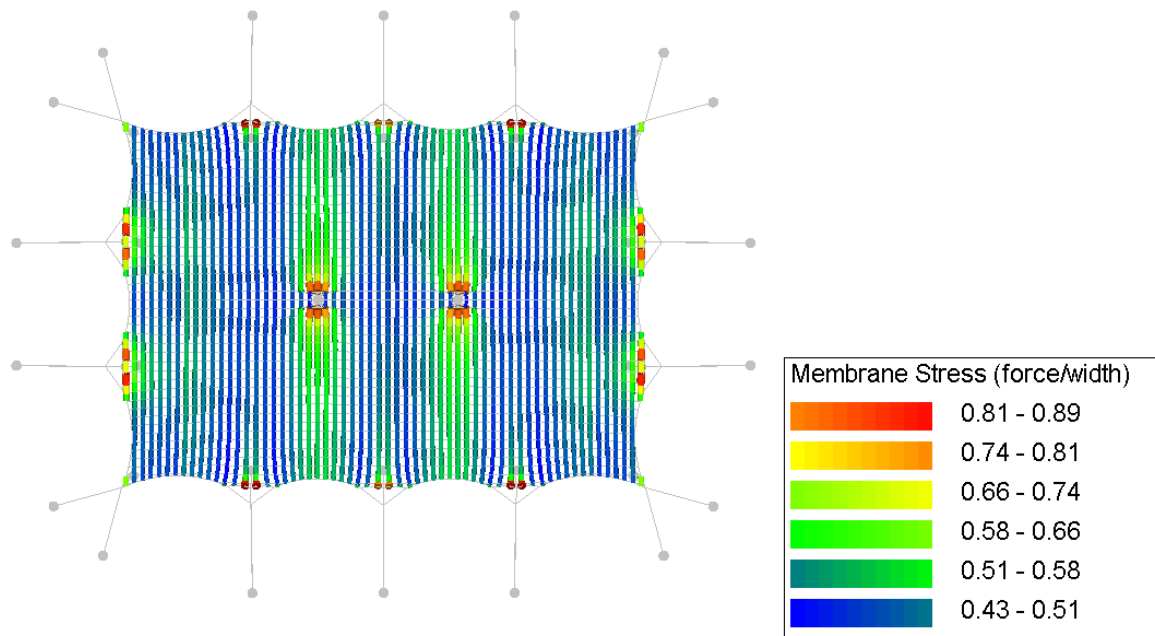
## Annex E. Results per load combination – 10x15m model

### Annex E.1. CO1. Own weight + pretension

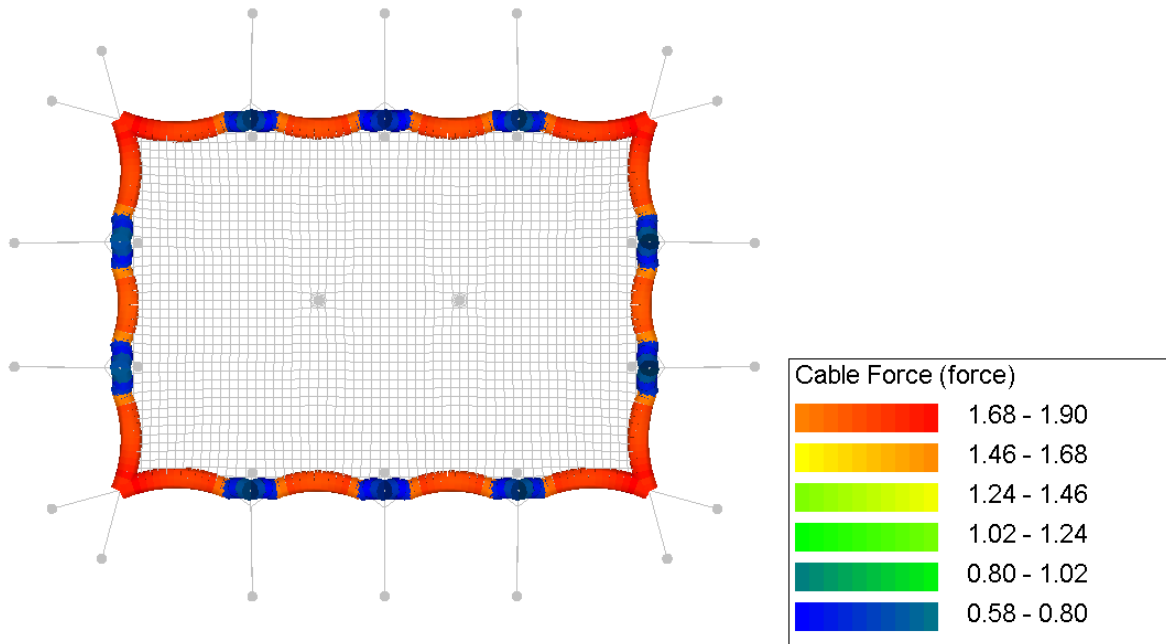
#### Annex E.1.1. Membrane stress (warp)



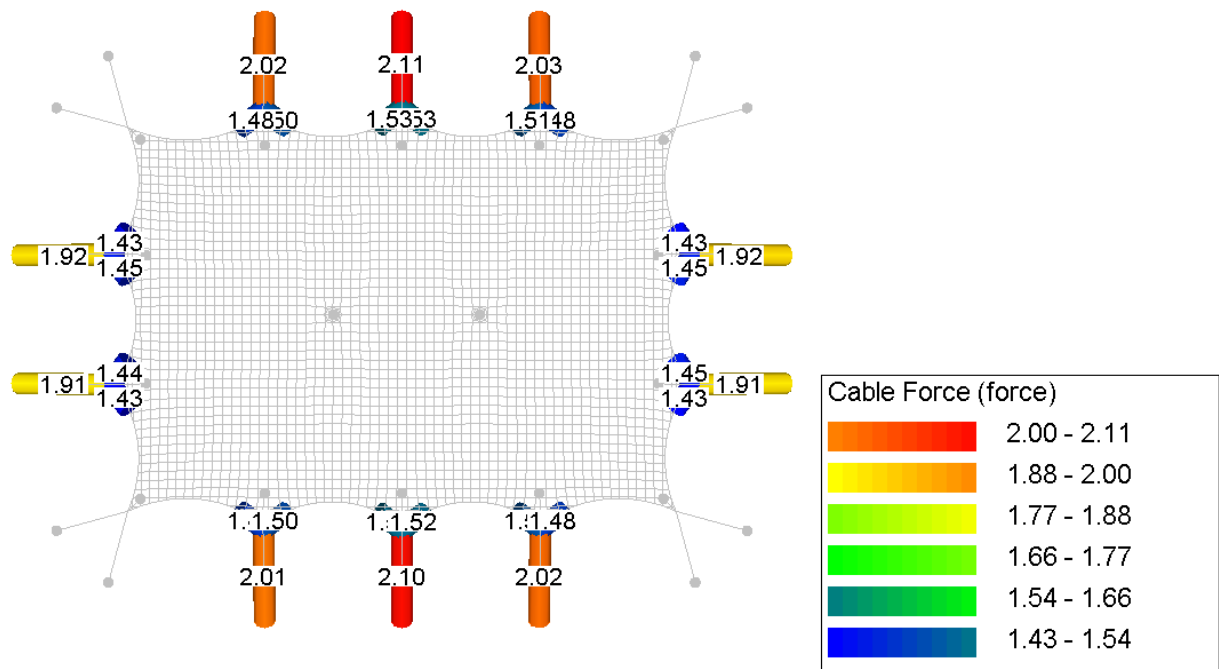
#### Annex E.1.2. Membrane stress (weft)



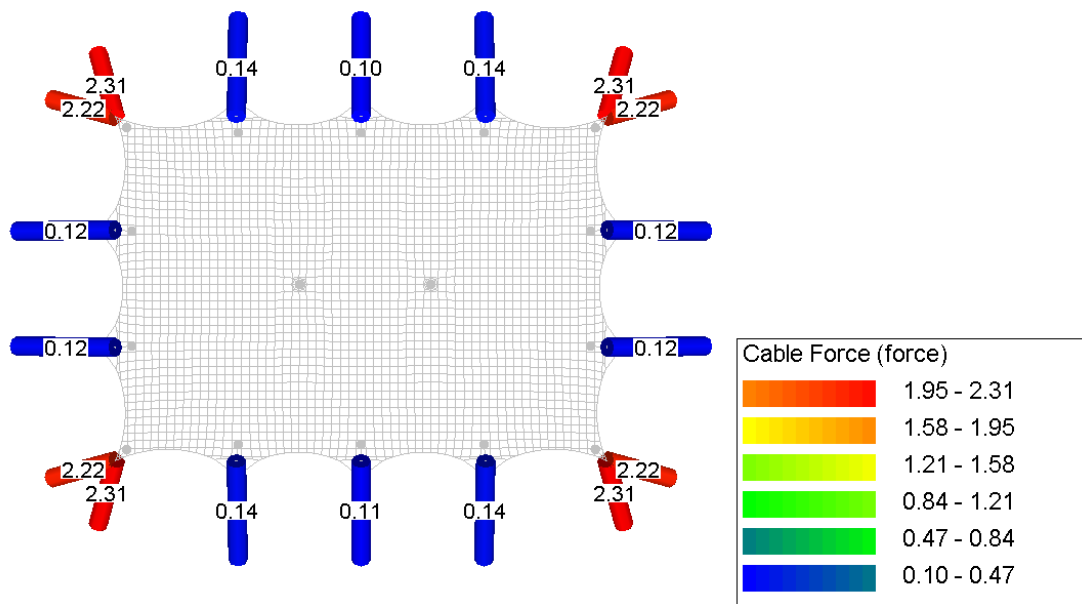
### Annex E.1.3. Circumference rope forces



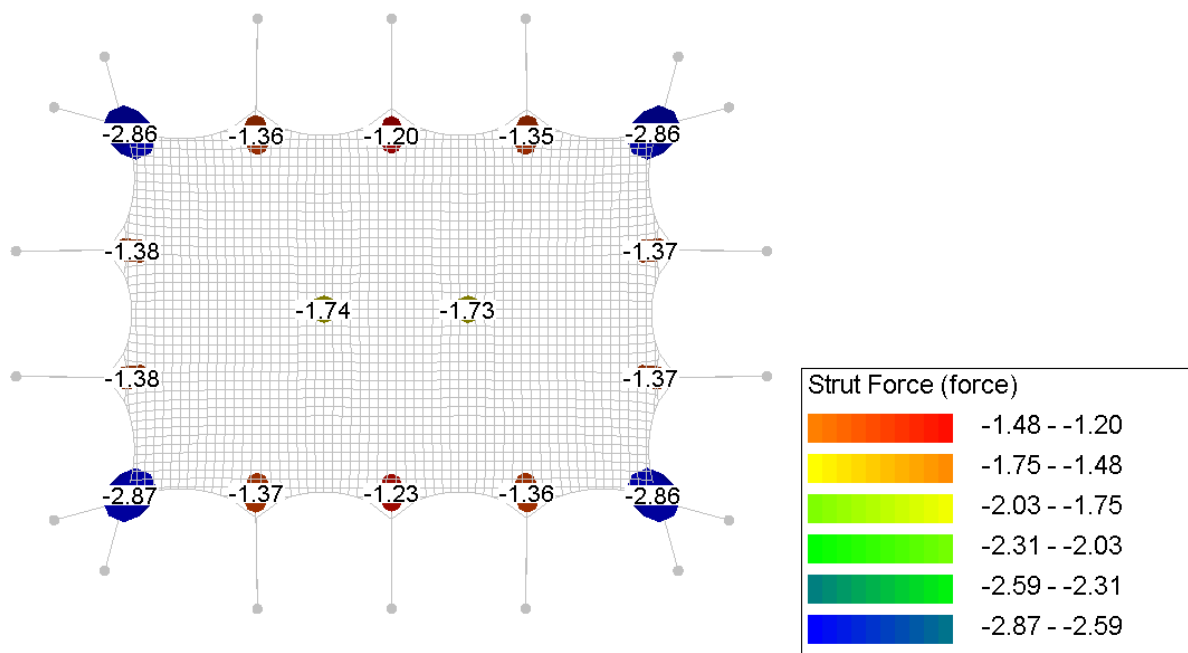
### Annex E.1.4. Rope forces



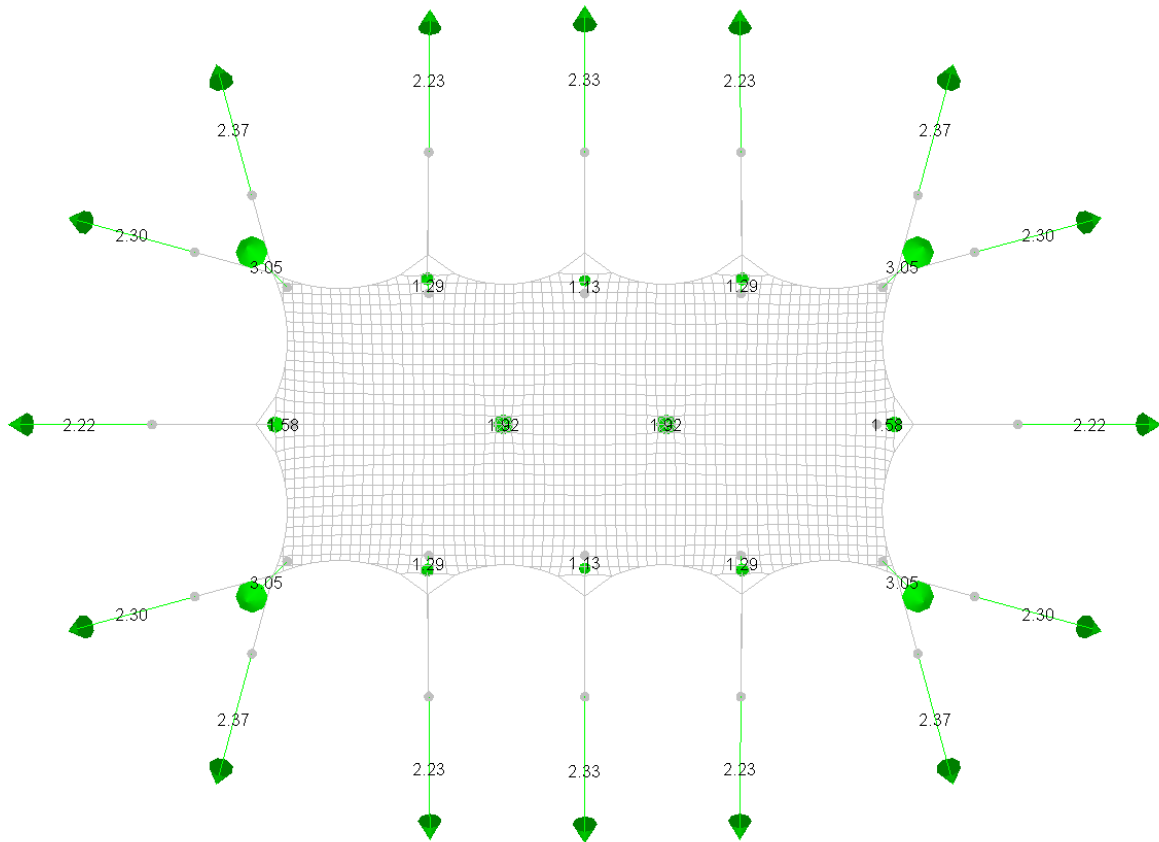
### Annex E.1.5. Belt forces



### Annex E.1.6. Pole forces

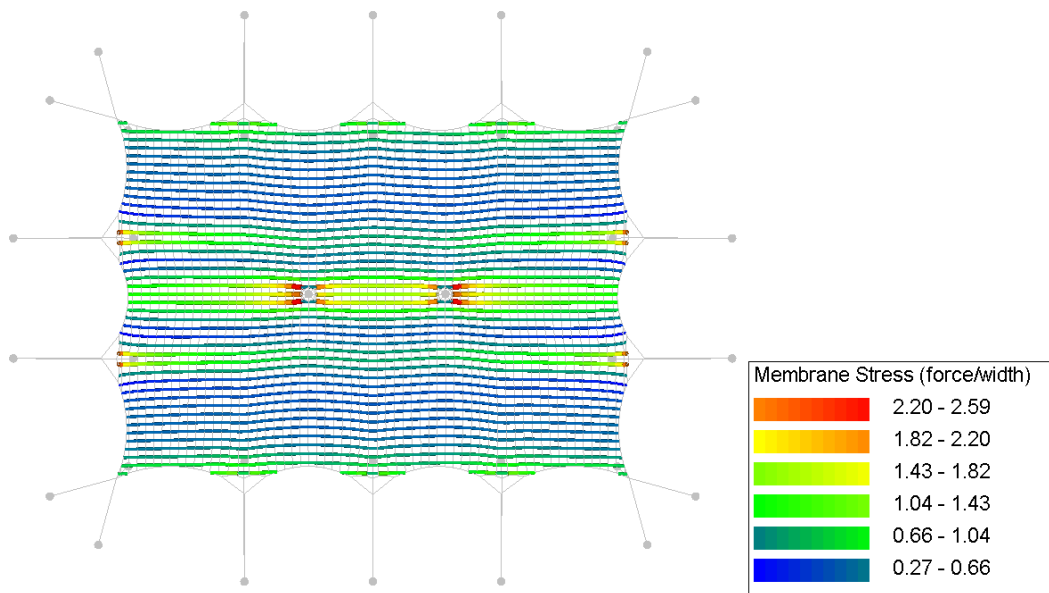


## Annex E.1.7. Reaction forces

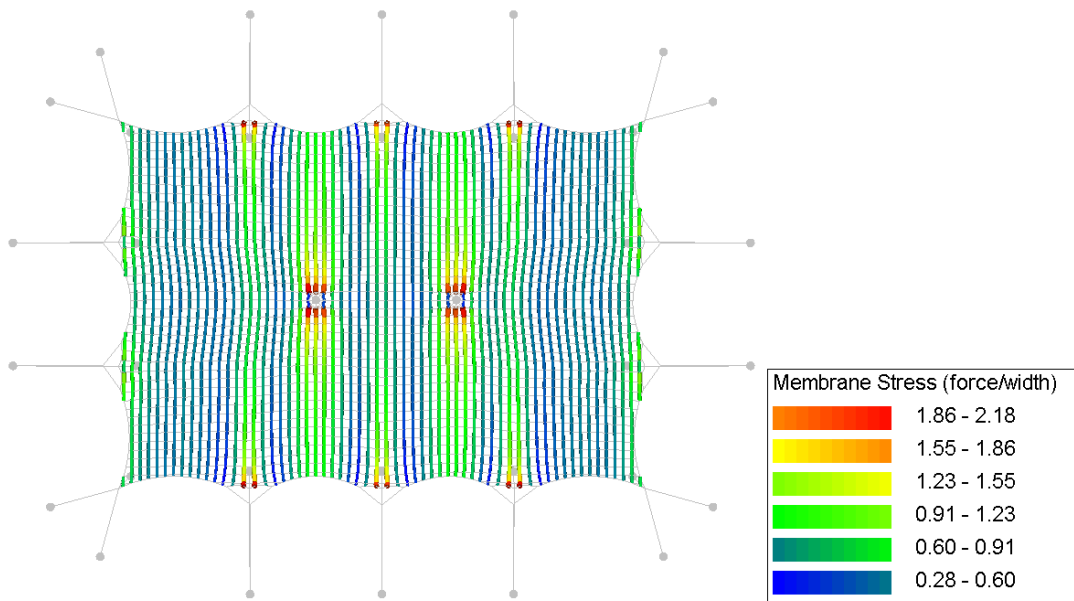


## Annex E.2. CO2. Own weight + pretension + snow

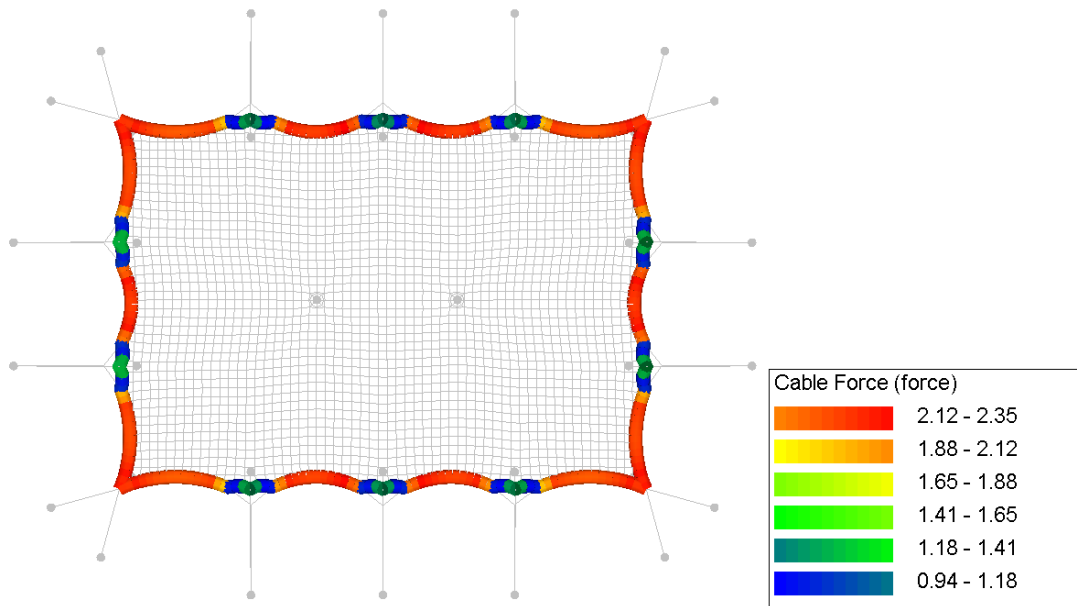
### Annex E.2.1. Membrane stress (warp)



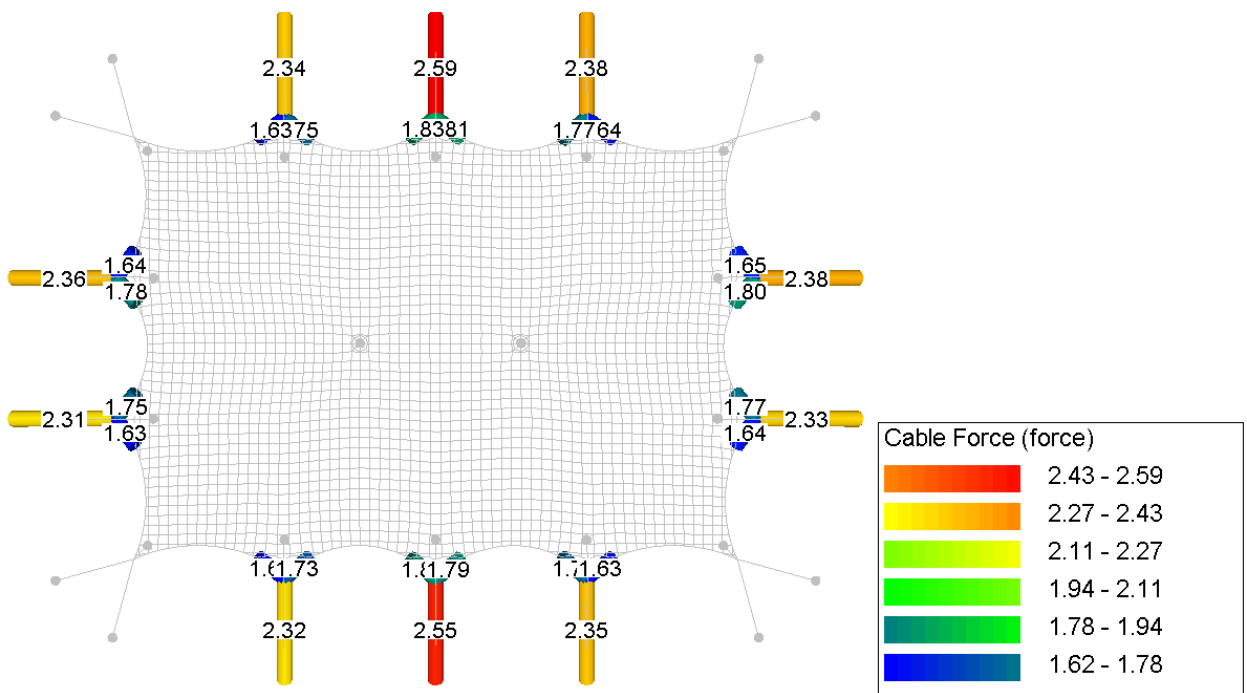
### Annex E.2.2. Membrane stress (weft)



### Annex E.2.3. Circumference rope forces

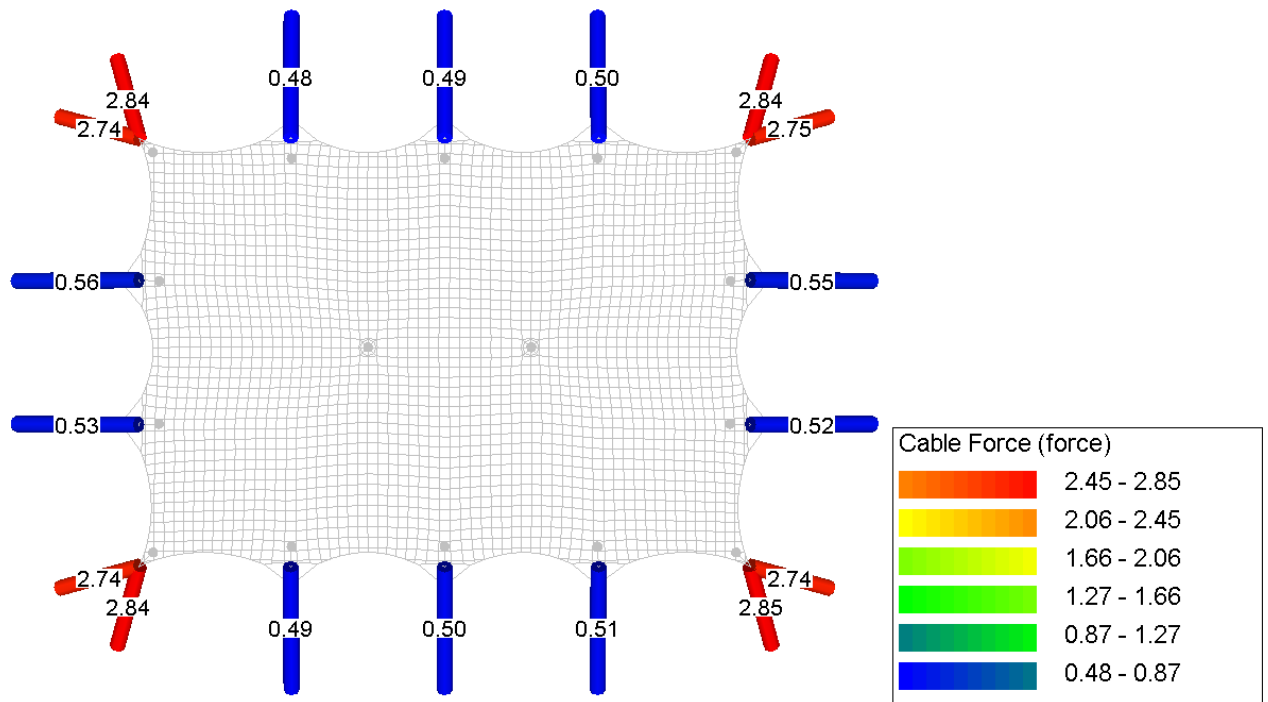


### Annex E.2.4. Rope forces

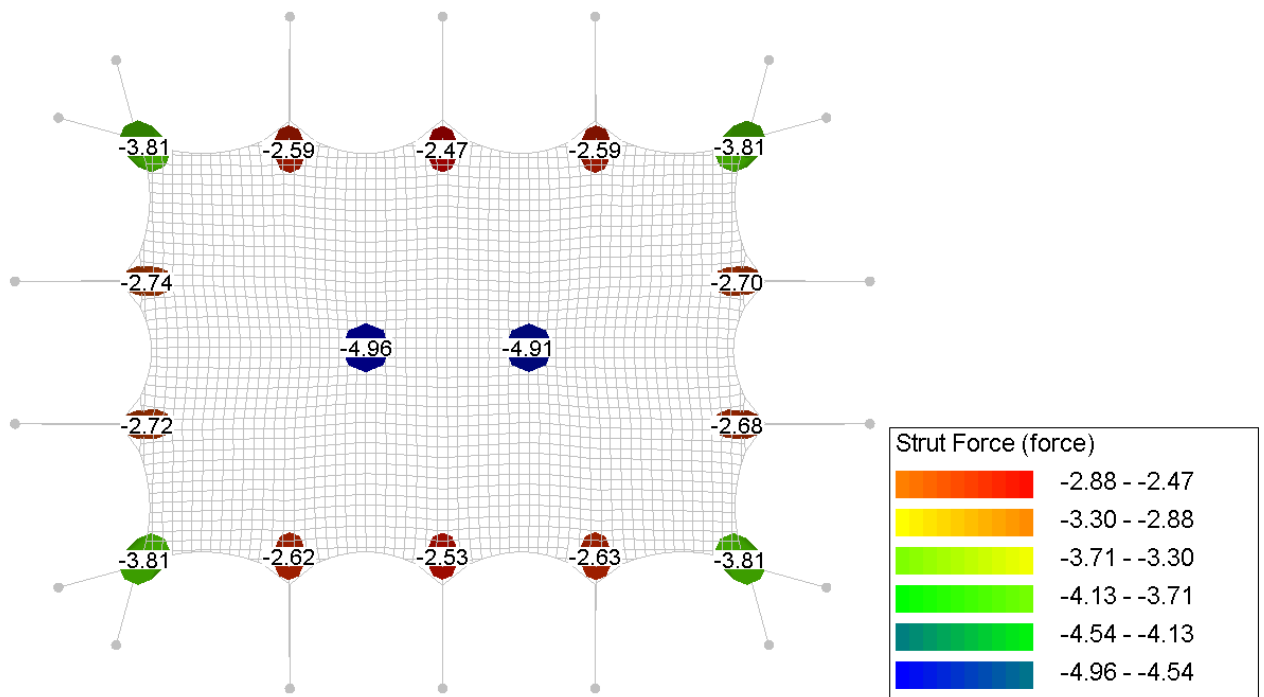




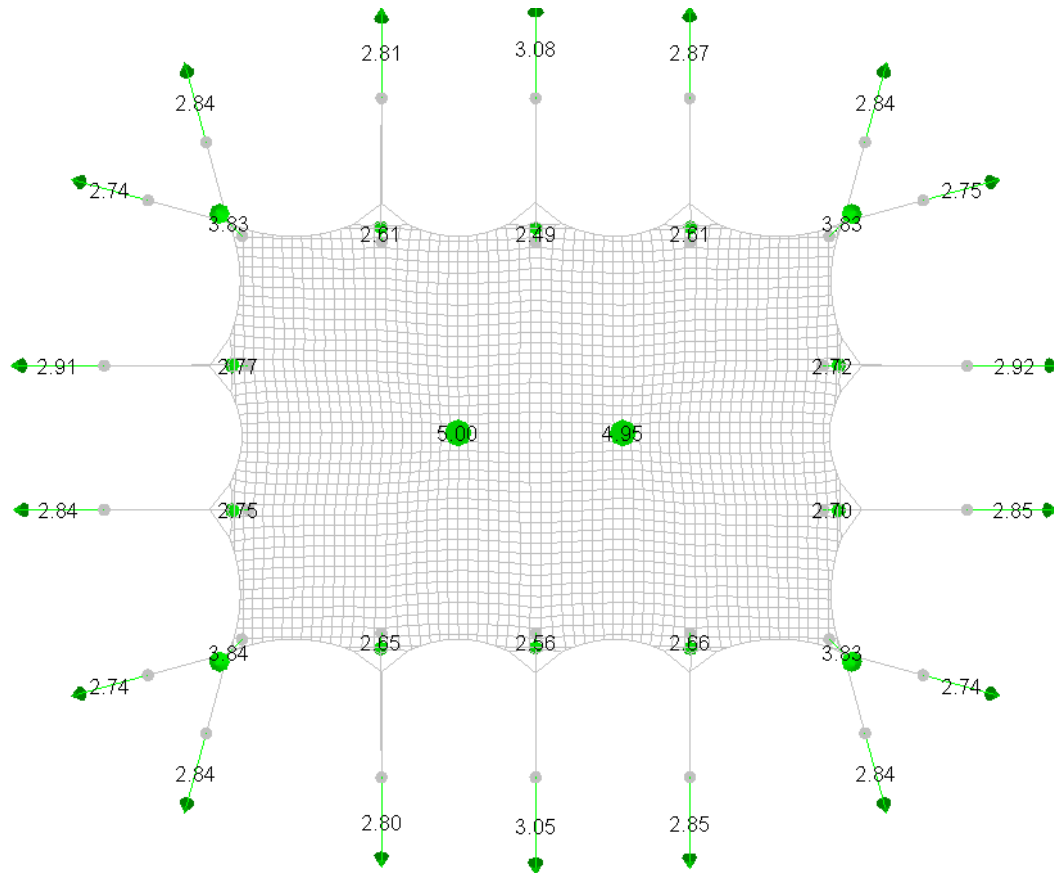
### Annex E.2.5. Belt forces



### Annex E.2.6. Pole forces

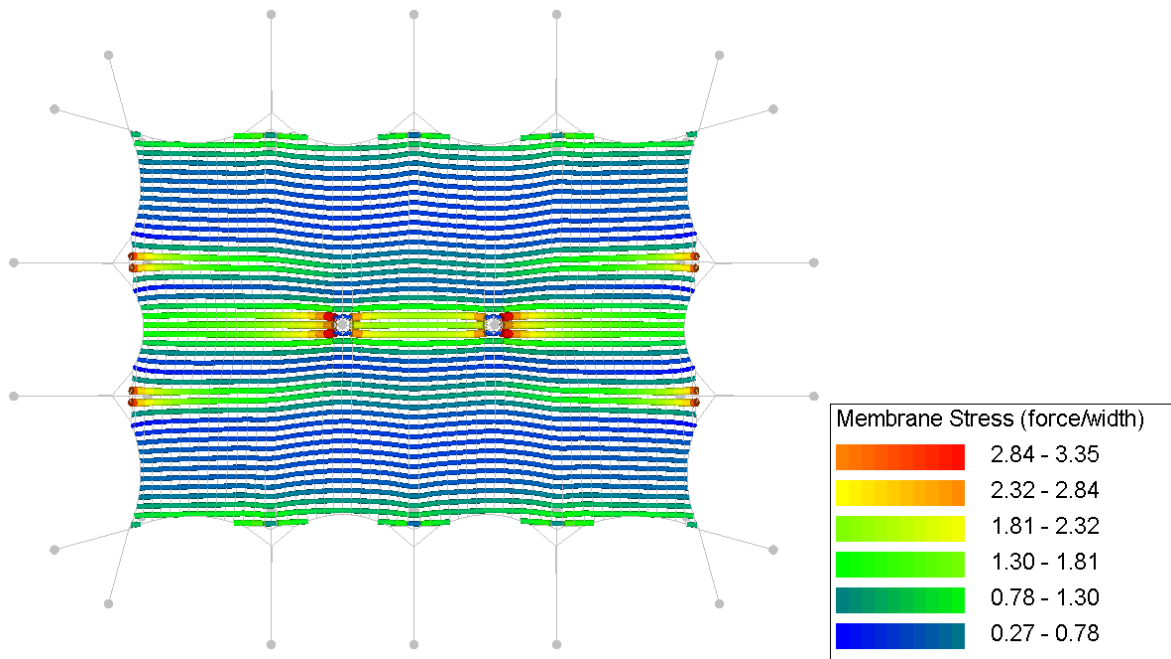


## Annex E.2.7. Anchor forces

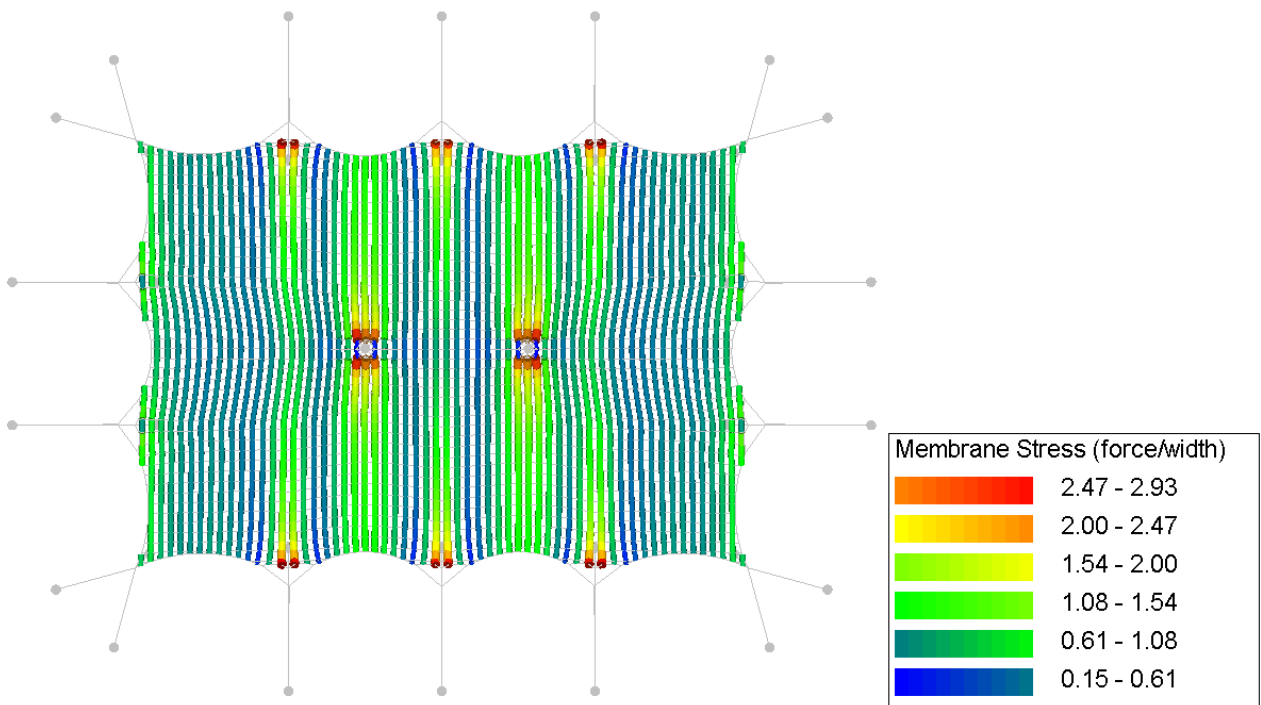


## Annex E.3. CO3. Own weight + pretension + wind pressure

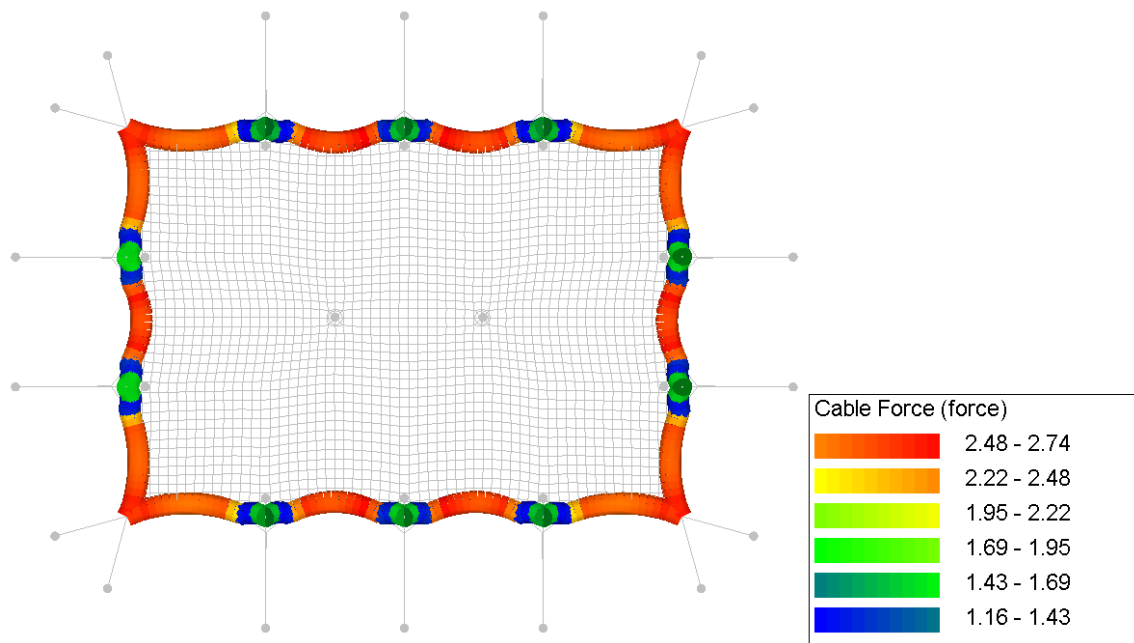
### Annex E.3.1. Membrane stress (warp)



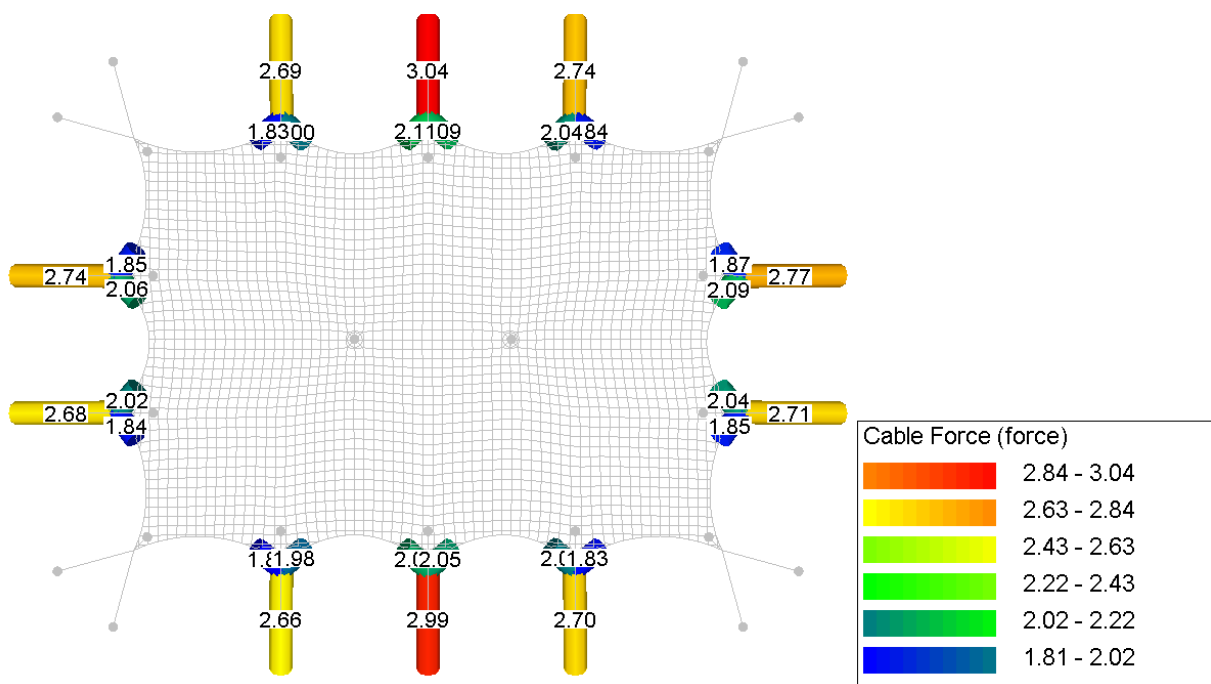
### Annex E.3.2. Membrane stress (weft)



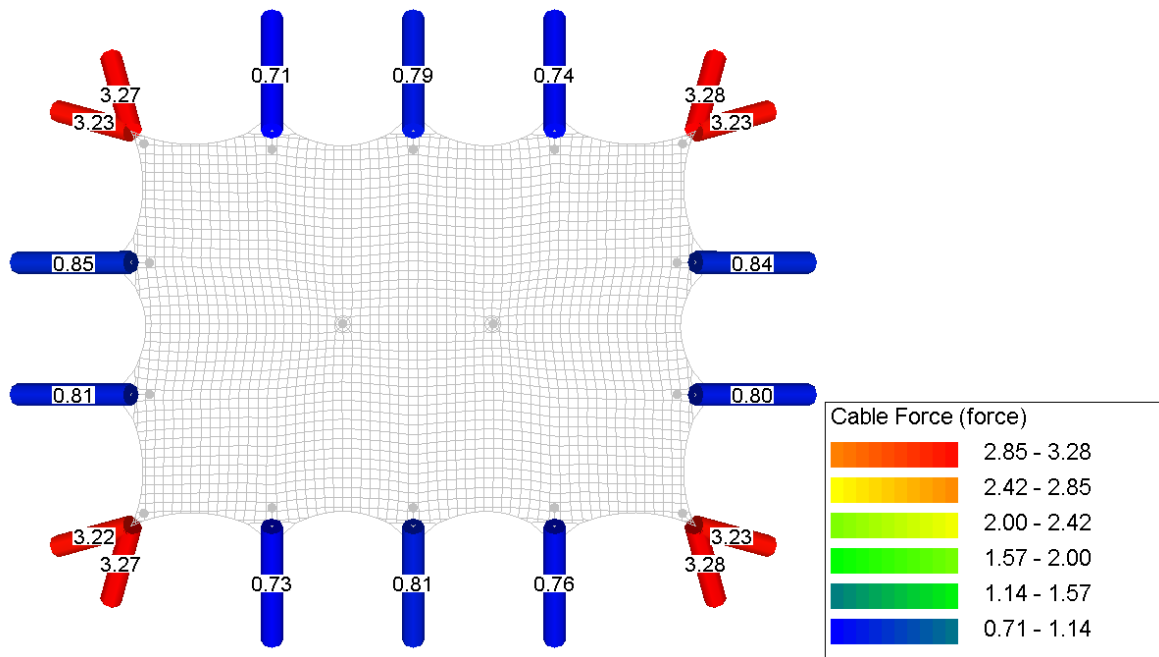
### Annex E.3.3. Circumference rope forces



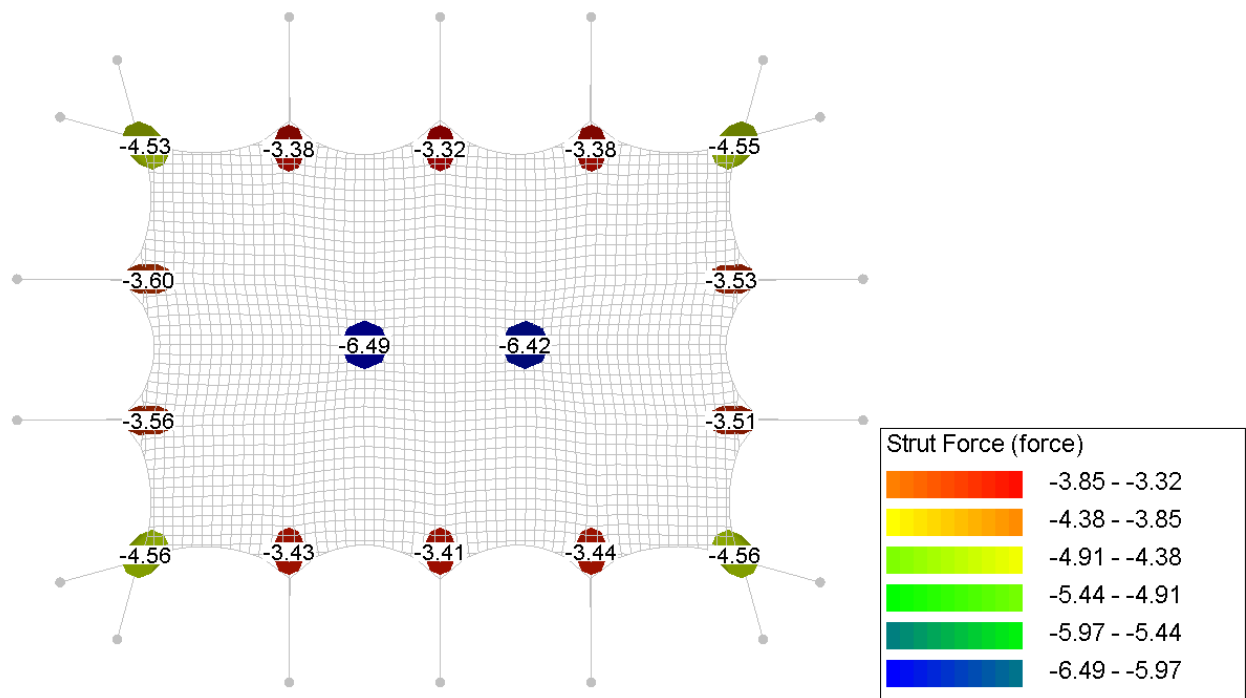
### Annex E.3.4. Rope forces



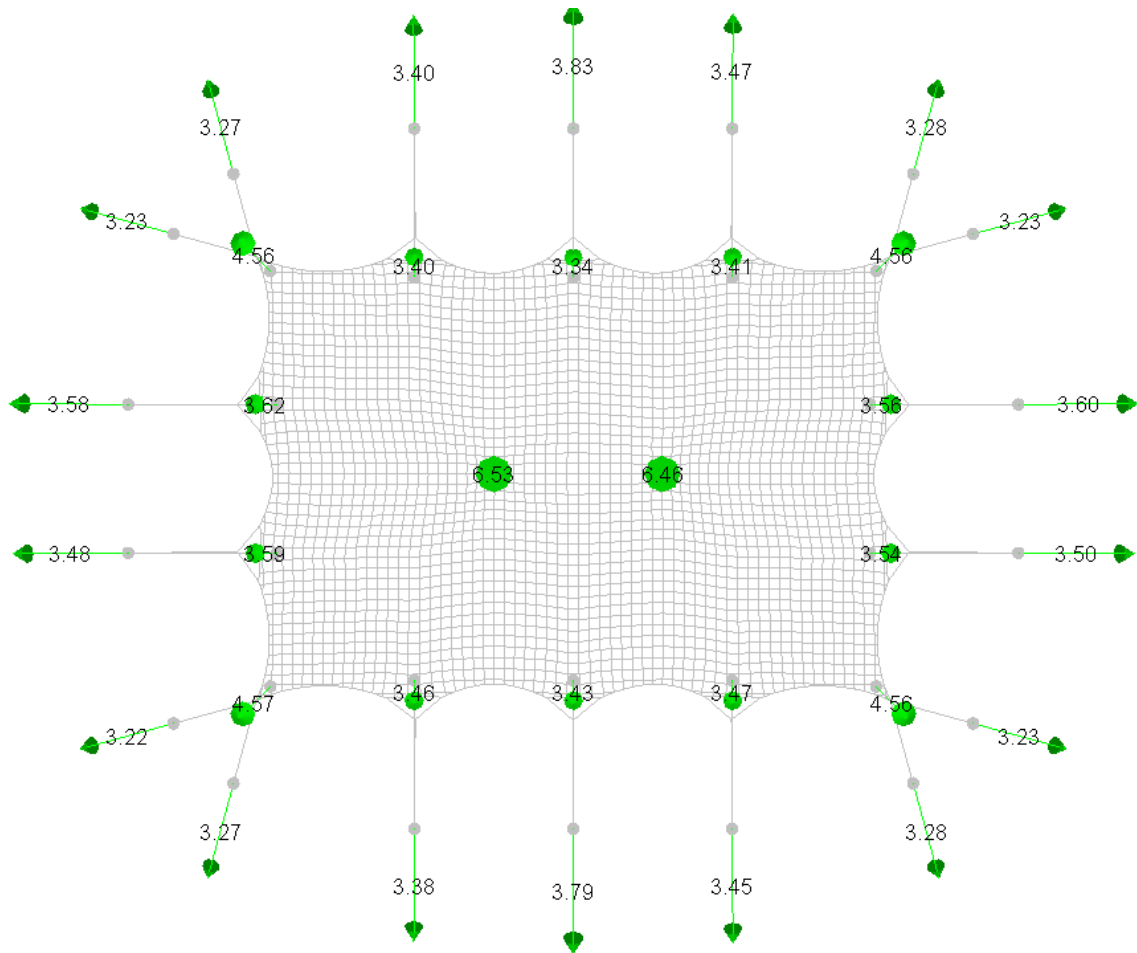
### Annex E.3.5. Belt forces



### Annex E.3.6. Poles forces

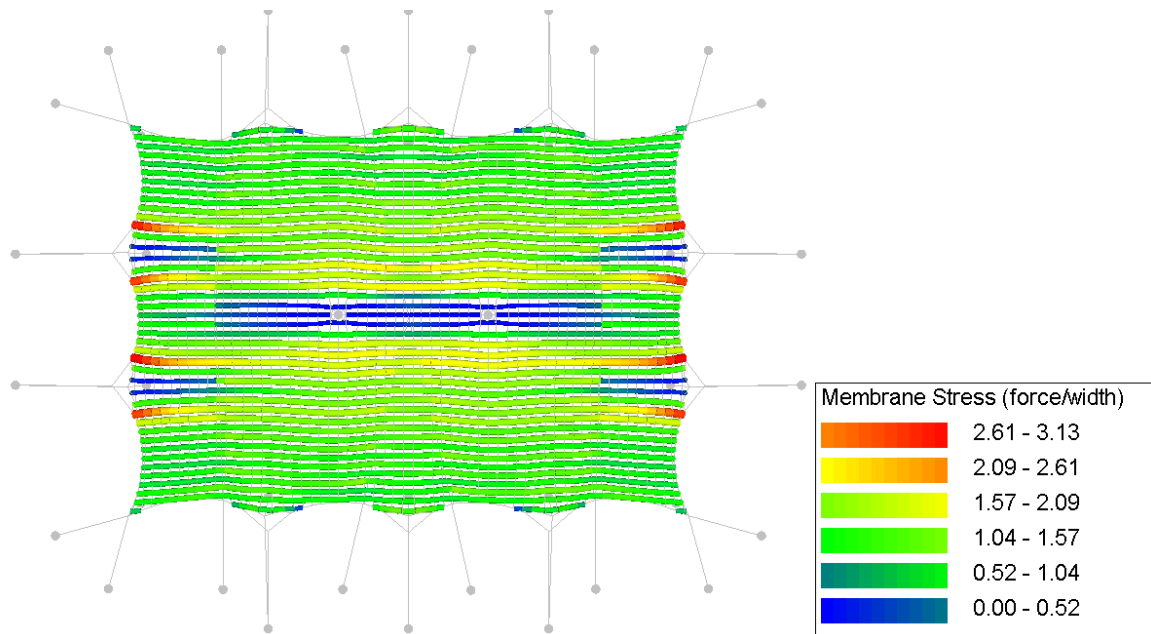


### Annex E.3.7. Reaction forces

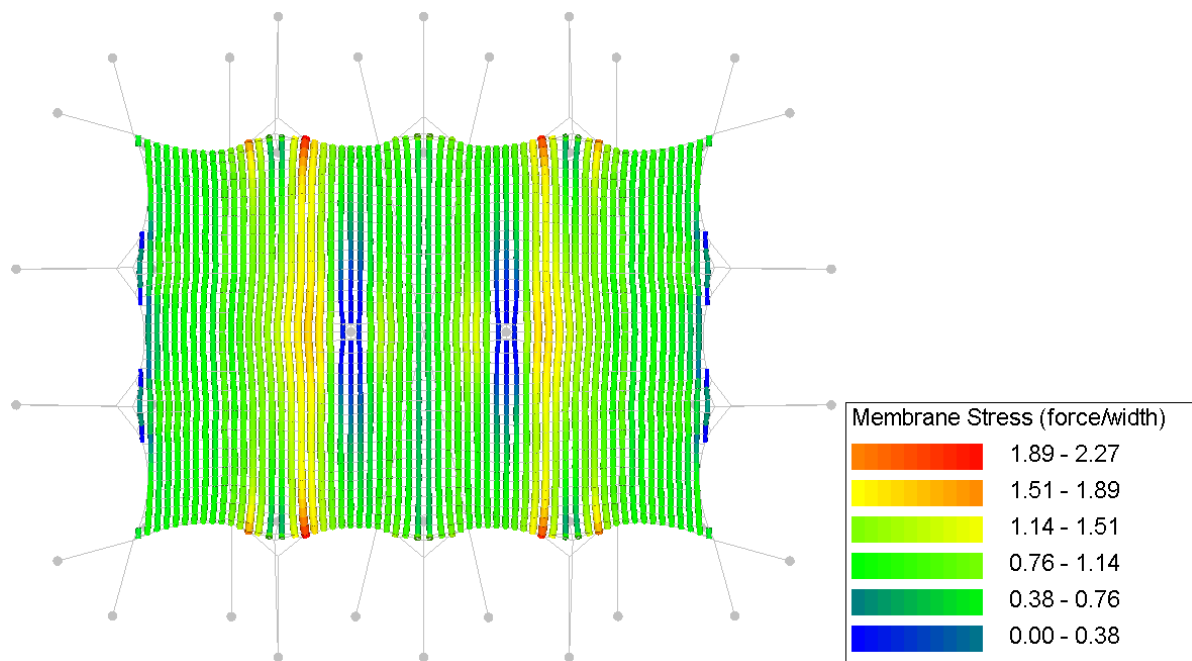


## Annex E.4. CO4. Own weight + pretension + wind suction

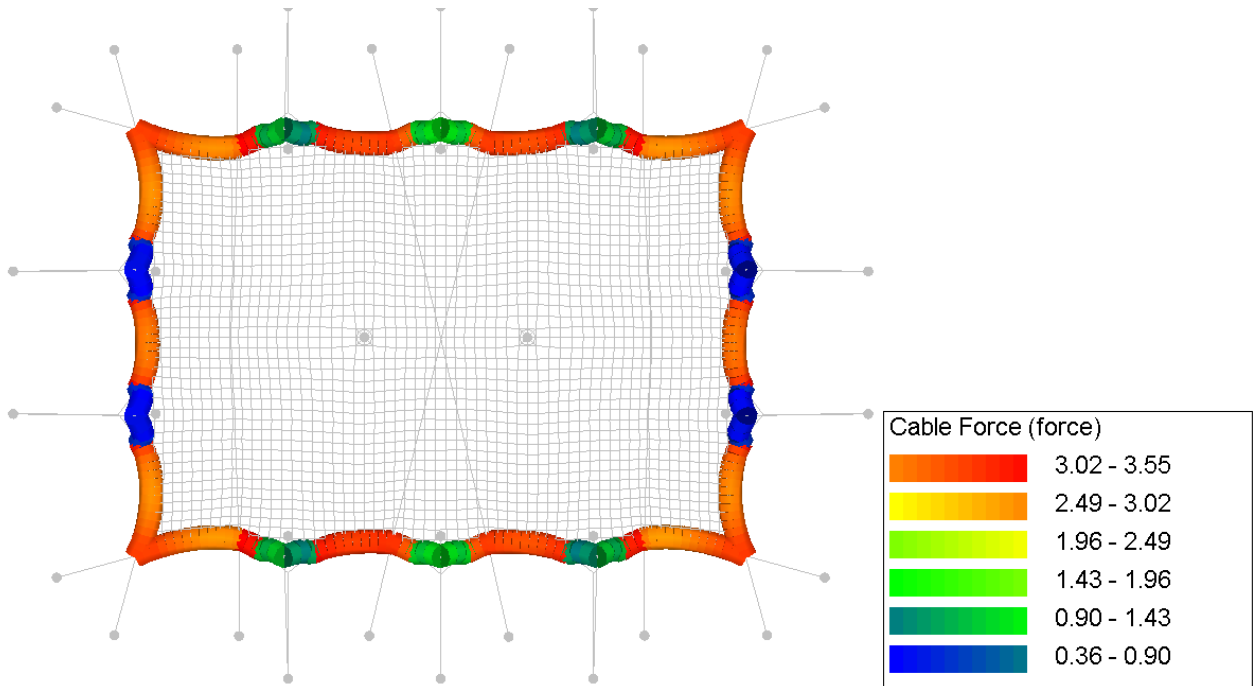
### Annex E.4.1. Membrane stress (warp)



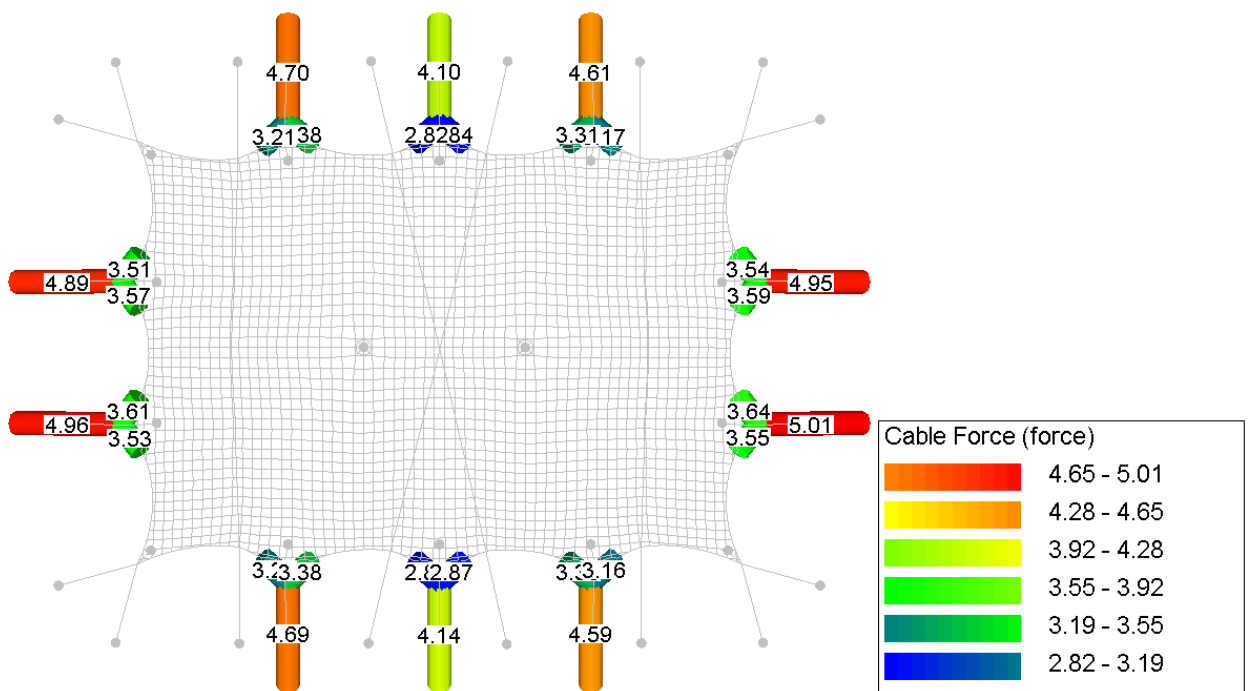
### Annex E.4.2. Membrane stress (weft)



#### Annex E.4.3. Circumference rope forces

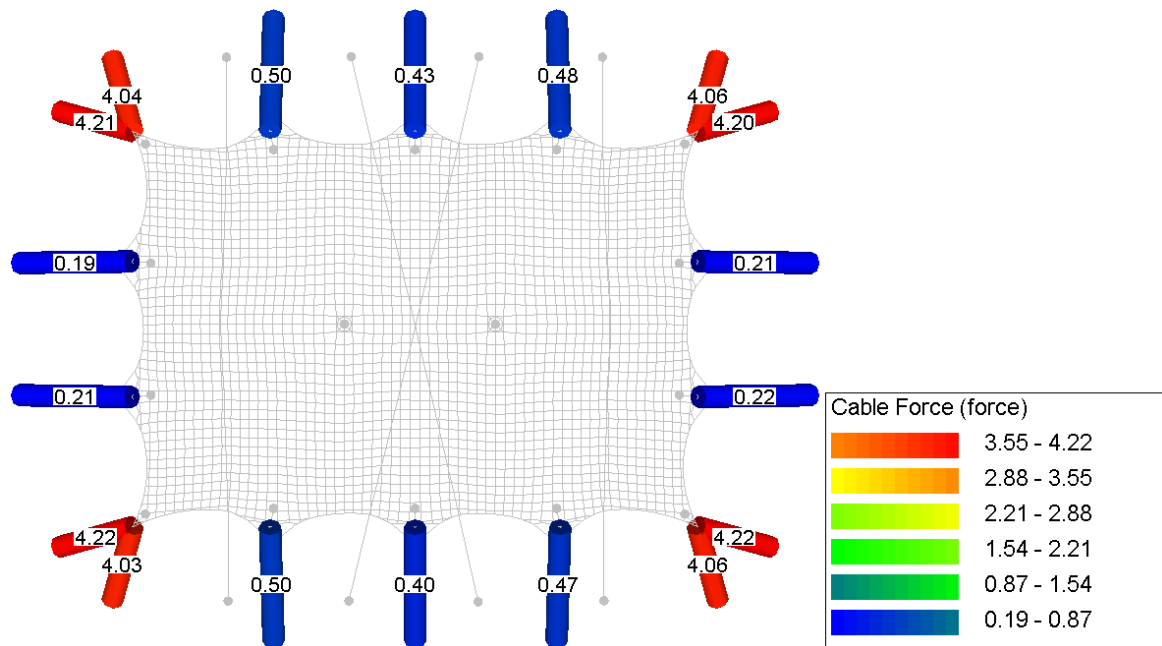


#### Annex E.4.4. Guy rope forces

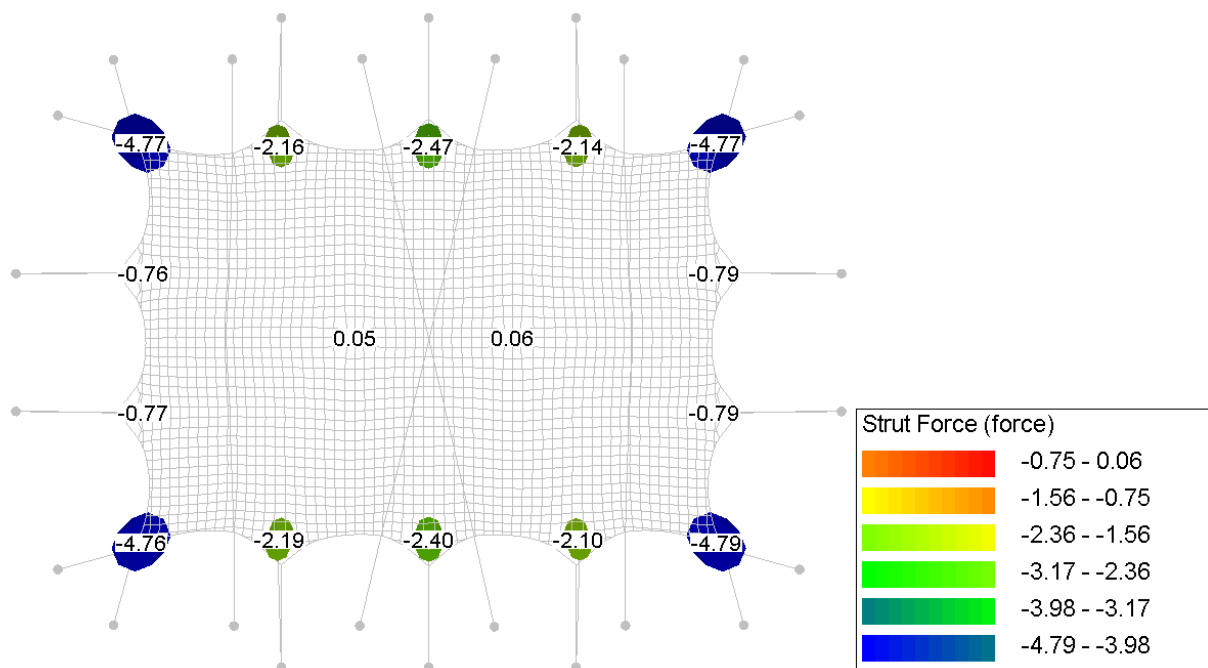




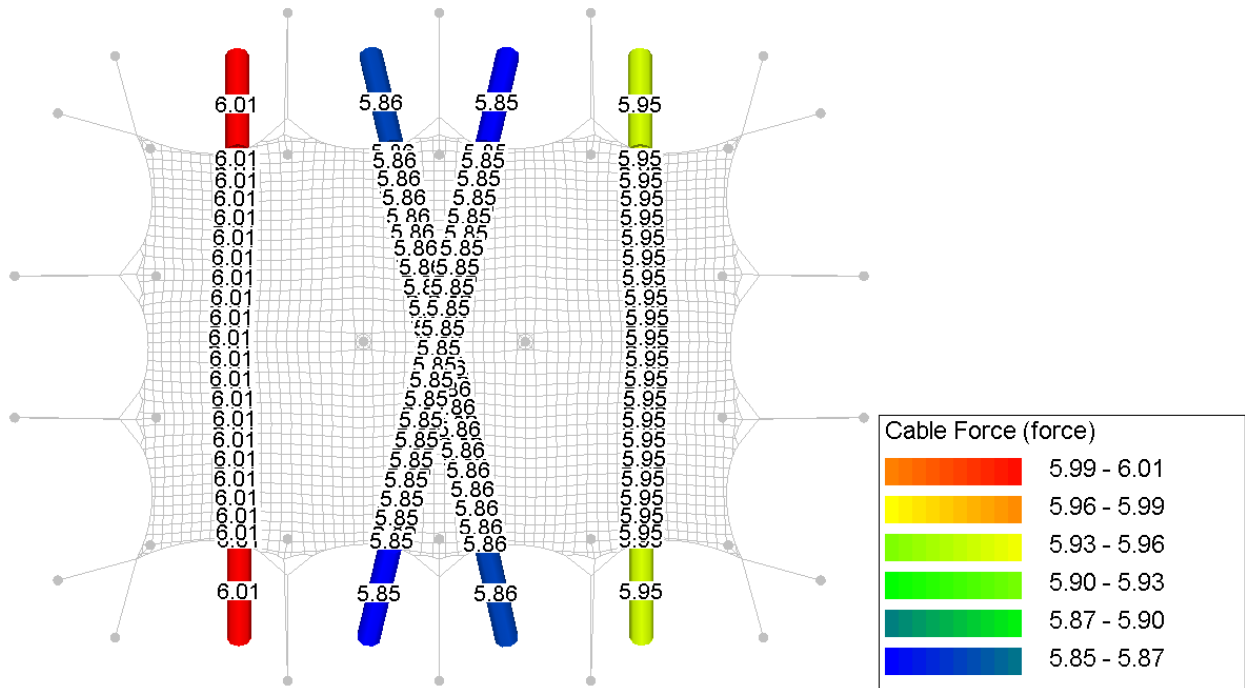
#### Annex E.4.5. Belt forces



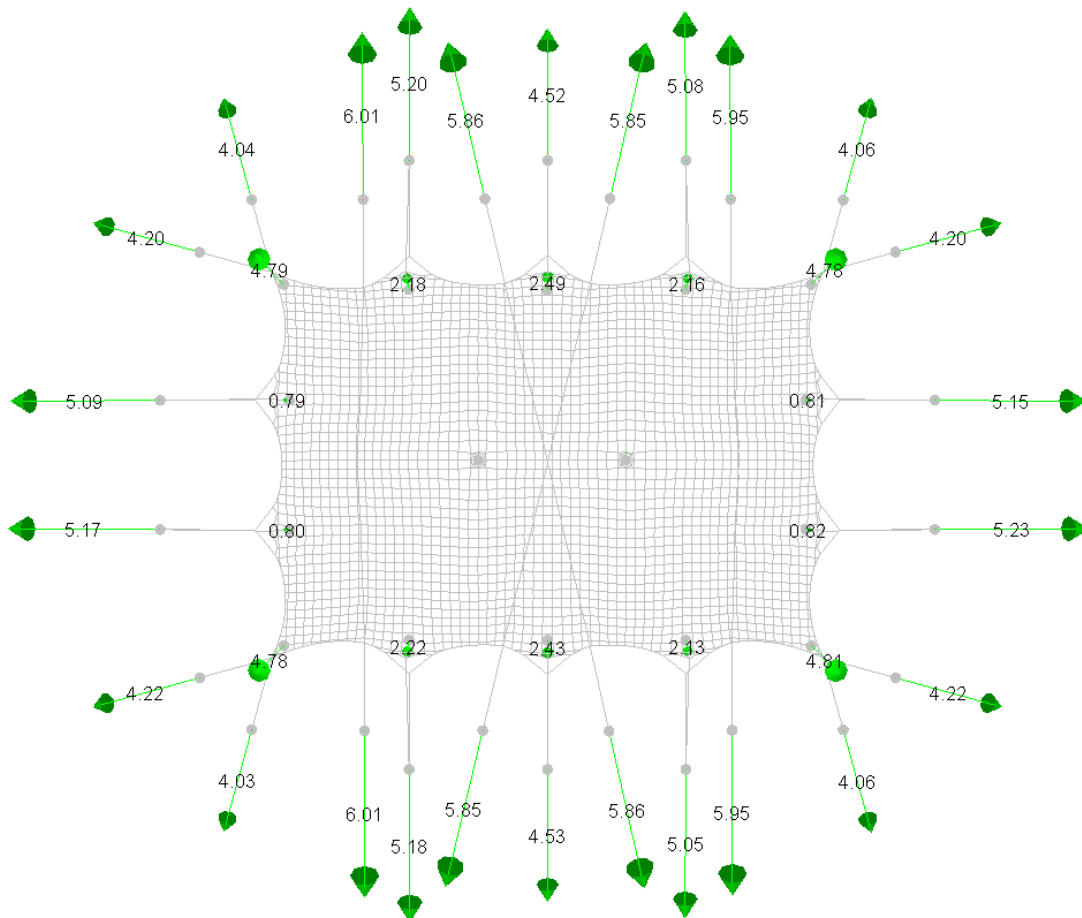
#### Annex E.4.6. Poles forces



#### Annex E.4.7. Storm belt forces

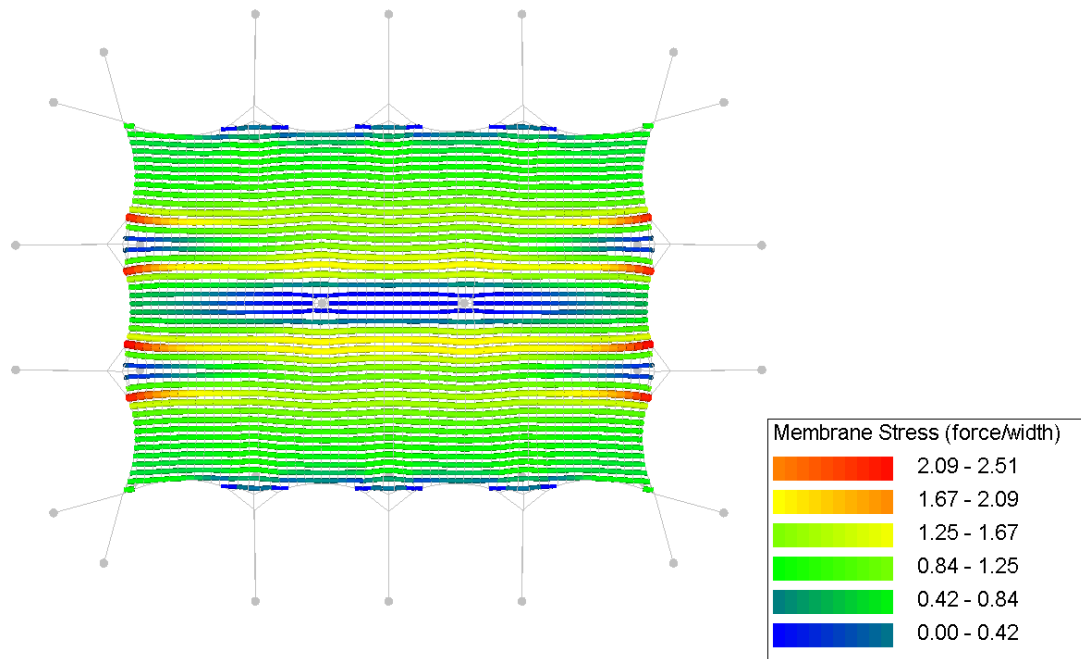


#### Annex E.4.8. Reaction forces

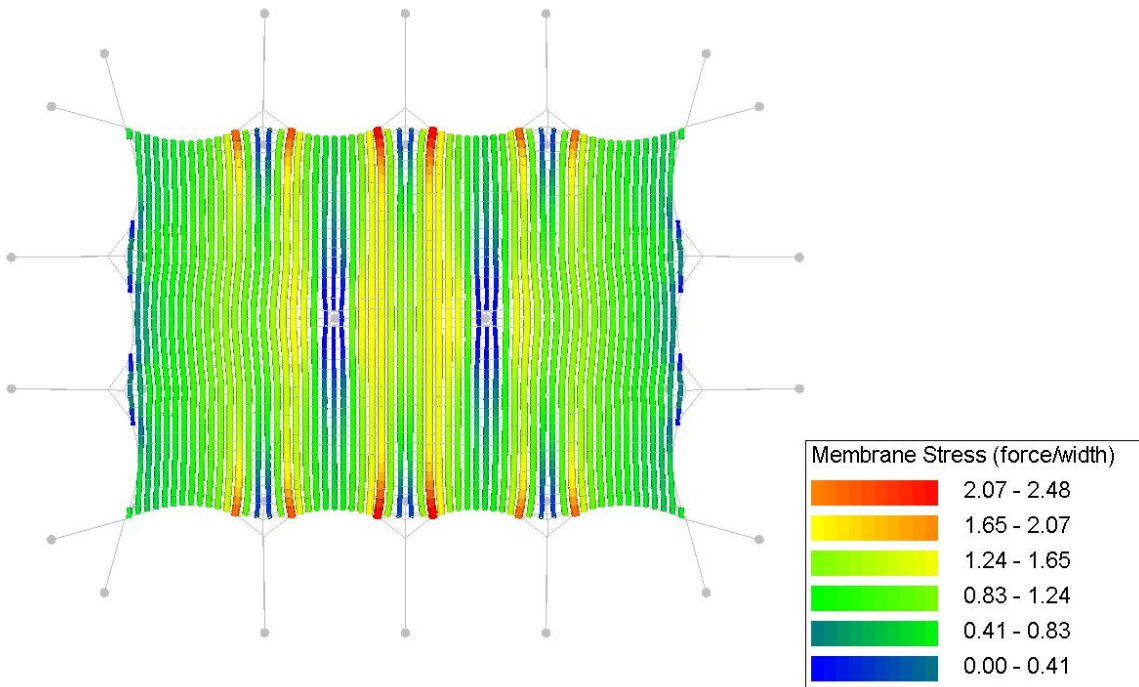


## Annex E.5. CO5. Own weight + pretension + wind suction reduced [NL]

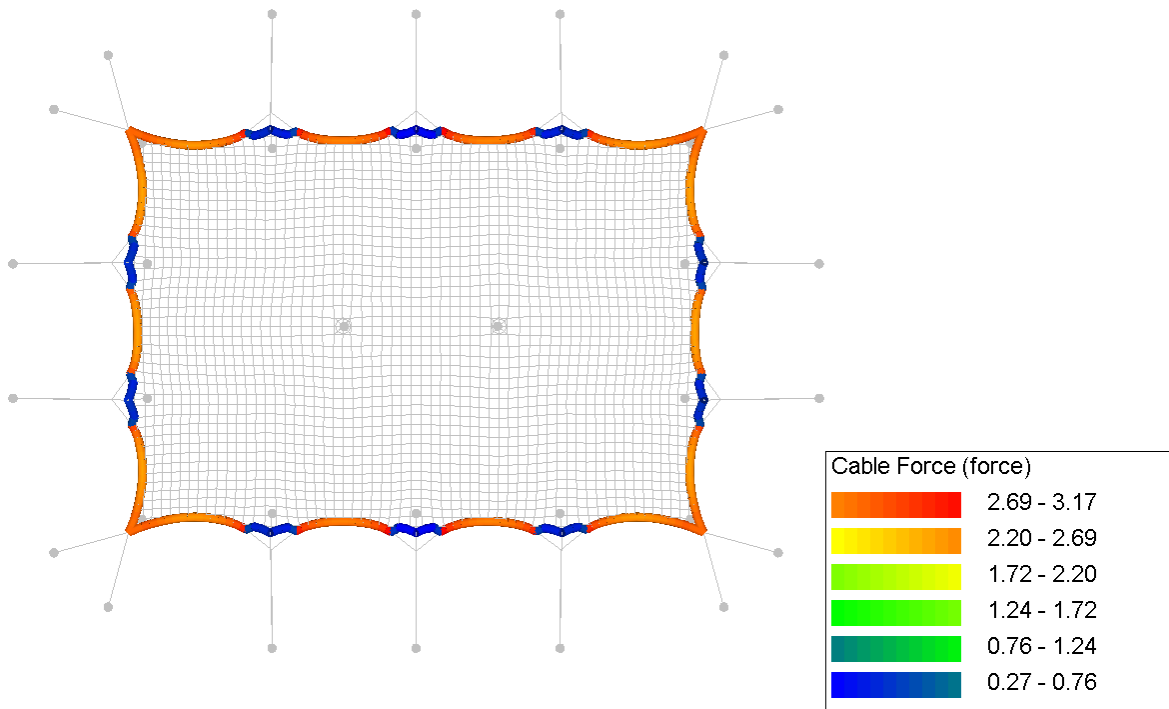
### Annex E.5.1. Membrane stress (warp)



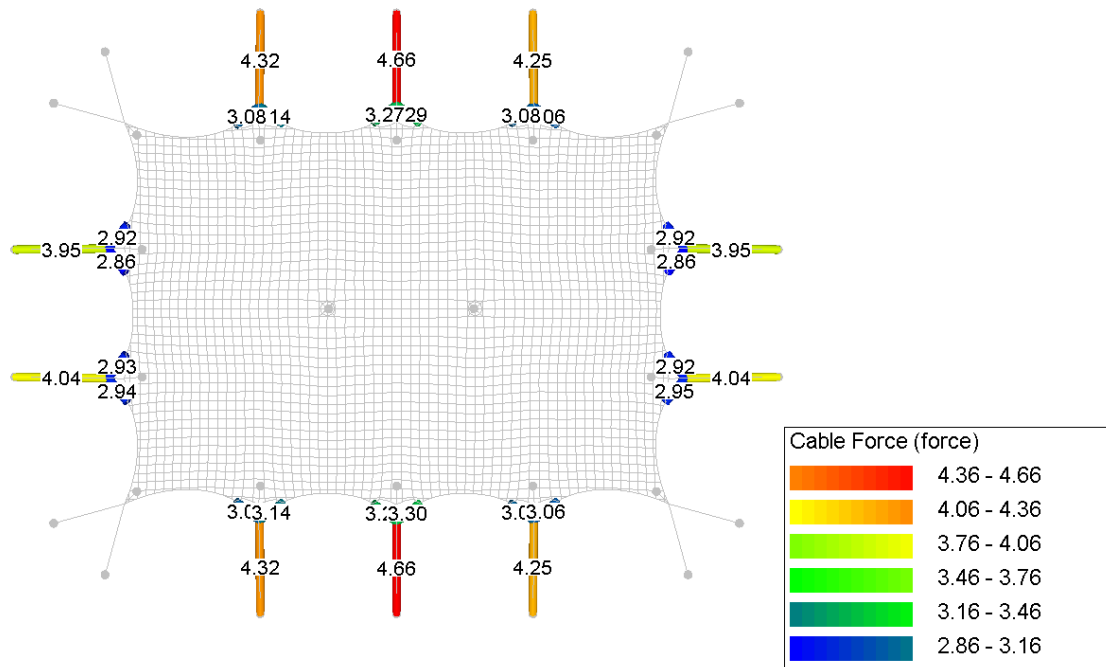
### Annex E.5.2. Membrane stress (weft)



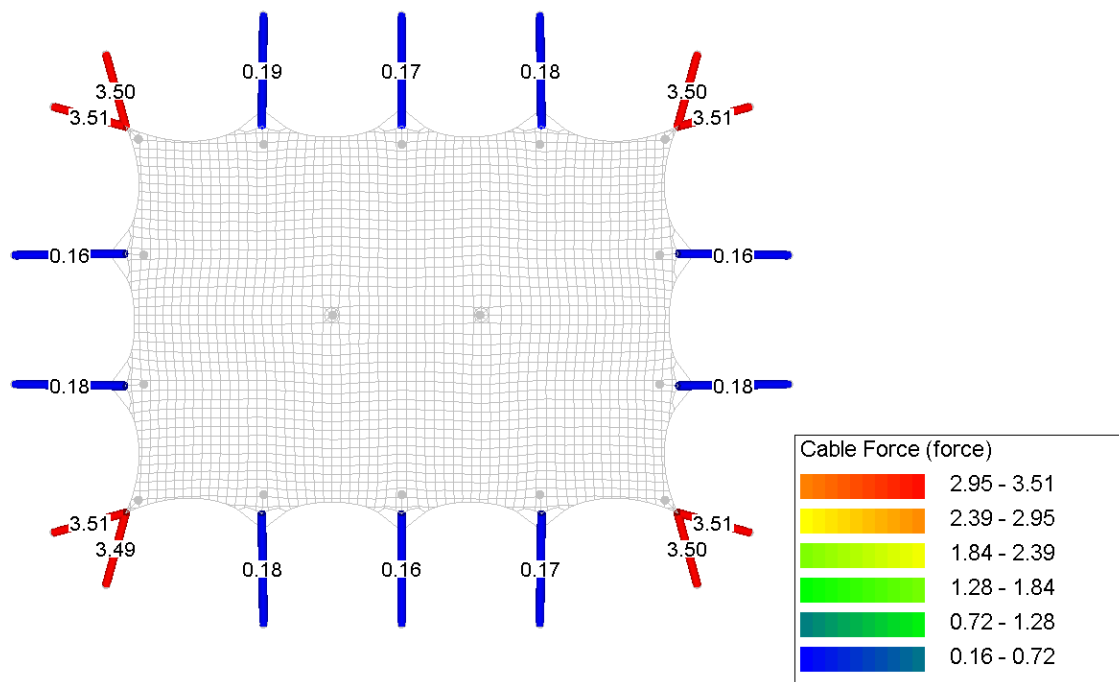
### Annex E.5.3. Circumference rope forces



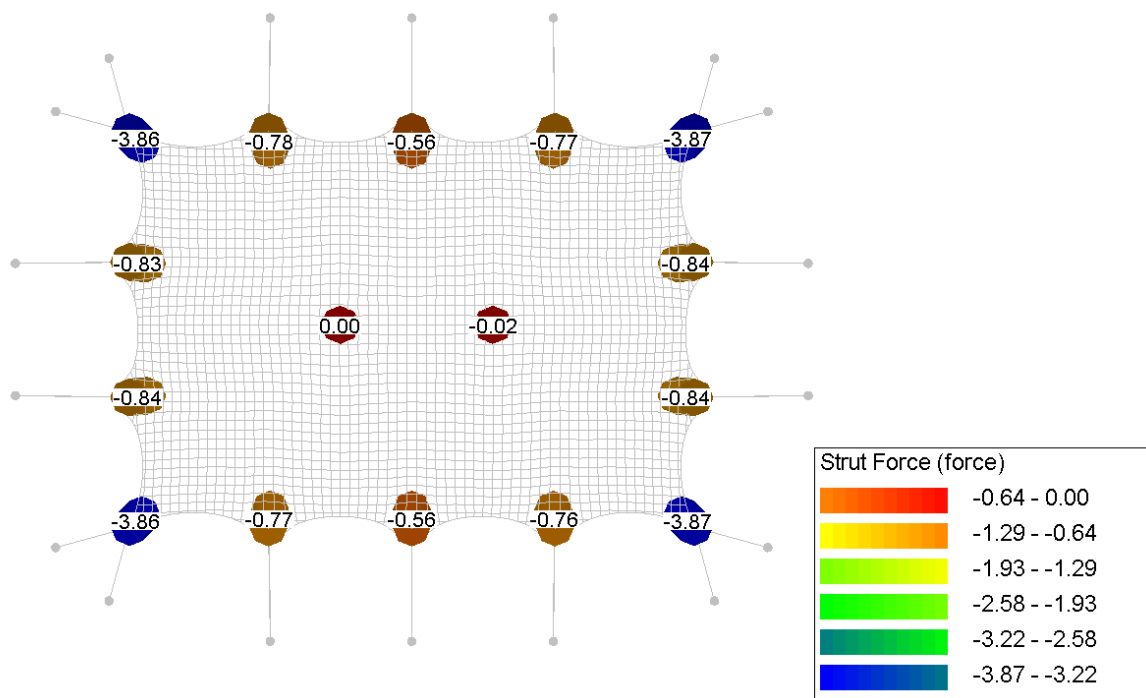
### Annex E.5.4. Rope forces



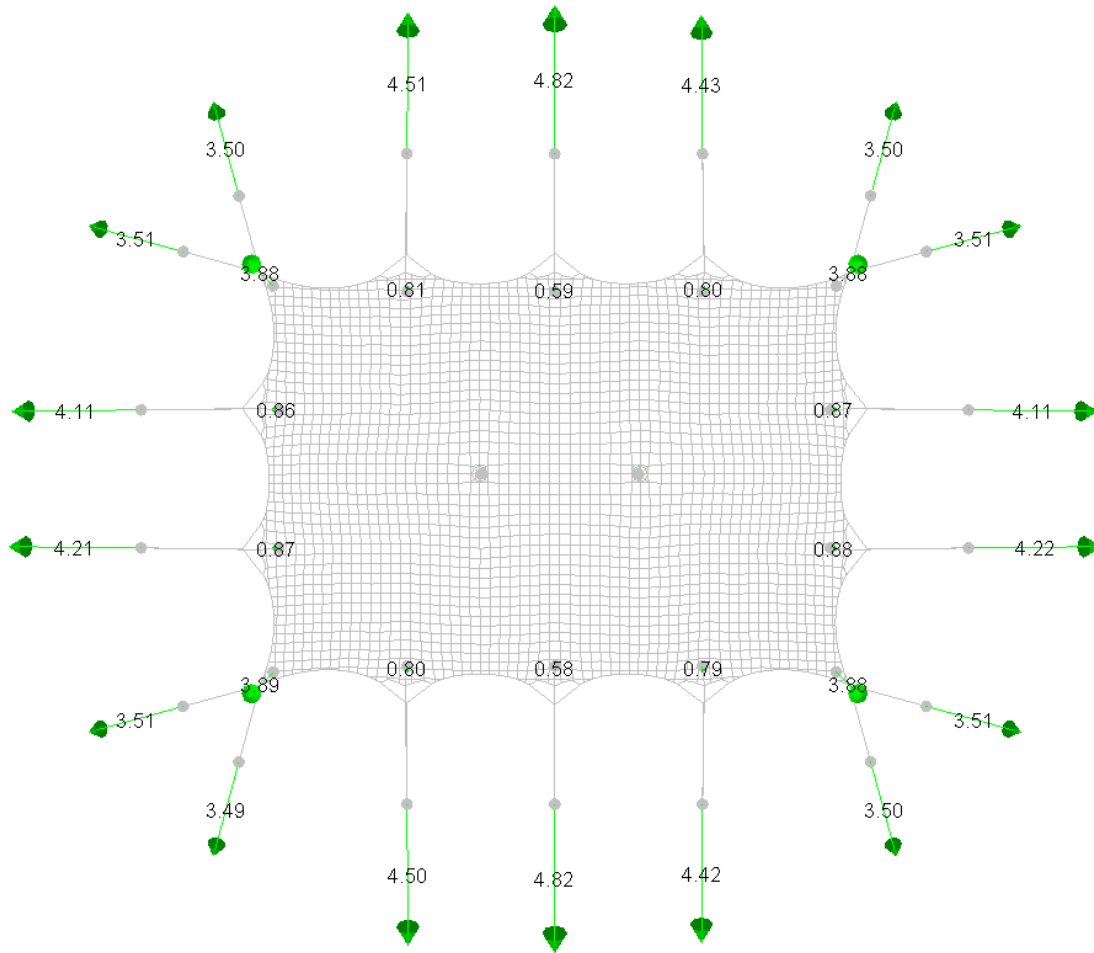
### Annex E.5.5. Belt forces



### Annex E.5.6. Poles forces

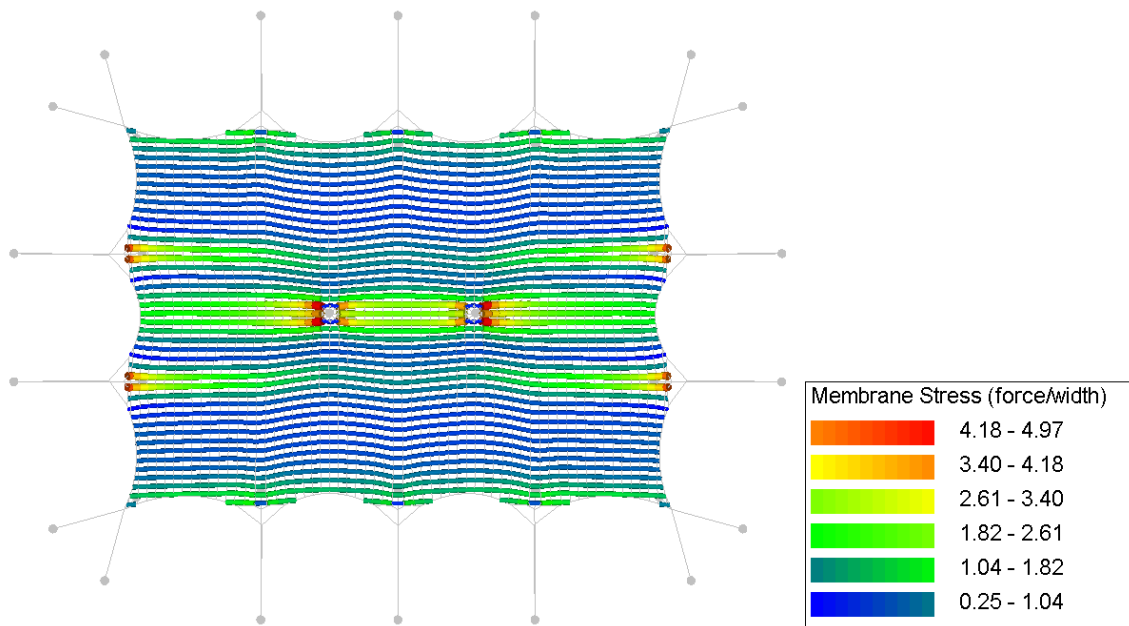


## Annex E.5.7. Reaction forces

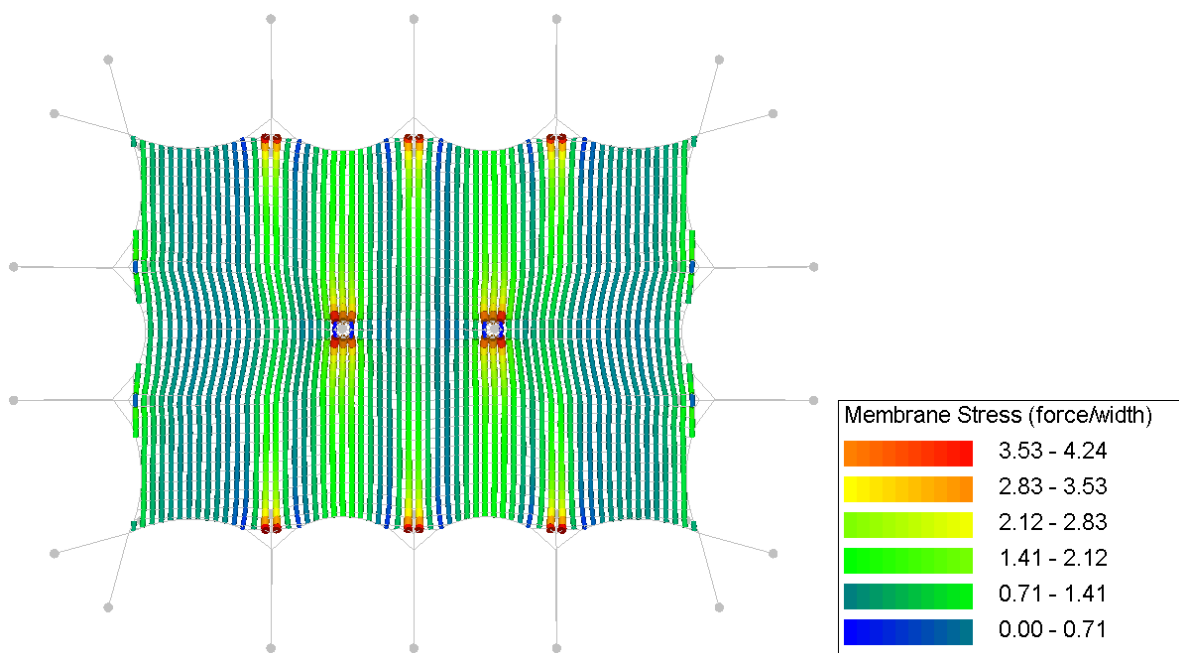


## Annex E.6. CO6. Own weight + pretension + wind pressure + snow [FR]

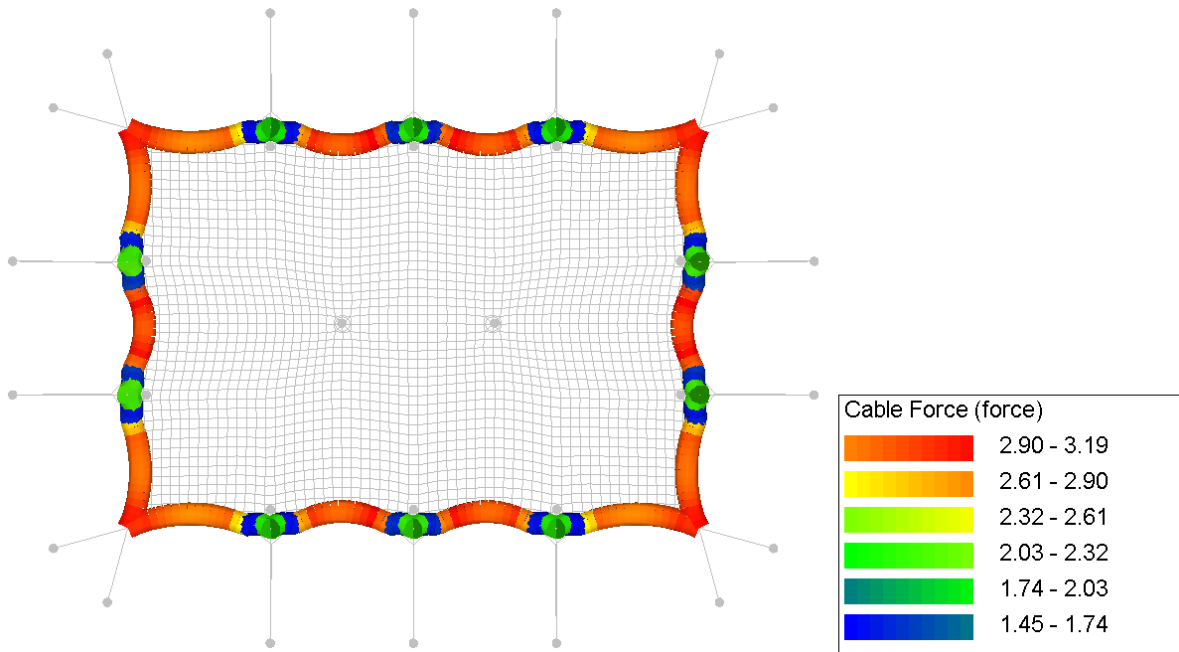
### Annex E.6.1. Membrane stress (warp)



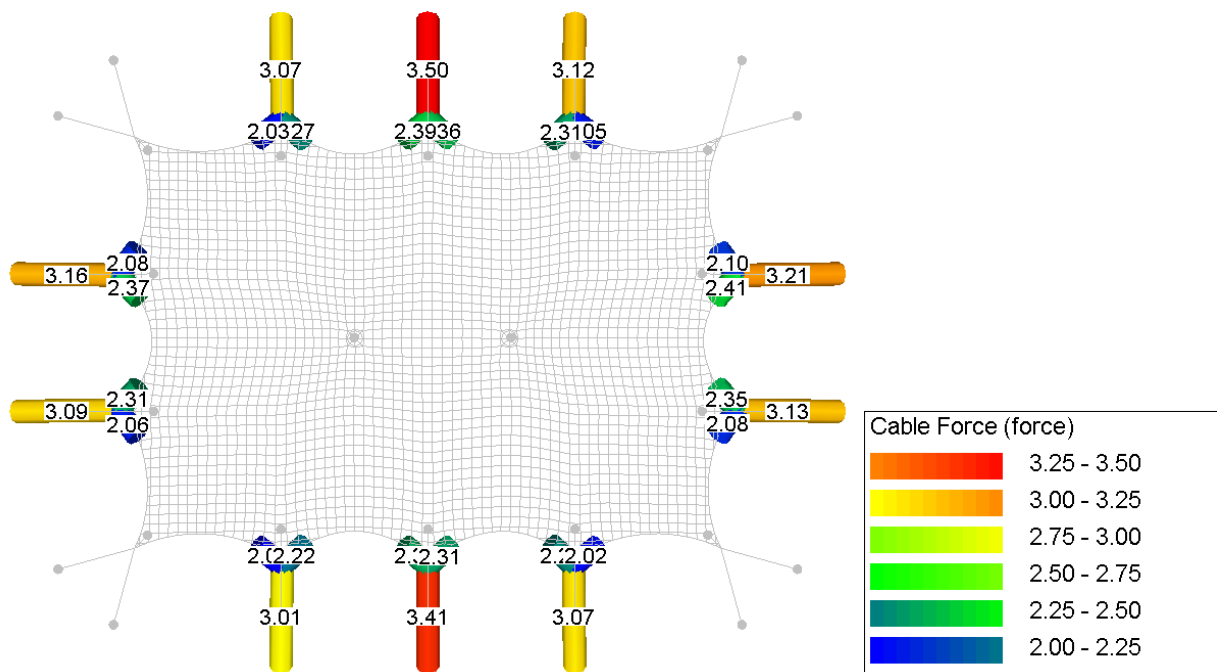
### Annex E.6.2. Membrane stress (weft)



### Annex E.6.3. Circumference rope forces

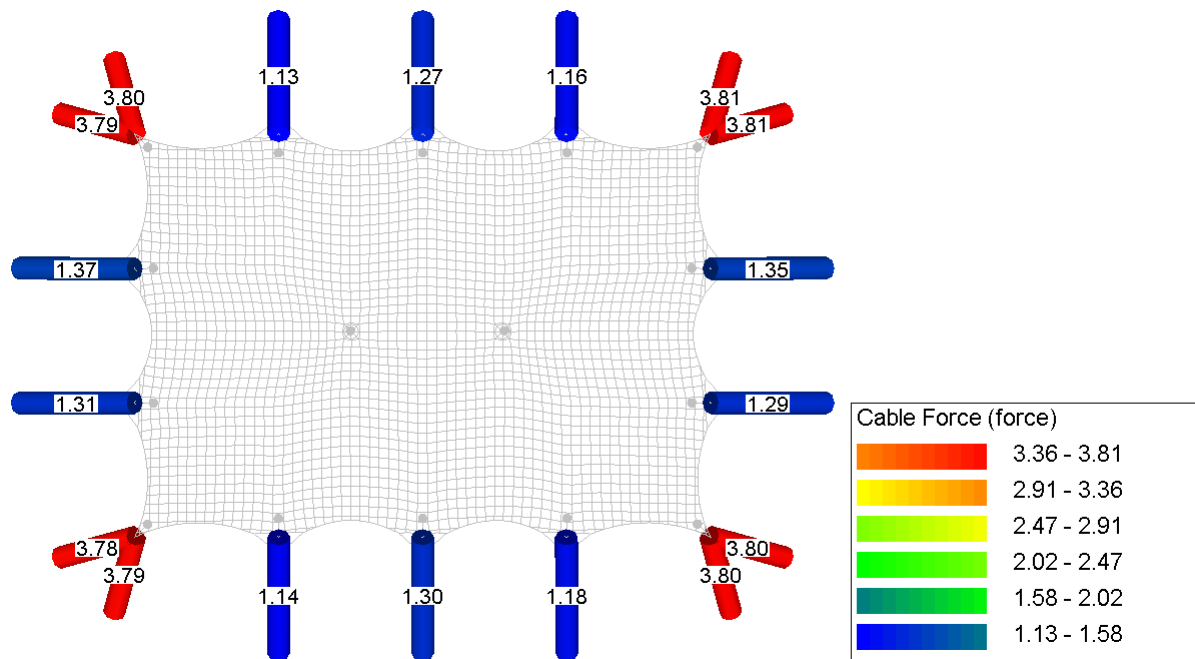


### Annex E.6.4. Guy rope forces

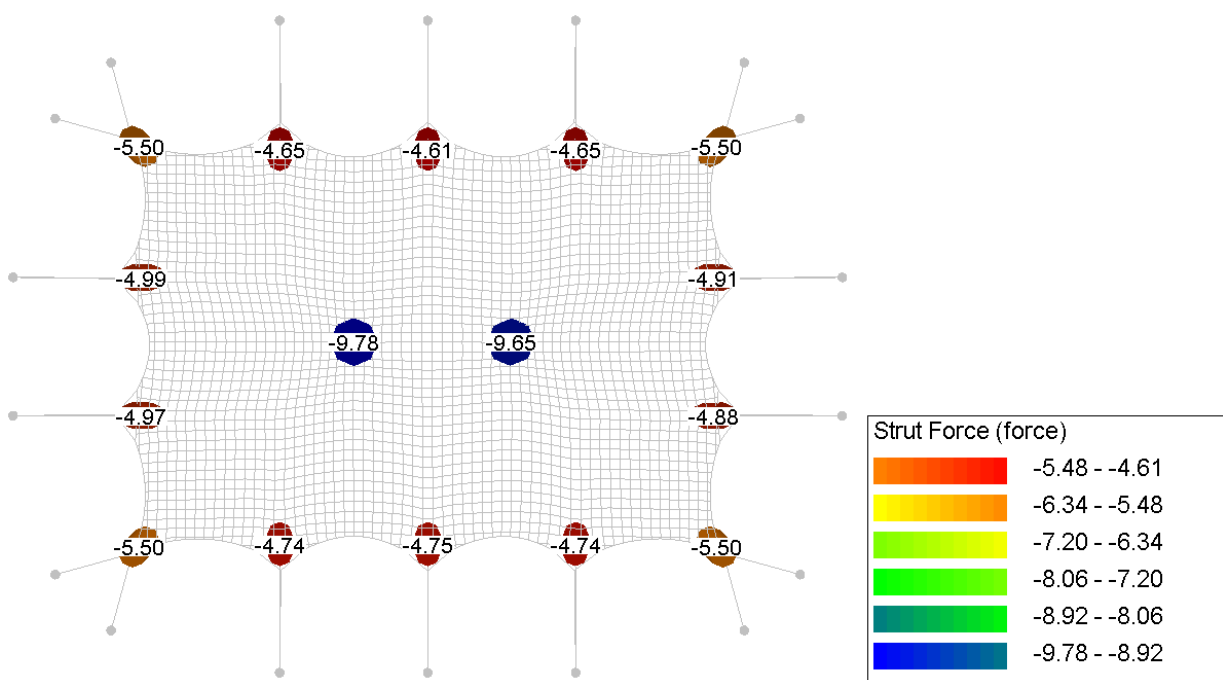




### Annex E.6.5. Belt forces

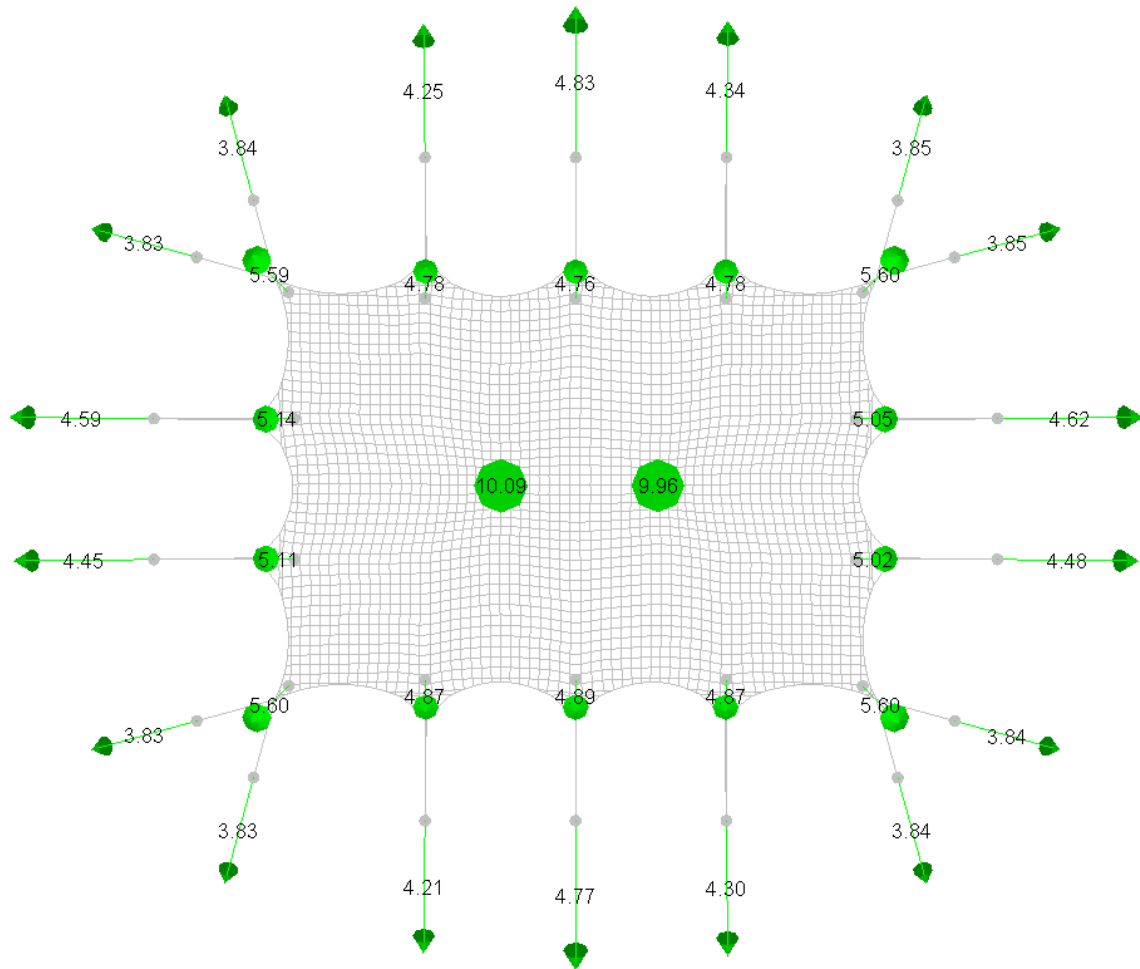


### Annex E.6.6. Poles forces



### Annex E.6.7. Reaction forces

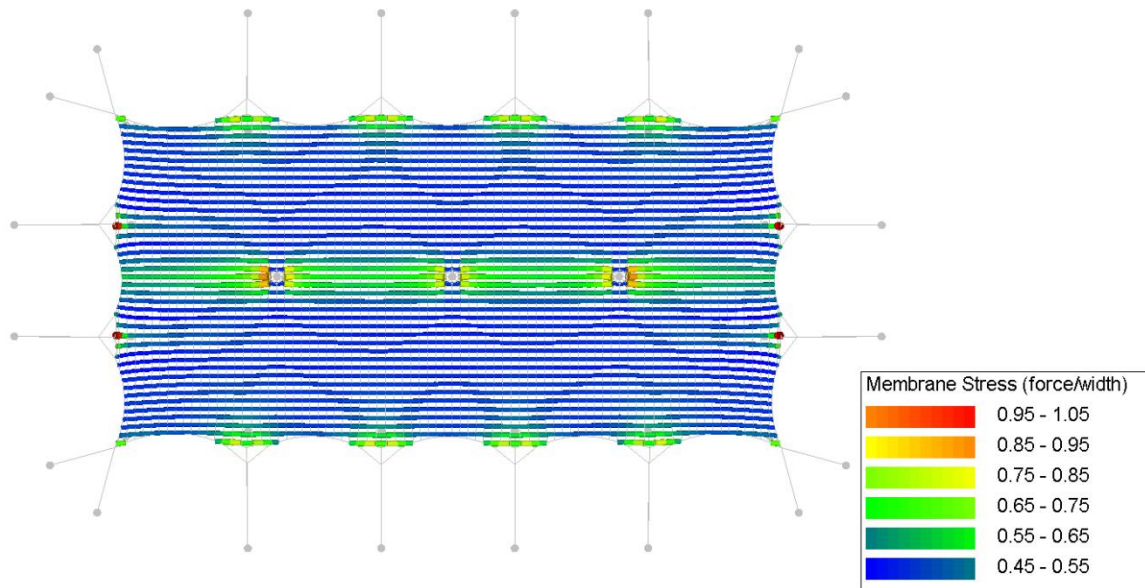
Increased snow load of  $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$  is applied for ballast and anchor calculation



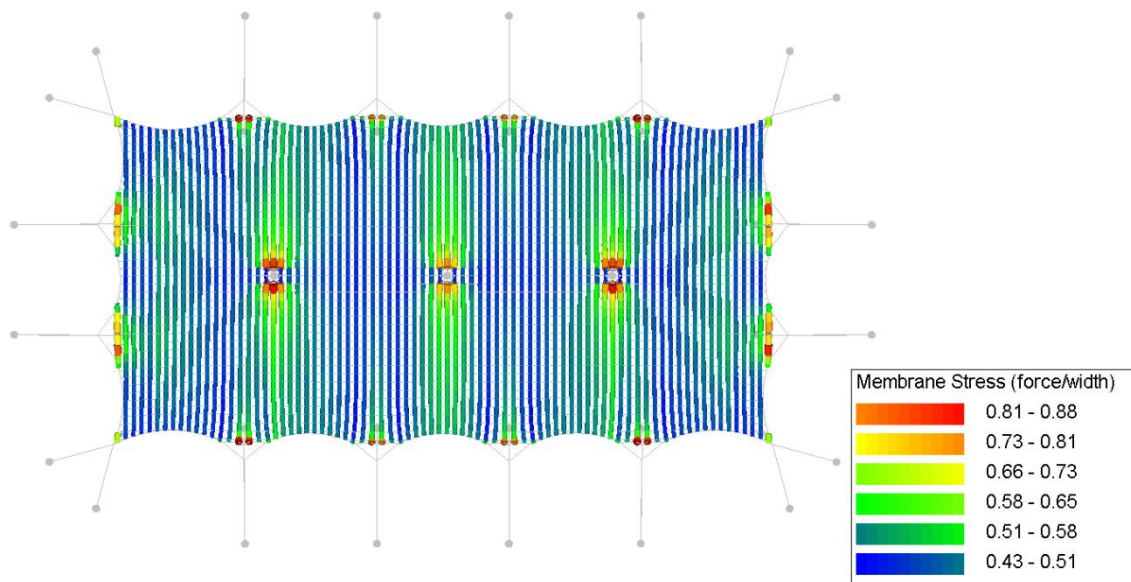
## Annex F. Results per load combination – 10x20m model

### Annex F.1. CO1. Own weight + pretension

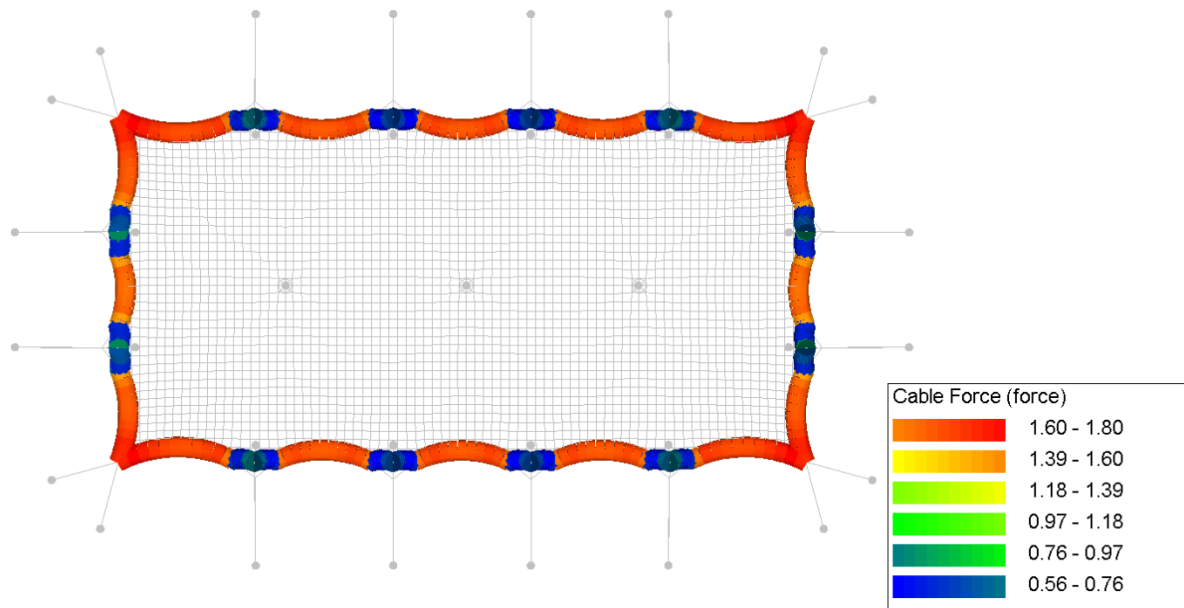
#### Annex F.1.1. Membrane stress (warp)



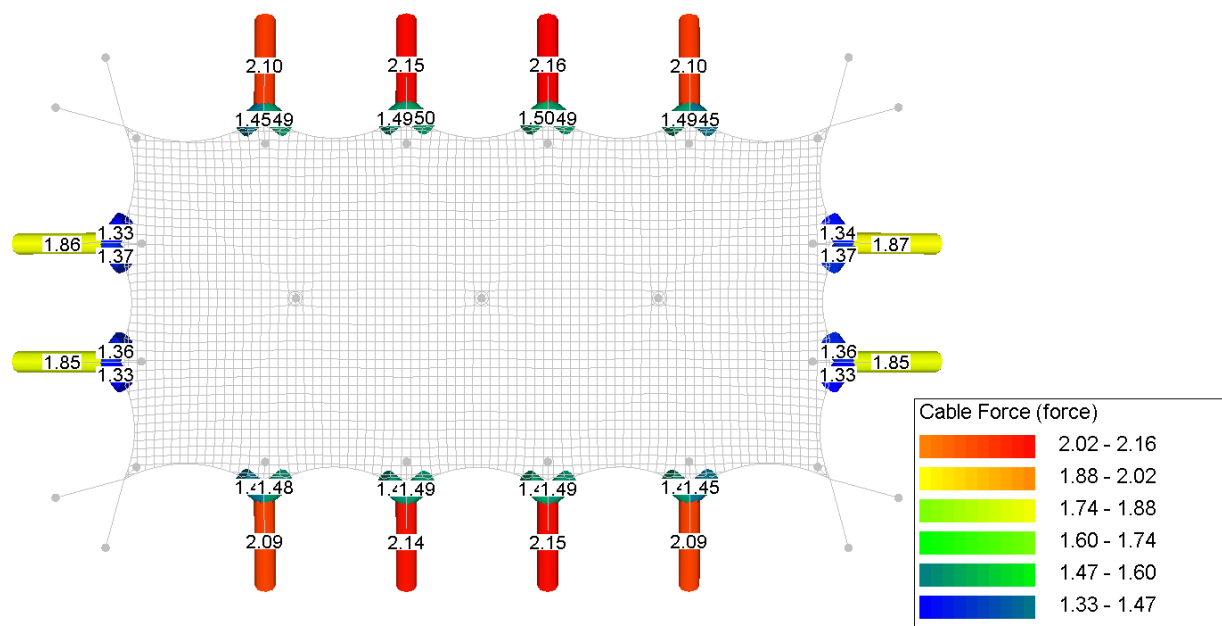
#### Annex F.1.2. Membrane stress (weft)



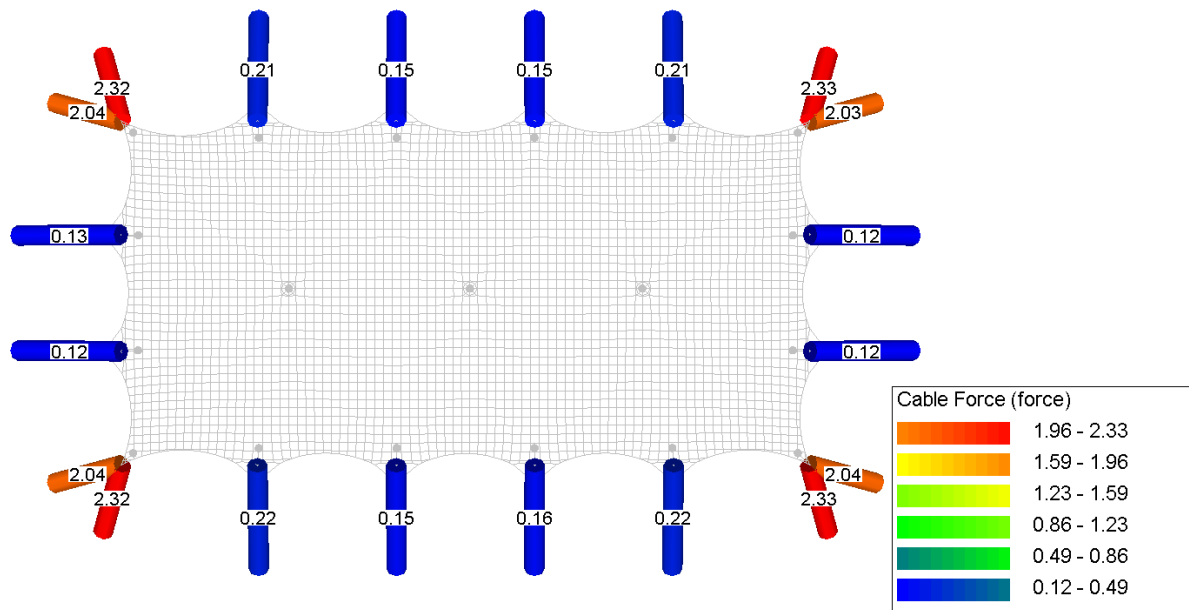
### Annex F.1.3. Circumference rope forces



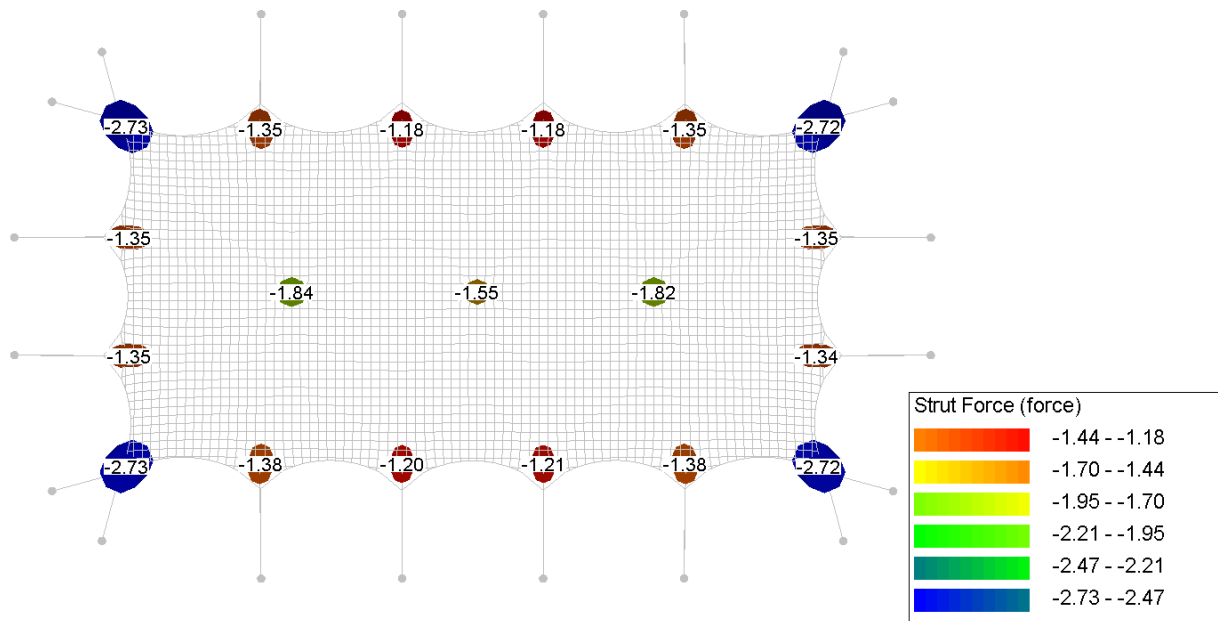
### Annex F.1.4. Rope forces



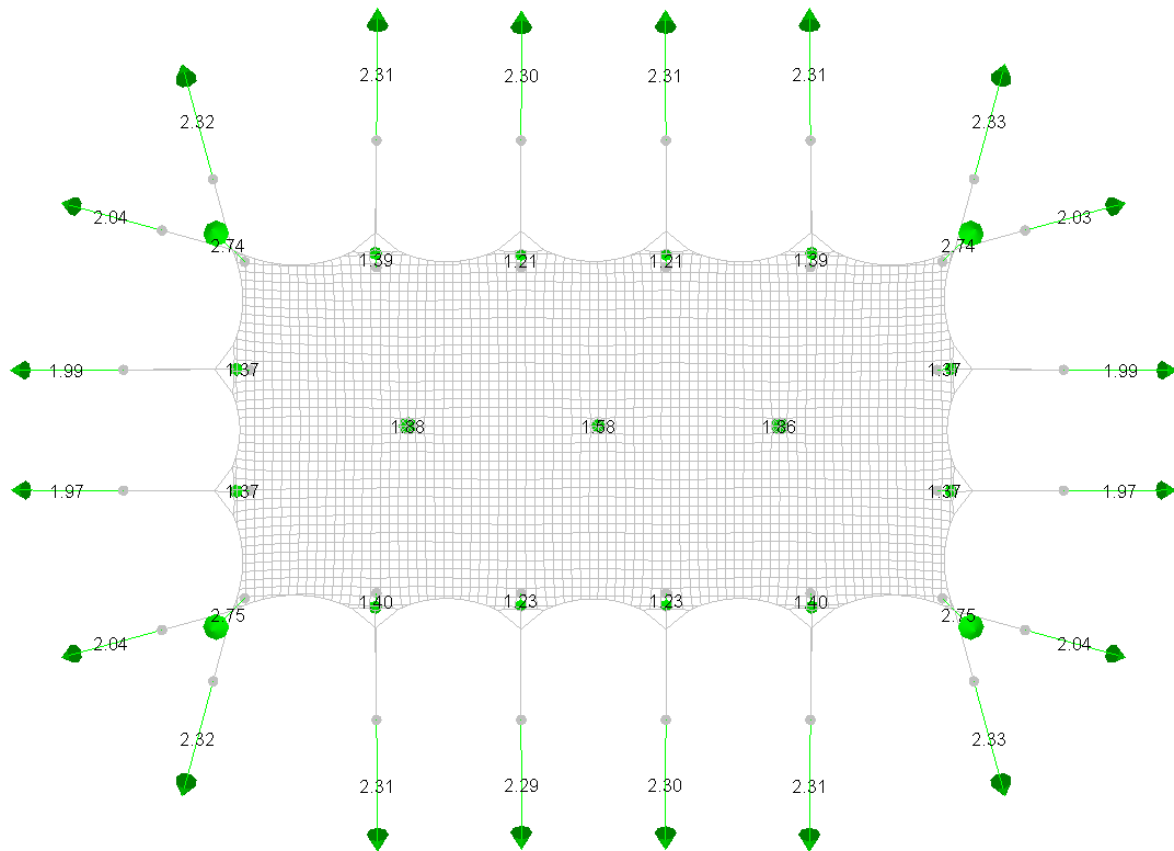
### Annex F.1.5. Belt forces



### Annex F.1.6. Pole forces

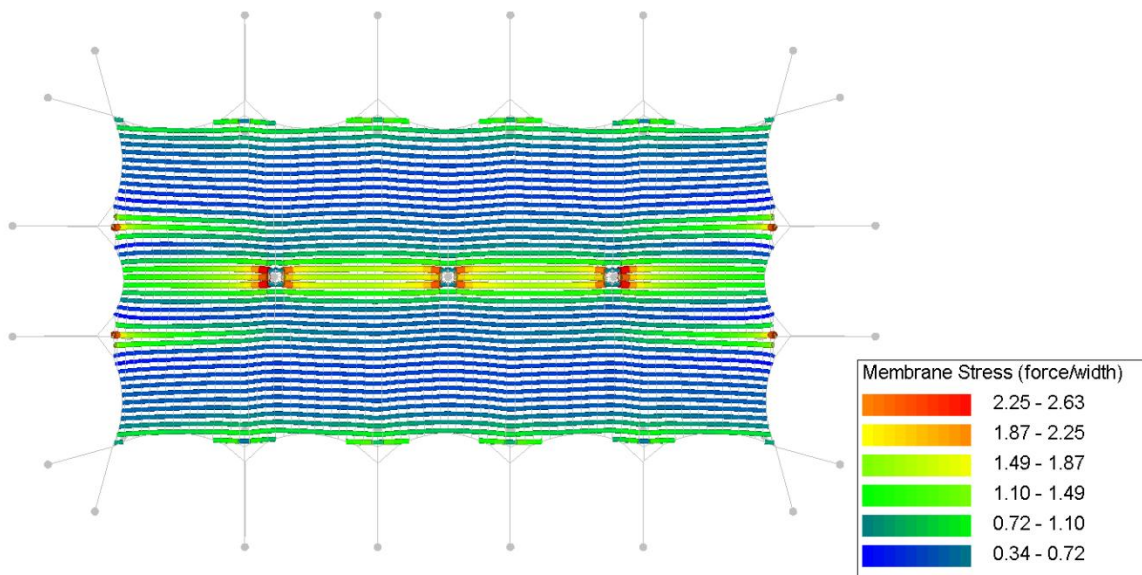


## Annex F.1.7. Reaction forces

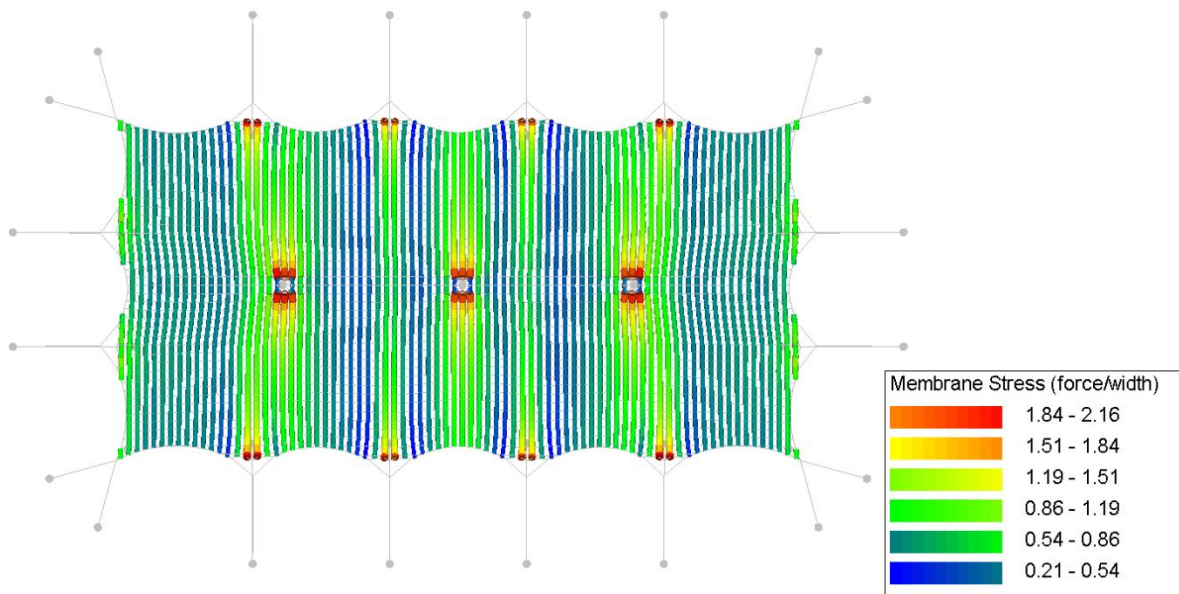


## Annex F.2. CO2. Own weight + pretension + snow

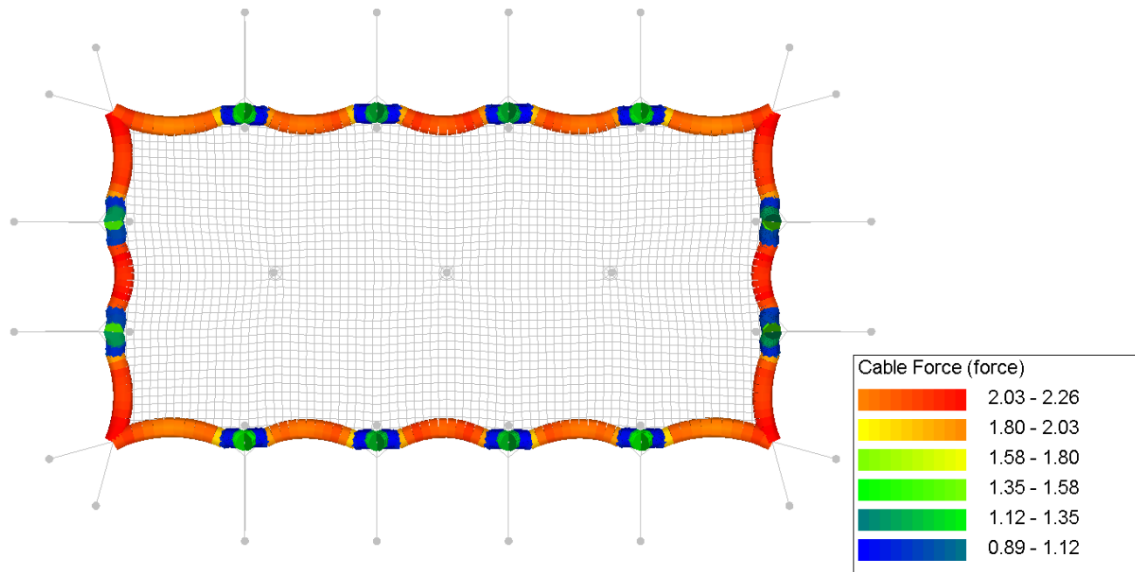
### Annex F.2.1. Membrane stress (warp)



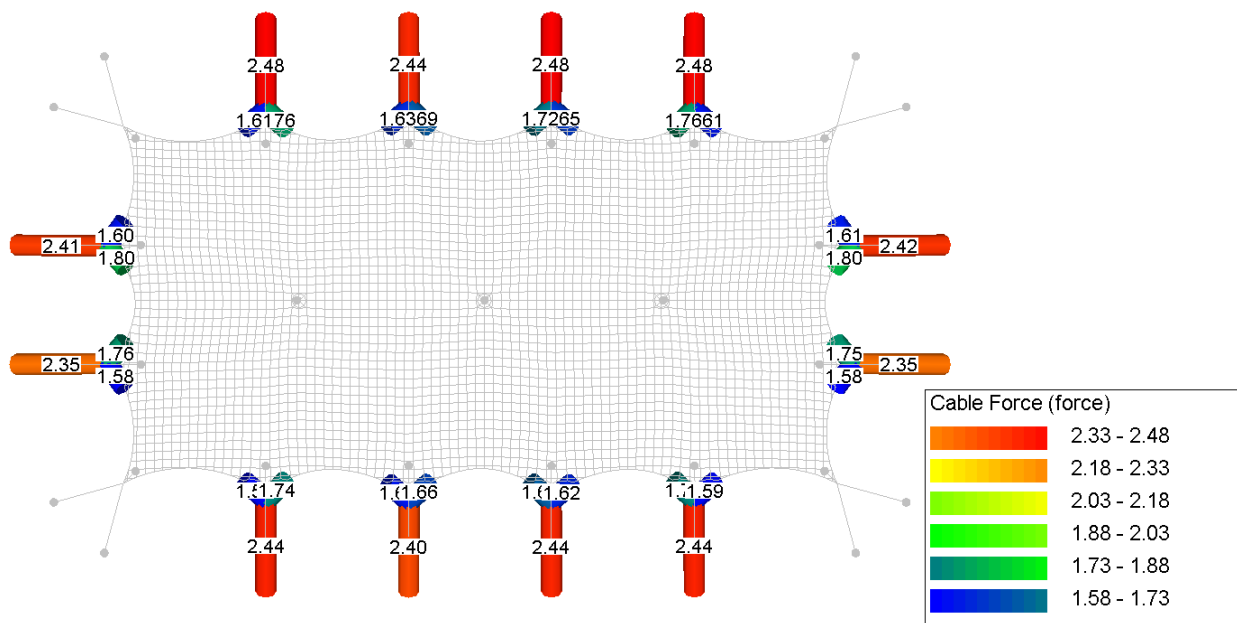
### Annex F.2.2. Membrane stress (weft)



### Annex F.2.3. Circumference rope forces

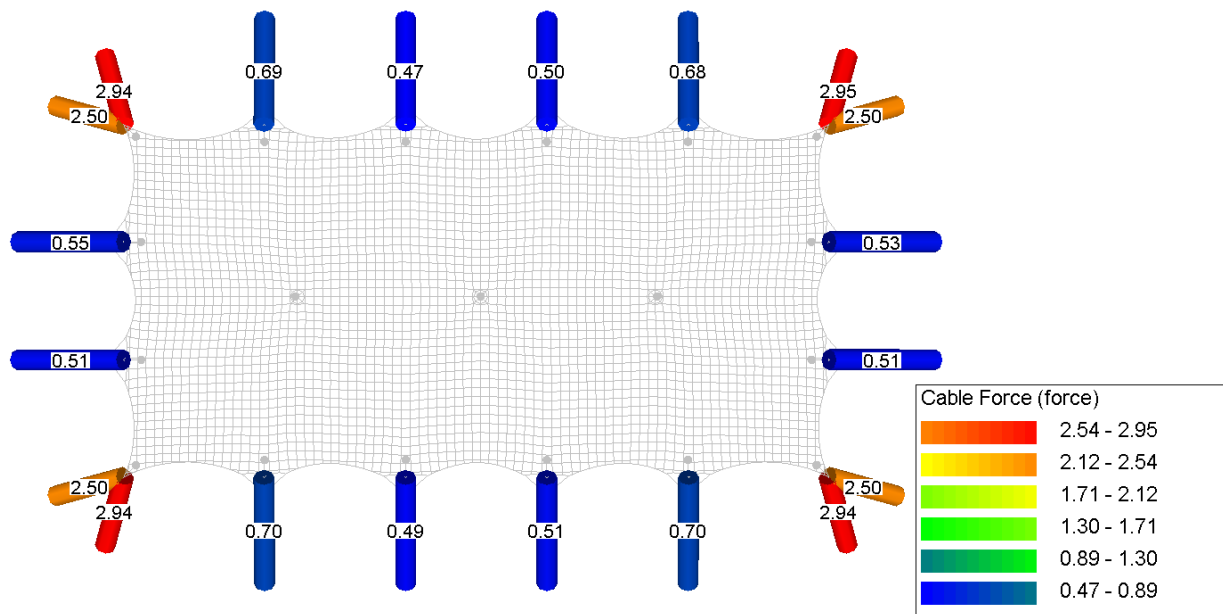


### Annex F.2.4. Rope forces

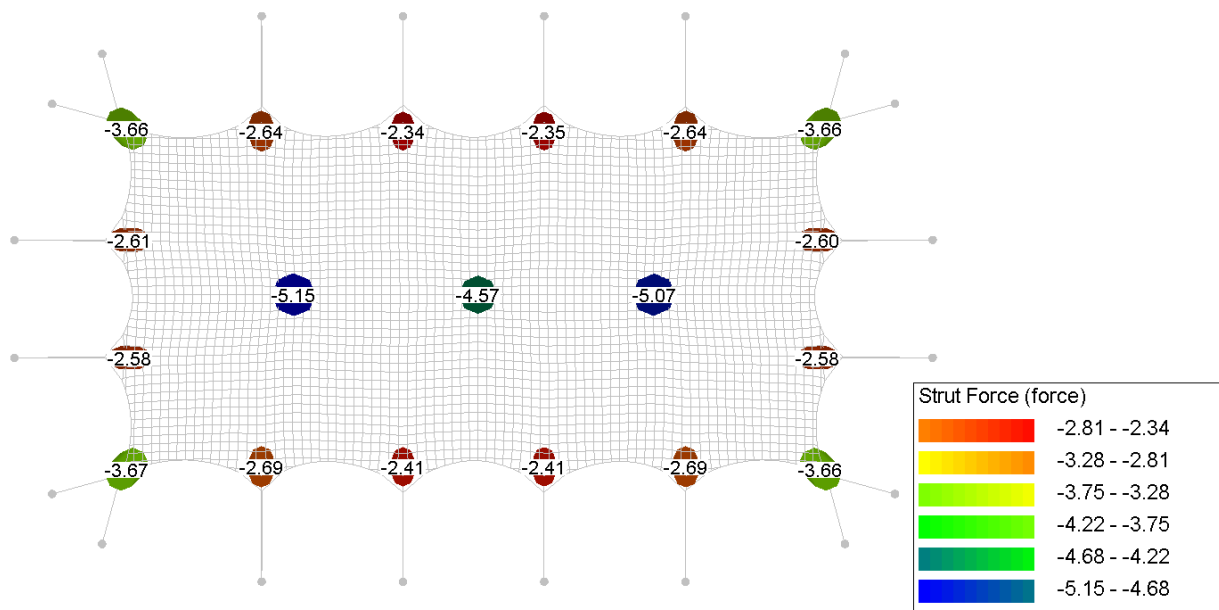




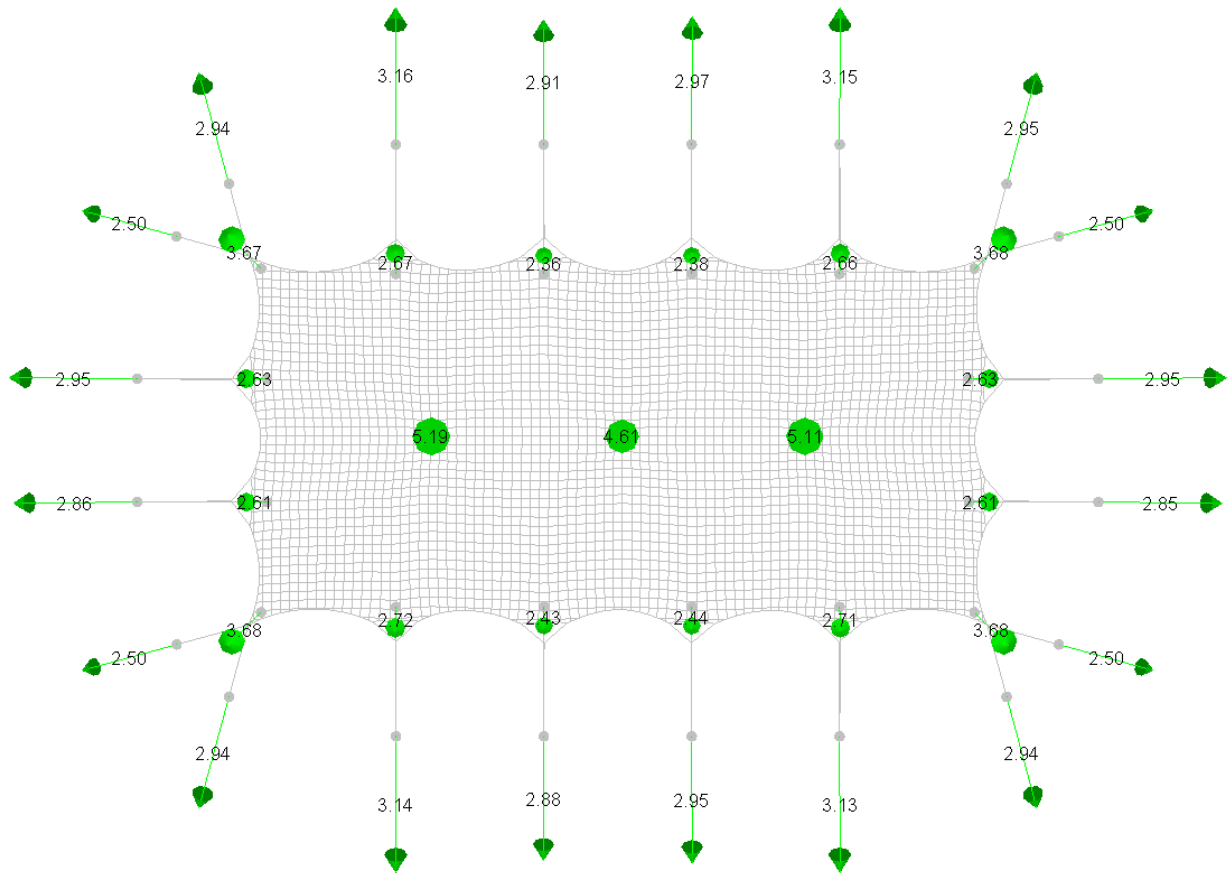
### Annex F.2.5. Belt forces



### Annex F.2.6. Pole forces

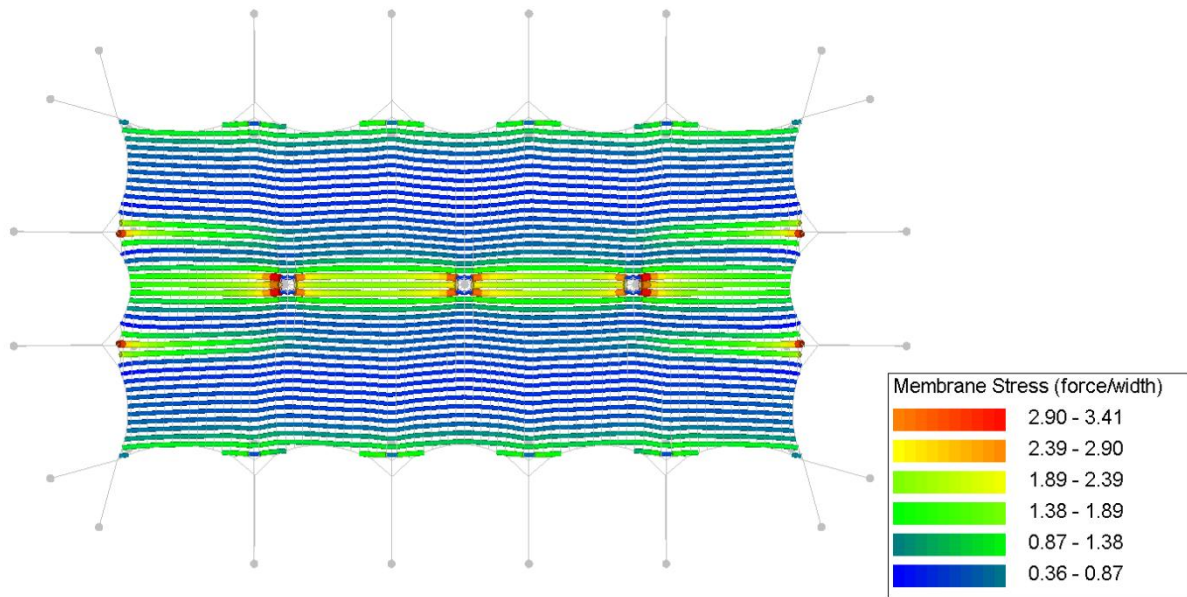


## Annex F.2.7. Anchor forces

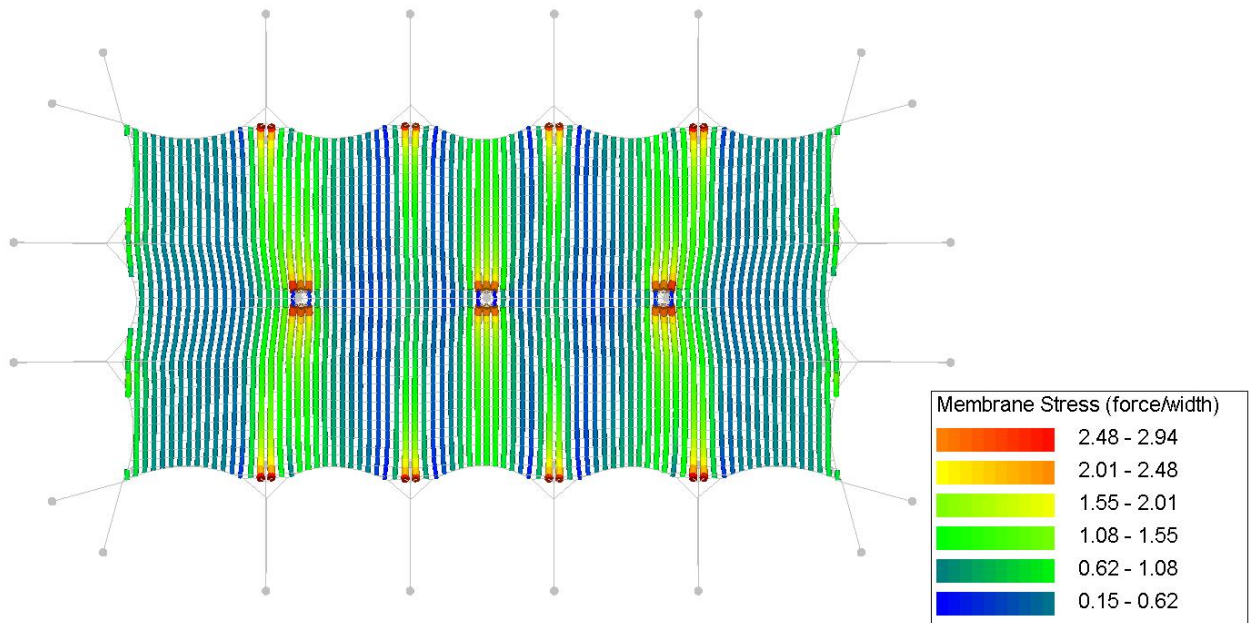


## Annex F.3. CO3. Own weight + pretension + wind pressure

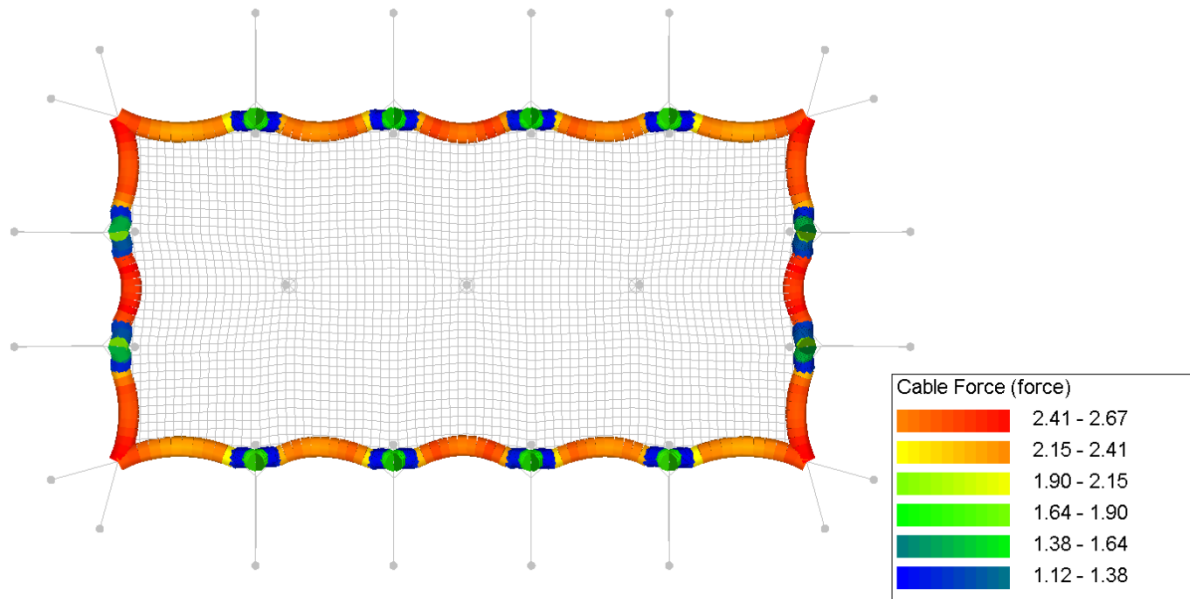
### Annex F.3.1. Membrane stress (warp)



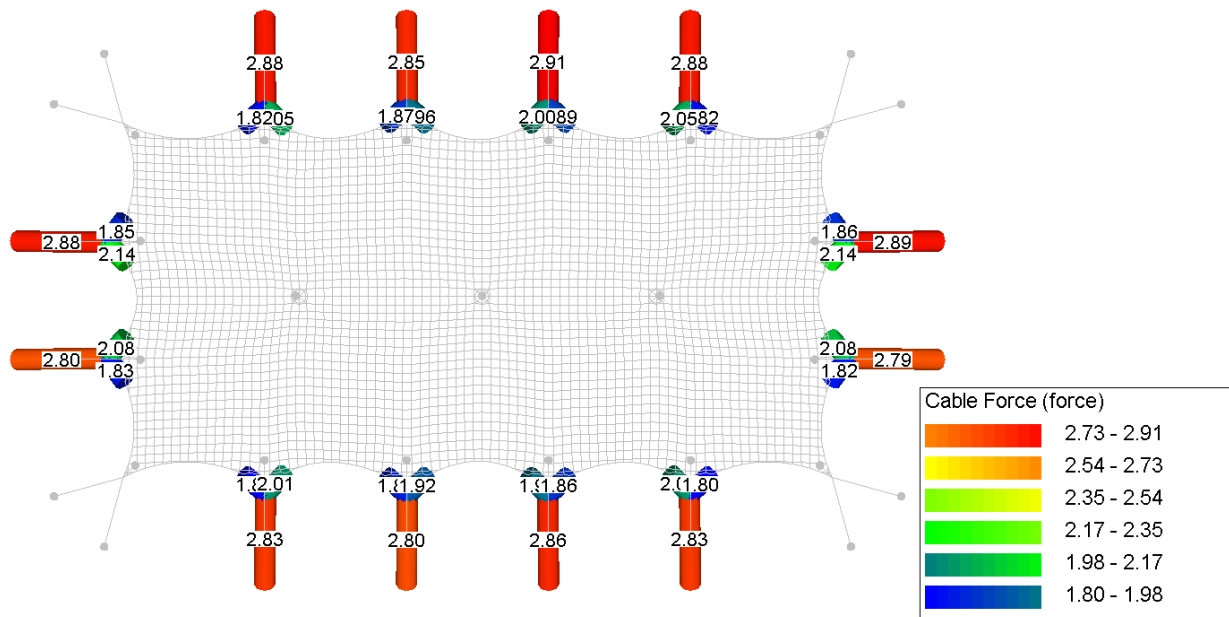
### Annex F.3.2. Membrane stress (weft)



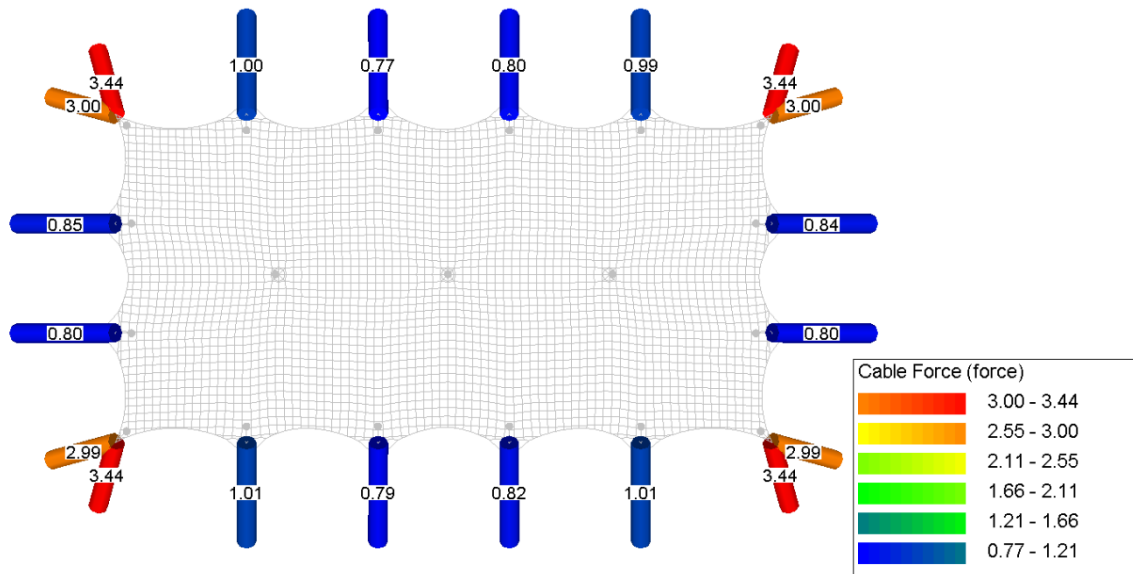
### Annex F.3.3. Circumference rope forces



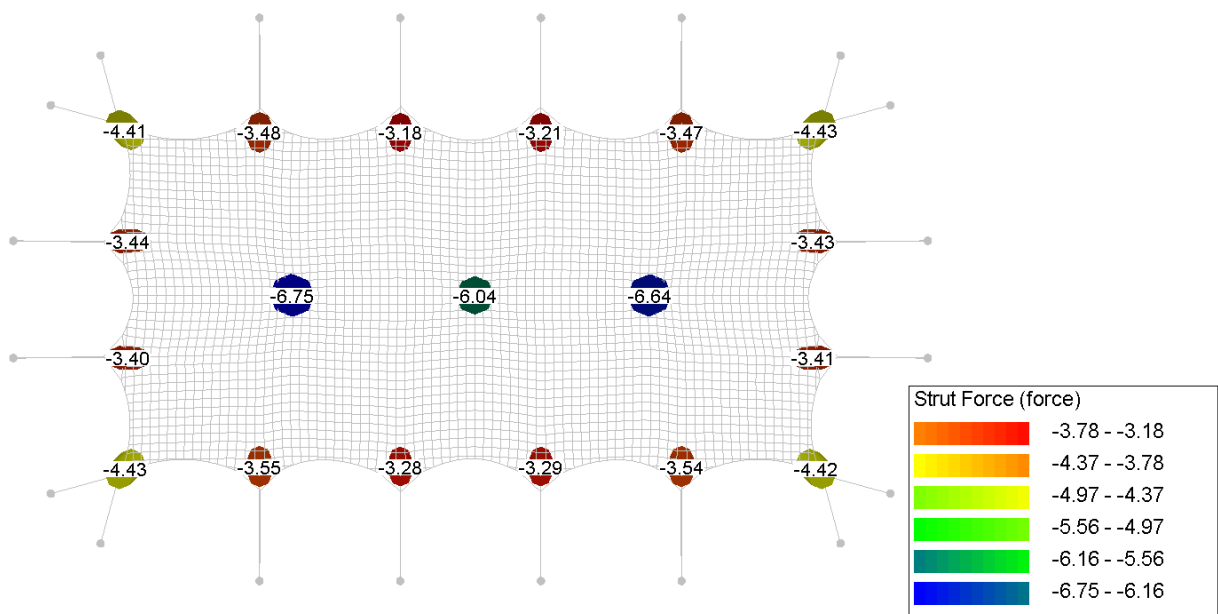
### Annex F.3.4. Rope forces



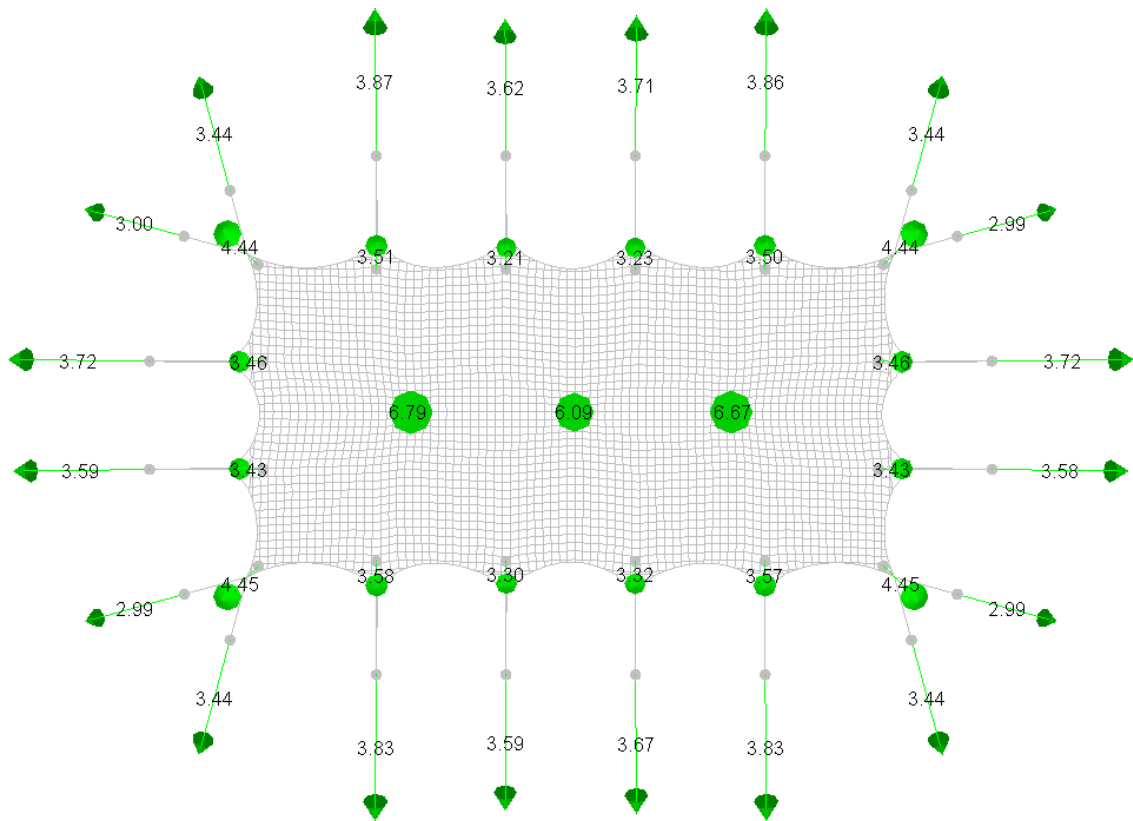
### Annex F.3.5. Belt forces



### Annex F.3.6. Poles forces

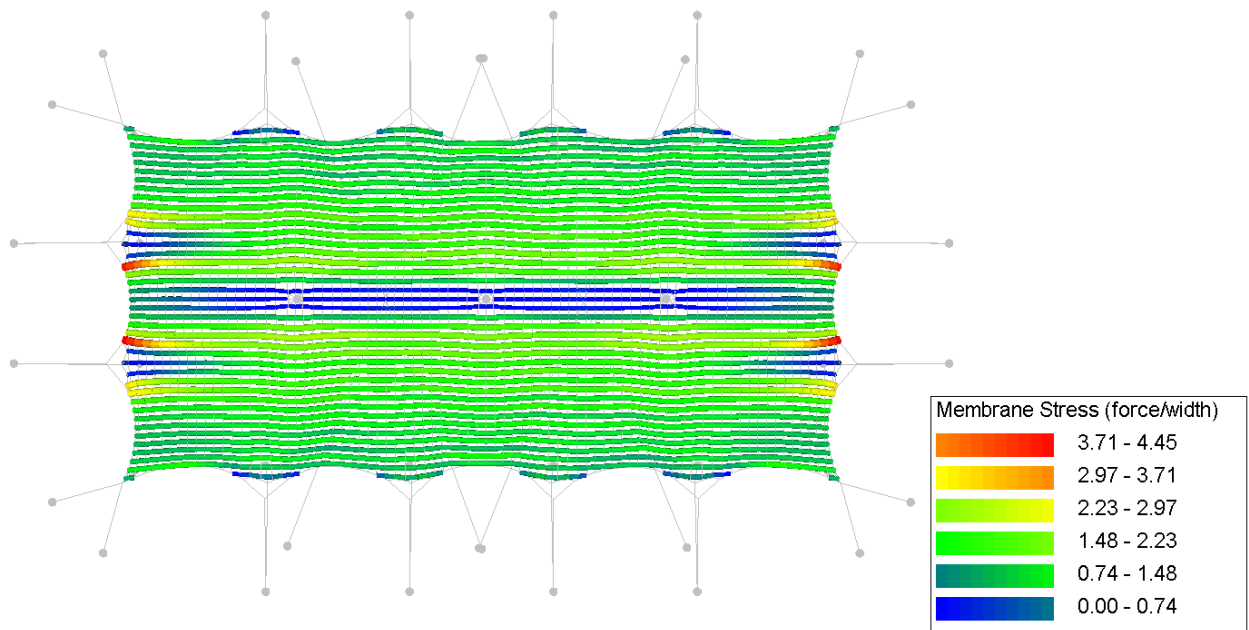


### Annex F.3.7. Reaction forces

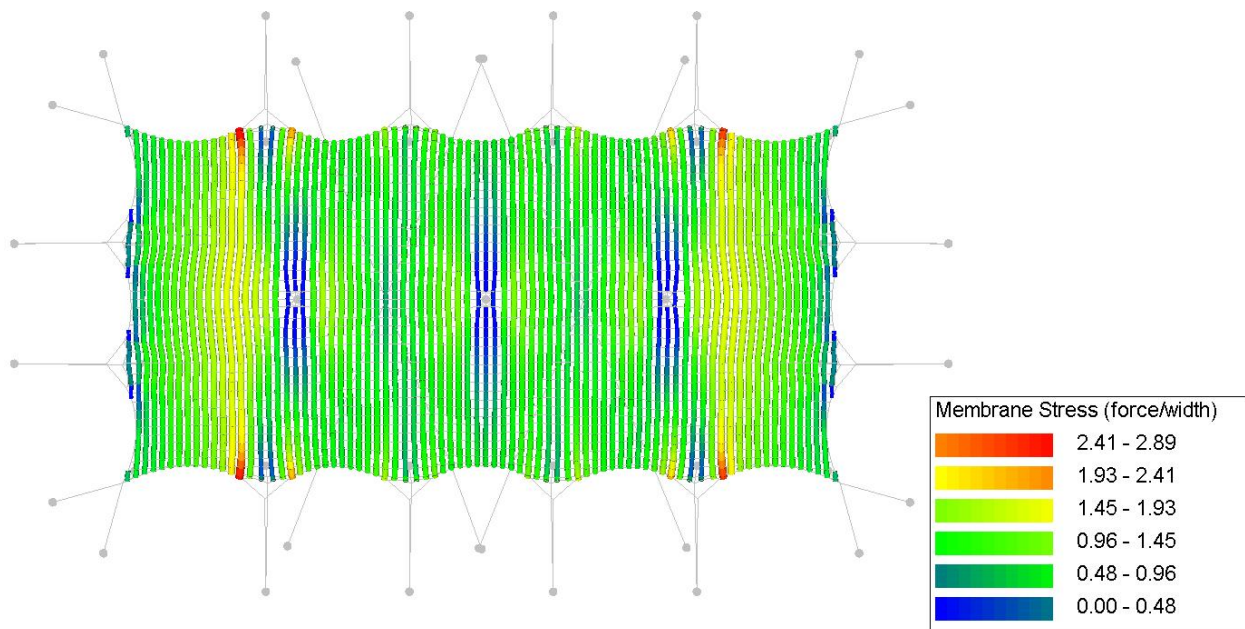


## Annex F.4. CO4. Own weight + pretension + wind suction

### Annex F.4.1. Membrane stress (warp)



### Annex F.4.2. Membrane stress (weft)

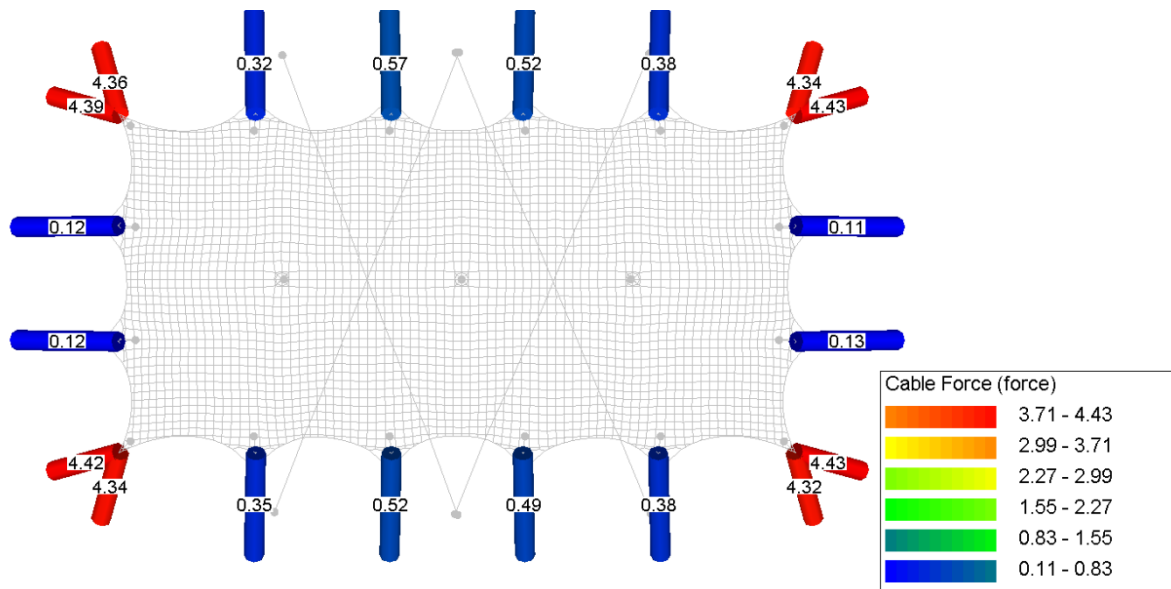




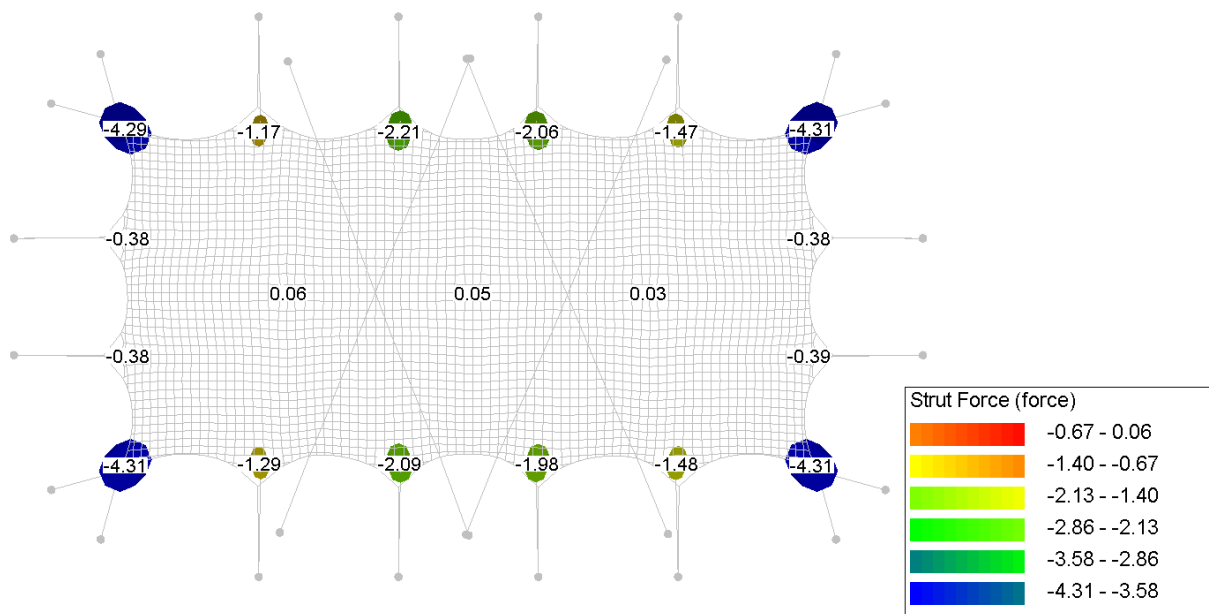




#### Annex F.4.5. Belt forces



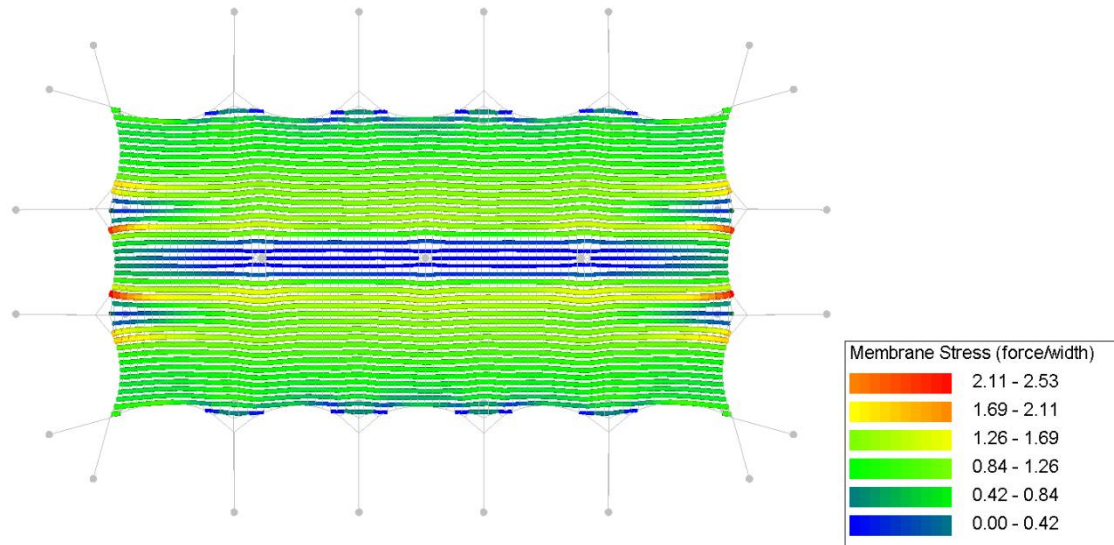
#### Annex F.4.6. Poles forces



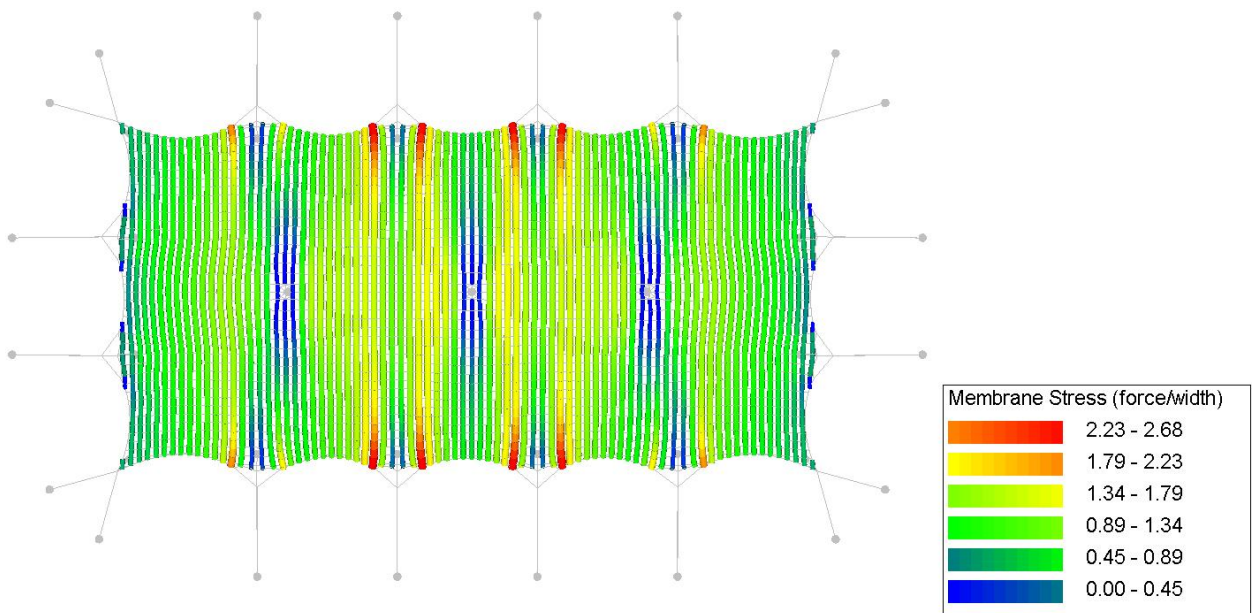


## Annex F.5. CO5. Own weight + pretension + wind suction reduced [NL]

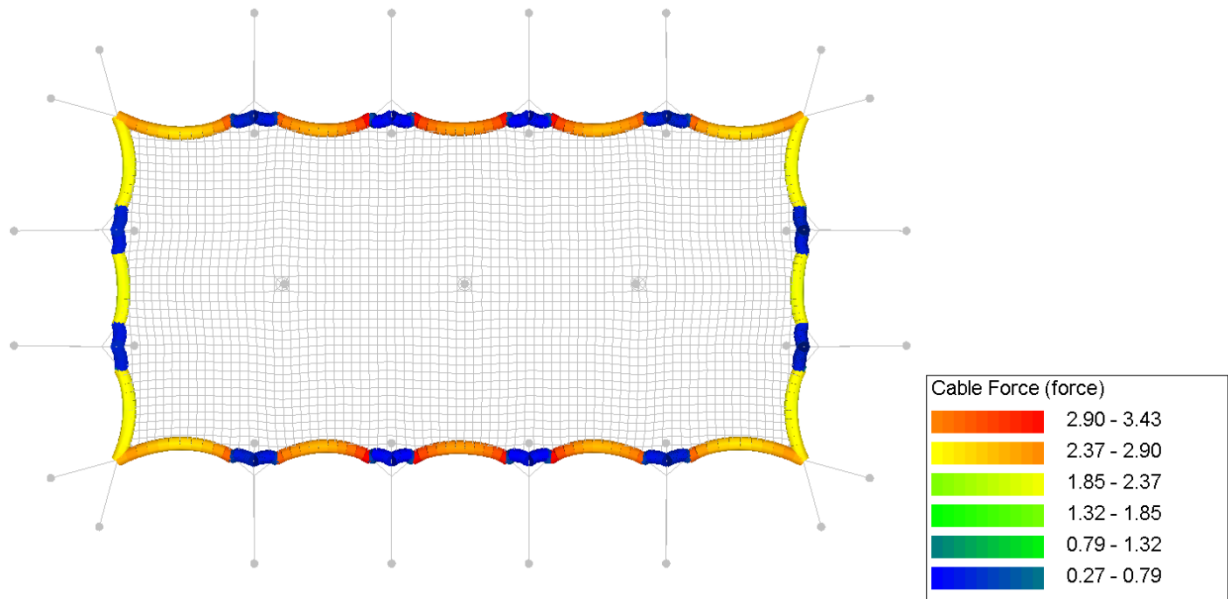
### Annex F.5.1. Membrane stress (warp)



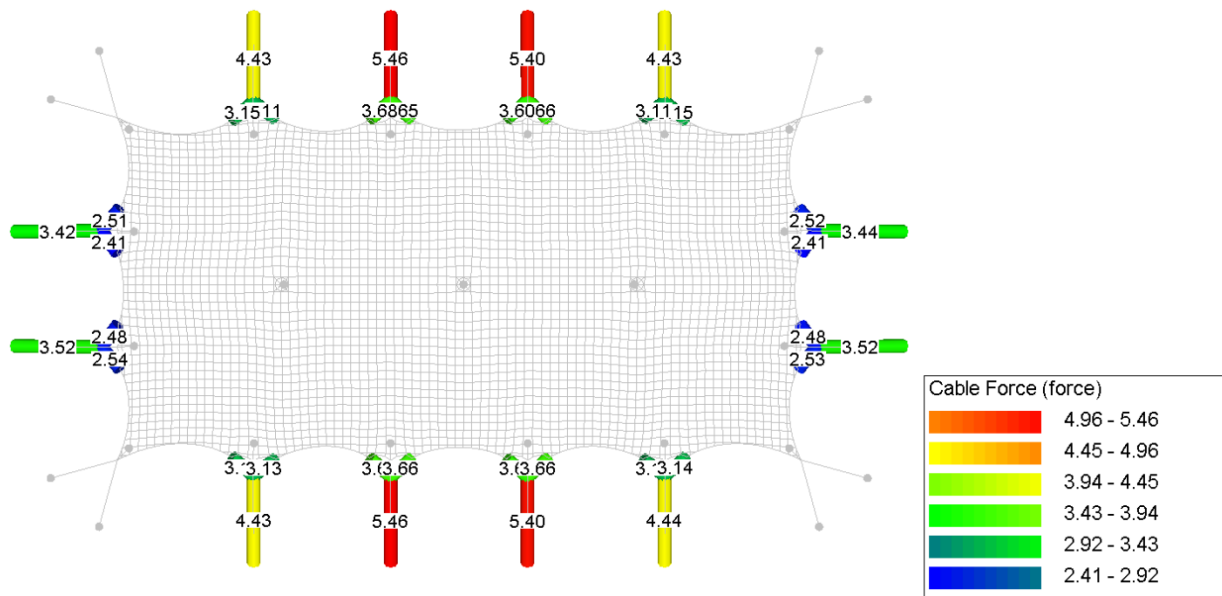
### Annex F.5.2. Membrane stress (weft)



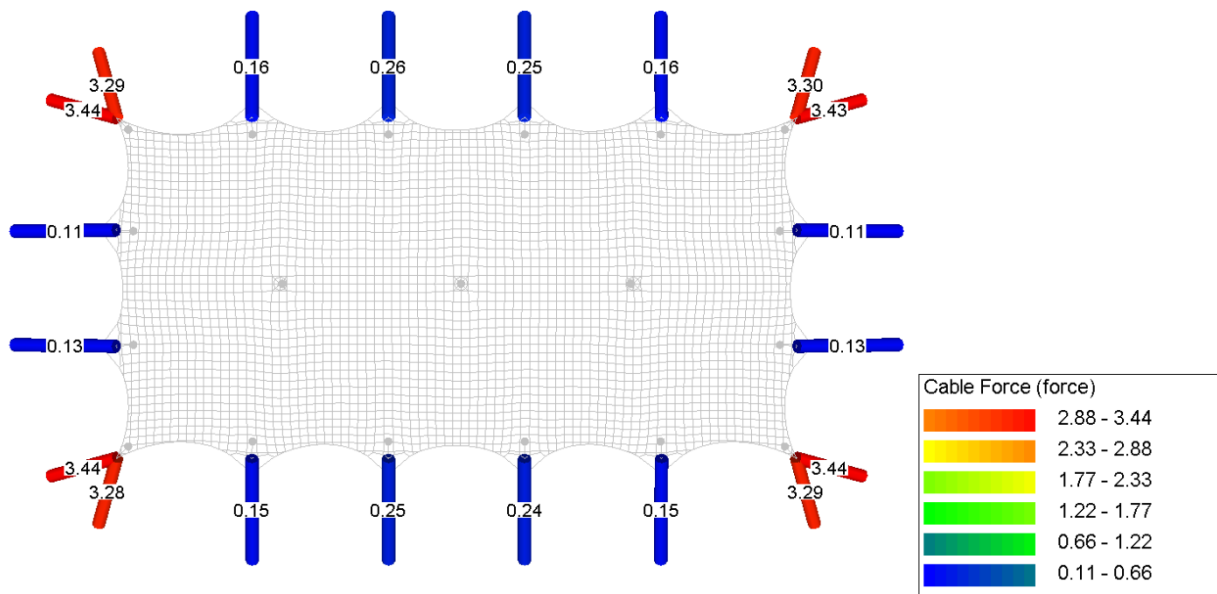
### Annex F.5.3. Circumference rope forces



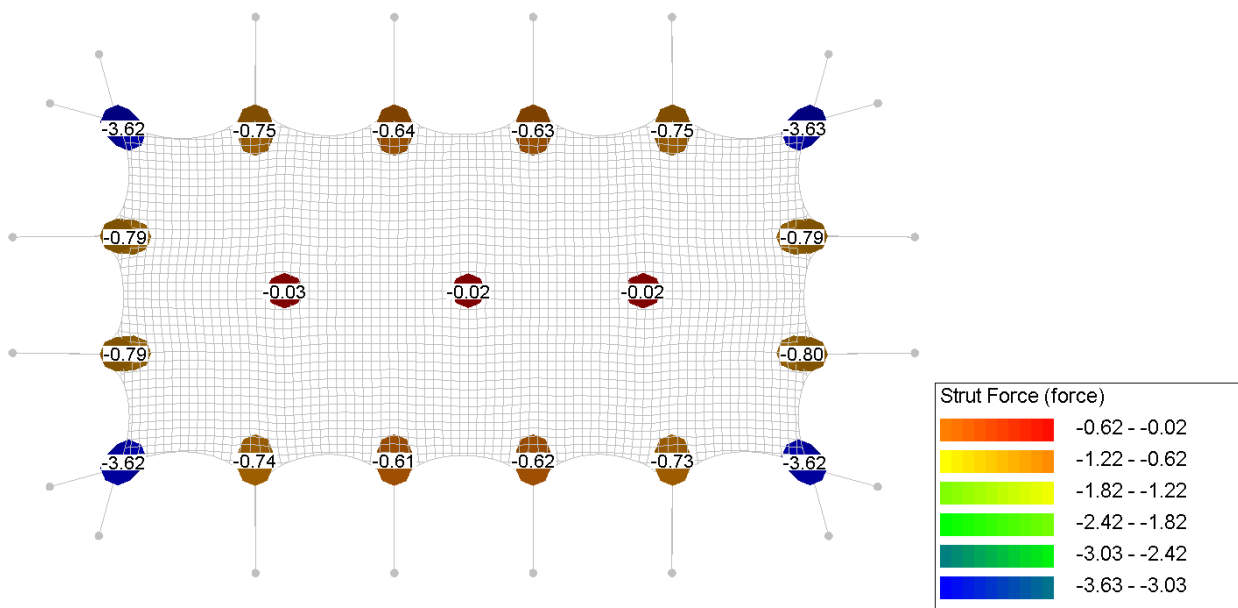
### Annex F.5.4. Rope forces



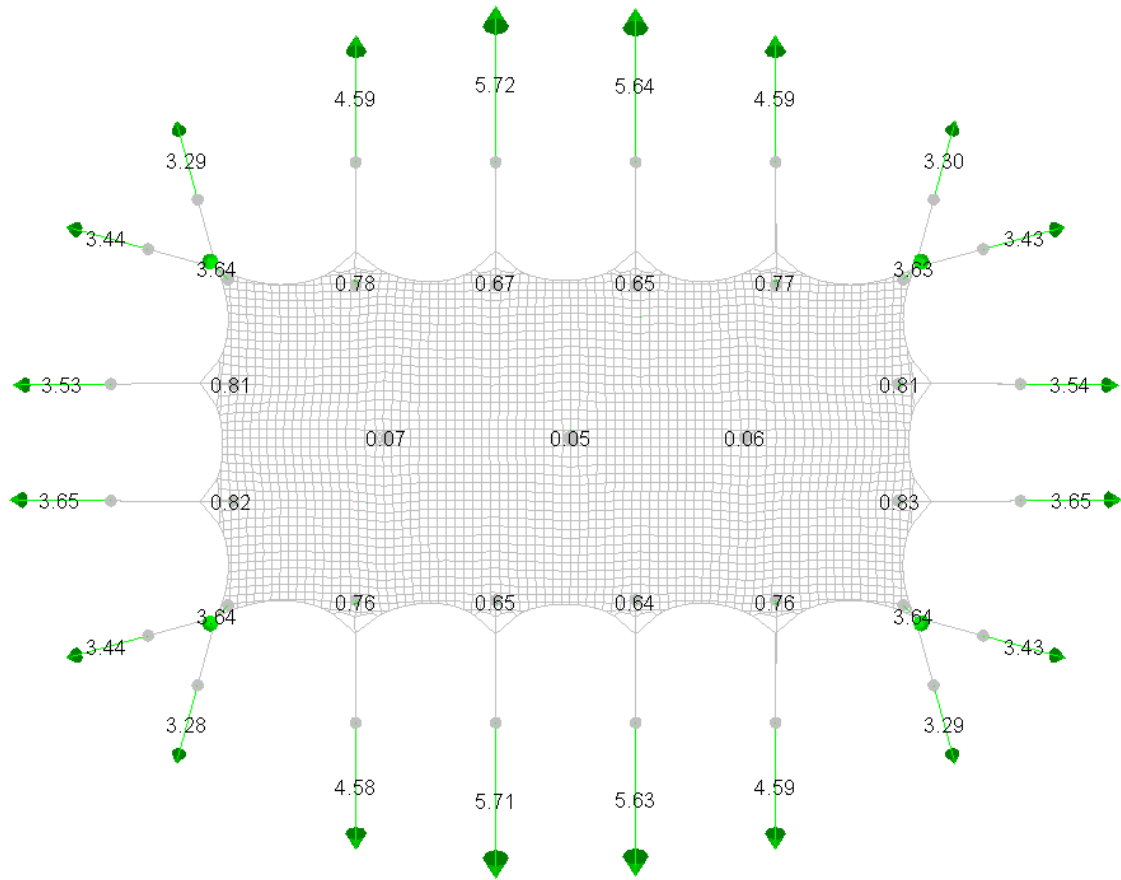
### Annex F.5.5. Belt forces



### Annex F.5.6. Poles forces

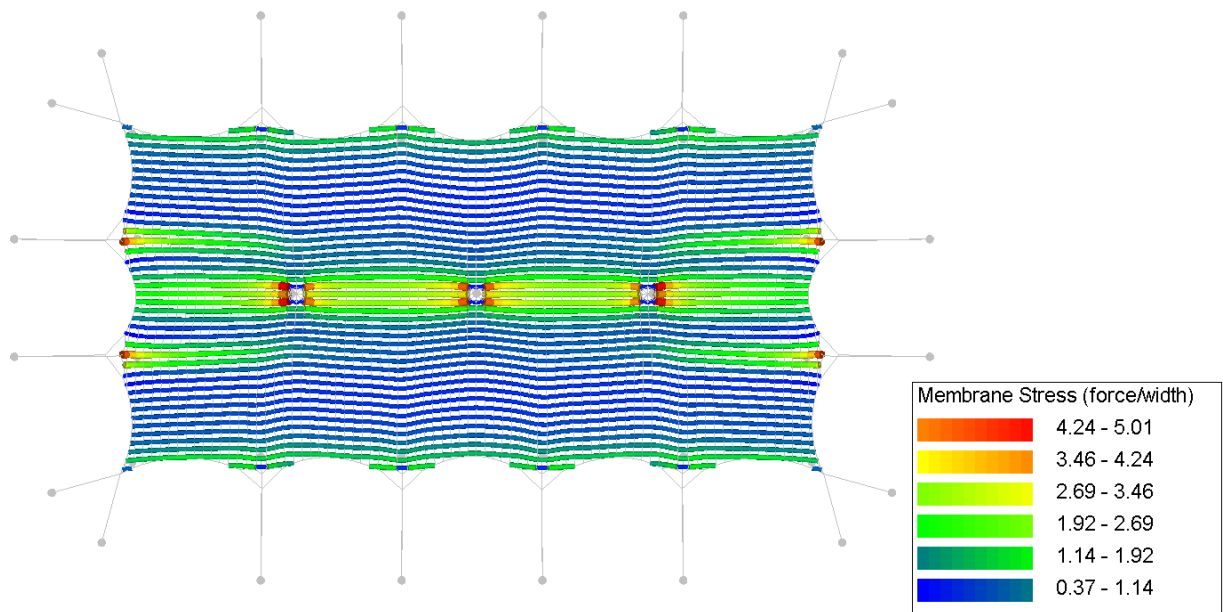


## Annex F.5.7. Reaction forces

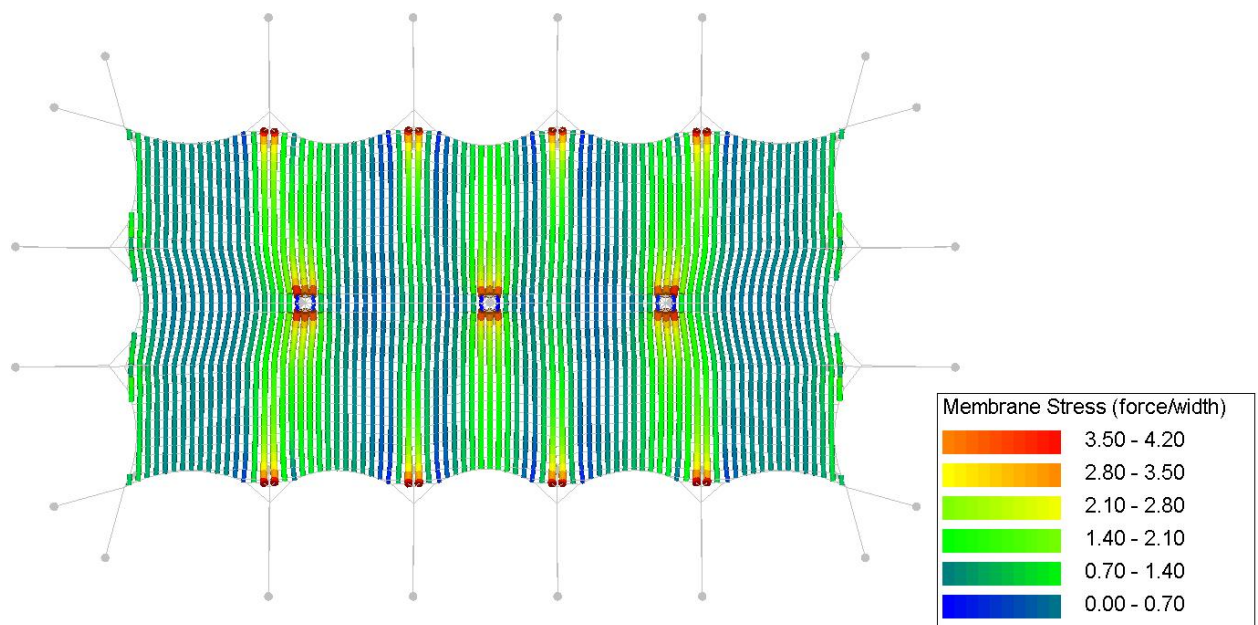


## Annex F.6. CO6. Own weight + pretension + wind pressure + snow [FR]

### Annex F.6.1. Membrane stress (warp)

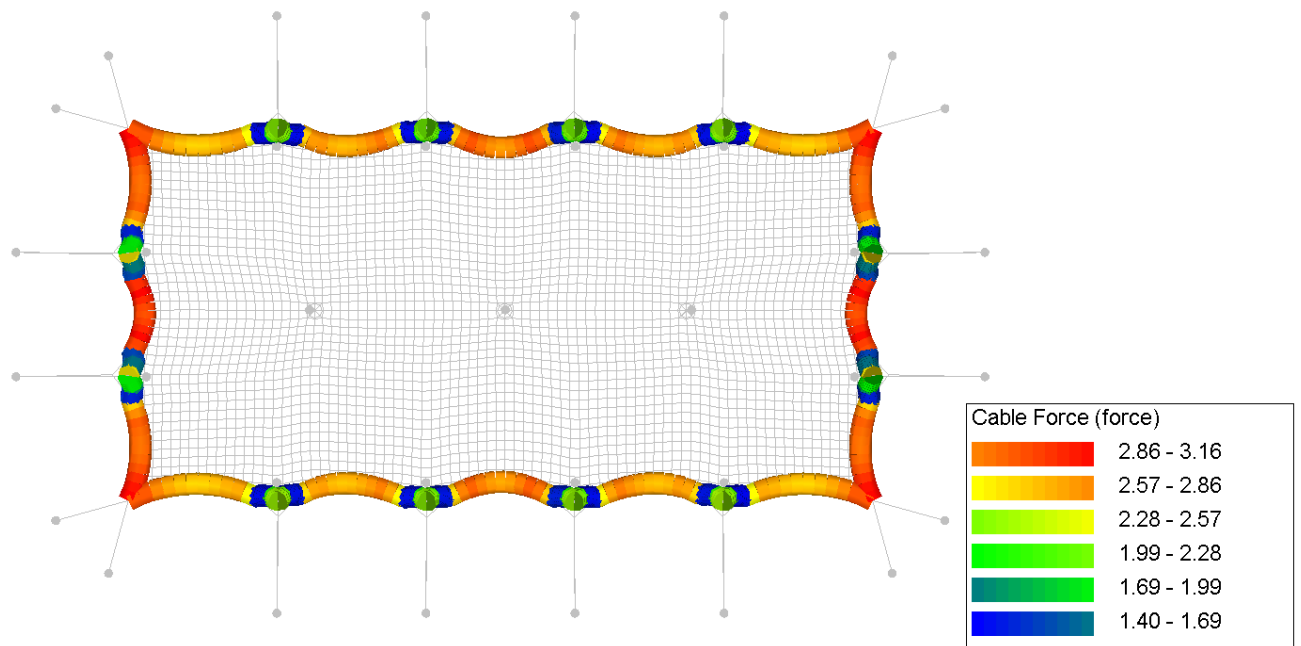


### Annex F.6.2. Membrane stress (weft)

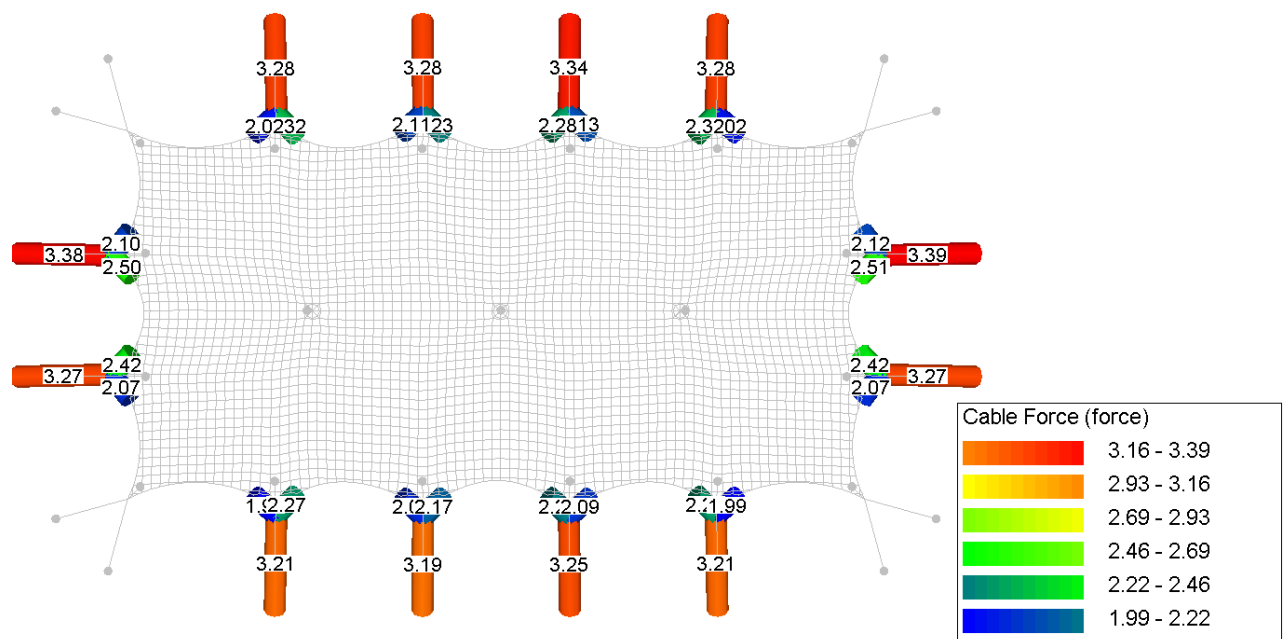




### Annex F.6.3. Circumference rope forces

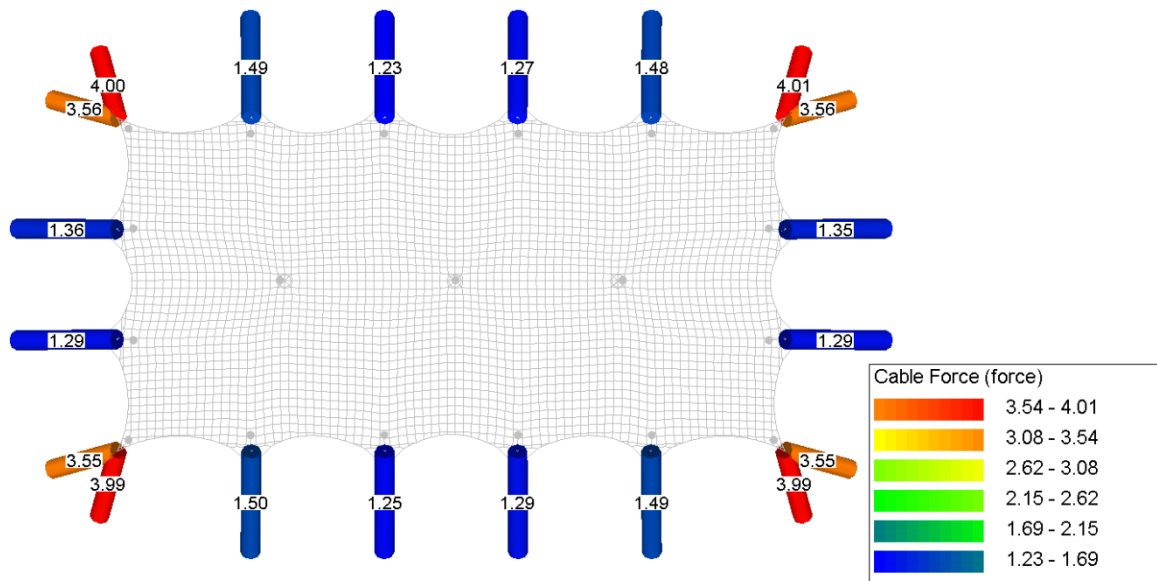


### Annex F.6.4. Guy rope forces

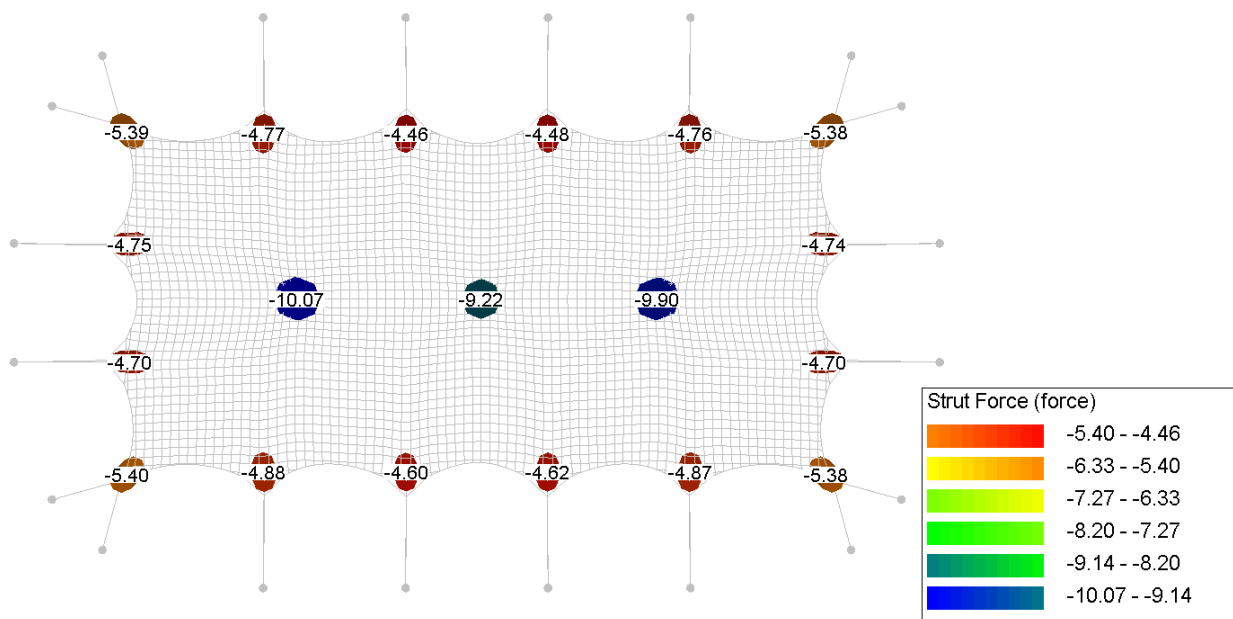




### Annex F.6.5. Belt forces

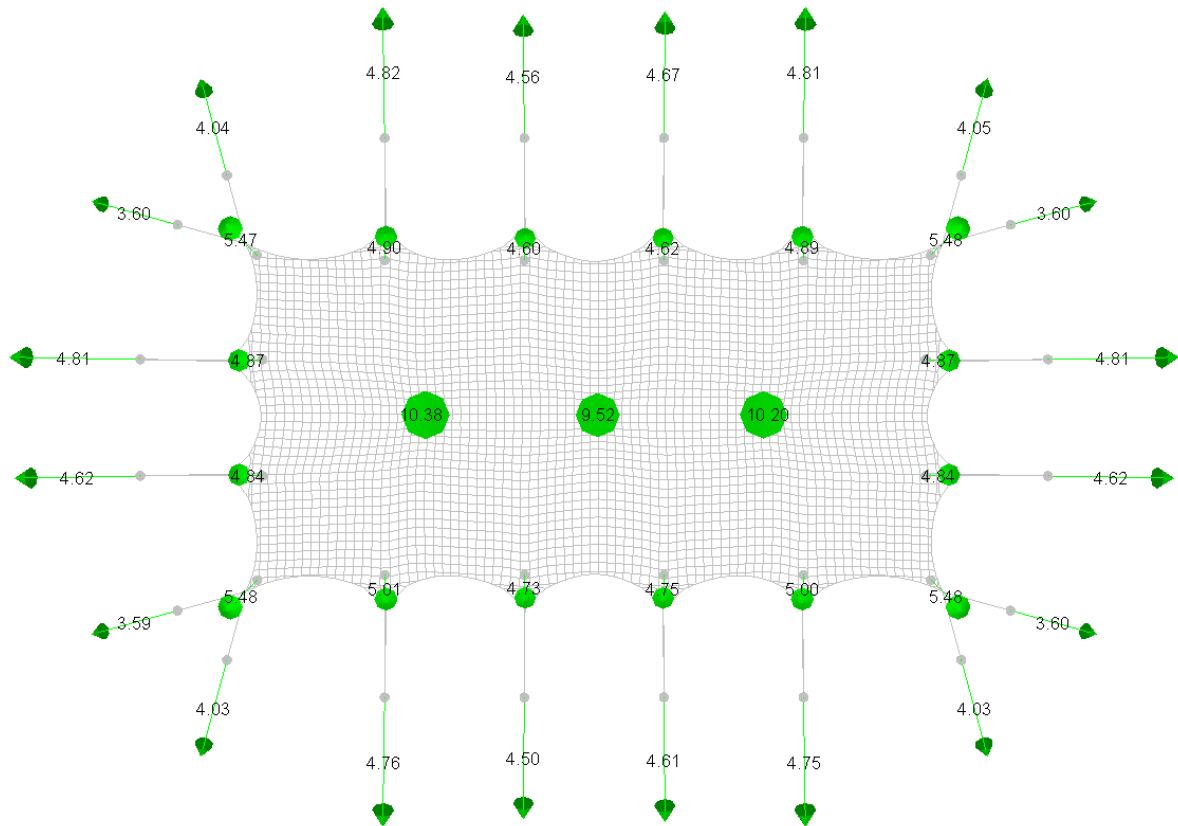


### Annex F.6.6. Poles forces



## Annex F.6.7. Reaction forces

Increased snow load of  $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$  is applied for ballast and anchor calculation



## Annex G. Member checks

### Annex G.1. Main pole 4.0m – Wood, Spruce C18

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 100 mm                      |                                      |
| Length (buckling)           | $l_{bucy}$        | 4 m                         |                                      |
| Effective area              | A                 | 7853.981634 mm <sup>2</sup> |                                      |
| Moment of innertia          | $I_y$             | 4908738.521 mm <sup>4</sup> |                                      |
| Elastic modulus             | $W_{ely}$         | 98174.77042 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modulus of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 25.0 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 160.0                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.79                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 4.64                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.120                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

#### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 10.76 kN                |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.37 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | short                   | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 0.92                    | NEN-EN 1995-1-1:2005 equ. 6.23 |

## Annex G.2. Side pole 3m – Wood, Spruce C18

### Annex G.2.1. 10x10m model

|                     |            |       |                                      |
|---------------------|------------|-------|--------------------------------------|
| Woodtype            | Spruce     |       |                                      |
| Strenght type       | C18        |       |                                      |
| Material factor     | $\gamma_M$ | 1.3   | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class       | 2          |       |                                      |
|                     |            |       | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor | $\beta_c$  | 0.2   | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height              | D          | 70 mm |                                      |

|                             |                   |                             |                                |
|-----------------------------|-------------------|-----------------------------|--------------------------------|
| Length (buckling)           | $l_{buck}$        | 3 m                         |                                |
| Effective area              | A                 | 3848.451001 mm <sup>2</sup> |                                |
| Moment of inertia           | $I_y$             | 1178588.119 mm <sup>4</sup> |                                |
| Elastic modules             | $W_{el,y}$        | 33673.94626 mm <sup>3</sup> |                                |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                |
|                             | $i_y$             | 17.5 mm                     |                                |
| Slenderness                 | $\lambda_y$       | 171.4                       |                                |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.99                        | NEN-EN 1995-1-1:2005 equ. 6.21 |
|                             | $k_y$             | 5.24                        | NEN-EN 1995-1-1:2005 equ. 6.21 |
| Buckling factor             | $k_{cy}$          | 0.105                       | NEN-EN 1995-1-1:2005 equ. 6.25 |

#### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 5.03 kN                 |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.31 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            | short         |                         |                                |
|                          |               |                         | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 1.00                    | NEN-EN 1995-1-1:2005 equ. 6.23 |

## Annex G.2.2. 10x15 and 10x20m model

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 75 mm                       |                                      |
| Length (buckling)           | $l_{buck}$        | 3 m                         |                                      |
| Effective area              | A                 | 4417.864669 mm <sup>2</sup> |                                      |
| Moment of inertia           | $I_y$             | 1553155.548 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{el,y}$        | 41417.48127 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{dk}$          | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{dk}$          | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 18.8 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 160.0                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.79                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 4.64                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.120                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 5.74 kN                 |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.30 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | short                   | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 0.87                    | NEN-EN 1995-1-1:2005 equ. 6.23 |

### Annex G.3. Corner pole 2.5m - Wood, C18

#### Annex G.3.1. 10x10 and 10x20m model

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 70 mm                       |                                      |
| Length (buckling)           | $l_{bucy}$        | 2.5 m                       |                                      |
| Effective area              | A                 | 3848.451001 mm <sup>2</sup> |                                      |
| Moment of innertia          | $I_y$             | 1178588.119 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{elky}$        | 33673.94626 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 17.5 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 142.9                       |                                      |
| Relative slenderness        | $\lambda_{relky}$ | 2.49                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 3.82                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.149                       | NEN-EN 1995-1-1:2005 equ. 6.25       |
| <u>Strength check</u>       |                   |                             |                                      |
| Pressure force              | $F_d$             | 7.12 kN                     |                                      |
| Bending moment              | $M_d$             | 0.00 kNm                    |                                      |
| Pressure stress             | $\sigma_{cdk}$    | 1.85 N/mm <sup>2</sup>      |                                      |
| Bending stress              | $\sigma_{cdk}$    | 0.00 N/mm <sup>2</sup>      |                                      |
| Load duration               |                   | short                       | NEN-EN 1995-1-1:2005 table 2.1       |
| Modificationfactor          | $k_{mod}$         | 0.90                        | NEN-EN 1995-1-1:2005 table 3.1       |
| Design pressure strenght    | $f_{cdk}$         | 12.46 N/mm <sup>2</sup>     | NEN-EN 1995-1-1:2005 equ. 2.14       |
| Design bending strenght     | $f_{mdk}$         | 12.46 N/mm <sup>2</sup>     | NEN-EN 1995-1-1:2005 equ. 2.14       |
| Strength check              |                   | 1.00                        | NEN-EN 1995-1-1:2005 equ. 6.23       |

### Annex G.3.2. 10x15m model

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $V_M$             | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 75 mm                       |                                      |
| Length (buckling)           | $l_{bucy}$        | 2.5 m                       |                                      |
| Effective area              | A                 | 4417.864669 mm <sup>2</sup> |                                      |
| Moment of inertia           | $I_y$             | 1553155.548 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{ely}$         | 41417.48127 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 18.8 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 133.3                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.32                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 3.40                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.170                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

#### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 7.49 kN                 |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.70 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | short                   | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 0.80                    | NEN-EN 1995-1-1:2005 equ. 6.23 |

#### Annex G.4. Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR]

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 105 mm                      |                                      |
| Length (buckling)           | $l_{bucy}$        | 4 m                         |                                      |
| Effective area              | A                 | 8659.014751 mm <sup>2</sup> |                                      |
| Moment of innertia          | $I_y$             | 5966602.352 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{ely}$         | 113649.5686 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 26.3 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 152.4                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.66                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 4.26                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.132                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

##### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 14.27 kN                |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.65 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | short                   | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 1.01                    | NEN-EN 1995-1-1:2005 equ. 6.23 |



## Annex G.5. Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR]

### Annex G.5.1. 10x10m model

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | <b>C18</b>                  |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | <b>2</b>                    | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | <b>75 mm</b>                |                                      |
| Length (buckling)           | $l_{bucy}$        | <b>3 m</b>                  |                                      |
| Effective area              | A                 | 4417.864669 mm <sup>2</sup> |                                      |
| Moment of inertia           | $I_y$             | 1553155.548 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{el,y}$        | 41417.48127 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 18.8 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 160.0                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.79                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 4.64                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.120                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

#### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | <b>6.18 kN</b>          |                                |
| Bending moment           | $M_d$         | <b>0.00 kNm</b>         |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.40 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | <b>short</b>            | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | <b>0.94</b>             | NEN-EN 1995-1-1:2005 equ. 6.23 |

#### Annex G.5.2. 10x15 and 10x20m model

|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | C18                         |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | 2                           | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | 80 mm                       |                                      |
| Length (buckling)           | $l_{bucy}$        | 3 m                         |                                      |
| Effective area              | A                 | 5026.548246 mm <sup>2</sup> |                                      |
| Moment of inertia           | $I_y$             | 2010619.298 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{ely}$         | 50265.48246 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 20.0 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 150.0                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.62                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 4.15                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.136                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

#### Strength check

|                          |               |                         |                                |
|--------------------------|---------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$         | 7.07 kN                 |                                |
| Bending moment           | $M_d$         | 0.00 kNm                |                                |
| Pressure stress          | $\sigma_{cd}$ | 1.41 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cd}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |               | short                   | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$     | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{md}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |               | 0.83                    | NEN-EN 1995-1-1:2005 equ. 6.23 |

## Annex G.6. Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR]

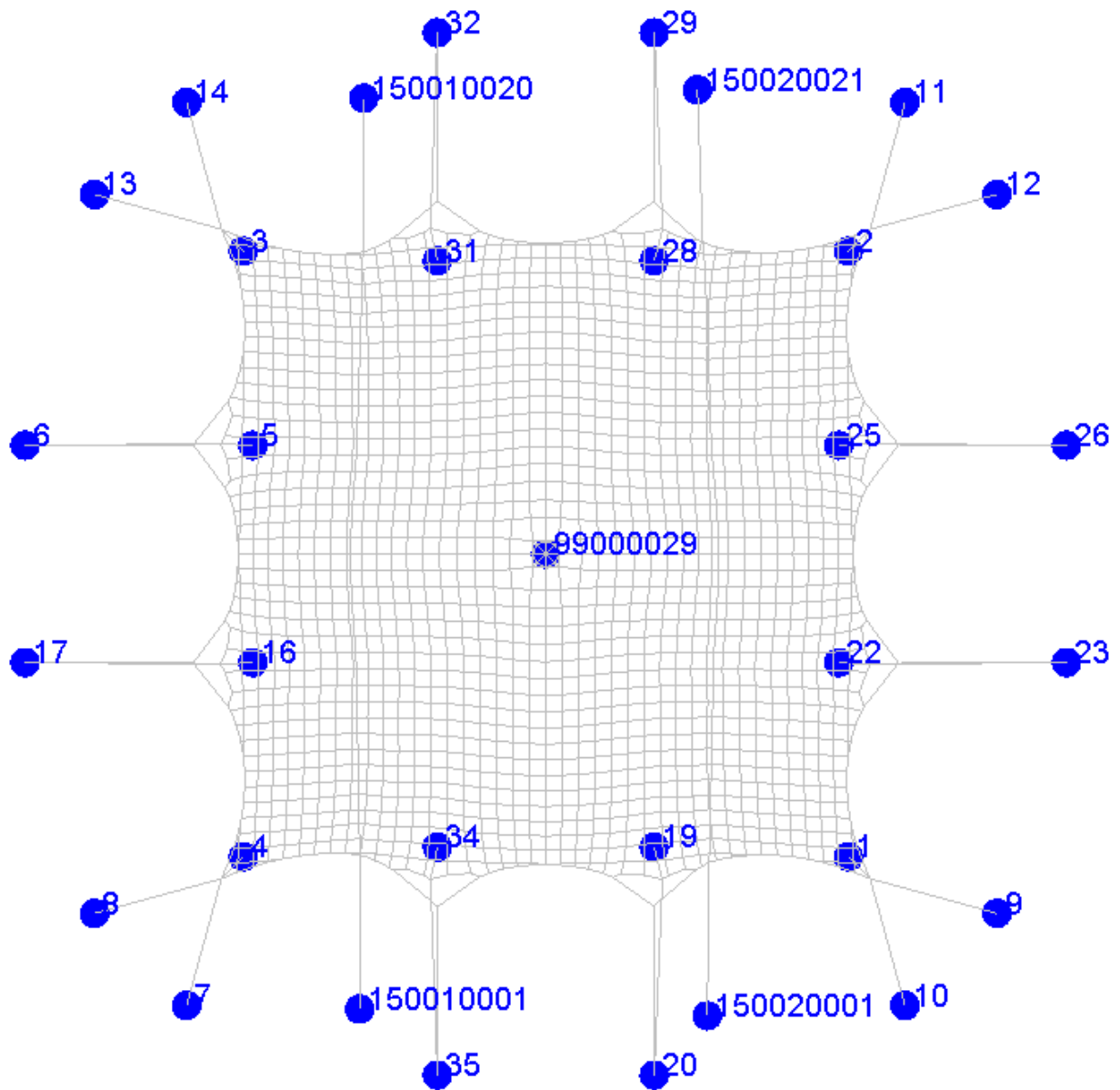
|                             |                   |                             |                                      |
|-----------------------------|-------------------|-----------------------------|--------------------------------------|
| Woodtype                    |                   | Spruce                      |                                      |
| Strenght type               |                   | <b>C18</b>                  |                                      |
| Material factor             | $\gamma_M$        | 1.3                         | NEN-EN 1995-1-1:2005 table 2.3       |
| Climate class               |                   | <b>2</b>                    | NEN-EN 1995-1-1:2005 article 2.3.1.3 |
| Straightness factor         | $\beta_c$         | 0.2                         | NEN-EN 1995-1-1:2005 equ. 6.29       |
| Height                      | D                 | <b>75 mm</b>                |                                      |
| Length (buckling)           | $l_{bucy}$        | <b>2.5 m</b>                |                                      |
| Effective area              | A                 | 4417.864669 mm <sup>2</sup> |                                      |
| Moment of inertia           | $I_y$             | 1553155.548 mm <sup>4</sup> |                                      |
| Elastic modules             | $W_{el,y}$        | 41417.48127 mm <sup>3</sup> |                                      |
| Charistic pressure strenght | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Charistic bending strenght  | $f_{cdk}$         | 18 N/mm <sup>2</sup>        |                                      |
| Modules of elasticity       | $E_{0.05}$        | 6 kN/m <sup>2</sup>         |                                      |
|                             | $i_y$             | 18.8 mm                     |                                      |
| Slenderness                 | $\lambda_y$       | 133.3                       |                                      |
| Relative slenderness        | $\lambda_{rel,y}$ | 2.32                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
|                             | $k_y$             | 3.40                        | NEN-EN 1995-1-1:2005 equ. 6.21       |
| Buckling factor             | $k_{cy}$          | 0.170                       | NEN-EN 1995-1-1:2005 equ. 6.25       |

### Strength check

|                          |                |                         |                                |
|--------------------------|----------------|-------------------------|--------------------------------|
| Pressure force           | $F_d$          | <b>7.76 kN</b>          |                                |
| Bending moment           | $M_d$          | <b>0.00 kNm</b>         |                                |
| Pressure stress          | $\sigma_{cdk}$ | 1.76 N/mm <sup>2</sup>  |                                |
| Bending stress           | $\sigma_{cdk}$ | 0.00 N/mm <sup>2</sup>  |                                |
| Load duration            |                | <b>short</b>            | NEN-EN 1995-1-1:2005 table 2.1 |
| Modificationfactor       | $k_{mod}$      | 0.90                    | NEN-EN 1995-1-1:2005 table 3.1 |
| Design pressure strenght | $f_{cd}$       | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Design bending strenght  | $f_{mod}$      | 12.46 N/mm <sup>2</sup> | NEN-EN 1995-1-1:2005 equ. 2.14 |
| Strength check           |                | <b>0.83</b>             | NEN-EN 1995-1-1:2005 equ. 6.23 |

## Annex H. Reaction forces model 10x10m

### Annex H.1. Point numbers



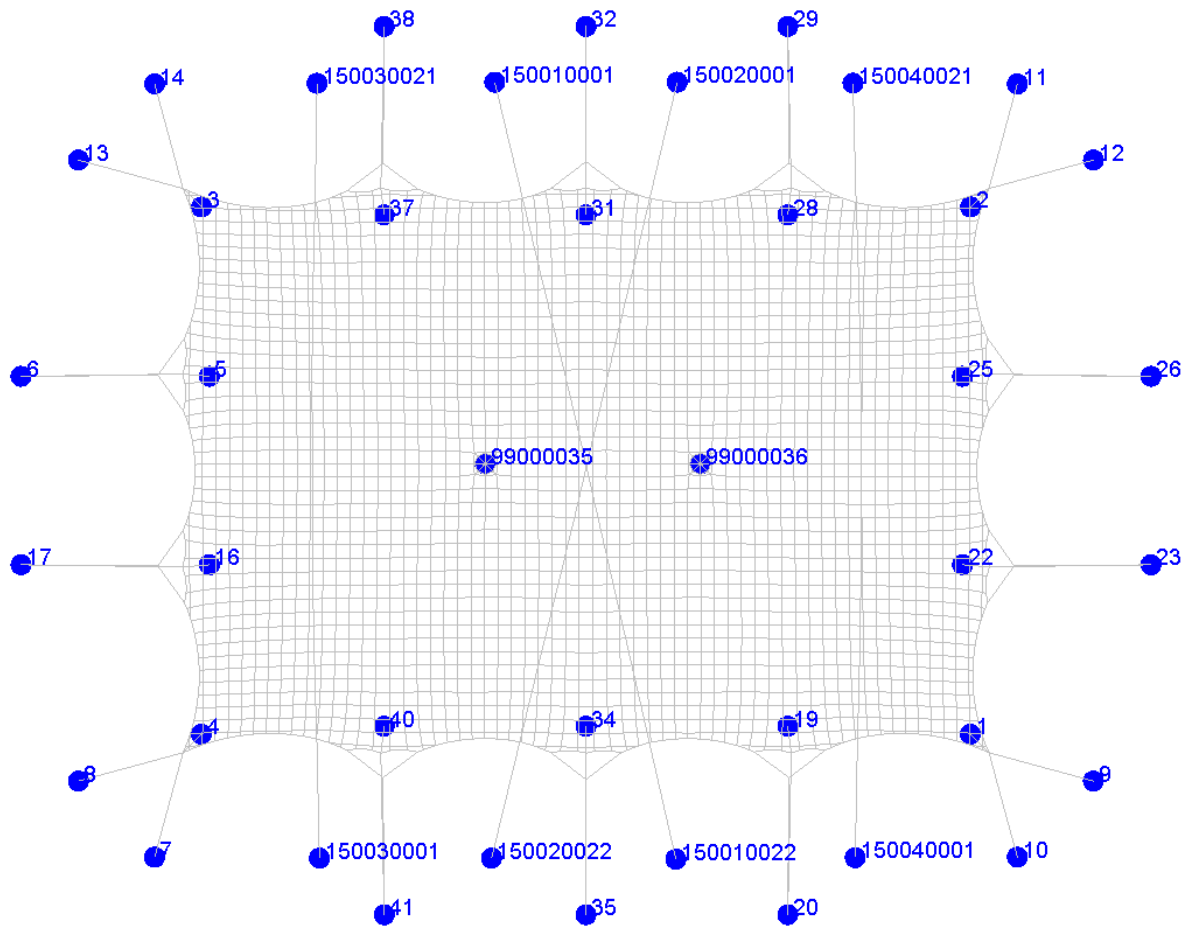
## Annex H.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

| Att       | Node      | Koor X Koor Y |       | CO1   |       |       | CO2   |       |       | CO3   |       |       | CO4   |       |       | CO5   |       |       | CO6   |       |       | CO6a  |       |       |
|-----------|-----------|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|           |           |               |       | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    |
| Tie       | 6         | -3.00         | 6.67  | -1.55 | -0.01 | -1.18 | -2.04 | 0.00  | -1.59 | -2.44 | 0.01  | -1.91 | -3.98 | -0.02 | -2.97 | -3.04 | -0.03 | -2.27 | -3.04 | 0.03  | -2.41 | -3.08 | 0.03  | -2.44 |
| Tie       | 17        | -3.00         | 3.33  | -1.55 | 0.01  | -1.18 | -2.04 | 0.00  | -1.59 | -2.44 | -0.01 | -1.91 | -3.98 | 0.02  | -2.97 | -3.04 | 0.03  | -2.28 | -3.04 | -0.03 | -2.41 | -3.08 | -0.03 | -2.44 |
| Tie       | 20        | 6.67          | -3.00 | -0.01 | -1.55 | -1.18 | 0.00  | -1.98 | -1.53 | 0.01  | -2.34 | -1.82 | 0.00  | -3.58 | -2.66 | -0.03 | -2.69 | -2.02 | 0.02  | -2.84 | -2.24 | 0.02  | -2.88 | -2.27 |
| Tie       | 23        | 13.00         | 3.33  | 1.55  | 0.01  | -1.18 | 2.04  | 0.00  | -1.59 | 2.44  | -0.01 | -1.91 | 4.03  | 0.02  | -3.01 | 3.04  | 0.03  | -2.27 | 3.04  | -0.03 | -2.41 | 3.08  | -0.03 | -2.44 |
| Tie       | 26        | 13.00         | 6.67  | 1.55  | -0.01 | -1.18 | 2.04  | 0.00  | -1.59 | 2.44  | 0.01  | -1.91 | 4.04  | -0.02 | -3.02 | 3.04  | -0.03 | -2.28 | 3.04  | 0.03  | -2.41 | 3.08  | 0.03  | -2.44 |
| Tie       | 29        | 6.67          | 13.00 | -0.01 | 1.55  | -1.18 | 0.00  | 1.98  | -1.53 | 0.01  | 2.34  | -1.82 | 0.00  | 3.60  | -2.67 | -0.03 | 2.69  | -2.02 | 0.02  | 2.84  | -2.24 | 0.02  | 2.88  | -2.27 |
| Tie       | 32        | 3.33          | 13.00 | 0.01  | 1.55  | -1.18 | 0.00  | 1.98  | -1.53 | -0.01 | 2.34  | -1.82 | 0.00  | 3.64  | -2.71 | 0.03  | 2.69  | -2.02 | -0.02 | 2.84  | -2.24 | -0.02 | 2.88  | -2.27 |
| Tie       | 35        | 3.33          | -3.00 | 0.01  | -1.55 | -1.18 | 0.00  | -1.98 | -1.53 | -0.01 | -2.34 | -1.82 | 0.00  | -3.63 | -2.70 | 0.03  | -2.69 | -2.02 | -0.02 | -2.84 | -2.24 | -0.02 | -2.88 | -2.27 |
| Tie       | 7         | -0.52         | -1.93 | -0.39 | -1.43 | -1.47 | -0.46 | -1.71 | -1.76 | -0.53 | -1.96 | -2.02 | -0.71 | -2.60 | -2.68 | -0.59 | -2.18 | -2.25 | -0.62 | -2.27 | -2.34 | -0.63 | -2.29 | -2.36 |
| Tie       | 8         | -1.93         | -0.52 | -1.43 | -0.39 | -1.48 | -1.71 | -0.46 | -1.76 | -1.96 | -0.53 | -2.02 | -2.67 | -0.73 | -2.75 | -2.04 | -0.56 | -2.11 | -2.28 | -0.62 | -2.34 | -2.30 | -0.63 | -2.37 |
| Tie       | 9         | 11.93         | -0.52 | 1.43  | -0.39 | -1.48 | 1.71  | -0.46 | -1.76 | 1.96  | -0.53 | -2.02 | 2.69  | -0.74 | -2.77 | 2.04  | -0.56 | -2.11 | 2.28  | -0.62 | -2.34 | 2.30  | -0.63 | -2.37 |
| Tie       | 10        | 10.52         | -1.93 | 0.39  | -1.43 | -1.47 | 0.46  | -1.71 | -1.76 | 0.53  | -1.96 | -2.01 | 0.72  | -2.61 | -2.68 | 0.59  | -2.18 | -2.25 | 0.62  | -2.27 | -2.34 | 0.63  | -2.29 | -2.36 |
| Tie       | 11        | 10.52         | 11.93 | 0.39  | 1.43  | -1.47 | 0.46  | 1.71  | -1.76 | 0.53  | 1.96  | -2.02 | 0.72  | 2.61  | -2.68 | 0.59  | 2.18  | -2.25 | 0.62  | 2.27  | -2.34 | 0.63  | 2.29  | -2.36 |
| Tie       | 12        | 11.93         | 10.52 | 1.43  | 0.39  | -1.48 | 1.71  | 0.46  | -1.76 | 1.96  | 0.53  | -2.02 | 2.68  | 0.74  | -2.76 | 2.04  | 0.56  | -2.11 | 2.28  | 0.62  | -2.34 | 2.30  | 0.63  | -2.37 |
| Tie       | 13        | -1.93         | 10.52 | -1.43 | 0.39  | -1.48 | -1.71 | 0.46  | -1.76 | -1.96 | 0.53  | -2.02 | -2.66 | 0.73  | -2.73 | -2.04 | 0.56  | -2.11 | -2.28 | 0.62  | -2.34 | -2.30 | 0.63  | -2.37 |
| Tie       | 14        | -0.52         | 11.93 | -0.39 | 1.43  | -1.47 | -0.46 | 1.71  | -1.76 | -0.53 | 1.96  | -2.02 | -0.71 | 2.60  | -2.67 | -0.59 | 2.18  | -2.25 | -0.62 | 2.27  | -2.34 | -0.63 | 2.29  | -2.36 |
| Pole      | 1         | 9.65          | 0.35  | 0.44  | -0.44 | 2.58  | 0.56  | -0.56 | 3.29  | 0.65  | -0.65 | 3.86  | 0.72  | -0.72 | 4.30  | 0.57  | -0.56 | 3.34  | 0.78  | -0.78 | 4.64  | 0.79  | -0.79 | 4.70  |
| Pole      | 2         | 9.65          | 9.65  | 0.44  | 0.44  | 2.58  | 0.56  | 0.56  | 3.29  | 0.65  | 0.65  | 3.86  | 0.71  | 0.71  | 4.25  | 0.57  | 0.56  | 3.34  | 0.78  | 0.78  | 4.64  | 0.79  | 0.79  | 4.71  |
| Pole      | 3         | 0.35          | 9.65  | -0.44 | 0.44  | 2.58  | -0.56 | 0.56  | 3.29  | -0.65 | 0.65  | 3.86  | -0.73 | 0.73  | 4.37  | -0.57 | 0.56  | 3.34  | -0.78 | 0.78  | 4.64  | -0.79 | 0.79  | 4.70  |
| Pole      | 4         | 0.35          | 0.35  | -0.44 | -0.44 | 2.58  | -0.56 | -0.56 | 3.29  | -0.65 | -0.65 | 3.86  | -0.74 | -0.74 | 4.41  | -0.57 | -0.56 | 3.34  | -0.78 | -0.78 | 4.64  | -0.79 | -0.79 | 4.71  |
| Pole      | 5         | 0.50          | 6.67  | -0.21 | 0.01  | 1.31  | -0.39 | 0.01  | 2.41  | -0.51 | 0.01  | 3.11  | -0.13 | 0.01  | 0.82  | -0.14 | 0.01  | 0.85  | -0.70 | 0.00  | 4.30  | -0.71 | -0.01 | 4.39  |
| Pole      | 16        | 0.50          | 3.33  | -0.21 | -0.01 | 1.31  | -0.39 | -0.01 | 2.41  | -0.51 | -0.01 | 3.11  | -0.13 | -0.01 | 0.82  | -0.14 | -0.01 | 0.85  | -0.70 | 0.00  | 4.30  | -0.71 | 0.01  | 4.39  |
| Pole      | 19        | 6.67          | 0.50  | 0.01  | -0.21 | 1.31  | 0.01  | -0.38 | 2.35  | 0.01  | -0.49 | 3.01  | 0.08  | -0.31 | 1.87  | 0.01  | -0.12 | 0.74  | 0.00  | -0.67 | 4.10  | -0.01 | -0.68 | 4.19  |
| Pole      | 22        | 9.50          | 3.33  | 0.21  | -0.01 | 1.31  | 0.39  | -0.01 | 2.41  | 0.51  | -0.01 | 3.11  | 0.13  | -0.01 | 0.79  | 0.14  | -0.01 | 0.85  | 0.70  | 0.00  | 4.30  | 0.71  | 0.01  | 4.39  |
| Pole      | 25        | 9.50          | 6.67  | 0.21  | 0.01  | 1.31  | 0.39  | 0.01  | 2.41  | 0.51  | 0.01  | 3.11  | 0.12  | 0.01  | 0.77  | 0.14  | 0.01  | 0.85  | 0.70  | 0.00  | 4.30  | 0.71  | -0.01 | 4.39  |
| Pole      | 28        | 6.67          | 9.50  | 0.01  | 0.21  | 1.31  | 0.01  | 0.38  | 2.35  | 0.01  | 0.49  | 3.01  | 0.09  | 0.33  | 2.02  | 0.01  | 0.12  | 0.74  | 0.00  | 0.67  | 4.10  | -0.01 | 0.68  | 4.19  |
| Pole      | 31        | 3.33          | 9.50  | -0.01 | 0.21  | 1.31  | -0.01 | 0.38  | 2.35  | -0.01 | 0.49  | 3.01  | -0.05 | 0.25  | 1.51  | -0.01 | 0.12  | 0.74  | 0.00  | 0.67  | 4.10  | 0.01  | 0.68  | 4.19  |
| Pole      | 34        | 3.33          | 0.50  | -0.01 | -0.21 | 1.31  | -0.01 | -0.38 | 2.35  | -0.01 | -0.49 | 3.01  | -0.05 | -0.24 | 1.48  | -0.01 | -0.12 | 0.74  | 0.00  | -0.67 | 4.10  | 0.01  | -0.68 | 4.19  |
| Pole      | 99000029  | 5.00          | 5.00  | 0.00  | 0.00  | 2.15  | 0.00  | 0.00  | 5.45  | 0.00  | 0.00  | 6.98  | 0.00  | 0.00  | -0.01 | 0.00  | 0.00  | 0.12  | 0.00  | 0.00  | 10.36 | 0.00  | 0.00  | 10.64 |
| Stormbelt | 150010001 | 2.14          | -1.99 |       |       |       |       |       |       |       |       |       | -0.01 | -3.80 | -3.35 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150010020 | 2.21          | 12.00 |       |       |       |       |       |       |       |       |       | 0.02  | 3.80  | -3.34 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150020001 | 7.48          | -2.09 |       |       |       |       |       |       |       |       |       | -0.04 | -4.22 | -3.72 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150020021 | 7.33          | 12.12 |       |       |       |       |       |       |       |       |       | -0.14 | 4.19  | -3.75 |       |       |       |       |       |       |       |       |       |

## Annex I. Reaction forces model 10x15m

### Annex I.1. Point numbers



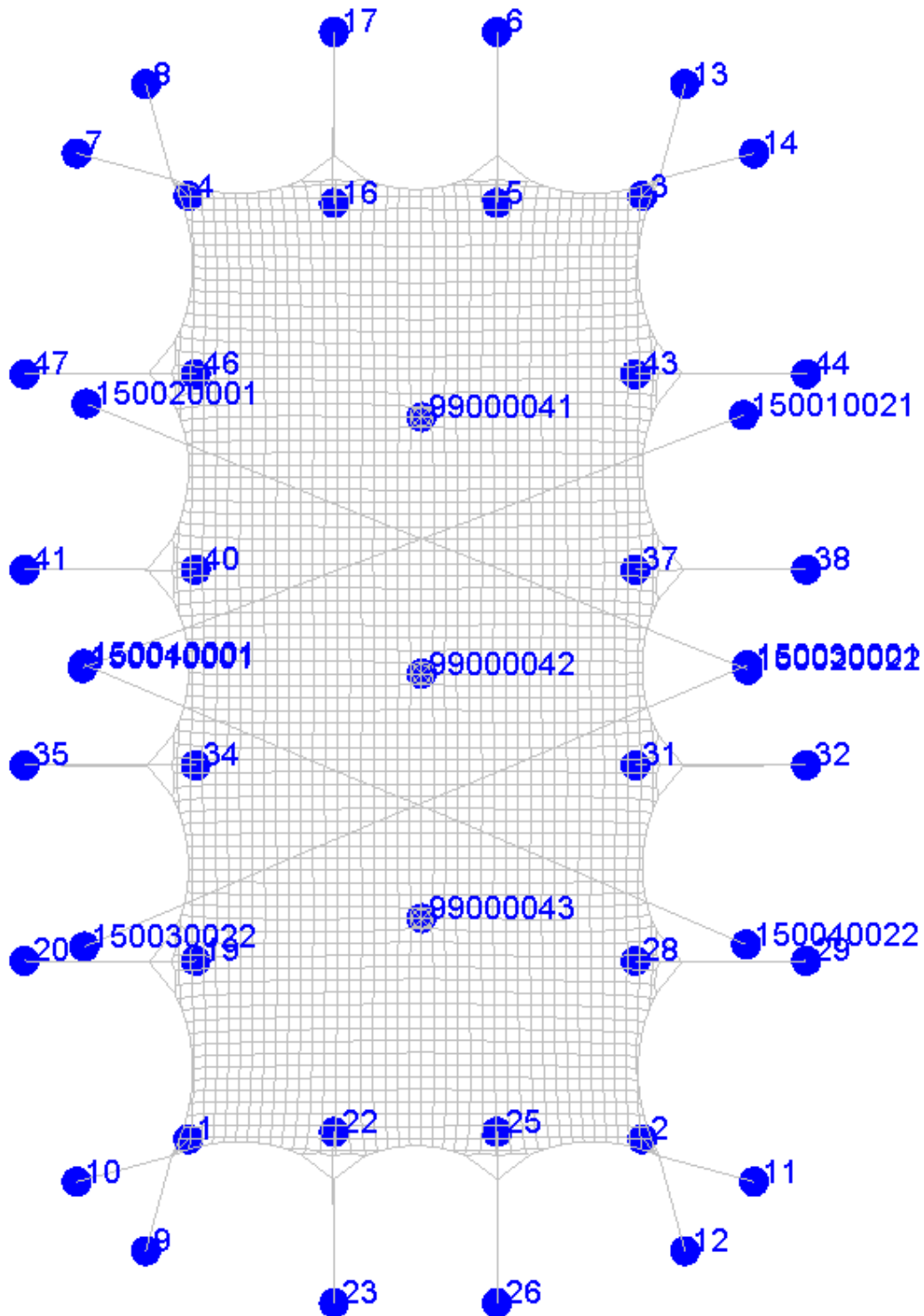
## Annex I.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

| Att       | Node      | Koor X | Koor Y | CO1   |       |       | CO2   |       |       | CO3   |       |       | CO4   |       |       | CO5   |       |       | CO6   |       |       | CO6a  |       |       |
|-----------|-----------|--------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|           |           |        |        | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    |
| Tie       | 6         | -3.00  | 7.00   | -1.62 | -0.01 | -1.24 | -2.30 | 0.00  | -1.79 | -2.82 | 0.02  | -2.21 | -4.07 | -0.02 | -3.05 | -3.29 | -0.04 | -2.47 | -3.54 | 0.04  | -2.80 | -3.60 | 0.04  | -2.84 |
| Tie       | 17        | -3.00  | 3.50   | -1.61 | 0.01  | -1.23 | -2.24 | 0.00  | -1.74 | -2.74 | -0.02 | -2.14 | -4.14 | 0.02  | -3.10 | -3.37 | 0.03  | -2.53 | -3.44 | -0.04 | -2.72 | -3.49 | -0.04 | -2.76 |
| Tie       | 20        | 11.25  | -3.00  | -0.02 | -1.71 | -1.31 | 0.00  | -2.25 | -1.75 | 0.02  | -2.72 | -2.13 | -0.02 | -4.05 | -3.02 | -0.03 | -3.52 | -2.67 | 0.04  | -3.32 | -2.62 | 0.04  | -3.37 | -2.66 |
| Tie       | 23        | 18.00  | 3.50   | 1.62  | 0.01  | -1.23 | 2.24  | -0.01 | -1.75 | 2.75  | -0.02 | -2.15 | 4.18  | 0.02  | -3.14 | 3.38  | 0.04  | -2.53 | 3.46  | -0.04 | -2.73 | 3.52  | -0.04 | -2.78 |
| Tie       | 26        | 18.00  | 7.00   | 1.62  | -0.01 | -1.24 | 2.31  | 0.01  | -1.80 | 2.84  | 0.02  | -2.22 | 4.12  | -0.02 | -3.09 | 3.29  | -0.04 | -2.47 | 3.57  | 0.04  | -2.82 | 3.63  | 0.04  | -2.87 |
| Tie       | 29        | 11.25  | 13.50  | -0.02 | 1.72  | -1.32 | 0.00  | 2.26  | -1.76 | 0.02  | 2.74  | -2.14 | -0.02 | 4.07  | -3.04 | -0.03 | 3.52  | -2.68 | 0.04  | 3.36  | -2.65 | 0.04  | 3.41  | -2.69 |
| Tie       | 32        | 7.50   | 13.50  | 0.00  | 1.74  | -1.37 | 0.00  | 2.41  | -1.92 | 0.00  | 3.00  | -2.38 | 0.00  | 3.59  | -2.75 | 0.00  | 3.78  | -3.00 | 0.00  | 3.72  | -2.97 | 0.00  | 3.77  | -3.02 |
| Tie       | 35        | 7.50   | -3.00  | 0.00  | -1.73 | -1.37 | 0.00  | -2.39 | -1.90 | 0.00  | -2.97 | -2.36 | 0.00  | -3.60 | -2.74 | 0.00  | -3.78 | -2.99 | 0.00  | -3.67 | -2.94 | 0.00  | -3.72 | -2.98 |
| Tie       | 38        | 3.75   | 13.50  | 0.02  | 1.71  | -1.31 | 0.00  | 2.22  | -1.73 | -0.02 | 2.68  | -2.09 | 0.02  | 4.16  | -3.11 | 0.03  | 3.59  | -2.73 | -0.04 | 3.29  | -2.59 | -0.04 | 3.33  | -2.63 |
| Tie       | 41        | 3.75   | -3.00  | 0.02  | -1.71 | -1.30 | 0.00  | -2.21 | -1.72 | -0.02 | -2.66 | -2.08 | 0.02  | -4.16 | -3.10 | 0.03  | -3.59 | -2.72 | -0.03 | -3.26 | -2.57 | -0.04 | -3.30 | -2.61 |
| Tie       | 7         | -0.52  | -1.93  | -0.43 | -1.58 | -1.63 | -0.53 | -1.94 | -2.00 | -0.61 | -2.24 | -2.30 | -0.76 | -2.76 | -2.84 | -0.65 | -2.39 | -2.46 | -0.71 | -2.59 | -2.67 | -0.72 | -2.62 | -2.70 |
| Tie       | 8         | -1.93  | -0.52  | -1.52 | -0.41 | -1.56 | -1.87 | -0.51 | -1.93 | -2.21 | -0.60 | -2.27 | -2.89 | -0.79 | -2.97 | -2.40 | -0.66 | -2.47 | -2.59 | -0.71 | -2.66 | -2.62 | -0.72 | -2.69 |
| Tie       | 9         | 16.93  | -0.52  | 1.52  | -0.41 | -1.56 | 1.88  | -0.51 | -1.94 | 2.21  | -0.60 | -2.28 | 2.89  | -0.79 | -2.97 | 2.40  | -0.66 | -2.48 | 2.60  | -0.71 | -2.68 | 2.63  | -0.72 | -2.71 |
| Tie       | 10        | 15.52  | -1.93  | 0.43  | -1.58 | -1.63 | 0.53  | -1.95 | -2.01 | 0.61  | -2.24 | -2.31 | 0.77  | -2.78 | -2.86 | 0.65  | -2.39 | -2.46 | 0.71  | -2.60 | -2.68 | 0.72  | -2.63 | -2.70 |
| Tie       | 11        | 15.52  | 12.43  | 0.43  | 1.58  | -1.63 | 0.53  | 1.95  | -2.01 | 0.61  | 2.24  | -2.31 | 0.76  | 2.78  | -2.86 | 0.66  | 2.40  | -2.47 | 0.71  | 2.61  | -2.68 | 0.72  | 2.63  | -2.71 |
| Tie       | 12        | 16.93  | 11.02  | 1.52  | 0.41  | -1.56 | 1.88  | 0.51  | -1.94 | 2.21  | 0.60  | -2.28 | 2.88  | 0.79  | -2.96 | 2.40  | 0.66  | -2.48 | 2.61  | 0.71  | -2.68 | 2.64  | 0.72  | -2.71 |
| Tie       | 13        | -1.93  | 11.02  | -1.52 | 0.41  | -1.57 | -1.88 | 0.51  | -1.94 | -2.21 | 0.60  | -2.28 | -2.88 | 0.79  | -2.96 | -2.40 | 0.66  | -2.47 | -2.59 | 0.71  | -2.67 | -2.62 | 0.72  | -2.70 |
| Tie       | 14        | -0.52  | 12.43  | -0.43 | 1.58  | -1.63 | -0.53 | 1.94  | -2.00 | -0.61 | 2.24  | -2.31 | -0.76 | 2.76  | -2.84 | -0.66 | 2.39  | -2.47 | -0.71 | 2.60  | -2.67 | -0.72 | 2.63  | -2.70 |
| Pole      | 1         | 14.65  | 0.35   | 0.48  | -0.48 | 2.81  | 0.63  | -0.63 | 3.73  | 0.75  | -0.75 | 4.44  | 0.78  | -0.78 | 4.68  | 0.63  | -0.64 | 3.78  | 0.90  | -0.90 | 5.38  | 0.91  | -0.91 | 5.45  |
| Pole      | 2         | 14.65  | 10.15  | 0.48  | 0.48  | 2.80  | 0.63  | 0.63  | 3.72  | 0.75  | 0.75  | 4.43  | 0.77  | 0.78  | 4.65  | 0.63  | 0.63  | 3.77  | 0.90  | 0.90  | 5.37  | 0.91  | 0.91  | 5.45  |
| Pole      | 3         | 0.35   | 10.15  | -0.48 | 0.48  | 2.81  | -0.63 | 0.63  | 3.73  | -0.75 | 0.75  | 4.44  | -0.78 | 0.78  | 4.66  | -0.63 | 0.63  | 3.78  | -0.90 | 0.90  | 5.37  | -0.91 | 0.91  | 5.44  |
| Pole      | 4         | 0.35   | 0.35   | -0.48 | -0.48 | 2.81  | -0.63 | -0.63 | 3.73  | -0.75 | -0.75 | 4.44  | -0.77 | -0.78 | 4.65  | -0.64 | -0.64 | 3.78  | -0.90 | -0.90 | 5.37  | -0.91 | -0.91 | 5.45  |
| Pole      | 5         | 0.50   | 7.00   | -0.23 | 0.02  | 1.39  | -0.45 | 0.02  | 2.73  | -0.58 | 0.01  | 3.57  | -0.12 | 0.01  | 0.78  | -0.14 | 0.01  | 0.85  | -0.80 | -0.01 | 4.96  | -0.82 | -0.01 | 5.07  |
| Pole      | 16        | 0.50   | 3.50   | -0.23 | -0.02 | 1.39  | -0.44 | -0.01 | 2.71  | -0.58 | 0.00  | 3.55  | -0.13 | -0.01 | 0.79  | -0.14 | -0.01 | 0.86  | -0.80 | 0.01  | 4.93  | -0.82 | 0.02  | 5.04  |
| Pole      | 19        | 11.25  | 0.50   | 0.02  | -0.22 | 1.37  | 0.02  | -0.43 | 2.62  | 0.01  | -0.56 | 3.42  | 0.06  | -0.34 | 2.10  | 0.01  | -0.13 | 0.78  | -0.02 | -0.76 | 4.70  | -0.02 | -0.78 | 4.81  |
| Pole      | 22        | 14.50  | 3.50   | 0.22  | -0.02 | 1.38  | 0.44  | -0.01 | 2.67  | 0.57  | 0.00  | 3.49  | 0.13  | -0.01 | 0.81  | 0.14  | -0.01 | 0.87  | 0.78  | 0.01  | 4.85  | 0.80  | 0.01  | 4.96  |
| Pole      | 25        | 14.50  | 7.00   | 0.22  | 0.02  | 1.38  | 0.44  | 0.02  | 2.69  | 0.57  | 0.01  | 3.51  | 0.13  | 0.01  | 0.80  | 0.14  | 0.01  | 0.86  | 0.79  | -0.01 | 4.88  | 0.81  | -0.01 | 4.99  |
| Pole      | 28        | 11.25  | 10.00  | 0.02  | 0.22  | 1.36  | 0.02  | 0.42  | 2.58  | 0.01  | 0.55  | 3.36  | 0.06  | 0.35  | 2.13  | 0.01  | 0.13  | 0.79  | -0.02 | 0.75  | 4.62  | -0.02 | 0.76  | 4.72  |
| Pole      | 31        | 7.50   | 10.00  | 0.00  | 0.20  | 1.21  | 0.00  | 0.40  | 2.46  | 0.00  | 0.54  | 3.30  | 0.00  | 0.40  | 2.46  | 0.00  | 0.09  | 0.59  | 0.00  | 0.74  | 4.59  | 0.00  | 0.76  | 4.69  |
| Pole      | 34        | 7.50   | 0.50   | 0.00  | -0.20 | 1.23  | 0.00  | -0.41 | 2.52  | 0.00  | -0.55 | 3.39  | 0.00  | -0.39 | 2.40  | 0.00  | -0.09 | 0.57  | 0.00  | -0.76 | 4.72  | 0.00  | -0.78 | 4.83  |
| Pole      | 37        | 3.75   | 10.00  | -0.02 | 0.22  | 1.36  | -0.02 | 0.42  | 2.58  | -0.01 | 0.55  | 3.36  | -0.06 | 0.35  | 2.15  | -0.01 | 0.13  | 0.80  | 0.02  | 0.75  | 4.61  | 0.02  | 0.77  | 4.72  |
| Pole      | 40        | 3.75   | 0.50   | -0.02 | -0.22 | 1.37  | -0.02 | -0.43 | 2.62  | -0.01 | -0.56 | 3.42  | -0.07 | -0.36 | 2.19  | -0.01 | -0.13 | 0.79  | 0.02  | -0.76 | 4.70  | 0.02  | -0.78 | 4.80  |
| Pole      | 99000035  | 5.63   | 5.38   | -0.01 | 0.00  | 1.77  | -0.01 | -0.01 | 5.00  | 0.00  | -0.02 | 6.53  | 0.00  | 0.00  | -0.01 | 0.00  | 0.00  | 0.04  | 0.01  | -0.05 | 9.82  | 0.02  | -0.05 | 10.09 |
| Pole      | 99000036  | 9.63   | 5.38   | 0.01  | 0.00  | 1.77  | 0.00  | -0.01 | 4.95  | -0.03 | -0.02 | 6.46  | 0.00  | 0.00  | -0.02 | 0.00  | 0.00  | 0.05  | -0.08 | -0.05 | 9.69  | -0.09 | -0.05 | 9.96  |
| Stormbelt | 150010001 | 5.80   | 12.46  |       |       |       |       |       |       |       |       |       | -1.00 | 4.25  | -3.91 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150010022 | 9.17   | -1.97  |       |       |       |       |       |       |       |       |       | 0.96  | -4.26 | -3.91 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150020001 | 9.19   | 12.46  |       |       |       |       |       |       |       |       |       | 0.99  | 4.24  | -3.90 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150020022 | 5.75   | -1.95  |       |       |       |       |       |       |       |       |       | -1.01 | -4.24 | -3.90 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150030001 | 2.55   | -1.95  |       |       |       |       |       |       |       |       |       | 0.00  | -4.54 | -3.95 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150030021 | 2.50   | 12.44  |       |       |       |       |       |       |       |       |       | -0.02 | 4.54  | -3.95 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150040001 | 12.50  | -1.94  |       |       |       |       |       |       |       |       |       | 0.01  | -4.48 | -3.91 |       |       |       |       |       |       |       |       |       |
| Stormbelt | 150040021 | 12.46  | 12.45  |       |       |       |       |       |       |       |       |       | -0.01 | 4.48  | -3.91 |       |       |       |       |       |       |       |       |       |

## Annex J. Reaction forces model 10x20m

### Annex J.1. Point numbers





## Annex J.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

| Att     | Node      | Koor X Koor Y |       | CO1   |       |       | CO2   |       |       | CO3   |       |       | CO4   |       |       | CO5   |       |       | CO6   |       |       | CO6a  |       |       |
|---------|-----------|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |           |               |       | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    | Fx    | Fy    | Fz    |
| 275     | 6         | -8.00         | 7.17  | -1.57 | -0.01 | -1.22 | -2.32 | 0.02  | -1.83 | -2.92 | 0.04  | -2.30 | -4.35 | -0.03 | -3.32 | -2.81 | -0.03 | -2.14 | -3.70 | 0.06  | -2.94 | -3.76 | 0.07  | -2.99 |
| 275     | 17        | -8.00         | 3.83  | -1.56 | 0.01  | -1.21 | -2.24 | -0.02 | -1.77 | -2.81 | -0.04 | -2.22 | -4.44 | 0.02  | -3.38 | -2.91 | 0.03  | -2.21 | -3.56 | -0.06 | -2.83 | -3.61 | -0.07 | -2.88 |
| 275     | 20        | 11.00         | -2.50 | -0.02 | -1.83 | -1.41 | 0.00  | -2.46 | -1.94 | 0.03  | -3.01 | -2.38 | -0.04 | -4.56 | -3.43 | -0.01 | -3.65 | -2.78 | 0.05  | -3.66 | -2.92 | 0.05  | -3.71 | -2.97 |
| 275     | 23        | 18.00         | 3.83  | 1.56  | 0.01  | -1.21 | 2.24  | -0.02 | -1.77 | 2.81  | -0.04 | -2.22 | 4.42  | 0.02  | -3.37 | 2.91  | 0.03  | -2.21 | 3.56  | -0.06 | -2.83 | 3.61  | -0.07 | -2.88 |
| 275     | 26        | 18.00         | 7.17  | 1.57  | -0.01 | -1.22 | 2.32  | 0.02  | -1.82 | 2.92  | 0.04  | -2.30 | 4.33  | -0.03 | -3.31 | 2.82  | -0.03 | -2.15 | 3.70  | 0.06  | -2.94 | 3.76  | 0.06  | -2.99 |
| 275     | 29        | 11.00         | 13.50 | -0.02 | 1.83  | -1.41 | 0.01  | 2.48  | -1.95 | 0.03  | 3.03  | -2.40 | -0.04 | 4.56  | -3.44 | -0.01 | 3.64  | -2.79 | 0.06  | 3.71  | -2.96 | 0.06  | 3.76  | -3.00 |
| 275     | 32        | 7.00          | 13.50 | 0.00  | 1.81  | -1.44 | 0.01  | 2.32  | -1.86 | 0.02  | 2.88  | -2.32 | 0.01  | 3.81  | -2.97 | 0.00  | 4.41  | -3.53 | 0.03  | 3.58  | -2.90 | 0.03  | 3.63  | -2.94 |
| 275     | 35        | 7.00          | -2.50 | 0.00  | -1.80 | -1.43 | 0.01  | -2.30 | -1.85 | 0.02  | -2.86 | -2.30 | 0.01  | -3.79 | -2.94 | 0.00  | -4.40 | -3.51 | 0.03  | -3.52 | -2.86 | 0.03  | -3.58 | -2.90 |
| 275     | 38        | 3.00          | 13.50 | 0.00  | 1.80  | -1.43 | -0.01 | 2.26  | -1.82 | -0.02 | 2.82  | -2.27 | -0.01 | 3.86  | -3.00 | 0.00  | 4.47  | -3.57 | -0.03 | 3.49  | -2.83 | -0.03 | 3.54  | -2.87 |
| 275     | 41        | 3.00          | -2.50 | 0.00  | -1.79 | -1.43 | -0.01 | -2.25 | -1.81 | -0.02 | -2.79 | -2.25 | -0.01 | -3.84 | -2.97 | 0.00  | -4.46 | -3.55 | -0.03 | -3.45 | -2.80 | -0.03 | -3.50 | -2.84 |
| 275     | 44        | -1.00         | 13.50 | 0.02  | 1.83  | -1.41 | -0.01 | 2.48  | -1.96 | -0.03 | 3.03  | -2.40 | 0.04  | 4.59  | -3.47 | 0.01  | 3.64  | -2.79 | -0.05 | 3.71  | -2.96 | -0.06 | 3.77  | -3.01 |
| 275     | 47        | -1.00         | -2.50 | 0.02  | -1.83 | -1.41 | 0.00  | -2.46 | -1.94 | -0.03 | -3.01 | -2.38 | 0.04  | -4.61 | -3.47 | 0.01  | -3.65 | -2.78 | -0.05 | -3.67 | -2.93 | -0.05 | -3.72 | -2.97 |
| 600     | 7         | -5.52         | -1.43 | -0.43 | -1.59 | -1.64 | -0.55 | -2.01 | -2.07 | -0.64 | -2.35 | -2.42 | -0.82 | -2.97 | -3.06 | -0.62 | -2.25 | -2.31 | -0.75 | -2.73 | -2.81 | -0.75 | -2.76 | -2.84 |
| 600     | 8         | -6.93         | -0.02 | -1.39 | -0.38 | -1.44 | -1.71 | -0.47 | -1.76 | -2.05 | -0.56 | -2.11 | -3.03 | -0.83 | -3.11 | -2.35 | -0.64 | -2.42 | -2.43 | -0.67 | -2.50 | -2.46 | -0.68 | -2.53 |
| 600     | 9         | 16.93         | -0.02 | 1.39  | -0.38 | -1.44 | 1.71  | -0.47 | -1.76 | 2.05  | -0.56 | -2.11 | 3.03  | -0.83 | -3.12 | 2.35  | -0.64 | -2.42 | 2.43  | -0.67 | -2.50 | 2.46  | -0.68 | -2.53 |
| 600     | 10        | 15.52         | -1.43 | 0.43  | -1.59 | -1.64 | 0.55  | -2.01 | -2.07 | 0.64  | -2.35 | -2.42 | 0.81  | -2.96 | -3.04 | 0.62  | -2.25 | -2.32 | 0.75  | -2.73 | -2.81 | 0.75  | -2.76 | -2.84 |
| 600     | 11        | 15.52         | 12.43 | 0.43  | 1.59  | -1.64 | 0.55  | 2.02  | -2.08 | 0.64  | 2.36  | -2.43 | 0.82  | 2.97  | -3.05 | 0.62  | 2.26  | -2.32 | 0.75  | 2.75  | -2.82 | 0.76  | 2.77  | -2.85 |
| 600     | 12        | 16.93         | 11.02 | 1.39  | 0.38  | -1.44 | 1.71  | 0.47  | -1.76 | 2.05  | 0.56  | -2.11 | 3.03  | 0.83  | -3.12 | 2.35  | 0.64  | -2.42 | 2.43  | 0.67  | -2.51 | 2.46  | 0.68  | -2.54 |
| 600     | 13        | -6.93         | 11.02 | -1.39 | 0.38  | -1.44 | -1.71 | 0.47  | -1.77 | -2.05 | 0.56  | -2.11 | -3.01 | 0.83  | -3.09 | -2.35 | 0.64  | -2.42 | -2.44 | 0.67  | -2.51 | -2.47 | 0.68  | -2.54 |
| 600     | 14        | -5.52         | 12.43 | -0.43 | 1.59  | -1.64 | -0.55 | 2.01  | -2.07 | -0.64 | 2.35  | -2.42 | -0.82 | 2.98  | -3.07 | -0.62 | 2.25  | -2.32 | -0.75 | 2.74  | -2.82 | -0.76 | 2.77  | -2.85 |
| 48160   | 1         | 14.65         | 0.85  | 0.46  | -0.45 | 2.67  | 0.61  | -0.60 | 3.58  | 0.73  | -0.73 | 4.33  | 0.70  | -0.70 | 4.22  | 0.60  | -0.60 | 3.54  | 0.89  | -0.88 | 5.26  | 0.90  | -0.89 | 5.33  |
| 48160   | 2         | 14.65         | 10.15 | 0.46  | 0.45  | 2.66  | 0.61  | 0.60  | 3.57  | 0.73  | 0.72  | 4.32  | 0.70  | 0.70  | 4.21  | 0.59  | 0.60  | 3.54  | 0.89  | 0.88  | 5.26  | 0.90  | 0.89  | 5.33  |
| 48160   | 3         | -4.65         | 10.15 | -0.46 | 0.45  | 2.66  | -0.61 | 0.60  | 3.57  | -0.73 | 0.72  | 4.32  | -0.70 | 0.70  | 4.20  | -0.59 | 0.60  | 3.54  | -0.89 | 0.87  | 5.25  | -0.90 | 0.89  | 5.33  |
| 48160   | 4         | -4.65         | 0.85  | -0.46 | -0.45 | 2.67  | -0.61 | -0.60 | 3.58  | -0.74 | -0.73 | 4.33  | -0.70 | -0.70 | 4.22  | -0.60 | -0.60 | 3.54  | -0.89 | -0.88 | 5.26  | -0.90 | -0.89 | 5.33  |
| 48160   | 5         | -4.50         | 7.17  | -0.22 | 0.01  | 1.35  | -0.42 | 0.00  | 2.60  | -0.56 | -0.01 | 3.42  | -0.06 | 0.00  | 0.40  | -0.13 | 0.01  | 0.80  | -0.76 | -0.03 | 4.71  | -0.78 | -0.03 | 4.81  |
| 48160   | 16        | -4.50         | 3.83  | -0.22 | -0.01 | 1.35  | -0.42 | 0.00  | 2.57  | -0.55 | 0.01  | 3.39  | -0.06 | 0.00  | 0.41  | -0.13 | -0.01 | 0.81  | -0.76 | 0.04  | 4.67  | -0.77 | 0.04  | 4.78  |
| 48160   | 19        | 11.00         | 1.00  | 0.02  | -0.22 | 1.38  | 0.01  | -0.44 | 2.68  | -0.01 | -0.57 | 3.53  | -0.03 | -0.24 | 1.49  | 0.00  | -0.12 | 0.75  | -0.04 | -0.78 | 4.83  | -0.04 | -0.80 | 4.93  |
| 48160   | 22        | 14.50         | 3.83  | 0.22  | -0.01 | 1.35  | 0.42  | 0.00  | 2.58  | 0.55  | 0.01  | 3.39  | 0.06  | -0.01 | 0.42  | 0.13  | -0.01 | 0.81  | 0.76  | 0.04  | 4.67  | 0.77  | 0.04  | 4.78  |
| 48160   | 25        | 14.50         | 7.17  | 0.22  | 0.01  | 1.35  | 0.42  | 0.00  | 2.59  | 0.56  | -0.01 | 3.41  | 0.06  | 0.00  | 0.41  | 0.13  | 0.01  | 0.80  | 0.76  | -0.04 | 4.70  | 0.78  | -0.04 | 4.81  |
| 48160   | 28        | 11.00         | 10.00 | 0.02  | 0.22  | 1.37  | 0.01  | 0.43  | 2.62  | -0.01 | 0.56  | 3.45  | -0.03 | 0.24  | 1.47  | 0.00  | 0.12  | 0.76  | -0.04 | 0.76  | 4.72  | -0.04 | 0.78  | 4.83  |
| 48160   | 31        | 7.00          | 10.00 | 0.00  | 0.19  | 1.19  | 0.00  | 0.38  | 2.35  | -0.01 | 0.52  | 3.19  | -0.05 | 0.34  | 2.06  | 0.00  | 0.10  | 0.65  | -0.02 | 0.72  | 4.45  | -0.03 | 0.74  | 4.56  |
| 48160   | 34        | 7.00          | 1.00  | 0.00  | -0.20 | 1.21  | 0.00  | -0.39 | 2.41  | -0.01 | -0.53 | 3.28  | -0.04 | -0.32 | 1.98  | 0.00  | -0.10 | 0.63  | -0.02 | -0.74 | 4.58  | -0.03 | -0.76 | 4.69  |
| 48160   | 37        | 3.00          | 10.00 | 0.00  | 0.19  | 1.19  | 0.00  | 0.38  | 2.33  | 0.01  | 0.52  | 3.17  | 0.04  | 0.36  | 2.21  | 0.00  | 0.10  | 0.66  | 0.02  | 0.72  | 4.43  | 0.03  | 0.73  | 4.54  |
| 48160   | 40        | 3.00          | 1.00  | 0.00  | -0.20 | 1.21  | 0.00  | -0.39 | 2.39  | 0.01  | -0.53 | 3.25  | 0.04  | -0.34 | 2.09  | 0.00  | -0.10 | 0.64  | 0.02  | -0.74 | 4.56  | 0.03  | -0.76 | 4.67  |
| 48160   | 43        | -1.00         | 10.00 | -0.02 | 0.22  | 1.37  | -0.01 | 0.43  | 2.63  | 0.01  | 0.56  | 3.46  | 0.02  | 0.19  | 1.18  | 0.00  | 0.12  | 0.77  | 0.04  | 0.76  | 4.73  | 0.04  | 0.78  | 4.84  |
| 48160   | 46        | -1.00         | 1.00  | -0.02 | -0.23 | 1.38  | -0.01 | -0.44 | 2.69  | 0.01  | -0.57 | 3.54  | 0.02  | -0.21 | 1.31  | 0.00  | -0.12 | 0.75  | 0.04  | -0.78 | 4.84  | 0.04  | -0.80 | 4.95  |
| 48160   | 99000041  | -0.12         | 5.63  | -0.01 | 0.00  | 1.88  | 0.06  | -0.01 | 5.19  | 0.15  | -0.02 | 6.78  | 0.00  | 0.00  | -0.02 | 0.00  | 0.00  | 0.07  | 0.32  | -0.05 | 10.10 | 0.34  | -0.05 | 10.38 |
| 48160   | 99000042  | 5.13          | 5.63  | 0.00  | 0.00  | 1.58  | -0.01 | -0.01 | 4.61  | -0.01 | -0.02 | 6.09  | 0.00  | 0.00  | -0.01 | 0.00  | 0.00  | 0.05  | -0.03 | -0.05 | 9.26  | -0.03 | -0.06 | 9.52  |
| 48160   | 99000043  | 10.13         | 5.63  | 0.01  | 0.00  | 1.86  | -0.05 | -0.01 | 5.11  | -0.14 | -0.02 | 6.67  | 0.00  | 0.00  | 0.01  | 0.00  | 0.00  | 0.06  | -0.30 | -0.05 | 9.93  | -0.32 | -0.05 | 10.20 |
| 5900001 | 150010001 | 5.01          | -1.32 |       |       |       |       |       |       |       |       |       | 1.52  | -4.17 | -3.82 |       |       |       |       |       |       |       |       |       |
| 5900001 | 150010021 | -0.16         | 12.22 |       |       |       |       |       |       |       |       |       | -1.51 | 4.10  | -3.90 |       |       |       |       |       |       |       |       |       |
| 5900002 | 150020001 | -0.40         | -1.24 |       |       |       |       |       |       |       |       |       | -1.59 | -4.10 | -3.94 |       |       |       |       |       |       |       |       |       |
| 5900002 | 150020022 | 5.05          | 12.30 |       |       |       |       |       |       |       |       |       | 1.61  | 4.18  | -3.85 |       |       |       |       |       |       |       |       |       |
| 5900003 | 150030001 | 4.95          | 12.30 |       |       |       |       |       |       |       |       |       | -1.68 | 4.17  | -3.87 |       |       |       |       |       |       |       |       |       |
| 5900003 | 150030022 | 10.71         | -1.28 |       |       |       |       |       |       |       |       |       | 1.67  | -4.08 | -3.96 |       |       |       |       |       |       |       |       |       |
| 5900004 | 150040001 | 4.93          | -1.29 |       |       |       |       |       |       |       |       |       | -1.66 | -4.14 | -3.85 |       |       |       |       |       |       |       |       |       |
| 5900004 | 150040022 | 10.67         | 12.27 |       |       |       |       |       |       |       |       |       | 1.67  | 4.05  | -3.94 |       |       |       |       |       |       |       |       |       |